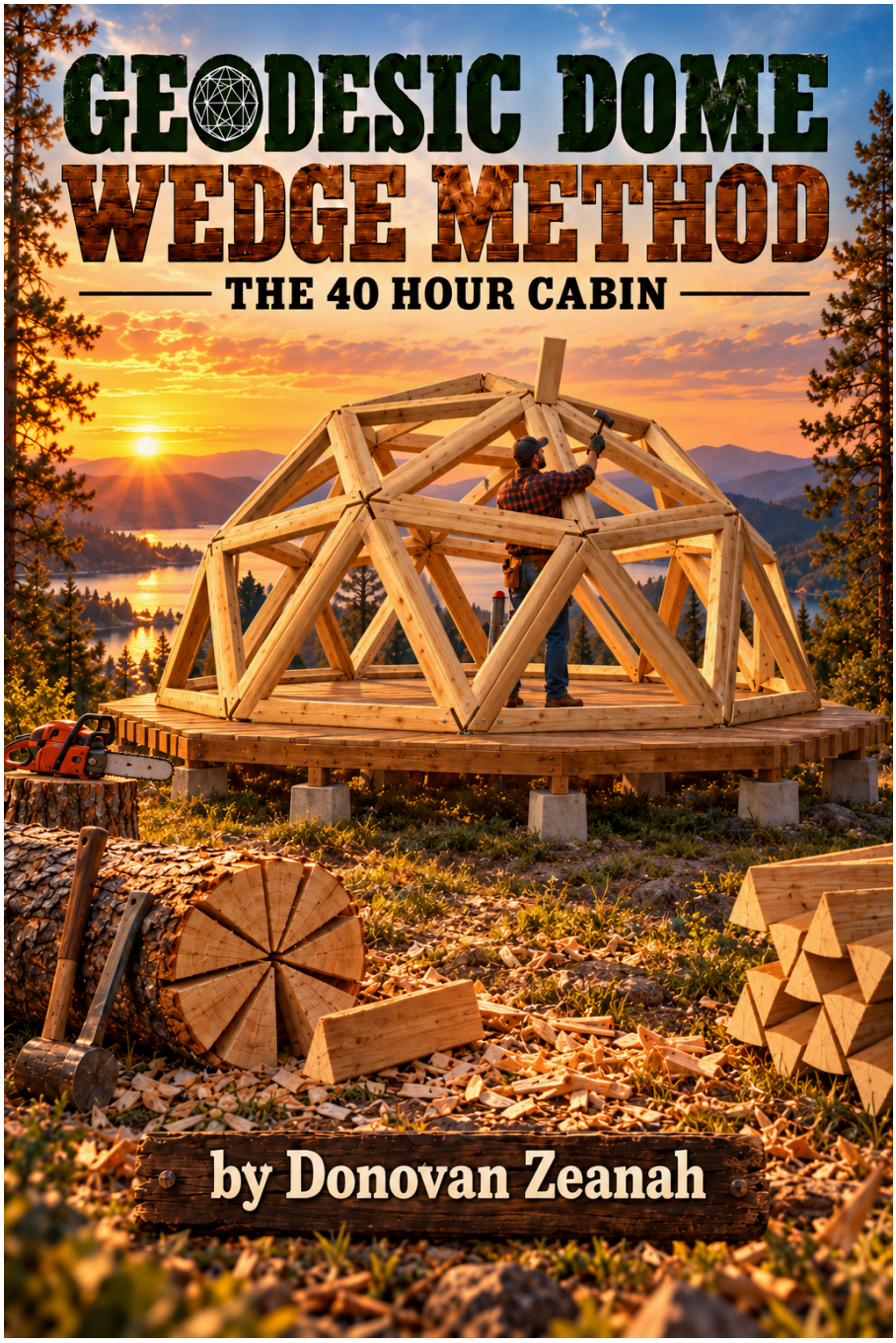
GEODESIC DOME
WEDGE METHOD
THE 40 HOUR CABIN
by Donovan Zeanah

The 40 Hour Cabin: Complete Edition

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Every figure in this book was computed from the geometry in the DomeSim project at the time this edition was built (October 2026). The software is open at github.com/dkzeanah/DomeSimulator.

Prices are United States retail as quoted in October 2026. They are there to show proportion -- which parts cost money and which do not -- and should be checked locally before anything is bought.

The illustrations are rendered by the project's own software.

This book describes methods, not rated structures. Read *The Engineer's Note* before building anything from it. The author and publisher accept no liability for injury, loss or damage arising from its use.

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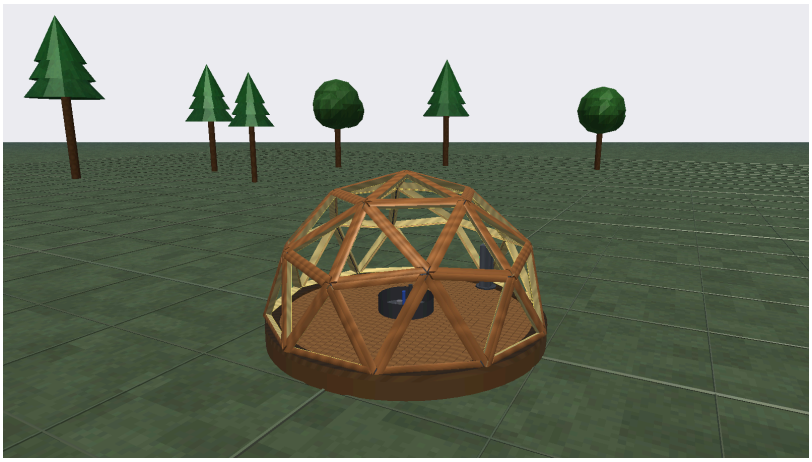


The frame, on its own land.

A dwelling of 120 split-log members, 40 panels and 277 square feet of floor — cut from standing timber with a chainsaw, with a frame that stands up in a working week.

The Engineer’s Note

Read this before anything else in the book, and before you pick up a saw.

This book is not engineering. Nothing in it is a structural rating. Snow load, wind load, the ground under the pad and the way a frame is fastened are specific to your site, and they are a licensed engineer’s numbers, not this book’s. Where a chapter reaches one of those questions it says so and stops. Before anybody sleeps in a building made this way, have an engineer look at the drawings and at the frame.

This book is not your building code. Permits, setbacks, fire, egress and occupancy rules are local. Ask your building department what they need before you fell the first tree, not after you raise the last panel.

Electricity and water are trades. The utility column carries a full-size electrical feed and real plumbing. Have a licensed electrician connect and inspect the electrical work and a licensed plumber the water and drain, and follow what your inspector asks for.

Felling is the most dangerous hour in this book. Chapter 22 lists it as the first process and, on purpose, does not teach it. Learn it from a qualified instructor, in person. Wear a helmet with a face screen, hearing protection, cut-resistant chaps, gloves and boots; never fell alone; plan your escape route before the first cut.

The rest of the work is heavy. Ripping logs throws chips and sawdust, so wear eye protection as well as the rest of your saw gear. Wedges and panels are lifted above your head, so work in pairs from a stable platform. The shell comes apart in pieces that are a job for several people or a crane, and the book gives their weights so nobody has to guess.

Some of this book is a design possibility, not a tested product. The book keeps three kinds of claim apart – known geometry, tested construction, and design possibility – and says which is which. The floating dome in Chapter 72 is the clearest example: it is priced, not rated.

You build at your own risk. The author and publisher accept no liability for injury, loss or damage arising from the use of this book.

Forty Hours

The goal is a house built so efficiently that one person can go from standing timber to a standing shell inside a single working week.

Not a kit. Not a shell delivered on a lorry. A tree, a chainsaw, and 40 hours.

That is not an aspiration bolted onto the method afterwards. It is where the number came from, and here is the arithmetic.

The clock

Splitting. One session at the log — 4 hours with a meagre chainsaw — produces 30 wedges. The frame needs 120, so every member in the building is **16 hours** at the log.

Everything else. Taken end to end, one member costs about 15 minutes from tree to standing: split, crosscut, drilled, joined, and up in the shell. That is an average over the fast work and the slow, not a stopwatch on one operation.

```
120 members x 15 minutes  =  30 hours
```

Splitting is 53 per cent of that, which is why Chapter 22 says the rip is the process worth improving first.

Getting it wrong. Add 25 per cent for the stick that splits off the radius, the panel laid down in the wrong order, and the hour spent looking for the drill.

```
30 hours + 25%  =  37.5 hours
```

Which is a 40 hour week, with a little in hand.

The honest part

You will not do it in five days.

This is hard physical work. Splitting a log is not typing; you get perhaps four good hours at it before the quality of your swing goes, and the quality of your swing is what keeps a split on the radius. Nobody does forty hours of this linearly, and anybody who tells you they did was counting the hours and not the days.

It comes in **spurts**. A weekend at the log. An evening on the bench. A day when three people turn up and the shell goes from nothing to closed.

That does not weaken the number — it is the reason the number is worth having. Forty hours of work is a thing you can *schedule* against a life: five Saturdays, or two weeks of evenings, or one very determined week off. A building measured in months is a building you never start, because months cannot be fitted into a year that already has things in it.

So: the forty hours is a **budget of effort**, not a calendar. The point of counting it is that it is small enough to count.

And the rest of the book

The shell is a week. The skin and the floor — Chapters 2 and 38, the two parts of this building that decide whether it is still standing in thirty years — are not, and this book spends more pages on them than on the frame for exactly that reason.

The frame is the part you can do in a week. It is also the part you are allowed to get wrong.

The whole building, counted the same way

So that nobody has to find it on the last page: the forty hours is the frame and its shell – 37.5 hours with error allowed for. The whole standard building, counted the same way – frame, panels, shell, utility column and the rest – is **67 hours** of labour, and Chapter 63 shows every one of them. The title is the part one person can do in a week. The rest is the part worth doing slowly.

PART 1

What Actually Has to Be Right

“A dome is a frame, a skin and a floor -- and the frame is the part you are allowed to get wrong.”

Three chapters that invert the usual advice, and one practical consequence: build the frame loose and the two layers that matter stay changeable for the life of the building.

CHAPTER 1

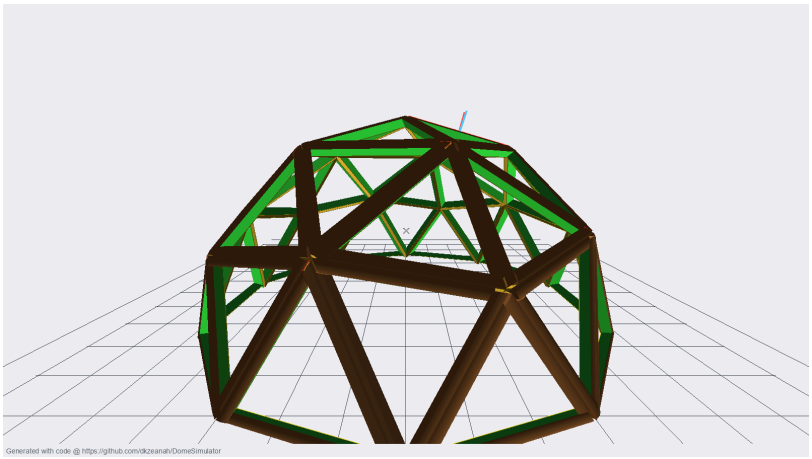
Your Skull Is Not Symmetrical Either

The frame is the part you are allowed to get wrong, and almost every dome book has this backwards

Every dome book I have read opens by telling you to be precise.

Measure twice. Hold your angles. A geodesic dome is a mathematical object and you are about to approximate one out of wood, so the closer you get to the maths the better your building will be.

I believed that for about a year, and it cost me most of a year.



The reference build. Nothing about it is as accurate as it looks.

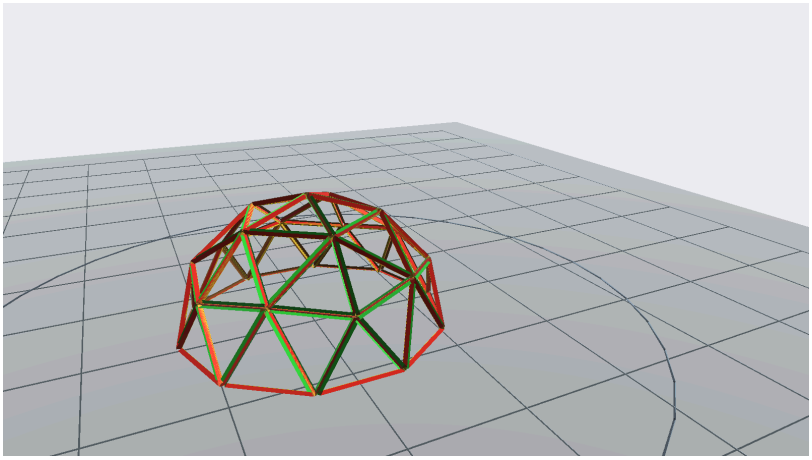
The afternoon I stopped believing it

I had a frame up. Not this one — an earlier one, made of whatever the chainsaw had produced that week, with no two sticks alike. I had built it to prove to myself that the method could not work, so that I could stop thinking about it and go and buy some lumber like a normal person.

It stood. It stood with a stick in it that was out by most of an inch, and a vertex where three members met that I could put my whole thumb into.

I pushed it. I hung off it. I put a ladder against it and climbed on it, which I do not recommend and did anyway.

It did not care.



How much shelter an hour buys, which is the question this book is an answer to.

Why it did not care

A dome carries its loads *axially*. The members are in compression along their own length — pushing end to end, the way a stack of bricks pushes — and a triangulated shell passes a load from one triangle to the next all the way round, so no single member is ever being asked to do very much on its own.

Put a stick in slightly out of position and two things happen. The triangle it is in is very slightly a different triangle. And the load path shifts by that same very slight amount.

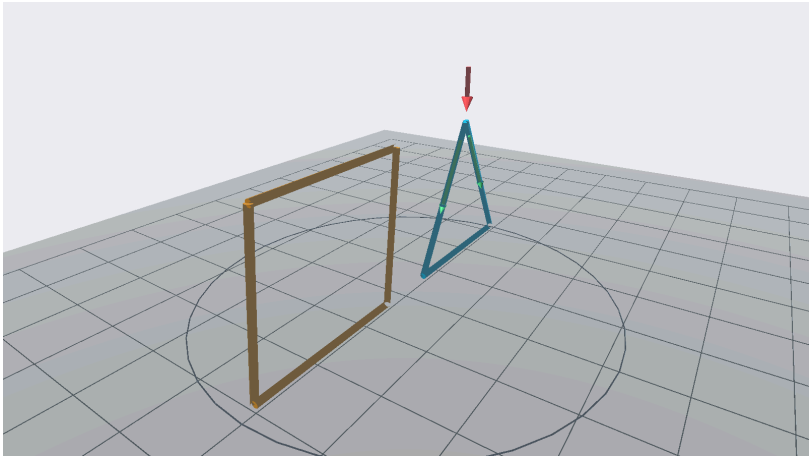
Neither of those is an error that grows. A frame is not a tower of tolerances stacking up toward a failure; it is a shell that absorbs the difference and carries on. The error goes into the seam — and there is already a key in the seam, because there was always going to be one.

Why a triangle, and not a square

Push the top corner of a square frame sideways and it leans. Nothing breaks: the four sides keep their lengths and the corners simply turn. A square is a mechanism, and it needs stiff corners – nailed, braced, glued – to stop being one.

Push the corner of a triangle and nothing moves unless one of its sides gets longer or shorter. Timber is very unwilling to do either along its length. So a triangle needs nothing from its corners at all.

A geodesic shell repeats that triangle across a curved surface, 40 times on this dome. The strength lives in the **network**, not in any one member. No single stick is asked to solve the whole problem; each one is asked to stay its own length, and that is all.



Where the strength actually lives.

That is what lets an odd-shaped member do the job. The structure cares about the member's section, its grain, its unsupported length and the path the load takes through it. It does not care whether the corner of an imaginary rectangle exists – which is the whole reason a split log, with no rectangle anywhere in it, can be a stick in a building.

The lumpy skull

You have an asymmetric skull.

Genuinely: measure it. One side of your head is further from the midline than the other, one eye socket is fractionally lower, your jaw is not centred. This is true of essentially everybody and it is why nobody's face looks right when you mirror it.

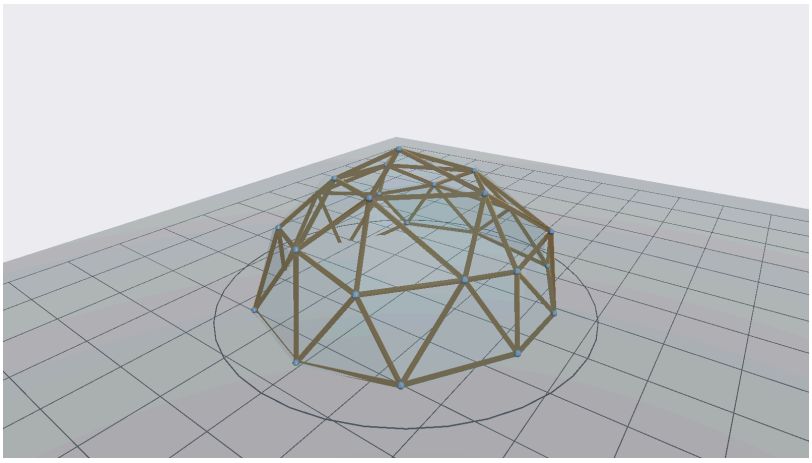
Nobody likes you any less for it, and your skull opposes a baseball bat exactly as well as a perfectly symmetrical one would — because the thing that stops the bat is the *shape*, a closed curved shell in compression, and not the precision.

That is a dome. The geometry does the work. The accuracy is decoration.

There are two reasons the lumpy one stays up, and both are worth knowing before you trust it.

The errors have nowhere to accumulate. A closed, triangulated shell is enormously redundant. Every triangle is rigid on its own, so a joint that is out is absorbed by the triangles around it instead of walking round a ring and adding up the way an error does in a precise, hubbed dome, where every strut has to land on the same point as five others. What you get is a frame that is near the sphere without being on it – a lumpy brain, not a ball.

And the skin covers the rest. One continuous layer bonded over the outside spans the slack between members and gives back the shell action the sloppiness cost. Which is the first hint of the argument the next chapter makes: the frame can be forgiving because the skin is not allowed to be.



The errors have nowhere to accumulate, and the skin covers the rest.

What this book does with that

It spends it.

If the frame does not need precision, then the frame does not need a mill. It does not need a planer, or a moisture-controlled shed, or lumber bought to a tolerance somebody else held for you. It can be made of logs split with a wedge, on the ground, in a field, by one person who is not very good at it yet.

That is the whole method. Everything else in this book — the 45.0-degree sector, the pinwheel joint, the one flat board that makes 40 identical panels — is a consequence of deciding that the frame is the part you are allowed to get wrong.

And what it does not spend

Here is the part that took me another year to learn, and it is the reason this book exists rather than being a pamphlet about splitting logs.

The permission is not general. It applies to the frame and it applies to nothing else.

The skin has to be right, because water does not negotiate and does not announce itself. The floor has to be right, because everything you own sits on it and because the ground sends its water up.

So the advice most dome books give is not merely unhelpful. It is aimed at the wrong layer. It asks for care where care is cheap to skip, and it treats as finishing work the two layers that decide whether the building is still there in thirty years.

The next two chapters are about those two layers. Then we can go and cut something.

CHAPTER 2

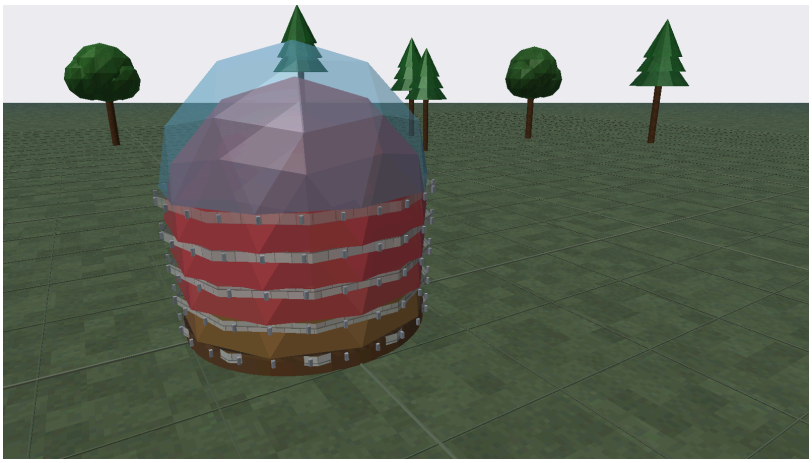
Water Is the Silent Killer

The skin is the part that has to be right, and it is the part every dome book treats as a finish

A frame that is out by an inch stands for fifty years.

A skin that leaks for one winter takes the frame with it.

That asymmetry is the most useful thing in this book, and almost every dome book has it the other way round.



One watertight layer over everything else.

Why water gets to decide

Structural failure is loud. Something cracks, something sags, something you can see from across the room. You get warning, and the warning arrives in a form that makes you do something about it.

Rot is not loud. Rot is a slow chemical argument between a fungus and your building, conducted at about twenty per cent moisture content, entirely out of sight, and it is usually two or three years old by the time anything shows on a surface you look at.

By then you are not repairing a leak. You are replacing members, and the members are the thing this whole method spent its cleverness on making cheap — so you are not even losing much money. You are losing the fortnight.

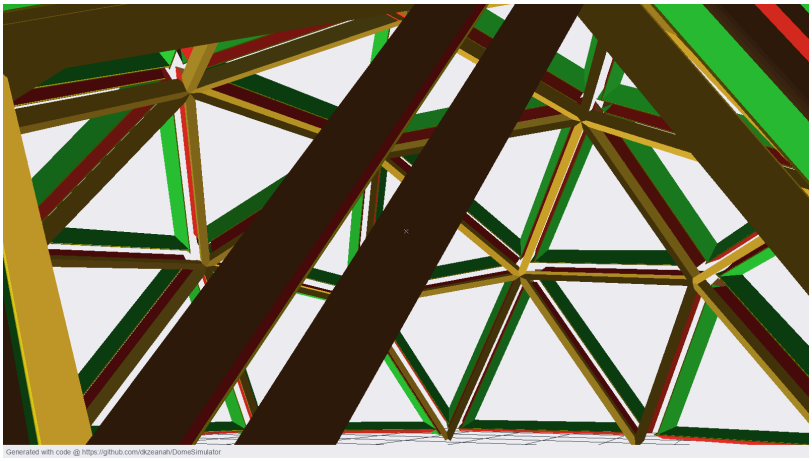
That is the trade nobody states plainly: the frame is cheap to build and expensive to replace, because replacing it means taking the building apart.

Where a dome actually leaks

Not through the panels. A panel is a large flat sheet with no holes in it and it is the easiest part of any building to keep water out of.

Domes leak at seams, at penetrations, and at the apex, in that order — and a 2V dome has a great many seams.

This one has 55 interior seams, which is 309 feet of joint. Every foot of that is a place where two panels meet and where somebody has to have done something deliberate.



Every seam in the dome, and the key in each one.

Three hundred feet of joint is not a detail. It is the single largest risk surface the building has, and it is spread out so evenly that no one part of it looks important enough to worry about.

The move that removes the category

You do not solve three hundred feet of joint by making each joint better.

You solve it by putting one continuous layer over all of it, so that the joints stop being the outside of the building.

That is what the shower cap is. One watertight membrane over the whole shell, strapped down, with nothing structural fastened through it and no seam in it that is not a welded one. The panels underneath do not have to be watertight. The seams underneath do not have to be watertight. They have to be dry, and they are dry because something above them is doing that job and only that job.

The chapter on the cap (28) has the physics and the arithmetic. What matters here is the ordering: **one layer whose only job is water, and it goes on last and comes off first.**

And the floor

The floor is the second thing, and for a different reason.

It is the surface you are in contact with all day. It carries everything you own. It is the only part of the building that is in permanent contact with the ground, which means it is where ground moisture arrives, and unlike a wall it cannot dry to the outside because there is no outside — there is dirt.

A floor that is cold, or damp, or that flexes, is a floor you notice every hour you are awake. A frame that is out of true is a frame you never think about again after the first week.

Get the floor right. Chapter 38 says what right means, and what the five ways of doing it cost on this footprint.

The hierarchy, stated once

1. **The skin.** Water. One continuous layer, no compromises, no penetrations you did not design.
2. **The floor.** Comfort, dryness, and everything you own.

3. **The frame.** Standing up, which it will do whether or not you were careful.

If you read nothing else in this book, invert your instincts in that order. It is worth more than every angle in the appendix.

The catch, and the next chapter

There is an obvious objection, and it is a good one.

The skin and the floor are the expensive parts. They are the parts you cannot cut out of a tree with a chainsaw. Telling somebody that the two layers that matter are also the two layers that cost money is not obviously helpful.

The answer is that they are also the two layers you can *defer* — and the next chapter is about the four things you do while building the frame that keep that option open for the life of the building.

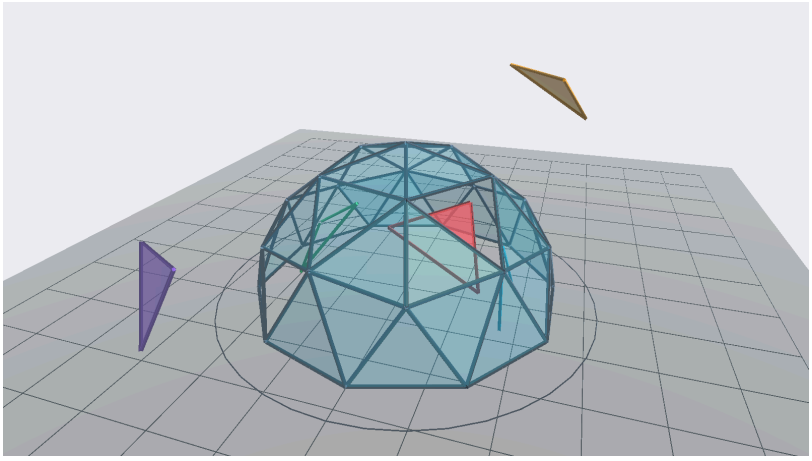
CHAPTER 3

Build It So You Can Change Your Mind

The skin and the floor are the two things that matter and the two things you can defer -- if the frame is built for it

The two layers that matter most are the two layers you can put off.

That sounds like a consolation and it is actually the design. A frame is a commitment: 120 members, cut to two lengths, and once the first panel is closed the shape of the building is decided. A skin is not. A floor is not. Those are things the building *wears*, and a building can change what it wears.



New insulation? Swap it. Solar panels? Add them.

The condition

It only works if you build for it, and the decisions are all made in the first week, before there is a building to look at.

None of them costs anything meaningful at the time. All of them are effectively impossible to add later.

The four things

Threaded inserts in the wedge faces, not screws into end grain.

A panel that will be taken off and put back needs somewhere to bolt to, and end grain is not it — the second time you drive a screw into the same hole it is holding about half of what it held the first time, and the fourth time it is holding nothing. An insert set into the flat sawn face of a wedge is a machine thread in a piece of metal. It does not care how many times you use it.

This is the single highest-leverage decision in the build, and it happens while the panel is still flat on the jig.

A cap sized a layer larger than it needs.

The outer skin goes over whatever is under it, so its size depends on how much is under it. A cap cut to fit a bare shell will not go over a shell wearing two quilted layers, and you find that out on the day you want to add the second layer, in November.

Chapter 30 has the arithmetic — each hat is a size up, and the sequence is not one number. Buy one size up from what you need on day one.

A floor that clamps rather than is built in.

A floor framed into the base ring is part of the structure. A floor that is a hub, radial spokes and decking, clamped to something in the middle, is a component — it can come out, go back, be replaced with a better one, or be left behind.

It also turns out to be the thing that makes a floating dome possible at all, which was not the reason for doing it.

Services in the seam channel, not in the wall.

The seam leaves a gap that has to be filled with something. Fill it with a hose and the building's plumbing is in its structure, reachable from outside by taking off one panel, and no pipe has ever been buried in anything.

An A seam's channel takes a line up to 2.878 inches across. That is not a tight fit for the things a small building actually needs.

What deferring costs

Honestly: some money, later, in exchange for less money now.

The standard article in this book's cost model lists at $12,036, built to $10,030. The shell inside that is $3,265. The bare version of the same cap — no quilted layers at all — is $3,265, and each layer you add afterwards is a few tens of dollars of material and an evening.

So the deferral is real and it is cheap. What you are buying with the four decisions above is the right to spend that money in the order your life actually allows, rather than in the order a building would prefer.

The failure this prevents

I have watched somebody build a beautiful frame, run out of money, put a tarpaulin over it for a winter, and lose the frame.

Not because the tarpaulin was a bad idea. A tarpaulin is a perfectly good temporary skin. They lost it because there was nowhere for a real skin to *attach* when the money came back — no inserts, no strap points, nothing but a frame that had been designed on the assumption that the skin would be figured out later by somebody with more time.

Later is not a plan. Later is a thing you design for.

Every answer is yes

Built that way, the questions people ask about a dome all get the same answer. New insulation? Swap it. Solar? Add it to a panel. A greenhouse wall? Sure. A workshop extension? Bolt it to the rim. Some technology nobody has invented yet? There should be a panel for that – and if the frame is a set of sockets (Chapter 21), there will be.

Where this leaves us

Three chapters in, the argument of this book is complete:

- the frame is allowed to be messy, which is what lets a split log be a structural member;
- the skin and the floor are not, because water and comfort do not forgive;
- and the two layers that are not allowed to be messy are the two you can postpone, upgrade and swap — if you spend four decisions on it in week one.

The rest of the book is how. It starts where the material does, with a tree.

CHAPTER 4

A House With a Parts List

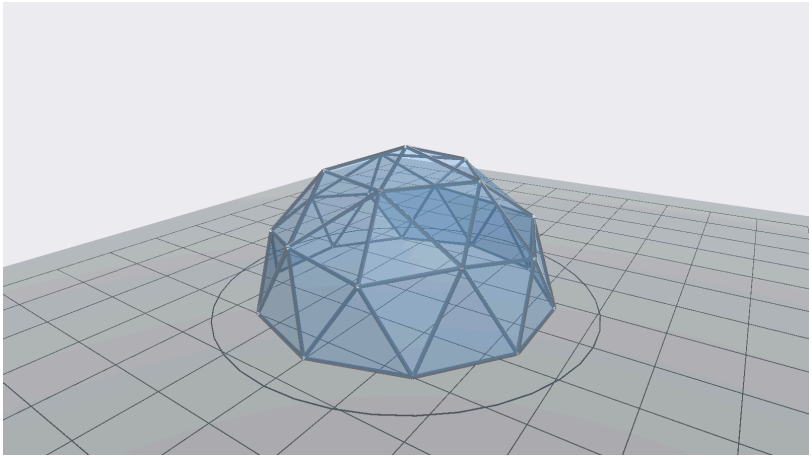
Count the model, and a building stops being a price opinion: the cut list makes itself, and the labour is a flat rate

The whole argument, start to finish

Here is everything this book is going to argue, said once, so you can watch it being proved.

A dome made this way is a shape that spends less on skin and gives that back every winter. It is made of two lengths of wood, cut on one flat board. It is built by a method that forgave a chainsaw, so it will forgive you. It has a price list with a receipt behind every line and an energy bill you can work out for yourself. And none of it is an opinion, because every number in it is counted off the same model of the building that draws its pictures.

That last part is the one this chapter is about.



Everything this project knows, in one film.

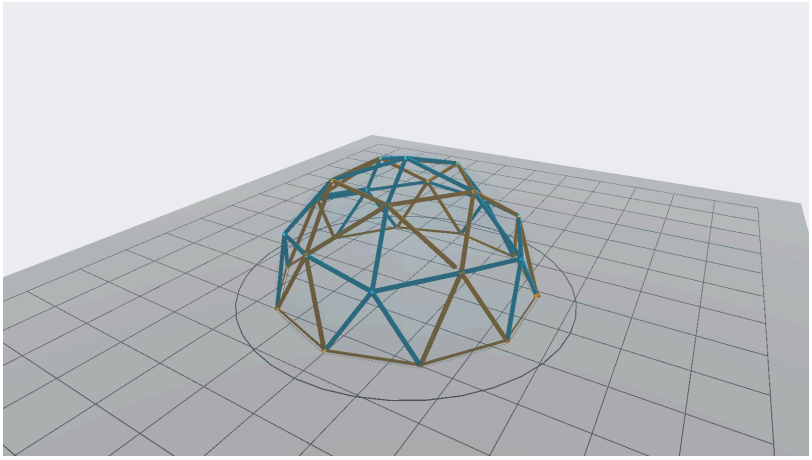
Nothing here is typed in

Ask a builder what a house will cost and you get an opinion, informed by experience and padded for the things that go wrong. Ask this dome, and it counts itself:

triangular panels	40
edges of the short length	30
edges of the long length	35
edges in all	65
corners where edges meet	26

Nobody wrote those numbers into this paragraph. The software that solves the dome and draws its pictures is the software that counts it, and the book fills those numbers in at the moment it is printed. Every figure in the chapters that follow – every angle, every dollar, every gallon – reaches the page the same way. Where something cannot be computed, a price or a bead size or how much oil rough wood drinks, the book says where it came from and marks it as an assumption.

That is not pedantry. It is the difference between a parts list and a price opinion. A parts list can be checked, and when one input changes, every figure that depends on it changes with it.



The panel on the right is counting the dome behind it.

The cut list makes itself

Here is the chain, as arithmetic.

Pick the size: one number, the radius of the sphere the dome is a slice of. This book's reference build has a radius of 116.4984 inches.

Two fixed factors turn that radius into the two chord lengths every geodesic reference publishes: 72.000 inches and 63.670. Chapter 16 explains why you do not cut either of them: the pinwheel joint bites 1.818 inches off a long member and 0.990 off a short one, and that bite is measured, never guessed. So the saw is set to 70.182 and 62.681.

Then the order falls out:

long members	60, each 70.182 in
short members	60, each 62.681 in
members in all	120
timber in the frame	664 ft
log sections	15, split 8 ways each

Change the radius and every line of that table recomputes. That is what it means for a house to have a parts list: the size is a decision, and everything after it is consequence.

Pick the radius; the cut list and the order fall out.

The flat rate, proved

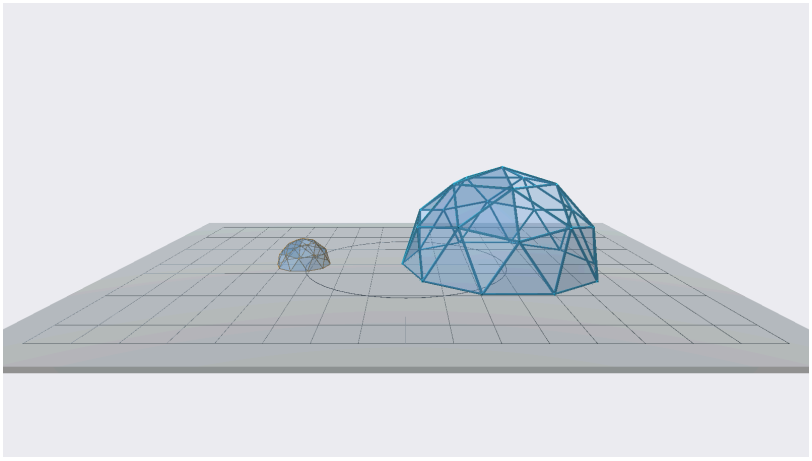
This is the number that turns a weekend project into a product, so prove it rather than repeat it.

Scale the dome from 10 feet across to 30 and count again:

```
10 ft dome      79 sq ft of floor
30 ft dome      707 sq ft of floor
members, either way            120
panels, either way             40
joints, either way             65
```

The floor grows 9 times over, because area grows with the square of the size. The parts list does not grow at all. The sticks get longer; the list of things you do to them stays exactly the same length – the same splits, the same two saw settings, the same panels on the same board, the same raising.

Labour is priced by operations, not by square feet. That is why the labour in a dome is close to a flat rate, and why the bigger house is the better deal. The honest limit is the stick: a member is a split log, and a log only comes so long, which is why Chapter 11 picks the size from the stick rather than from the floor plan.

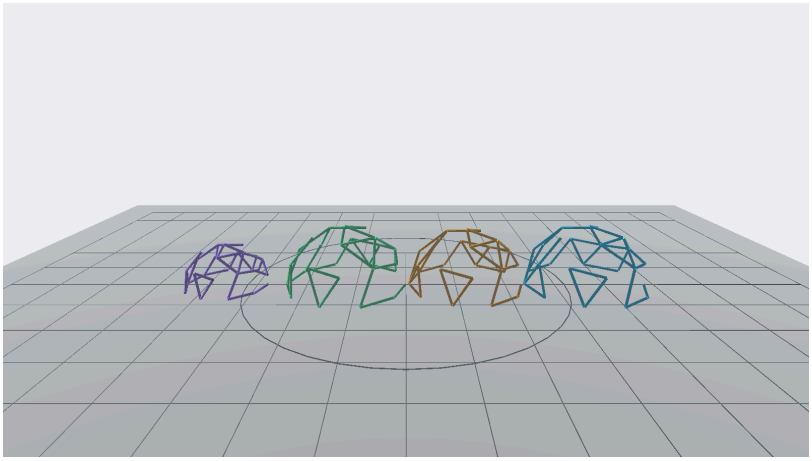


Scale the dome and count what changes.

One skeleton, four buildings

The same frame is already four products in the software that wrote this book: a dome home, a storage shed, a greenhouse and a storm shelter. Four buildings at four price points, and the same triangles under every one of them.

That is the pattern the rest of the book keeps finding. The skeleton is the constant and everything that makes it a particular building – the skin, the floor, the openings, the services – is a choice made on top of it. Chapter 65 takes that all the way, to 15 structures from one body.



Four buildings, one skeleton.

CHAPTER 5

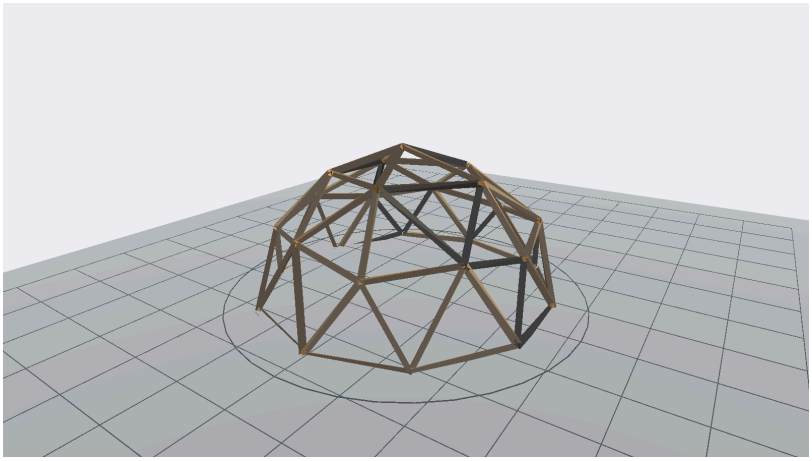
Why I Built the Ugly One

The beautiful maths needed an ugly test: salvage, a folded bracket, and nothing landing where it should

Everything in the last chapter assumes care. Measured deductions, a jig proved before it is trusted, a cut list worked out to the thousandth of an inch.

The first dome that taught me anything was the opposite experiment, and I built it on purpose.

I took whatever the chainsaw had made that week. I skipped the compound cuts entirely. Where two members met at a seam I folded a flat piece of steel into a bracket instead of mitring anything. I let nothing land where it was supposed to, and then I went to see what the geometry would forgive.



Take whatever the chainsaw made, and see what the geometry forgives.

Here is why that was the right test.

A careful hubless dome replaces the hub with **precision**. Every strut is cut exactly so that it lands exactly where five or six others land, and the hub that would have hidden the difference is no longer there to hide it.

The ugly one replaced the hub with **tolerance**. Nothing landed exactly, and the triangles absorbed it, the way Chapter 1 describes.

That difference is the whole experiment. If triangulation only works for careful people with good timber and a workshop, it is a luxury: a hobby for people who already own the tools. If it works for salvage and a folded bracket, it is a way of housing people. The ugly one stood. So the rest of this book is about how to do on purpose, and well, what that frame did by accident.

Who is behind this

For anyone wondering who is behind this unreasonable quantity of triangles: a Navy veteran, a programmer and an avionics bench technician, now using the G.I. Bill to study toward aerospace engineering. The longer version is at the back of the book. The short version is that a dome, to me, is a home written the way I would write software – modular, reusable, built from parts with clean interfaces so one can change without disturbing the rest – and this book is that idea in timber.

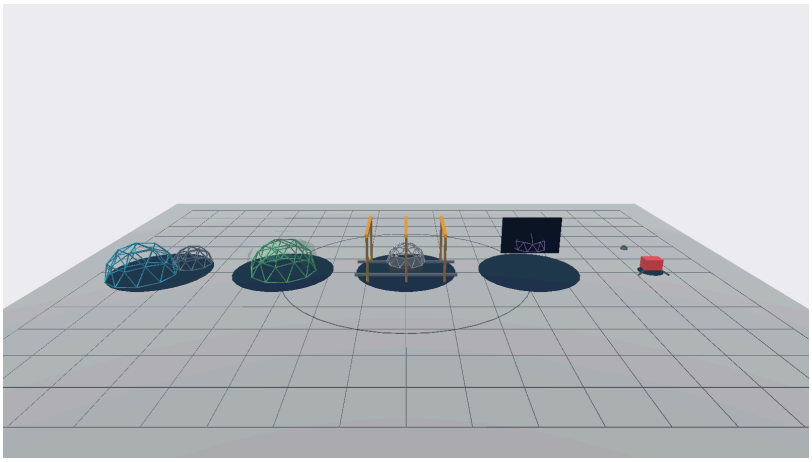
Build one. Then build the next one faster.

So that is the argument, before any of it is proved.

A shape that encloses the most room for the least material, and so spends less on skin and gives it back every winter. Two lengths of wood and one flat board. A method that forgave a chainsaw. A price with a receipt on every line, and a parts list that does not grow when the house does.

Every number in this book came out of code you can run, and the parts that are borrowed are named out loud. A sphere encloses the most room with the least material whether you are a developer or a widow. The geometry does not check your credit.

Build one. Then build the next one faster.



The geometry does not check your credit.

PART 2

The Geometry, From Scratch

“One irrational number places twelve points in perfectly even space. Everything else in this book is consequences.”

The whole derivation, on screen: why triangles, why an icosahedron, where the golden ratio really sits, and the two lengths that come out of it.

CHAPTER 6

Why It Is Triangles

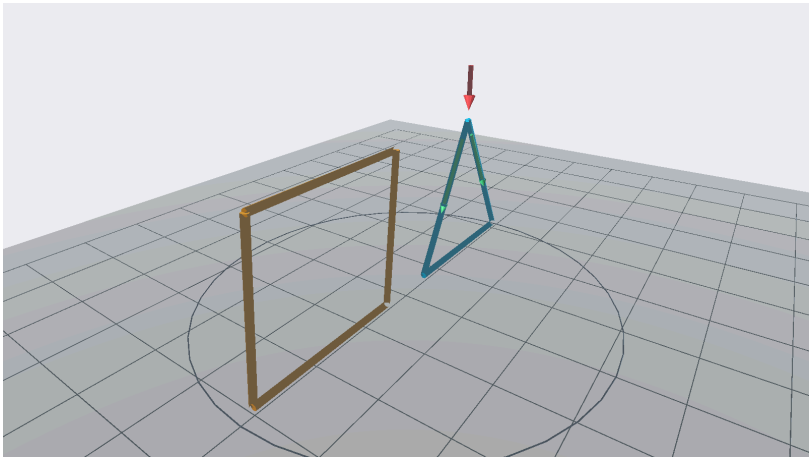
Every other polygon is a mechanism. A triangle is the only shape that cannot change without something breaking

Take four sticks and pin them at the corners into a square. Push one corner.

It folds. You now have a rhombus, and you can keep pushing until it is flat, and at no point did anything break or bend or come loose. A four-bar linkage is a **mechanism**. It is a machine for changing shape, and you have built one by accident.

Take three sticks and pin them at the corners. Push one corner.

Nothing happens. Nothing *can* happen — the only way that triangle changes shape is if a stick changes length or a pin fails. There is no motion available to it.



Why the shape is triangles at all.

That is the entire structural argument

A square frame needs a rigid corner, because the corner is what stops it folding. Rigid corners are welds, gussets, plates, moment connections — and they are where every framed building spends its engineering.

A triangulated frame does not. The geometry is holding the shape, so the corner only has to stop the sticks sliding apart. A pin will do. Two flat faces pressed together will do, which is Chapter 20 and is the whole reason this method can use a joint you could make with a handsaw.

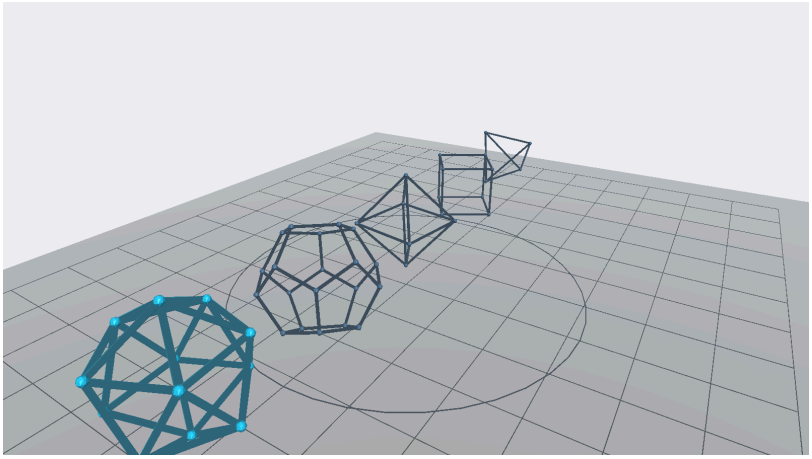
And it is why the frame can be lumpy

Chapter 1 claimed a dome forgives error. This is why.

A mechanism amplifies error — push a linkage slightly out and it moves a lot. A triangle cannot move at all, so the error has nowhere to go and nothing to grow into. A triangle whose sides are each a shade off is simply a slightly different triangle, sitting a shade differently in the shell, and its neighbours absorb the difference at the seam.

The shape is doing the work. That is the sentence this whole book keeps returning to.

And why an icosahedron



The solid everything starts from.

If triangles are the unit, the question becomes: how do you wrap a ball in them?

The answer has been known since Plato. There are exactly five solids whose faces are all the same regular polygon meeting the same way at every corner, and of those, three are made of triangles: the tetrahedron with 4 faces, the octahedron with 8, and the icosahedron with **20**.

Twenty is the most triangles you can get with every one identical and every corner the same. So it is the roundest starting point available, and starting rounder means less distortion when you subdivide.

Everything after this is subdivision. Cut each of those twenty faces into smaller triangles, push the new points out onto the sphere, and you have a geodesic dome. Cut each edge once and you have the 2V of this book. Cut it twice and you have a 3V.

But before any of that, you have to *place* the twenty faces — which means placing twelve points in perfectly even space, which is the next chapter and is the one piece of this derivation that looks like magic.

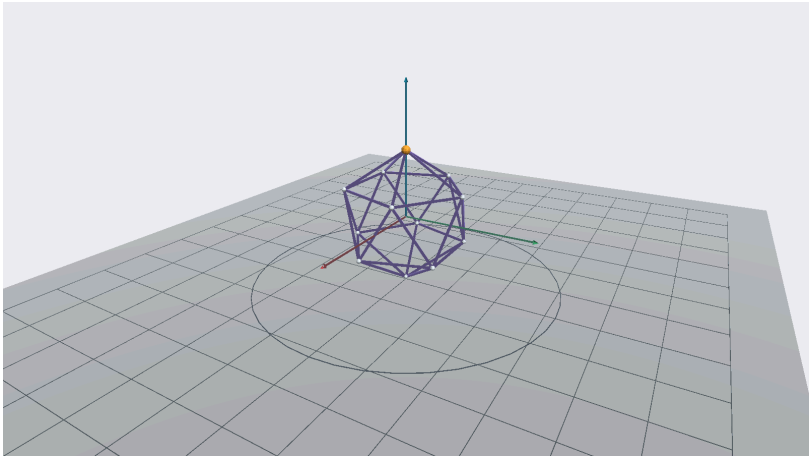
CHAPTER 7

Twelve Points From One Number

Where the golden ratio actually sits in this method -- and the place everybody expects it and it is not

The golden ratio has a real job in this method and it is not the one people expect.

Here is the job.



Twelve points placed by one irrational number.

The number

φ — phi — is **1.618033989**.

It is defined by one property: it is the number whose square is itself plus one. $\varphi^2 = \varphi + 1$. Solve that and you get $(1 + \sqrt{5}) / 2$, which is where the decimal comes from.

That is the whole definition. No rectangles, no seashells, no Parthenon.

What it does here

Write down three numbers: 0, 1 and φ.

Now make points by putting them in every order with every combination of signs:

```
(0, ±1, ±φ)
(±1, ±φ, 0)
(±φ, 0, ±1)
```

Three families, four sign choices each. **Twelve points.**

Measure the distance from each point to its nearest neighbours: exactly 2.000, every time. Measure each point's distance from the origin: **1.902113**, every time, because it is $\sqrt{(1 + \varphi^2)}$ and every point has the same coordinates in a different order.

Twelve points, all the same distance from the centre, all the same distance from their neighbours. That is an icosahedron — the most even way to place twelve points on a sphere — and it fell out of writing one irrational number into a list of coordinates.

Nobody measured anything. Nobody iterated toward a solution. The evenness is a consequence of φ's one property, and this is the only place in the whole derivation where something arrives for free.

Why that property does it

Briefly, because it is worth knowing rather than accepting.

The icosahedron's twelve vertices can be seen as three golden rectangles standing in three perpendicular planes, interlocked. A golden rectangle is one whose sides are in the ratio 1 : φ — and the reason *that* ratio works is $\varphi^2 = \varphi + 1$, which is exactly the relationship needed for the three rectangles' corners to land the same distance apart.

So the defining property of φ is not decorative here. It is the thing being used.

Where it survives to

Now subdivide. Cut every edge of those twenty faces in half, push the new points out to the sphere, and measure what you get.

Two lengths, and only two — Chapter 9 proves there are no others. Expressed as multiples of the radius:

short	0.546533 R
long	0.618034 R

Look at the long one. **0.618034 is exactly 1/φ.**

0.618033989, to nine places, and it is not an approximation — it is φ − 1, which is the same number by φ's own defining property.

So the golden ratio places the twelve starting points, and then it comes all the way through the subdivision and the projection and lands in your cut list. When you crosscut a member at 72.000 inches, you are cutting the radius divided by the golden ratio.

And the place it is not

Here is the part that surprises everybody, including me.

The **ratio between the two struts is not golden.**

$$long / short = 1.130826$$

Not 1.618. Not 0.618. 1.130826, which is not φ, not 1/φ, not φ^2, and not any simple expression of φ at all.

People assume it is. It feels like it should be — the whole thing came out of φ, the long factor *is* 1/φ, so surely the two lengths are in a golden relationship. They are not, and a dome cut on that assumption is a dome with sixty wrong sticks in it.

The short factor 0.546533 is its own number. It comes out of the subdivision arithmetic and it does not simplify.

The lesson, which is about more than phi

A pattern that holds four times is not a rule.

φ places the points. φ gives the long chord. Both are exact and both are checkable. The third thing — the ratio — looks like it must follow and it does not, and the only reason this book knows that is that something computed it rather than assuming it.

That is why every number in these pages is derived rather than typed, and it is why the next chapter computes the same two factors four separate ways and prints the difference between them.

CHAPTER 8

Halve Every Edge, Then Push It Out

One division puts the icosahedron on a unit sphere; halving its edges and pushing the midpoints back out is the whole of 2V

Chapter 7 placed twelve points in perfectly even space, and twenty triangles strung between them make an icosahedron. That is a ball, of a kind: a very faceted one, with flat faces the size of a door.

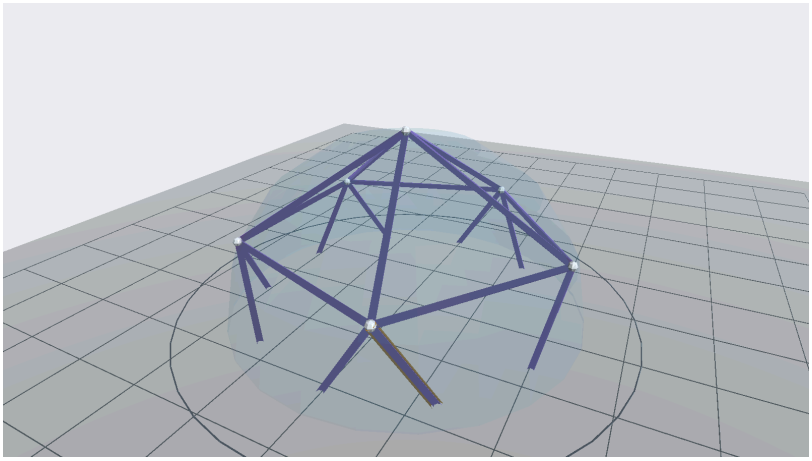
Two moves turn it into a geodesic dome. Halve every edge. Then push the new points out to the sphere. This chapter is those two moves, and the reason the second one is the whole idea.

Put it on a sphere of radius one

The twelve points come out of the golden ratio at an awkward size: every one of them sits 1.902113 units from the centre. Not one, and not anything you would choose.

So divide every point by its own distance from the centre. Each point keeps its direction and lands at exactly one. The icosahedron now sits on a sphere of radius one, and its edge, which measured two units, measures 1.051462.

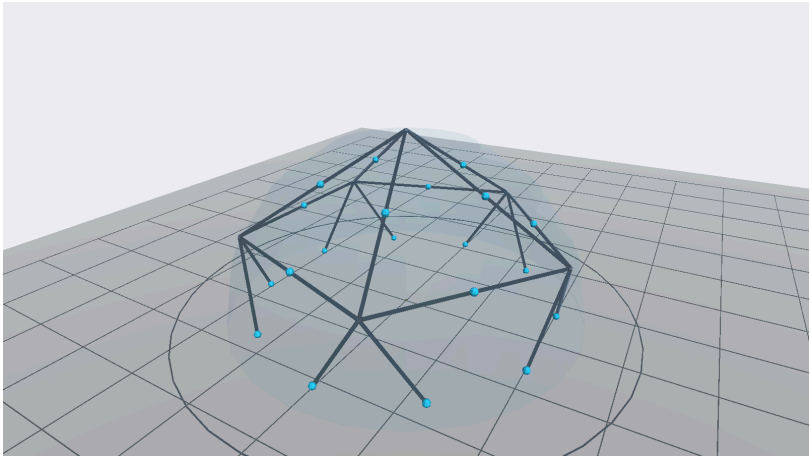
That number is not a length. It is a **chord factor**: a multiplier. Every distance in the model is now a multiple of a radius you have not chosen yet. Choose a radius in inches and multiply, and you have inches; choose one in metres and you have metres. The same numbers serve a playhouse and a hangar, which is why the size of this book's dome is decided last (Chapter 11) rather than first.



The icosahedron on a sphere of radius one.

Halve every edge

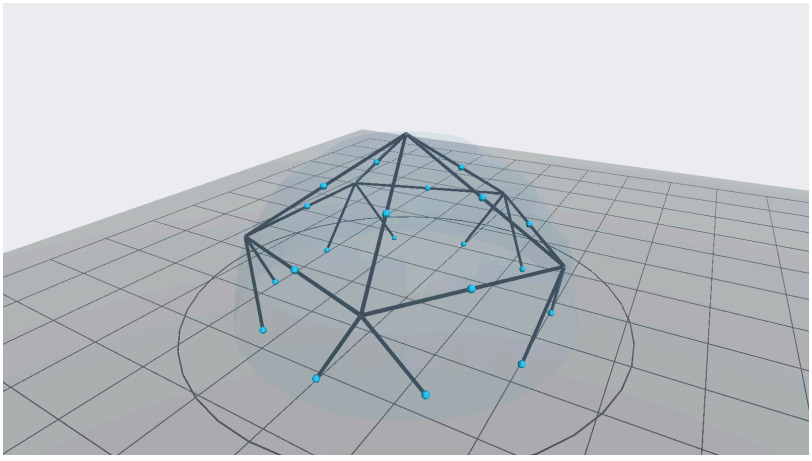
The icosahedron has 30 edges. Find the halfway point of each one – add the two end points and halve them – and you have 30 new points. Join them up and every one of the 20 faces is cut into four smaller triangles.



Thirty parent edges, thirty midpoints.

But look where the new points are. Both ends of an edge are on the sphere; the straight line between them cuts *inside* the ball, the way a chord cuts inside a circle. Every midpoint sits only 0.850651 of the way out: 14.9 percent of the radius short of the sphere. On this book's dome that is 17.4 inches.

Leave the midpoints there and you have an icosahedron with more triangles. Flat faces, just smaller ones. Not a dome.

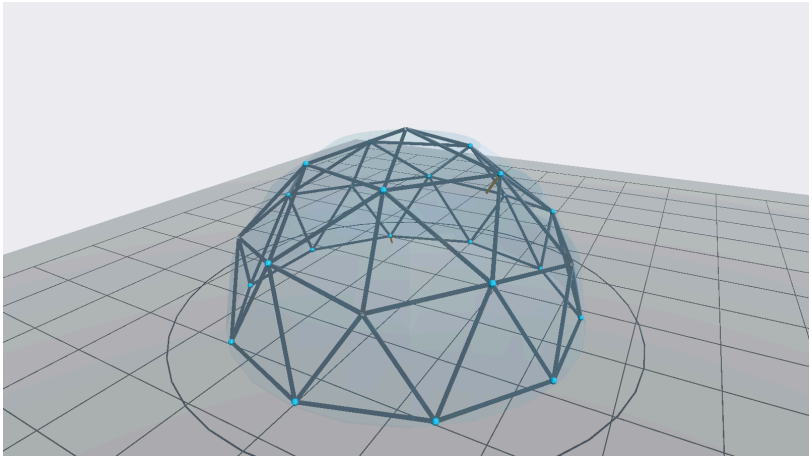


The straight line cuts the corner.

Push the midpoints out

So push every midpoint straight out from the centre until it reaches the sphere. And here is the elegant part: that is the same division as before. Divide the point by its own distance from the centre. Its direction does not change – it slides out along its own ray – and its distance becomes one.

Each midpoint travels 0.149349 of a radius outward. Each face keeps its layout and simply bulges.



Same direction, distance set to the radius.

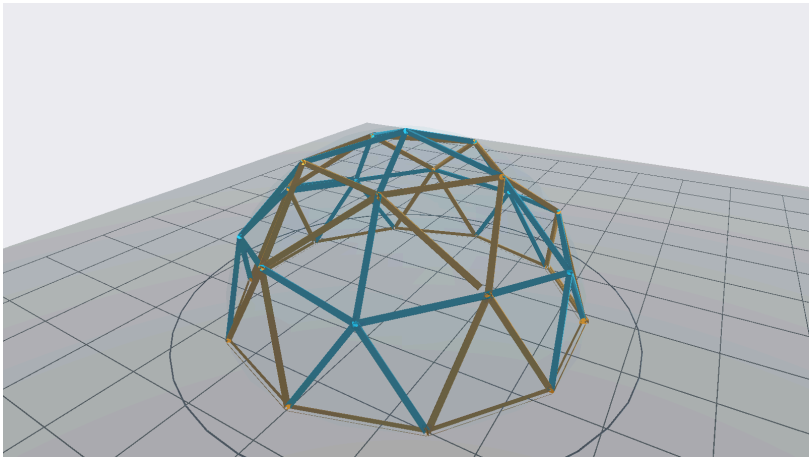
This one division is the entire difference between a faceted ball and a geodesic dome. And it does something the halving did not: it breaks the equal edges. Before the push, the four small triangles in each face were identical. After it, all three corners of the middle triangle have moved out, so it grows more than the three at the corners of the face, each of which has only two of its three points moved. The middle triangle comes out equilateral in the long length; each corner triangle keeps two short sides from its parent corner and gains one long one. The dome now has edges of two lengths.

Two lengths come out

Measure every edge of the whole sphere again: 120 of them. They collapse into exactly two numbers:

short edge	0.546533 x the radius
long edge	0.618034 x the radius

Those are the two chord factors of a 2V dome, and they are why this book cuts two lengths of member and only two. Chapter 9 derives them four independent ways and prints the difference between the answers. The long one is exactly one over the golden ratio; the short one is not golden at all (Chapter 7).



A hundred and twenty edges, exactly two lengths.

Euler's check

How do you know the shape in the computer really closed up, with no missing face and no doubled corner? One addition.

For any closed surface without holes, corners minus edges plus faces is two.

the icosahedron	12 - 30 + 20 = 2
after 2V	42 - 120 + 80 = 2

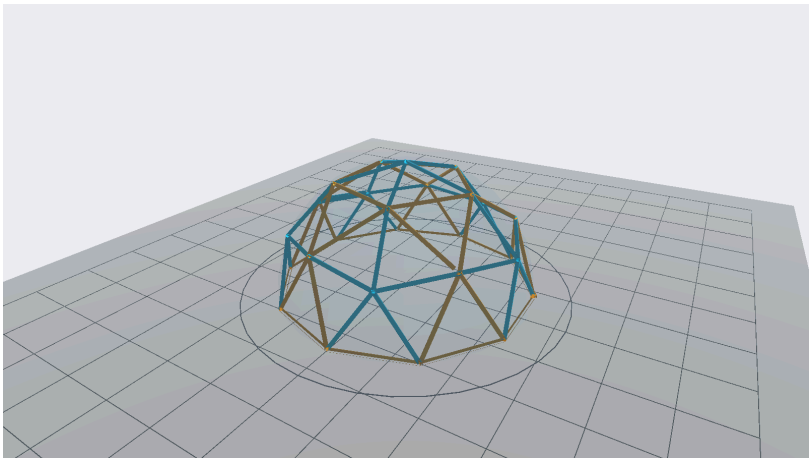
Anything other than two is a bug: a hole, a missing triangle, a corner counted twice. It is the cheapest proof in geometry, and the software that drew every picture in this book runs it.

CHAPTER 9

Two Lengths, Four Ways

Four independent routes to the same chord factor, and the residual between them printed rather than promised

Never trust one calculation.



The two chord factors, derived four ways.

What is being checked

After subdivision and projection, the finished sphere has 120 edges. Measure all of them and they fall into exactly two lengths — no more, no fewer:

short	0.546533 R
long	0.618034 R

Those two numbers decide every stick you will ever cut. If either is wrong, every member is wrong, and the frame will not close — but it will not fail *obviously*. It will be a bit out, everywhere, and you will spend the assembly blaming your sawing.

So it is worth more than one derivation.

Route one: subtract the coordinates

The two points are known — they came out of Chapter 7's twelve, then a midpoint, then a projection. Subtract one from the other and take the length.

$$|u - v| = 0.546533$$

Direct, and it depends on nothing but arithmetic.

Route two: the central angle

Two points on a unit sphere subtend an angle at the centre. Take the dot product of the two unit vectors, take the arc cosine, and you have that angle. Then the chord across it is $2 \cdot \sin(\theta/2)$.

Completely different arithmetic — trigonometry rather than subtraction — and it uses the fact that the points are on a sphere, which route one never assumed.

Same answer.

Route three: the law of cosines

The same triangle, solved the way a surveyor would: two sides of length one and a known included angle.

Same answer again, by a third road.

Route four: measure it

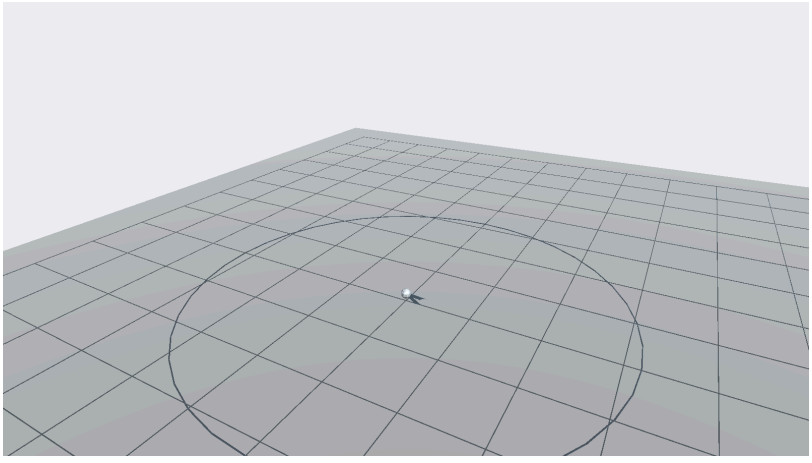
Build the thing in CAD and put a dimension on it.

This one matters more than it looks. The first three are all the same person's reasoning in three costumes; if the underlying model is wrong they will agree with each other and all be wrong together. Measuring the drawn object catches an error in the *model* rather than in the algebra.

What the residual is

Routes one and two differ by about $\mathbf{1.1 \times 10^{-16}}$.

That is not "close". That is the gap between two different sequences of floating-point operations on numbers that are mathematically identical — the smallest disagreement a computer can express at this magnitude. It is the answer you get when two methods agree completely.



Two ways of computing the same number, and the residual between them.

Why the book does this everywhere

This is the pattern, not a one-off.

Where two ways of computing something exist, this project computes both and prints the difference. The strut table in Chapter 16 does it for the pinwheel's bite. The seam angles do it for the fold. And the audit described in Chapter 79 does it for the whole book, checking every printed figure against what the code can produce.

The reason is not rigour for its own sake. It is that **a wrong number in a building method does not announce itself.** It becomes a pile of sticks that nearly fit, two weeks in, with the tree already down.

Four routes and a residual of 10^{-16} costs an afternoon once. Finding out the other way costs the tree.

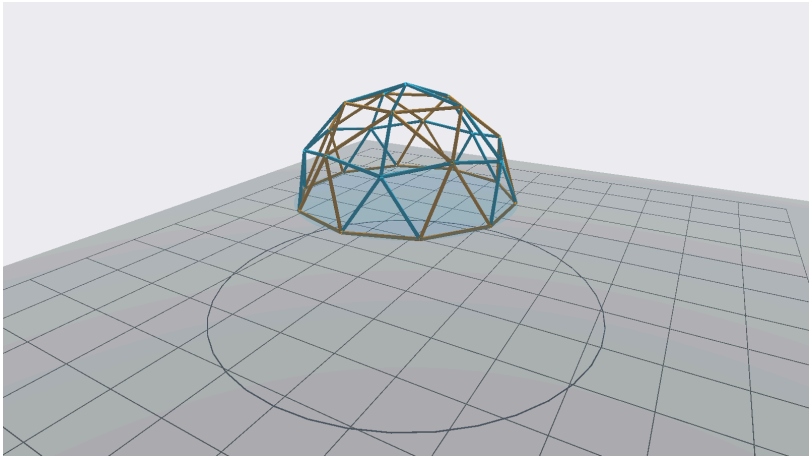
CHAPTER 10

Half a Sphere Is a Building

Where the sphere gets cut, what survives the cut, and how every count in this book falls out of it

A sphere is not a building. It has no floor and no door and half of it is underground.

So you cut it, and where you cut it is a design decision with consequences.



Where a sphere gets cut to become a building.

Cutting at the equator

This book cuts at the equator: a hemisphere, exactly half.

The 2V geometry cooperates. There is a ring of vertices exactly on the equator, so the cut lands on existing points and the base comes out as a clean regular polygon — 10 sides, each one a long chord, sitting flat on the ground with nothing to trim.

That is not true of every frequency and it is not true of every cut. Take a 3V dome and the equator falls between vertex rings, so you either move the cut or accept a base that is not level. Take a 2V and cut it above or below the equator and you get a shallower or deeper shell, a smaller or larger floor, and a base ring that is no longer made of whole chords.

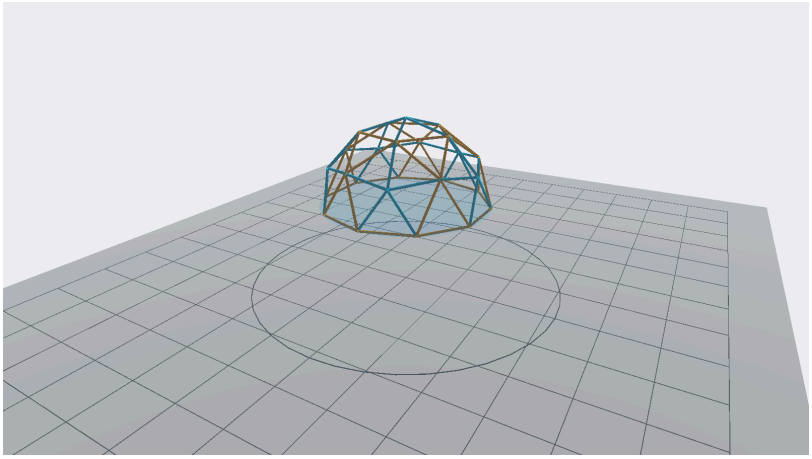
The equator is where the geometry is kindest, and this book takes the kindness.

What the cut gives you

```
across the base      19.42 ft
to the apex          9.71 ft
floor inside the ring 277 sq ft
shell surface        550 sq ft
```

Apex height equals the radius, because a hemisphere's apex *is* the radius. That one identity saves more arithmetic than any other fact in this book.

Counting the building



Every part of the building, counted.

Everything else falls out of the topology rather than being counted by hand:

vertices	26
geodesic edges	65
triangular panels	40
equilateral (AAA)	10
isosceles (BAB)	30
structural members	120
interior seams	55
base edges	10

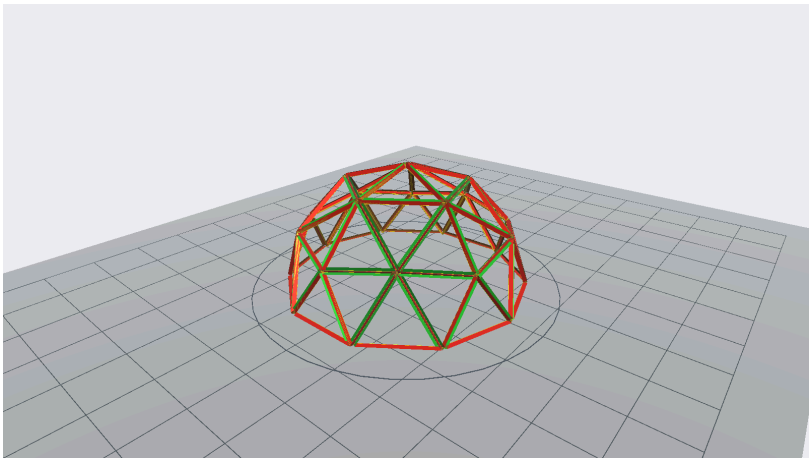
Two of those are worth a second look.

120 members for 65 edges. The duplication of Chapter 20: each panel brings its own three sticks, so every interior edge has two.

55 interior seams, not 65. The 10 base edges are the rim and have nothing on the other side of them, so they are not seams. That distinction matters the moment you start counting keys, or hose, or feet of gasket — and it is the kind of thing that is obvious once stated and silently wrong in a bill of materials otherwise.

This is the actual solved shell

Every picture of the frame in this book is the output of one solver, not an artist's impression of it. It takes the hemisphere above and builds each of the 120 members as a real split-log wedge: its own compound butt cut, its own trim at the vertex, its own place in a pinwheel. It closes the 55 interior seams, each with the key that fills it. When a number about a joint or a seam appears later in this book, it was measured off that model, not typed in beside it.



Not a drawing of the dome. The dome.

What actually moves when the log changes

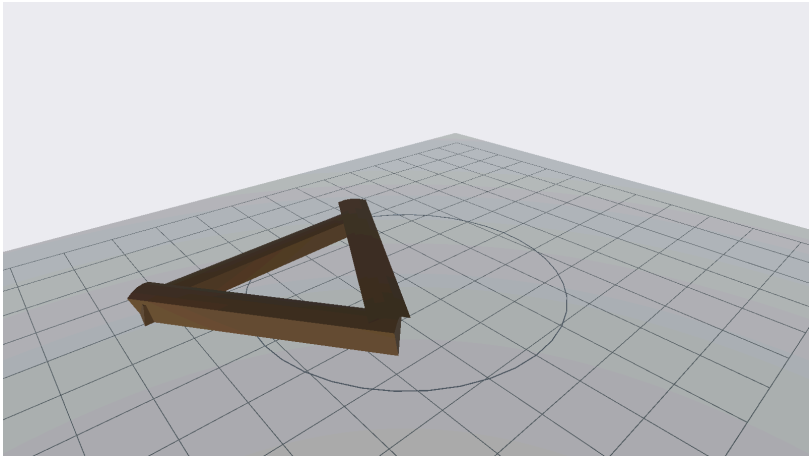
The rest of this book assumes one log diameter. Real trees do not sort themselves; a stand of pine gives you a spread of trunks. So solve the same dome for a thin log and a fat one, 10 inches and 15, and see what changes.

fold angles, 10 in log	18.029 and 22.459 deg
fold angles, 15 in log	18.029 and 22.459 deg
key base, 10 in log	1.95 to 2.33 in
key base, 15 in log	2.93 to 3.50 in

The fold angles are identical. Not one angle in the shell depends on the size of the log, because the angle between two panels is a property of the subdivision, not of the wood. The geodesic solution does not know what a log is.

What does move is the key, which gets wider as the log gets fatter, and the member's offset from its ideal edge. The key was already absorbing the spread between the dome's two fold angles, so this is one more job it was already shaped for.

And the strength question is settled by the thinnest member on the building, not the average one. Even at 10 inches, a wedge's weakest section modulus is 3.20 cubic inches, against 3.06 for a two-by-four on edge. Every log in that range makes a member stiffer than the stud it replaces.



The same shell solved at both ends of the range.

Why it is a decagon and not a circle

The floor figure above is the **decagon** inside the base ring, not the circle it sits in.

The circle is bigger — about seven per cent bigger — and the dome does not reach it. Quoting the circle is the single most common way a dome's floor area gets overstated, and it is overstated by exactly the amount of the ten little segments between the chords and the arc.

This book quotes the decagon everywhere. It is the honest number and it is the one you can put furniture on.

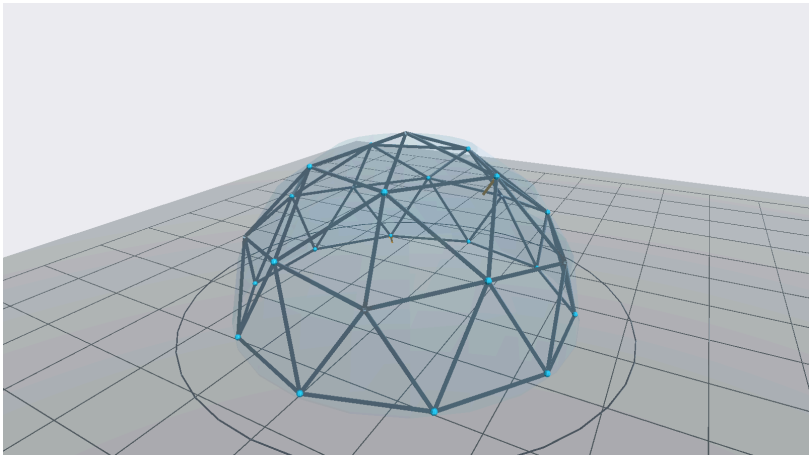
CHAPTER 11

Choosing a Size, At Last

One number decides the whole building, and this book picks it from the stick rather than from the floor plan

Everything up to here has been ratios. Nothing has had a unit on it.

That is deliberate: the geometry is scale-free, so it is worth solving once and then choosing a size — rather than choosing a size and re-solving the geometry inside it, which is how most people do it and is why most people only ever build one dome.



The push that makes it geodesic.

One number does it

Pick any one of these and the rest follow:

- the radius
- the long member's chord
- the diameter
- the apex height (which is the radius)
- the floor area

They are all the same number wearing different units. Multiply the radius by 0.618034 and you have the long chord; by 0.546533 and you have the short one; everything else is those two and the counts from Chapter 10.

This book picks the stick

The reference build is sized from the **member**, not from the floor plan.

72.000 inches — six feet — because:

- it is what comes comfortably out of a twelve-foot log section, with the crosscut in the middle;

- it is what one person carries without thinking about it;
- it fits in a pickup bed, diagonally, with the tailgate up.

The dome that falls out of that is 19.42 feet across with 277 square feet of floor. **The diameter is the answer to the member, not the other way round.**

That is backwards from how buildings are normally sized and it is the right way round for a method whose whole argument is about what one person can make and move.

What it costs to size from the member

One real inconvenience, stated plainly.

No panel fits a four-foot sheet. The narrowest face is 52.516 inches across its shortest altitude and a sheet is 48. Every panel of the reference build needs a seam in its sheathing or a wider sheet.

That is a direct consequence of sizing from the stick, and anybody who would rather have the sheets than the six-foot member should build a different diameter. Chapter 16's table will give them one.

What a bigger radius buys

Pick the size for a reason, because the radius does not scale everything the same way. Double it and you get four times the floor, four times the skin you have to buy and keep watertight, and eight times the air you have to heat. Floor and skin grow with the square of the radius; volume with its cube.

And headroom at the wall is the thing people underestimate. At the very edge of a hemisphere there is none at all: the shell meets the floor at the rim. The first ring of a 19.42-foot dome is the band you sit against, not the band you stand in, and a bigger dome only pushes that band outward.

Audit the boards you have

A chord factor turns a radius into a length. It also runs backwards: a length you have already cut implies the radius it was cut for. That makes it a way to check boards you already own.

Say you have two members measured at 72.0 and 63.5 inches. Each one, on its own, implies a radius:

from the long board	72.0 / 0.618034 = 116.498 in
from the short board	63.5 / 0.546533 = 116.187 in

They disagree, because tapes and saws are not exact. The fair answer is the radius that misses both by the least – a least-squares fit:

best radius	116.362 in
long board misses by	+0.084 in
short board misses by	-0.096 in

Two lessons come with it. Keep the centre-to-centre geometry separate from the physical cut length: what you measure on a board is the cut, which is the chord less what the joint takes (Chapter 16). And the ratio of the two boards, 1.1339, should be close to the geometry's own 1.1308; if it is far off, one of the two lengths is wrong, and the fit will not tell you which.

If you would rather size from the floor

Work backwards.

Floor area for a decagon inscribed in a circle of radius R is 5·R²·sin(36°), so a target floor of A square feet wants

$$R = sqrt(A / (5 \cdot sin\ 36°))$$

and then the long chord is 0.618034 R and the cut is that minus 1.818 inches. A 400-square-foot floor wants a radius of about 11.7 feet and a long member a shade over seven feet — which is past comfortable one-person handling, which is the trade.

Five sizes worth knowing

The reference designs, from a demonstration you build to get it wrong on, up to the point where a 2V starts to strain. Each one is a full cut list in the tables at the back, and each one is the same 120 members and the same nine processes.

The only thing that changes between them is two numbers.

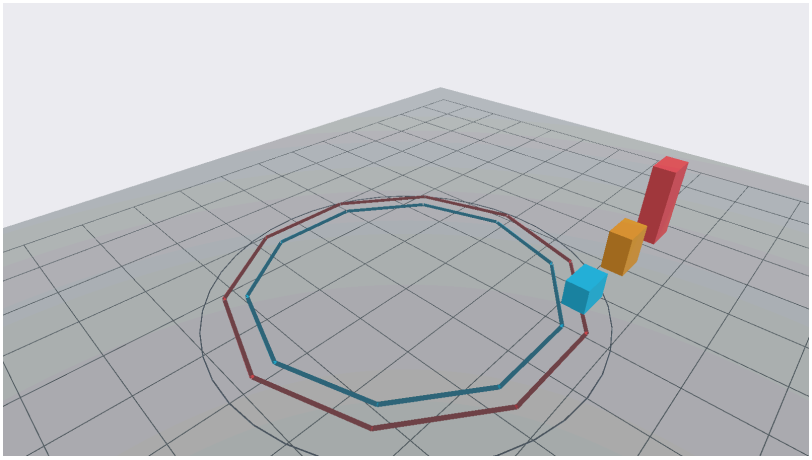
What an eighth of an inch does

Here is a number worth carrying to the site.

The base ring is a regular ten-sided polygon, and the radius of a regular polygon is its side divided by a fixed constant. For ten sides, that constant is exactly one over the golden ratio. So every error in a base member reaches the foundation multiplied by 1.618034:

each base member long by	0.125 in
the base radius grows by	0.202 in
the diameter by	0.405 in

The same golden ratio that placed the twelve parent corners now governs how your mistakes grow. It is not a large factor, but it is a multiplier, and it is why a hand-built dome drifts bigger rather than smaller.



The base ring multiplies your mistakes by phi.

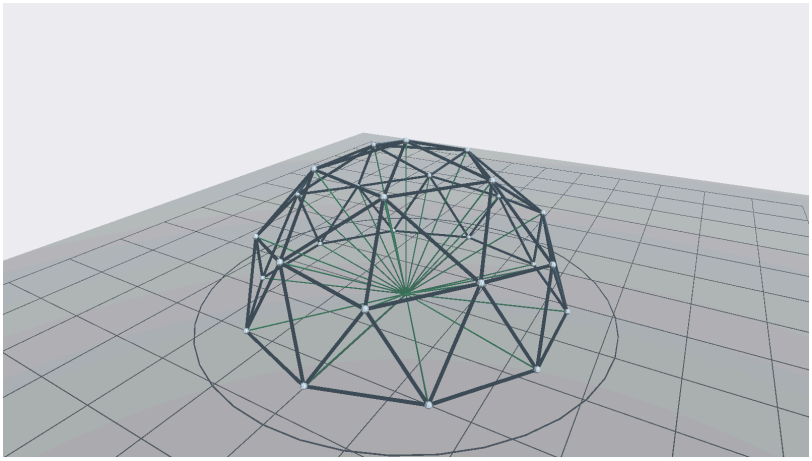
Close the measurement loop

So measure in a fixed order, every time, and do not go on until each check passes:

1. **The member**, against a master gauge stick.
2. **The triangle**, on all three corners, on the jig.
3. **The ring**, on its diameter, three ways across.
4. **The radius**, from a centre pin to every corner.

5. **The height**, from the floor to the apex: on a true hemisphere it is the radius, so the two numbers have to match.

Each check catches what the one before it could hide. A member can be right and a triangle wrong; a ring can be round and the wrong size. Doing them in order is what stops a small error becoming a structural one, and it is why the ring is checked before the next ring goes up. For anything you will live in or load, your local code and an engineer have the last word.



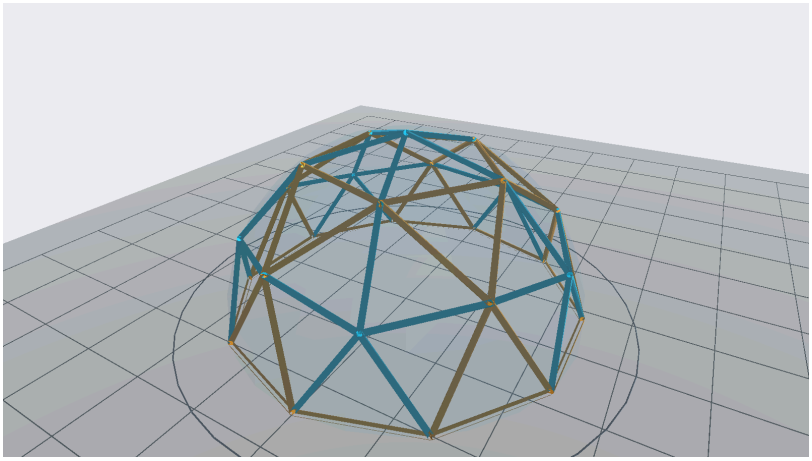
Member, triangle, ring, radius, height.

The whole transformation

That is the whole derivation, and it is short enough to say in one breath.

The golden ratio places twelve corners. One division puts them on a sphere of radius one. Halving every edge and pushing the midpoints back out leaves exactly two edge lengths, as multipliers. One radius turns those into two chords; the joint's bite turns the chords into two saw settings; one flat board turns the settings into 40 identical panels; the rings turn panels into a shell; and a skin turns the shell into a room.

Every number on the way came from the geometry, and every one of them can be recomputed rather than trusted.



From the golden ratio to a building you can stand inside.

PART 3

The Stick

“Every radius of a round log is the same as every other. That is the whole trick, and the tree held the tolerance.”

How a standing tree becomes 120 structural members, what a 45-degree sector is, the two lengths you actually cut, what drying does to them, and the one face that gets a finish.

CHAPTER 12

The Woodpile Is Standing Up

What one tree holds, how it comes down, and why the trees decide the size of the building rather than the other way round

This is a house frame before it is cut.

Not a pile of lumber from a yard, and not a drawing. A tree, standing on the land you are going to build on, already the right shape for something. The question this Part asks is what happens if the tree gets a vote in the design: if instead of forcing it into rectangles, you ask what it is already good at and build around that.

The answer, it turns out, is a frame. Split a round log through its heart and every piece is a structural member, and nothing but the width of the saw blade leaves the log. The frame is already on your land. It is standing up.

A whole house frame, before it is cut.

> **NOT ON THE CLOCK**
>
> Felling and bucking are not on the 40-hour clock. The clock counts the frame, from the first log on the splitting stand to the last bolt in the shell – Chapter 27 adds it up. Budget the trees on their own: the clock starts with a log already on the stand.

What is standing there

Before anything is cut, measure what you have. A trunk is a cone with its tip cut off, so its wood is a formula in its two end diameters and its length. This is the tree the films that come with this book work through – one pine, 15 inches at the butt, 5.5 where it stops being useful, 60 feet between:

```
one pine, measured before anything is cut:
  15.0 in across at the bottom of the usable length
  5.5 in across where it stops being useful
  60 ft of usable trunk between them
a trunk is a cone with its tip cut off, so its volume is
  V = pi L (r1^2 + r1 r2 + r2^2) / 3
  r1 and r2 are the two end radii, L the length
  one board foot is 144 cubic inches, so divide by 144:
  solid wood = 442.1 board feet
buck it into 5 lengths of 12 ft:
  section 1: 15.0 -> 13.1 in, 155.3 bf
  section 2: 13.1 -> 11.2 in, 116.2 bf
  section 3: 11.2 -> 9.3 in, 82.8 bf
  section 4: 9.3 -> 7.4 in, 55.0 bf
  section 5: 7.4 -> 5.5 in, 32.9 bf
```

That is the starting balance. Every way of turning the tree into a building, in this chapter and the next, is measured against it.

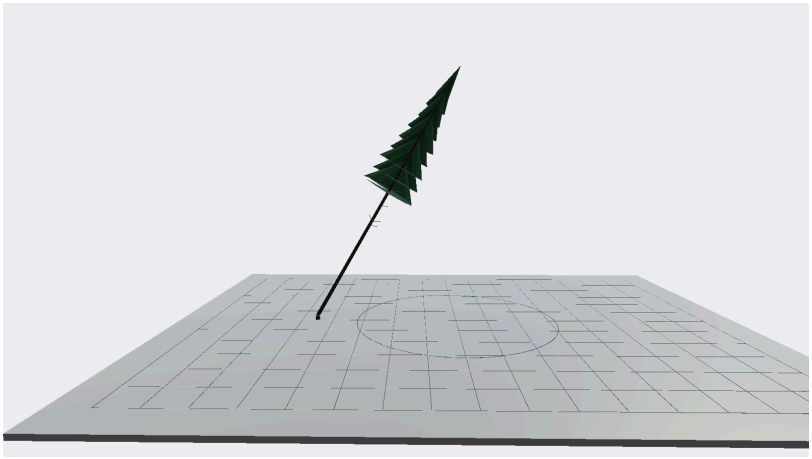
The fell line

Felling gets its own section because nothing after it is safe until it is right, and because the part of it that does the work is not the saw.

1. **Cut the notch** on the side you want the tree to fall toward: a wedge of wood taken out of the trunk, about a quarter of the way in.
2. **Cut the back cut** from the other side, a little higher than the floor of the notch.
3. **Stop before the cuts meet.** The strip of wood left between them is the **hinge**. It steers the tree all the way to the ground; cut through it and the tree goes wherever it likes.

The hinge steers the tree. The saw only sets it free.

That is the whole technique in a paragraph, and it is not enough to learn it from. Fell your first trees with somebody who has done it, wear the gear, and clear an escape path at an angle behind you before the first cut. This book's saw is an ordinary hardware-store chainsaw; the skill is not in the machine.



The hinge steers the tree.

All at once

Then everything happens at once. The crown comes off. The trunk is bucked into sections the length of the longest member you need. And every section splits the same way: in half, in half again, and once more, into 8 sectors. One tree's whole contribution to the frame comes off the saw in the same few operations, repeated, and the arithmetic of it is short:

```
one trunk: 15.0 in at the butt, 5.5 in at the top, 60 ft usable
bucked into 5 sections of 12 ft
solid wood in those sections = 442.1 board feet
SPLIT RADIALLY:
  8 sectors per section, 45 deg each = 40 full-length sectors from one tree
  the saw passes remove 51.2 bf of kerf
  wood kept = 442.1 - 51.2 = 390.9 bf = 88.4 %
  measured across whole trees on site: 88 %
SAWN INTO TRUE TWO-BY-FOURS, same sections:
  rectangles that grid-pack inside the round = 25 pieces
  25 x 8 bf = 200.0 bf = 45.2 %
  square milling in practice recovers 50-65 %, losing wood to slabs, edgings, trim, drying degrade and planer shavings
crosscut once, one tree gives 80 members; the dome needs 120 and 2 trees give 160
```

The honest cost is the stick itself, not the wood: from a small log, a single wedge is the weaker member in bending, and it is the triangulated shell, not the wedge, that makes it enough. Chapter 15 puts numbers on exactly where that stops being true.

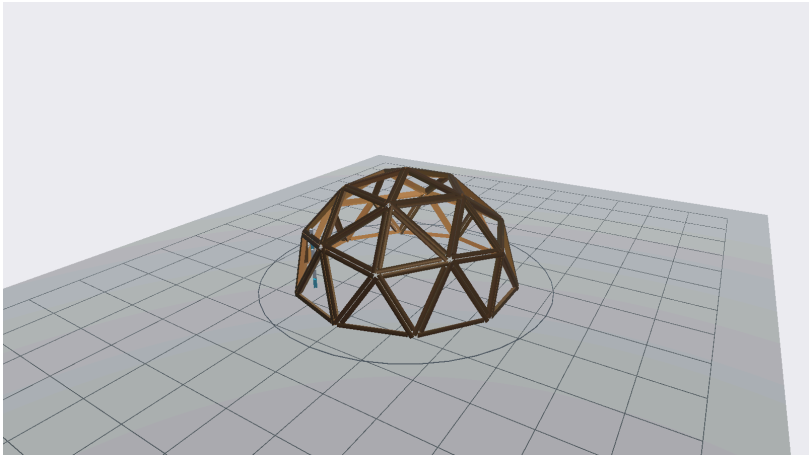
The trees size the building

Here is the inversion that runs through this book. A normal building is designed first and the timber is bought to suit. Here the timber comes first, and the building is solved **backwards from the longest member the wood can give**.

Buck a section, cut it once across the middle, and that half-length is your longest member. The dome that member makes is then fixed by geometry – a size nobody chose:

```
the tree sets the strut length, so it also sets the dome:
  12 ft section, cut once = 6 ft members
  the longest pinwheel member is 72.0 in
  solve for the radius that makes that true:
  R = 132.01 in = 11.00 ft
  span   22.00 ft across
  height 11.00 ft at the crown
  floor  380 sq ft
and the stock:
  2 trees x 80 struts = 160 members
  the frame needs 120
  160 - 120 = 40 left for the floor and the fit-out
  timber standing in the frame = 672 ft of member
```

That is the worked tree of these films: a different, slightly larger dome than this book's reference build, because those two trees gave a slightly longer member. The method is the same; only the tree changed. And that is the point: **the number that depends on the tree is the trunk's diameter**, because it sets every member's section and so its strength. Everything else – the angles, the two lengths, the panel count – belongs to the geometry and does not care what tree you started with (Chapter 10).



The dome two trees make.

One log at a time, 8 rip cuts to a section, and the frame of a house comes off the land it will stand on. The next chapter is why splitting beats squaring, and what it gives up.

CHAPTER 13

Stop Squaring. Start Splitting.

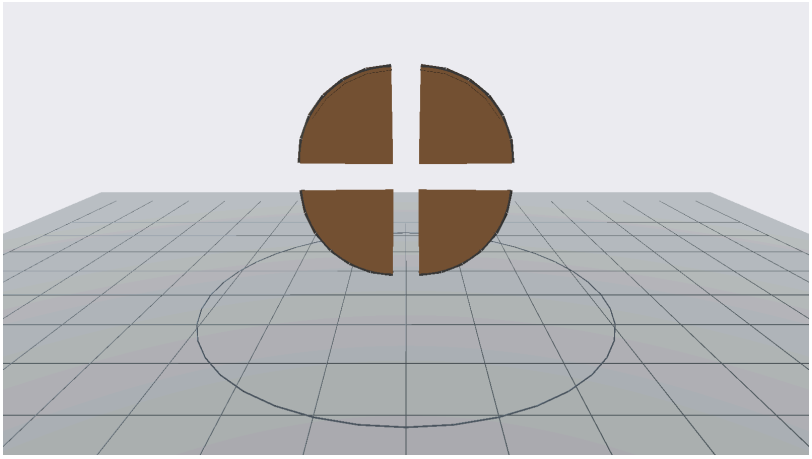
A round log is already rotationally symmetric, and three splits give you eight identical sticks nobody had to measure

I was sitting on a log when this stopped being difficult.

Not thinking about it. Sitting on it, having spent the morning on connectors.

The connector question does have an answer, and it is a flat V bracket: two flaps meeting at the panel's angle, four holes in each flap, and each flap run out to as much as half the member's length so the screws are spread along the stick instead of clustered at its end. Chapter 20 has the detail. It works, it is cheap, and you can make forty of them on a bench brake in an afternoon.

What it does not do is make the sticks alike. It joins dissimilar sticks by absorbing the difference, which means the connector is now doing that work, in every joint, forever.



The same log, opened a different way.

THE 40-HOUR CLOCK

16.0 hours of splitting – the owner's measured rate. The clock goes from 0.0 to **16.0 of 40**, leaving 24.0 hours of the week. Splitting is the biggest single block of the week, and the only one timed at the log rather than worked out.

The wrong question

The question I had been asking was: *how do I join sticks that are all different?*

Everybody's answer to that is the same. Mill them. Square them up, run them through a planer, and now they are all the same because you made them the same.

That answer costs a mill, a planer, a shed to keep the mill dry, the fuel to run it, and — this is the part nobody mentions — about half the tree. A round log does not contain a rectangle. Squaring one is the act of cutting a rectangle out of a circle and calling the rest waste.

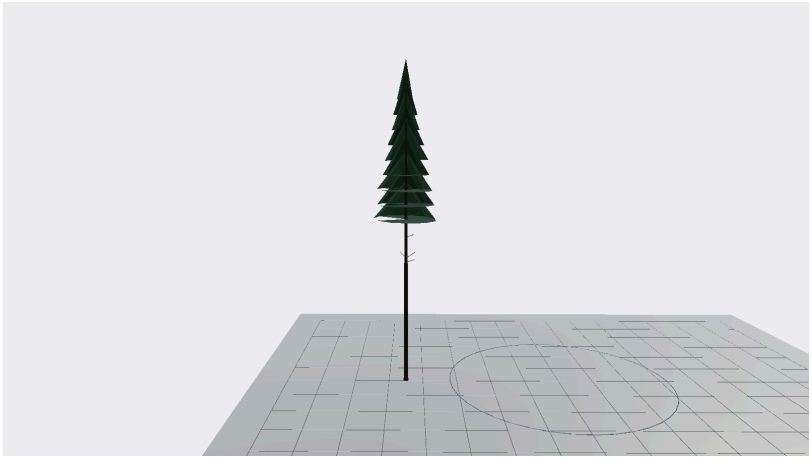
The better question

Sitting on the log, I noticed the log.

A round trunk is rotationally symmetric. Every radius is the same as every other radius. That is not a thing you achieve; it is a thing the tree did while you were not there.

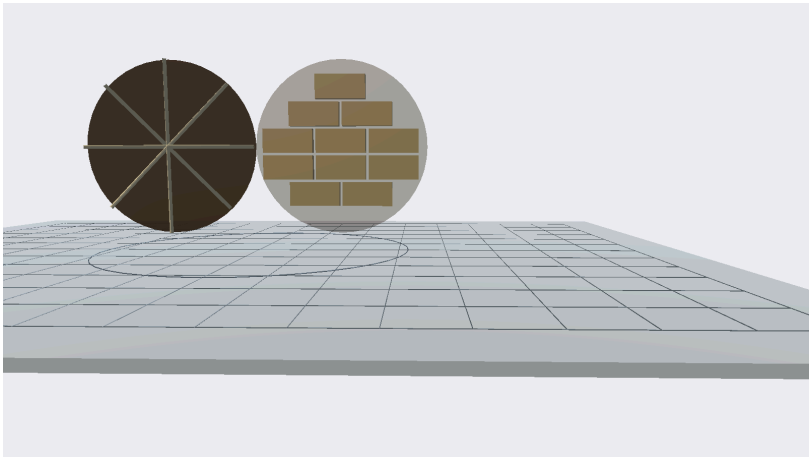
So split it. Split it through the middle and you get two identical halves — identical not because you were careful but because the log was round before you touched it. Split those and you get four. Split those and you get 8.

Three passes. Eight sticks. Every one of them a 45.0-degree sector of the same circle, to whatever tolerance the tree was round to, which is a tolerance nobody had to hold and nobody had to check.



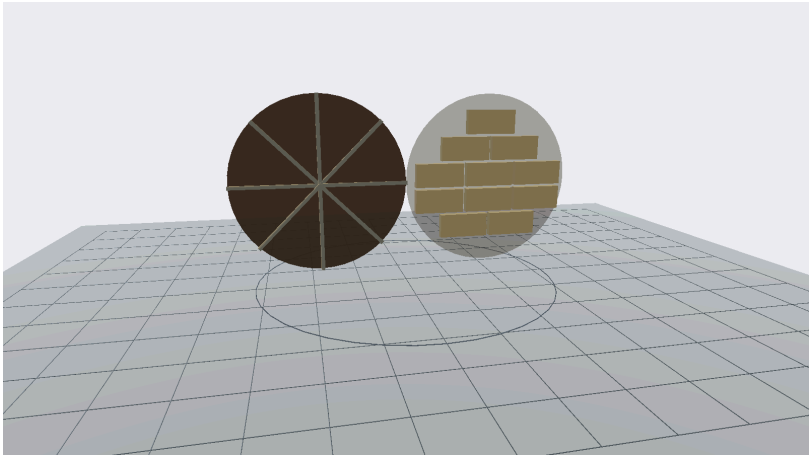
The whole woodpile, standing up.

That was the afternoon. It is not a clever idea and it took me a year to have it, because I was busy asking a question that had no answer.



How much of the stem survives splitting, against sawing.

What the mill is actually for



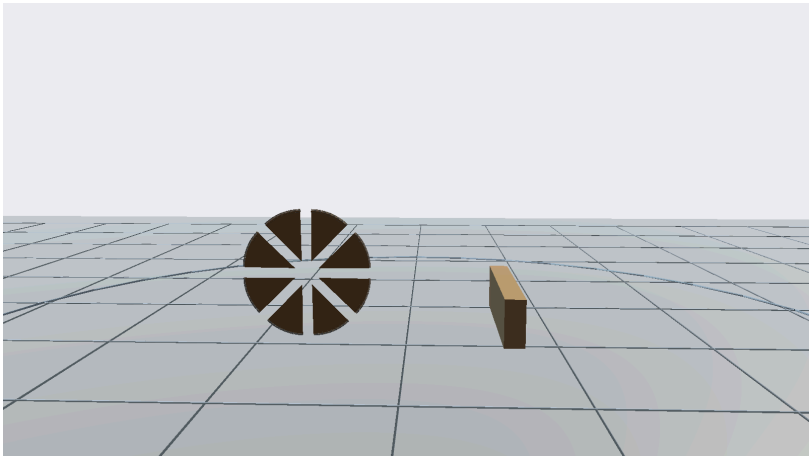
The corners of a round log are not bad wood. They are wood of the wrong shape.

One thing to be careful about, because the recovery argument is easy to overstate.

The corners a mill throws away are not bad wood. They are perfectly good wood of the wrong *shape*. A sawmill exists to turn trees into a standard rectangular product that can be stacked, graded, priced, shipped and sold by people who will never see the tree. It is extremely good at that, and the waste is the price of the standardisation, not of incompetence.

Splitting does not beat sawing at making lumber. It beats sawing at making *this* — a structural member for a frame that does not care about rectangles.

That distinction matters because it tells you when the method stops working. Build something with right angles in it and you want a mill. Build a triangulated shell and you do not.



The tree is the product.

How many rectangles fit in a circle?

Conventional milling squares the log first: four slabs come off the outside to make a rectangular cant. Then the cant is sawn into boards, each board edged, trimmed, dried and graded. Every step removes wood, and every one costs a machine, a setup and a handling. What survives is beautifully standard. What is on the floor is the curved outside of the tree – which is also the densest wood it grew.

How much survives is a packing problem: true two-by-fours, laid out in rows inside each section's small end, where a row can only be as wide as its narrowest edge. For the worked tree of the last chapter:

```
to get true 2 in x 4 in sticks, rectangles have to be packed
into a circle. A mill saws parallel rows, and a row is only as
wide as its narrowest edge, because a stick has to be full
width along its whole length:
  section 1 (13.1 in small end): 0 + 2 + 3 + 3 + 2 + 1 = 11
  section 2 (11.2 in small end): 0 + 2 + 2 + 2 + 1 = 7
  section 3 (9.3 in small end): 0 + 1 + 2 + 1 = 4
  section 4 (7.4 in small end): 0 + 1 + 1 = 2
  section 5 (5.5 in small end): 0 + 1 = 1
  total = 25 true 2x4x12 per tree
  each one holds 8 bf, so 25 x 8 = 200 bf
  out of 442.1 bf standing, that is 45.2% recovery
the brief this lesson works from estimates 32 pieces
  (256 bf, 57.9%),
  which is kinder to the mill than this packing. Both numbers
  are carried through to the comparison.
```

Note what the working does with its own brief. The brief it started from estimated a kinder figure for the mill than honest packing gives, and both are carried forward rather than the convenient one.

The same two trees, both ways

Put the two conversions side by side, for the same two trees:

```
the same two trees, converted both ways:
  radial wedges      782 bf   160 struts
  packed 2x4s        400 bf   100 struts
  ratio = 782 / 400 = x1.95
against the brief's kinder estimate for the mill:
  brief's 2x4s       512 bf   128 struts
  ratio = 782 / 512 = x1.53
  = 53% more wood, which is the figure the brief quotes
either way the direction is the same, and the member count
  moves with it:
  160 wedge struts against 128 sawn ones = 32 more
```

Somewhere between those two ratios more structural wood comes out of the same two trees split than sawn – and one fewer machine was needed to get it.

What this does not claim

One honest paragraph, because the argument does not need overstating.

This is not the claim that any eighth of any tree equals a graded two-by-four. It does not. Species, moisture, knots, checks, decay, grain running off the axis, taper and the connections all still matter, and a building people live in has to be designed and checked around the members actually being used. The figures in this chapter are geometry and volume: what the tree contains and what the frame needs. They are not a structural calculation, and they are not a substitute for one. Chapter 15 is where the structural numbers start.

What comes off the log

At the reference build's 12.0-inch trunk, each sector is 4.592 inches across its bark face and 6.0 inches deep from point to bark. The frame needs 120 of them, which is 664 feet of stick.

Nothing in that sentence required a tape measure on the tree. It required a chainsaw, a steady rip along the radius, and a willingness to let the log decide the section.

The next chapter is what that section actually is, and which way round to put it — which is the first decision in this build you cannot take back.

The round tree is not defective lumber

So, plainly: a wedge from a small tree is not a stronger stick than a stud. It is a cheaper stick, a faster one, one that leaves far more of the tree standing as structure instead of shavings, and one you can make without owning a factory. From a big enough tree it is also simply the stronger stick, and Chapter 15 gives the diameter where that flips.

What makes it enough, either way, is the building. A triangulated shell puts its members mostly in compression along their length, doubles every interior edge, and asks only that the ends can be made to meet. Efficiency is not always a better machine for making a conventional material. Sometimes it is a structure that no longer needs the conventional material at all.

The round tree is not defective lumber. The wedge is not an unfinished two-by-four. It is the member, and the dome is the geometry that lets you use it.

CHAPTER 14

What the Saw Actually Does

Two timed sittings, the fuel written down, a prediction the geometry makes without being told -- and the cuts counted both ways

Everything in the last two chapters could be argued about. This chapter was measured.

Measured, and guessed

Before any rate, here is what the figures rest on, sorted into the two piles this book always keeps apart. Geometry, board feet and angles are computed. The rest was either timed, weighed and read off a rack on site, or it is an estimate, and it says which:

```
geometry, board feet and angles in this film are computed. The rest
splits into two piles, and the difference matters:
MEASURED ON SITE -- timed, weighed or read off a price rack:
  session_a_wedges                 12.00 wedges
  session_a_hours                   2.00 h
  session_a_consumables_usd        10.00 USD
  session_b_rip_feet               42.00 ft
  session_b_hours                   1.00 h
  session_b_depth_in                6.00 in
  session_b_fuel_tanks              2.00 tanks
  session_b_oil_tanks               2.00 tanks
  planning_wedges                  30.00 wedges
  planning_hours                    6.00 h
  stud_shelf_usd                    8.88 USD
  stud_with_tax_usd                10.00 USD
  measured_log_diameter_in         12.00 in
  builder_volume_multiplier         2.50 x
  used_volume_multiplier            2.00 x
  wedge_length_ft                   6.00 ft
  wedge_recovery_share              88.0 %
  traditional_recovery_low          50.0 %
  traditional_recovery_high         65.0 %
ASSUMED -- overheads nobody times, so they are estimates:
  store_round_trip_miles           40.00 mi
  truck_usd_per_mile                0.55 USD/mi
  store_run_hours                   1.50 h
  studs_per_load                   40.00 studs
  cull_share                        10.0 %
  transit_damage_share               3.0 %
  labour_usd_per_hour              25.00 USD/h
```

Every estimate sits on the store-bought side of the comparisons that come later, which is the side it flatters least to guess about.

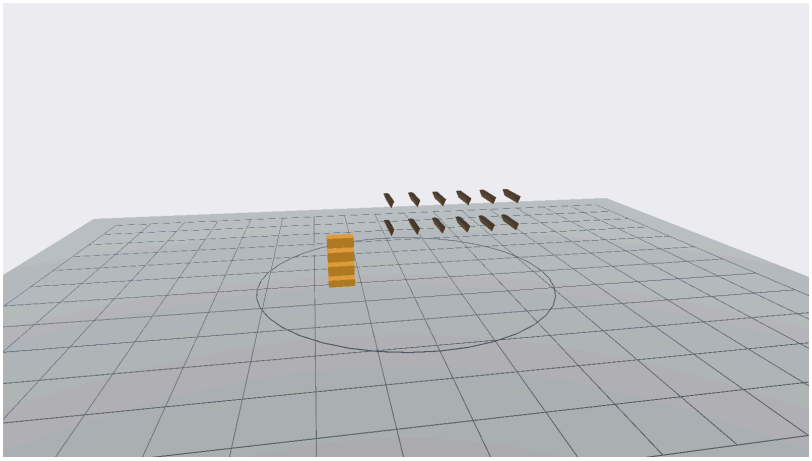
Two sessions, and do they agree?

Two separate sittings with the saw, each measuring something different. One counted finished wedges against the clock. The other measured feet of rip cut in an hour at full depth, with the fuel and bar-oil tanks counted and one chain sharpening included.

Because they counted different things, they can be divided – feet of cut per finished wedge – and the geometry can predict that same number without being told it:

```
SESSION A, timed:
  12 wedges in 2.0 h = 6.0 wedges/hour
  $10.00 of fuel and bar oil = $5.00/hour, the only cash spent
SESSION B, timed separately:
  6 passes at 7 ft = 42 ft of rip in 1 h, at 6 in+ of depth
  2 tanks of fuel and 2 of bar oil, one sharpening
the two sessions measured different things, so divide them:
  42 ft/h / 6.0 wedges/h = 7.00 ft of rip per wedge
and the geometry predicts the same number independently:
  8 sectors takes 7 passes down a 7 ft section
  7 x 7 / 8 = 6.12 ft per wedge
the pure cutting is 88 % of the measured time; the other 12 % is rolling the log, setting up and sharpening
planning uses 5.0 wedges/hour, 17 % below the measured rate
```

Two independent measurements and a geometric prediction agree closely enough for the rate to be worth planning on, and the planning figure is set deliberately below the measured one.



Two sittings, timed.

What the harvest took

At the planning rate, ripping the frame's 120 members takes 24 hours of saw time. The whole harvest is laid out as 8 days: 2 of felling, 2 of bucking and 4 of ripping.

Ripping is the part that was timed with the fuel written down: 2 tanks an hour for 24 hours, 48 tanks. The saw's tank is an estimate – the listings disagree, between 0.28 and 0.37 litres – so the fuel is a range: somewhere between **3.6 and 4.7 US gallons** for all the ripping. Felling and bucking were never metered, and they are left out rather than guessed.

A few gallons of fuel for the frame of a house. That is the figure the whole method is built on, and it is a measured one.

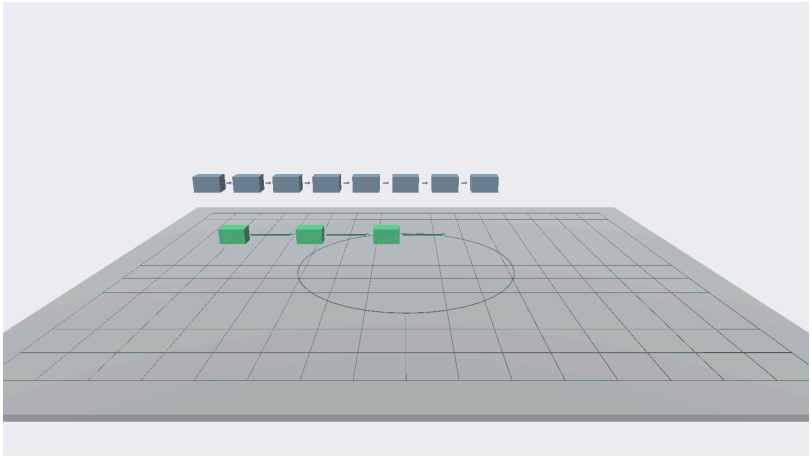
Counting the cuts

The real measure of a system like this is not money. It is the number of actions per piece of finished building: measuring, marking, setting up, cutting, edging, planing, sorting, stacking, hauling and handling the waste. Standard lumber is only convenient because somebody else already paid for all of those. When the tree is standing next to the site, the question becomes: what is the shortest honest path from that tree to a finished member?

Count it both ways, for one tree:

```
count operations instead of money. One tree, to strut:
  splitting:
    5 sections x 6 rip cuts = 30
    40 crosscuts, one per wedge
    total 70 cuts for 80 struts
    = 0.88 cuts per member
  milling to true 2x4:
    30 rips to open the cant and part the rows
    50 edging passes, two per board
    25 crosscuts
    total 105 cuts for 50 struts
    = 2.10 cuts per member
and that is only the cuts: not the setting out, the handling,
  the sorting, the stacking or the hauling between machines
```

And that is only the cuts. Every other action on the milling side – the setting out, the sorting, the hauling between machines – comes on top. The shorter chain is also the one that keeps more of the tree.



One tree to a finished member, counted both ways.

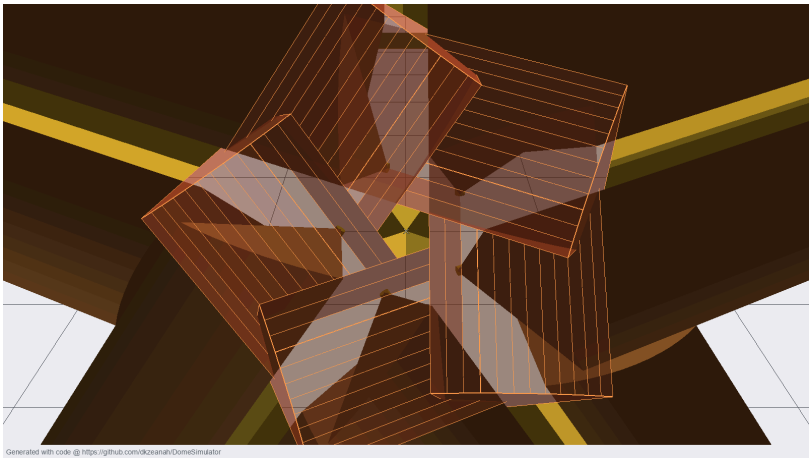
CHAPTER 15

One Eighth of a Tree, Used as a Stick

What a 45-degree sector actually is, and why its shape is an advantage rather than something to apologise for

A member in this frame is a 45.0-degree slice of a 12.0-inch log. It has a point, two flat sawn faces, and a curved bark face.

That is the whole part. Nothing is machined.



The end of a member: two sawn faces and a bark face, and nothing machined.

What the shape gives you for free

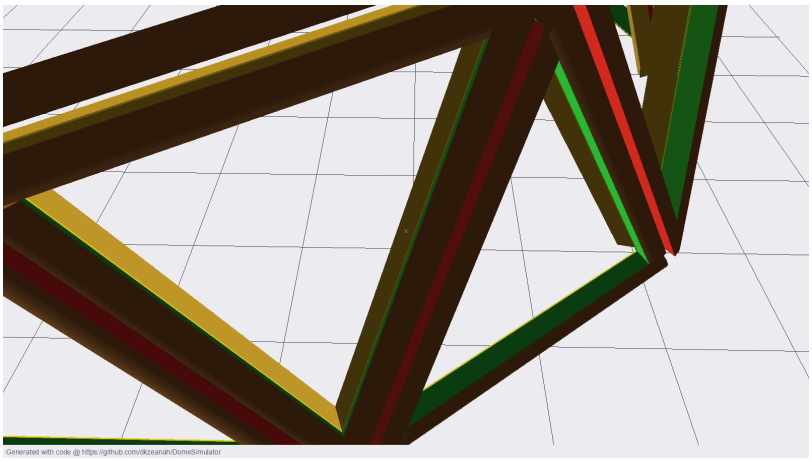
Two reference surfaces. The split faces are flat and they are at a known angle to each other — 45.0 degrees, because that is what a 8-way split means. Every jig in this book locates off those two faces. You never reference the bark, which is lumpy, and you never reference a measurement, which is a thing that can be wrong.

A point that tells you which way round it is. A rectangular stick has a four-fold ambiguity: drop it in rotated ninety degrees and it still looks fine. A sector has one obvious orientation, and — as Chapter 24 shows — the fixture is built so a stick dropped in the wrong way round physically does not seat.

Depth where you want it. The section is deep from point to bark (6.0 inches here) and narrower across the face (4.592 inches). Stand it point-inward and the depth is in the direction the shell wants stiffness.

Which way the point faces

This is the first decision in the build you cannot take back, because it changes the seam, the key, and the shape of every panel.



Point inward: the reference build.

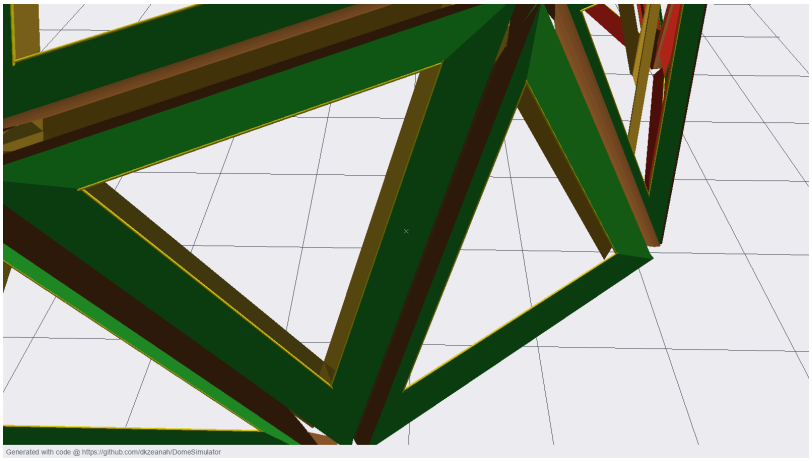
The reference build faces the point **into the dome**. The bark faces are outside, the flat sawn faces look at each other across every seam, and the leftover sector angle opens toward the inside.

The consequences, from the solve: an A seam folds 18.029 degrees and leaves 26.971 degrees of sector angle over; a B seam folds 22.459 and leaves 22.541.

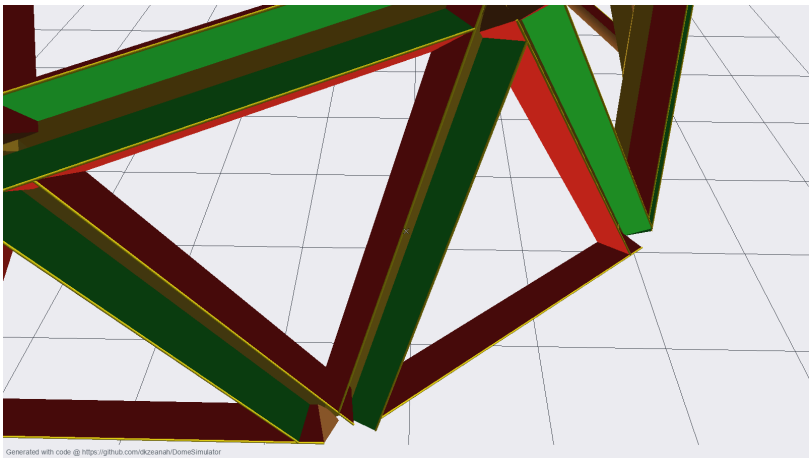
Each pair sums to 45.0. That is not a coincidence and it is the closest thing this method has to a theorem: **the sector you split out of the tree is exactly the seam budget the dome needs.**

The other three

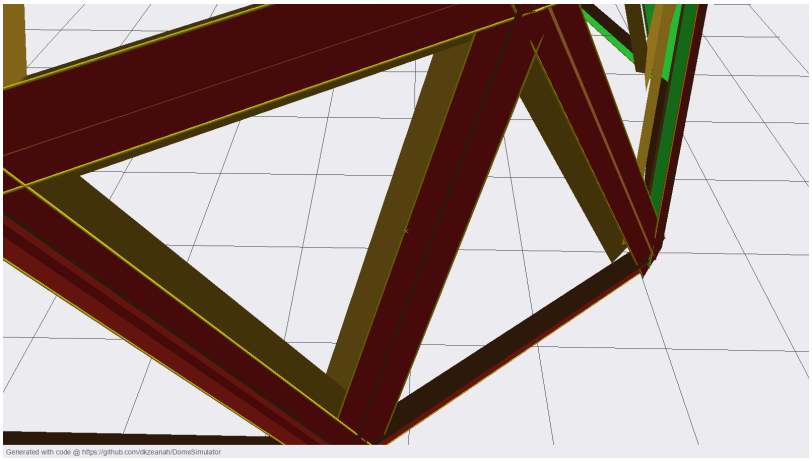
The solver will build all four. They are not equally good and they are not equally bad, and the book shows them because a reader who wants something different should be able to see what they are choosing.



Points apart, each into its own triangle.



Both points outward, at the sky.



Points toward each other, across the seam.

Point outward puts the flat faces where the weather is, which is a real argument for it: a flat outer face is easier to seal a panel against than a curved one. It costs you a key more than twice the size, and it puts the narrow point where the shell wants depth.

Points apart, and points across, both produce a seam that wants a key around eleven and a half inches across its base. That is not a key. That is a beam you are now also making.

The member, measured

A sector is exactly an eighth of the round, so its size is a formula in the log's diameter: its area is the circle's over eight, its depth is the radius, its width across the bark face is a chord. At the films' design diameter:

```
one sector is 45 degrees of the round:
  area  = pi r^2 / 8
  width = 2 r sin(180 deg / 8)   across the bark face
  depth = r                      pith to bark
at the design diameter of 12.5 in:
  area  = 15.34 sq in
  width = 4.78 in
  depth = 6.25 in
a true 2x4 is 2 x 4 = 8 sq in, so this
  sector is worth 1.92 of them
and the depth is the useful part: bending stiffness grows with
  the cube of depth, and this section is deep and narrow
the break-even diameter, where one eighth equals one 2x4:
  pi r^2 / 8 = 8  ->  diameter = 9.03 in
  check: sector at 9.0 in = 8.00 sq in
```

The depth is the useful part. Bending stiffness grows with the cube of depth, and this section is deep and narrow – the right shape for a member whose job in the shell is mostly to stay its own length.

Section properties, and the crossover

Area is not strength. What decides how a member bends is its second moment and its section modulus, and those grow faster than area as the log gets bigger. So there is a diameter where the sector overtakes the stud, and it is worth knowing exactly:

```
a raw sector is literally 1/8 of the tree, and how strong that is depends entirely on how big the tree was.
AT THE SIMULATOR'S 8 IN TRUNK:
  area 6.28 in2, S 2.05 in3
  = 120 % of a dressed 2x4's area but only 67 % of its bending
AT THE 12 IN TRUNKS ACTUALLY BEING CUT:
  area 14.14 in2, I 26.98 in4, S 6.92 in3
  = 269 % of a dressed 2x4's area and 226 % of its bending
  and 177 % / 130 % against a FULL-dimension 2x4
section grows with the square of the radius and bending with the
  cube, so there is a diameter where the sector overtakes the stud:
  more area than a dressed 2x4 above 7.31 in
  stiffer than a dressed 2x4 above 9.14 in
  stiffer than a FULL 2x4 above 11.00 in
and neighbouring panels never share a stick, so all 55 interior edges carry TWO sectors:
  539 % of a stud's area, 452 % of its bending
```

> **THE NUMBER THAT DOESN'T HELP**
>
> **Under that crossover, the wedge is the weaker member and this argument fails.** That is the number that does not help, and it is the reason this book names a minimum trunk rather than pretending any tree will do. Above it, every member is stiffer than the stud it replaces – and every interior edge carries two of them.

What the rotation is worth

The wedge is not symmetric, so turning it about its own long axis keeps the same wood but changes how deep it is in the direction the shell bends. Measured for all four ways round, at the simulator's smaller trunk:

```
the wedge is not symmetric, so which way it faces is a decision.
turn it about its own long axis and the SHAPE stays the same
  while the STIFFNESS does not, because bending in a shell is
  radial: in and out along the dome's own radius.
  DOME-IN    I  5.32 in4   S  2.05 in3
             a pair at one seam: both points inward, toward the centre of the dome
  PANEL-IN   I  2.50 in4   S  1.63 in3
             a pair at one seam: points apart, each into the middle of its own triangle
  DOME-OUT   I  5.32 in4   S  2.05 in3
             a pair at one seam: both points outward, away from the centre, at the sky
  PANEL-OUT  I  2.50 in4   S  1.63 in3
             a pair at one seam: points toward each other, across the seam
area is 6.28 in2 in all four -- rotation moves no wood
but S runs 1.63 to 2.05 in3, a factor of 1.25
```

The same stick, turned before it goes down: the reference build takes the free stiffness and points every member at the centre of the dome.

Why the bark stays on

It is free, it is already there, and taking it off is an operation on 120 sticks.

The honest version: bark is not a finish and it will come off in its own time on the outside faces, in patches, over years. On a frame that is inside a weatherproof skin — which is the argument of Chapter 2 — that is a cosmetic event rather than a structural one.

If you want it off, take it off green, with a drawknife, in the first week when it is still wet and comes away in strips. Do not plan to do it later.

Why six feet is the useful length

The members are short on purpose. A knot or a bend does not ruin wood; it ruins whatever length has to be thrown away around it, and that length is set by the piece you were trying to make:

```
a knot or a bend does not ruin wood. It ruins whatever length has
  to be thrown away around it -- and that length is the piece you
  were trying to make.
this dome's members run 5.13 to 5.90 ft, cut from 6 ft stock
a 16 ft stick from the same trunk is 2.7 times longer
one defect costs, out of 60 usable ft:
  short members  6 ft = 10.0 % of the tree
  long sticks   16 ft = 26.7 % of the tree
and the trunk gives 10 short pieces against 3 long ones, so there are more places for a
  defect to fall harmlessly between
a real bend is cut out at its middle: go six feet clear either side
  and the sections above and below it are both still good
in square milling the same bend follows the board down its whole
  length, because the rectangle has to be straight before it exists
```

Short members are also easier to carry, season, sort, store and replace, and a modest tree can produce them – nobody waits a century for old growth. The geodesic frame turns all of that into an advantage, because it is built from many short members by design.

The one number to hold on to

A member's section is set by the log, and the log is set by what you have. Everything downstream — the seam key's size, the depth of the wall, how many trees you need — follows from the trunk diameter and the split count, and Chapter 57 prices what happens when you change either.

What does *not* change is the two lengths you cut them to, and that is the next chapter.

CHAPTER 16

Two Lengths, and Neither Is the Chord

72.000 and 63.670 inches are what the geometry says; what you cut is those minus a bite that does not scale

A 2V dome has two member lengths. This is the page you come back to.

THE 40-HOUR CLOCK

4.9 hours of cutting and drilling – this book's estimate. The clock goes from 16.0 to **20.9 of 40**, leaving 19.1 hours of the week. The owner timed fifteen minutes a member for everything after the split; how those minutes divide between cutting, panels and raising is this book's split of them, not a stopwatch's.

The two chords

A chord is the straight-line distance between two vertices of the sphere. It is the number every geodesic reference publishes, and it is a property of the shape rather than of your building:

A chord	=	0.618034 x radius
B chord	=	0.546533 x radius

At this book's radius of 116.4984 inches, that is 72.000 and 63.670 inches.

The A chord being a round 72 is not luck. The dome was sized from the stick: a six-foot member is what comes comfortably out of a twelve-foot log section, what one person carries, and what fits in a pickup. The diameter is the answer to the member, not the other way round.

The chord is not what you cut

Neither end of a member reaches the vertex it is named after.

One end butts into the *side* of its neighbour, short of the corner. The other runs past the mathematical vertex so that the previous member can butt into its side. Chapter 20 is why. The consequence is that the joint takes a bite out of each end, and the cut length is:

$$cut = chord - bite$$

For this frame:

A members	72.000	-	1.818	=	70.182 in
B members	63.670	-	0.990	=	62.681 in

The bite does not scale

This is the part that catches people.

The bite is set by **how wide the member is**. A member does not get wider because the dome gets bigger — you are still splitting the same size of log — so the bite is the same number at every diameter.

Solve the same pinwheel at a four-foot radius and at a fifteen-foot one with the same stick and the bite comes out identical to four decimal places.

So: **scale the chord, subtract the bite.** Not chord times a ratio.

If you take one thing from this chapter, take that.

The bite used here

A members	1.818 in
B members	0.990 in

Both at this book's 4.592-inch member. A wider member takes a bigger bite, roughly in proportion, so a frame cut from heavier stock wants its own solve rather than this number.

Why the distinction is worth a page

Scaling the bite with the dome is the natural mistake, because everything else about a dome does scale.

It is also invisible at the reference build, where the two arithmetics agree exactly. It only shows up at other sizes: at eight feet across, scaling makes every stick more than an inch too long; at twenty-four feet, half an inch short. Every stick, all 120 of them, in the two sizes a first-time builder is most likely to pick.

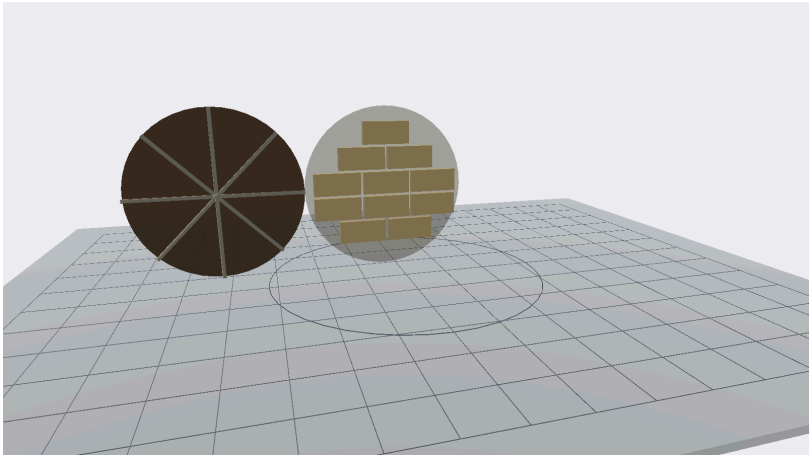
That is why every figure in this book is generated rather than typed, and checked against the solver on each build — Chapter 79. An error that is exactly right in the one case you test is the kind that survives proofreading.

Two models, and they disagree

This repository solves the pinwheel twice.

The raw-wedge solver treats a member as a real 45.0-degree log sector. That is where the bites above come from, it is what the simulator draws, and it is what the reference build is cut to.

A second, simpler model treats a member as a rectangular band lying with its face on the edge line. It gives a bite of 9.253 inches for the same stick — several times larger.



How much of the tree survives, measured.

Neither is a mistake. They are answers about differently shaped sticks.

The films that come with this book carry a third answer, and it should be named rather than quietly dropped. An earlier pinwheel model in them gives **four** stick lengths rather than two – it lets where an end stops depend on which corner it stops in, and so splits each edge length in two:

```
a 2V hemisphere has two edge lengths, SHORT and LONG.
the pinwheel gives four stick lengths, because where an end
stops depends on the corner it stops in:
  SHORT x30   62.523 in   (edge 72.146 in)
  SHORT x30   63.194 in   (edge 72.146 in)
  LONG  x30   71.082 in   (edge 81.584 in)
  LONG  x30   72.000 in   (edge 81.584 in)
  120 sticks in 4 lengths, three per panel
cut them in that order and the saw stop moves four times for
  the whole building
  longest  72.000 in
  shortest 62.523 in
  spread   9.477 in
```

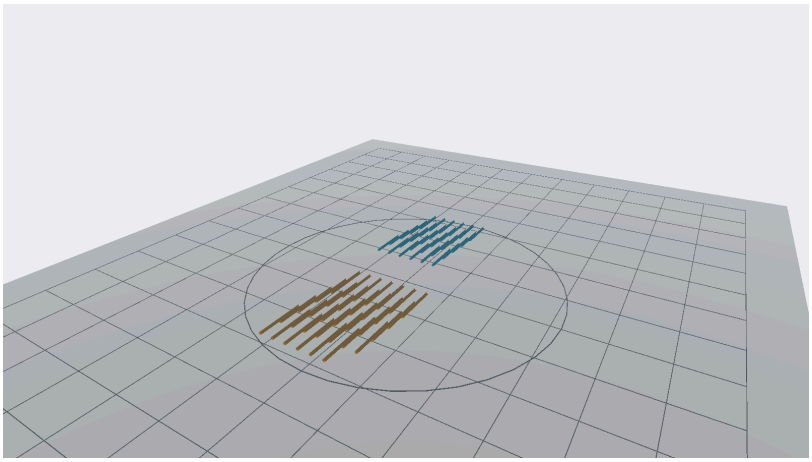
That model also sized a larger dome. The raw-wedge solver, which this book is cut to, gives two blank lengths, 60 of each, because this method never asks the saw to land the head of a member exactly: each head is left long and cut flush in place on the jig (Chapter 24), so the corner-to-corner difference is taken up there, not by a third and fourth saw setting. Where the two disagree, this book follows the solver, and the four-length table is printed here only so that a reader who has seen it in a film knows which one to cut to.

The numbers in this book are the first. **If you are cutting rectangular stock rather than log sectors, this table does not describe your frame**, and the difference is inches on every member.

The row to check yourself against

19.42 feet across: A chord 72.000, A cut 70.182.

If your own arithmetic gives that, the rest will behave.



Chord factors turning into a cut list.

The order

So the whole order for the frame of a 19.42-foot dome with 277 square feet of floor fits on a card:

long members	60 at 70.182 in
short members	60 at 62.681 in
in all	120, from 15 log sections

Add spares. Two or three of each length costs very little and saves an entire afternoon the first time a piece splits or a cut goes wrong.

CHAPTER 17

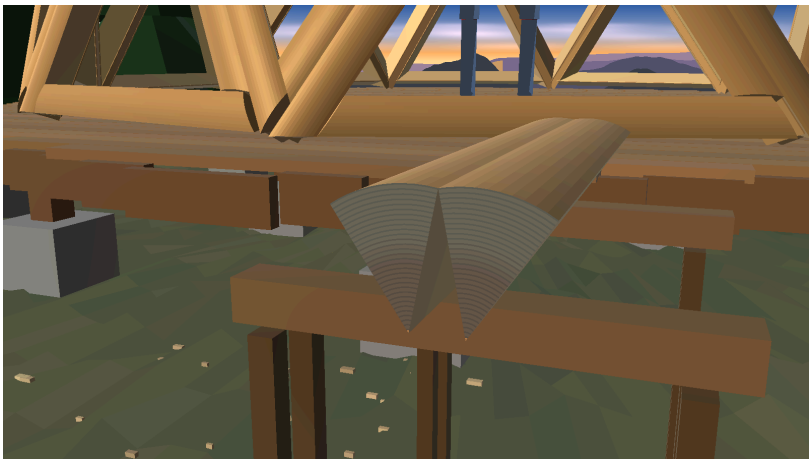
Green Wood Moves, and a Split Moves Least

A wedge closes its own angle as it dries -- which is why it checks less than a squared timber, and why the key is cut last

A tree is mostly water. The frame of this dome comes off the saw weighing 2,478 pounds in green pine, and a good part of that weight is going to leave through the surface of the wood over the next few months, whether you plan for it or not.

When water leaves wood, the wood gets smaller. Not evenly: wood shrinks about twice as much going round the growth rings as it does going across them. For the pine this book prices, green to bone dry, that is 7.4 percent round the rings and 4.8 percent across them.

That one fact explains two things about the method. It explains why a split wedge checks less than a sawn, squared timber. And it explains why the key is the last thing you make.



Two sawn faces meeting at the ridge and opening to the room: the V the key fills.

Why a boxed heart tears and a wedge does not

Picture the end of a squared beam cut with the pith in the middle: a boxed heart. As it dries, every ring wants to get shorter round its circumference, and it cannot, because the ring is a closed loop held at full size by the wood inside and outside it. The stress builds until the wood gives, and it gives along a radius. That split running in from the surface is a check.

Now picture one of this book's wedges. It was split along a radius to begin with. Its rings are not closed loops any more; they are arcs, each one free at both ends. When an arc wants to get shorter it simply does, and the two sawn faces swing toward each other a little. The wedge **closes its own angle** instead of tearing.

The tree held the tolerance when you split it. It keeps holding it while the wood dries.

What the key sees

Here is the consequence you have to plan for.

A wedge dried from green down to 12 percent moisture closes from 45.0 degrees to 44.28: 0.72 degrees. The V in every seam is the sector angle less the fold between the two panels (Chapter 20), and each of the two members turns its face away by half of that. So the V opens by the whole of it:

the tighter V, cut green	22.54 deg
the same V, once dry	23.26 deg
width at the room side, green	2.35 in
width at the room side, dry	2.35 in

The width at the room side barely changes, because the faces also get shorter as the wood shrinks across the rings: 6.0 inches deep becomes 5.83. What changes is the **tilt**. Each face swings 0.037 inches at the room side, so a key cut to fit the green V now touches along one edge and rocks.

That is small, and it is enough. A key is only worth its space if it bears on both faces for its whole depth.

The rule: dry the frame, then make the keys. Split and build green if you must, because green wood cuts and splits more easily. But print or cut the keys last, to the V the dried frame actually has, and bolt the seams up after the frame has come down to its working moisture. The channel's own air is what gets it there.

How long, and how much water

Drying time goes roughly with the square of the thickness the water has to travel through. A wedge's thickest point is the biggest circle that fits inside its sector, 3.3 inches across, against the 12.0-inch log it came from. So:

- a wedge dries about **13 times faster** than the round log would;
- and about **5 times slower** than a two-by, which is thinner still.

Splitting is, among other things, a way of drying a log.

The water it gives up, green down to 12 percent, is about 563 pounds: 67 gallons, from the frame alone. All of it has to leave through the surface of the wood and then leave the building. If the frame is already standing when it does, that water is in the seams, and Chapter 32 is about moving it.

Wood below 19 percent is what lumber grading calls dry: under that line, rot cannot get going. Above it, it can. Framing indoors settles to about 12.

Drying the stack

1. **Stack the wedges off the ground**, on bearers, in the shade, with the round face up so the rain that gets in runs off the bark side.
2. **Sticker every layer.** Put thin spacers across the stack at every layer so air reaches all three faces. A wedge laid sawn face on sawn face dries from one side and cups.
3. **Cover the top, not the sides.** A roof over the stack keeps rain off; open sides let the wind through, and the wind does the drying.
4. **Seal the end grain.** Water leaves end grain many times faster than it leaves the side, which is why logs check from the ends. A coat of wax or paint on each end slows it to match the rest.
5. **Buy a moisture meter.** It is the cheapest tool in the book and the only one that tells you when to make the keys.

The figures in this chapter use handbook shrinkage for pine, and they are declared as such:

fibre saturation	0.3 kg/kg	standard
shrink radial	0.048 -	standard
shrink tangential	0.074 -	standard
board thick in	1.5 in	nominal
char mm	3 mm	assumption
oil m2 per l	8 m2/L	estimate
oil coats	3 coats	assumption

CHAPTER 18

Char the Outside, Oil It, and Leave the Faces Bare

Yakisugi and linseed on the round face only, because the two sawn faces carry the key, the bolts and the seam's air

Every wedge has three faces, and only one of them wants a finish.

The two sawn faces are a joint. One of them butts into the side of the next member in the panel (Chapter 20); both of them meet their neighbours' faces across a seam; and they are the walls of the channel the seam's air runs down (Chapter 29). Anything you put on them changes the fit of the key and slows the one thing the channel is there to do, which is carry water out of the wood.

The third face is the round one: the bark face, turned out toward the weather. That is the face that stands in the sun and rain for the weeks the frame is up before its cap goes on, the face that lives in the vented gap under the cap afterwards, and the face a frame without a cap shows the world. It gets the finish.

Round face out, point in: the only face the weather ever sees.

The whole frame has 24 square metres of bark face and 62 of sawn face. The finish in this chapter goes on the smaller number.

> **NOT ON THE CLOCK**
>
> Charring and oiling are not on the 40-hour clock. The clock counts the frame, from the first log on the splitting stand to the last bolt in the shell – Chapter 27 adds it up. They fit in the weeks the members dry, which is waiting time anyway.

Charring the round face

Charring wood is old. The Japanese name for it, yakisugi, means burnt cedar, and the idea is simple: a layer of char on the surface has nothing left in it for fungus or insects to eat, sheds water, and does not break down in sunlight the way bare wood does. It is also a finish that costs a tank of gas.

Char goes on wood, not on bark. Chapter 15 lets the bark stay on if you want it, and that is a fine choice for a frame inside a skin – but then skip the char. To char the round face, take the bark off first, green, with a drawknife, in the first week while it still comes away in strips. The steps below assume that face is bare wood.

1. **Char before you build**, with the wedges on bearers on bare ground, well away from the stack, the chips and the sawdust. Never char a standing frame.
2. **Use a propane torch with a wide head**, and work along the bark face in one steady pass, holding the flame until the surface cracks into a fine alligator pattern. Stop there; a deep char just crumbles away.
3. **Keep the flame off the sawn faces and the ends.** Stop short of the two sawn edges so they stay crisp. The key, the butt joint and the bolts all bear on those faces.
4. **Brush it.** A stiff brush along the grain takes off the loose soot and leaves the hard char. Brush more for a lighter, textured face; less for a darker one.
5. **Let it cool and blow it off** before any oil goes near it.

> **THE NUMBER THAT DOESN'T HELP**
>
> A single pass of about 3 millimetres of char takes 3.9 percent off each member's cross-section, all of it from the bark face, which is the part of the section furthest from the seam. That is the number that does not help, and it is small: the frame's strength argument in Chapter 1 does not rest on the last few millimetres of the bark side.

Char is not fireproofing. A charred surface is slower to catch than a bare one, and that is all it is. Do not count it in anything to do with fire.

Oiling it

Linseed oil is pressed from flax seed, and it does not dry by evaporating: it takes up oxygen from the air and sets into a tough film inside the surface of the wood. Over char, it binds the char so it stops dusting off on your hands, and it deepens the colour.

- **Raw or boiled.** Raw linseed soaks in deeper and sets slowly, over days. Boiled linseed has driers added so it sets in a day or so. Over char, either works; raw is the traditional choice and the more forgiving.
- **Thin coats, wiped off.** Brush or rag it on, let it soak for twenty minutes, then wipe off everything that has not gone in. A thick coat that sits on the surface stays tacky for weeks and collects dust.
- **3 coats**, a day or more apart.

How much the frame takes, at a declared 8 square metres a litre on rough or charred wood:

bark faces only, 3 coats	9 litres
every face, 3 coats	32 litres

Oiling everything would take more than three times the oil to finish surfaces that should not be finished at all.

What oil will not do. Linseed is a food for mildew in wood that stays damp. On a bark face that dries between wettings, that does not matter. On a sawn face inside a seam, where the whole point of the channel is to move moisture, a film of oil slows the drying and feeds the mould. Leave the faces bare until the frame is dry, and then leave them bare anyway.

Oily rags catch fire on their own

A rag soaked in linseed oil can set itself alight. The same reaction that sets the oil gives off heat, and a crumpled rag holds that heat in until it reaches the point where cloth burns. It happens hours later, in a bin, with nobody there.

Lay used rags out flat, one layer thick, outdoors, until they are stiff. Or drop them in a metal can full of water with a lid on. Never put them in a pile, a bag or a bin with anything else, and never leave them in the shop or in the frame.

The figures in this chapter are declared, not measured:

fibre saturation	0.3 kg/kg	standard
shrink radial	0.048 -	standard
shrink tangential	0.074 -	standard
board thick in	1.5 in	nominal
char mm	3 mm	assumption
oil m2 per l	8 m2/L	estimate
oil coats	3 coats	assumption

CHAPTER 19

Linseed Oil, Where It Belongs

Read the can, set up the rag station, oil the faces people touch, and do not ask a finish to be a roof, a seal or a repair

A video that this project's research pipeline ingested – *20 LOST Linseed Oil Tricks Old-Timers Swore By That Outperform Anything at the Hardware Store, by The Lost Handyman* – offers twenty uses for linseed oil, and some of them are good ones for a dome workshop. This chapter sorts them by what survives contact with the building: exposed interior wood, shelves and trim, and hand tools on one side; seams, weather and windows, where a finish has to work with the rest of the assembly, on the other.

The source is a list of claims, not a comparison test, so this chapter shows the working, the quantities and the limits before it calls anything a saving. Chapter 18 already covers the round face of the frame; this one is everything else.

The rag station comes first

Before the can is open, set up for the used cloths. Linseed oil cures by taking up oxygen, and that reaction gives off heat. A heap of oily cloths traps the heat and can ignite with no spark at all.

Follow the label and your local fire service. The usual practice: cover used rags completely with water and an oil-breaking detergent in a metal container with a tight lid, stored outside and away from anything that burns, and arrange proper disposal. Keep them out of pockets, sawdust and ordinary bins. A wet rag that dries out in a heap is dangerous again.

Read the can

Raw, boiled and polymerised linseed oil are not interchangeable.

- **Raw** oil cures slowly, over days.
- **Boiled** oil, as sold in hardware stores, is usually raw oil with drying additives; it is not boiled, and you should never try to boil oil yourself.
- **Polymerised** oil has been heat-treated by the maker and cures differently again.

Read the ingredients, the safety sheet and the intended uses. A name like *natural*, *boiled* or *Danish* does not identify a formulation. For anything that touches food, choose a product sold for that use and follow its cure instructions – the source's rule that only raw oil is food-safe is too broad in both directions.

Two labels, two schedules

Two products from one maker show why a generic timing cannot be applied to every can. These are the manufacturers' claims, marked as such, not measurements:

Danish oil	rub dry after at least 5 min, cure at least 8 h
oil-and-wax	rub dry after at least 60 min, cure at least 24 h

Both are minimums. In cold, damp conditions neither guarantees the surface is ready. Use the current label of the product you actually bought.

Prepare the member, then apply thinly

1. **Inspect.** Reject decay; deal with splits and loose joints through the repair the design calls for. Oil cannot restore lost strength.
2. **Dry.** Let the wood reach the moisture the finish needs, and check it with a meter (Chapter 17).
3. **Clean.** Remove dirt and unsuitable old coatings, smooth splinters, clear the dust.
4. **Mask.** Mark the places that must bond to glue, tape, sealant or a gasket, and keep the oil off them.
5. **Test an offcut** for colour and cure before the real piece.
6. **Apply a very thin coat**, evenly, with the applicator the label specifies. After the product's soaking interval, rub away everything that has not gone in until the surface is dry to the touch.
7. **Cure** with the ventilation and temperature the label asks for, and look at the surface before the next coat. Never bury a sticky layer under more oil.

Watch the end grain, which drinks more, but stop drips running into joints. And never close an uncured finish inside a panel or an air passage people breathe.

Apply thinly, then wipe dry.

How much, and how long

Measure wood, not floor area. The solver's stock schedule is 664.3 feet of member, and each flat radial face is 6.0 inches deep. The film works an allowance from declared planning inputs, marked by kind:

coats	2 coats	assumption
faces	2 flat faces/member	assumption
coverage	350 sq ft/US gal/coat	estimate
handling	15 % extra volume	assumption
minutes	2 min/member/coat	estimate
danish_wait	5 min minimum	claimed
danish_cure	8 hours minimum	claimed
wax_wait	60 min minimum	claimed
wax_cure	24 hours minimum	claimed
label_coverage	1000 sq ft/US gal	claimed

That gives a planning envelope of 664 square feet, and:

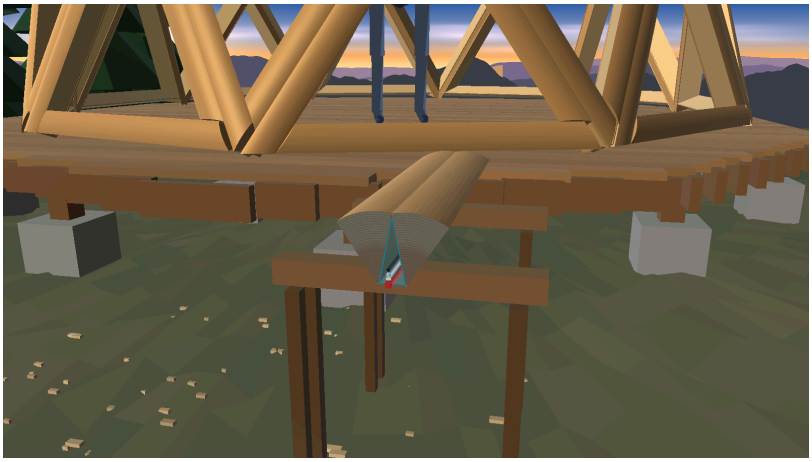
the declared coats	3.80 US gal
with the handling allowance	4.37 US gal
at the label's best coverage	1.33 US gal

Two cautions about those numbers. First, the envelope counts *both sawn faces* of every member as a reference area. It is not an instruction to oil them – Chapter 18 argues they should stay bare, and its own allowance for the round faces alone is 9 litres. The two figures answer different questions; the difference between them is the sawn faces. Second, coverage is the input that decides everything, and the label's best figure is a claim made on smooth wood, not a rough-sawn wedge. Measure how much you use on a known offcut through the whole coat schedule, and scale that.

> **THE NUMBER THAT DOESN'T HELP**
>
> And the number that argues against it: hand application is labour. At an estimated 2 minutes a member a coat, the coats above are **8.0 hours of active work**, before any preparation, access, inspection or waiting for cure. A cheap can can still make an expensive job.

Where it does not belong

Not in the seam's working surfaces. Oil on an exposed interior face is a finish; oil on a sealing land is a contamination. Adhesives and tapes need clean, compatible surfaces. Keep the printed key, the gasket lands, the drainage path and the services clean, check both makers' compatibility before coating wood beside a gasket or sealant, and never count an oil coat as an air seal, a rain seal or a cure for condensation.



Finish decisions follow the assembly.

Not as the weather skin. The source describes linseed oil as outdoor protection and waterproofing. A dome still needs its watertight layer (Chapter 28), flashed openings, drained joints and a way to dry. Plain oil establishes none of those, proves no resistance to sunlight, rot or insects, and damp, linseed-rich wood can grow mildew. Outdoors, use a finish rated for the exposure.

Not as glazing in a dome roof. Linseed-based putty has a real place in traditional sash windows, with mechanical retainers holding the glass – putty alone must never carry it. It is not a design for sloped or overhead dome glazing, which needs a complete system specified for that position, and the video's homemade chalk-and-oil putty does not belong in the dome's weather seams.

The parts people touch, and keeping them

Inside, linseed oil is at its best: shelves, trim and sound furniture. Test for yellowing and for compatibility with any old coating. An oil-and-wax blend gives a low sheen, but wax complicates later gluing and recoating. Floors need a floor-rated system; kitchen surfaces need a product sold for food contact.

Tools. The handle trick transfers well when the handle is sound: inspect the hammer, mallet and jig handles for cracks and loose heads, then finish them thinly and let them cure before gripping. Oil does not repair a broken handle. A drying oil can protect clean, non-moving steel in storage, but keep it out of bearings, hinges, the saw chain and brake, electrical contacts and fastener threads, which need their own products.

Maintenance. Keep a finish record: product, batch, date, preparation and where it went, with a sample piece in the building's notes. Inspect for wear, tackiness, mildew and water marks; find the cause before renewing. Grey wood darkens beautifully under oil, but colour is not strength – oiling does not reverse decay, and it earns no credit for waterproofing a foundation.

So: use linseed oil where a maintainable wood finish suits the job – accessible interior faces, trim and sound tool handles. Keep seams and bonding surfaces clean, give the roof its own weather system, choose glazing for the actual opening, base the quantity on measured wood, budget the hand labour, and end every day with every oily cloth accounted for.

PART 4

The Panel

“Forty panels come off one board, and nothing on the board is adjusted between them. Panel forty is panel one.”

The joint that removes the hub, the nine processes that make the whole shop, the one fixture that makes forty panels the same, and where the door goes.

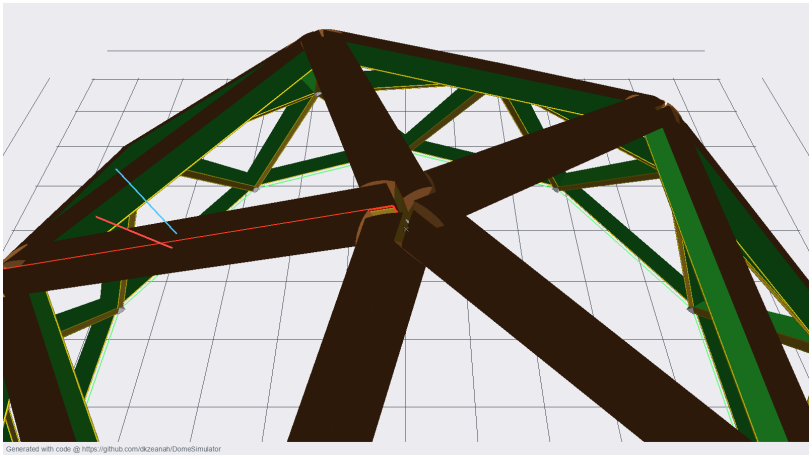
CHAPTER 20

Nothing Meets at the Corner

Three sticks, each butting into the side of the next, and not one of them reaching the vertex it is named after

In a classical geodesic frame, five or six struts arrive at a point and something has to hold them there. That something is a hub, and hubs are why most people give up on domes.

This frame has no hubs, because nothing arrives at the point.



Five triangles meeting, and an empty corner.

The pinwheel

Each panel is built complete before it goes anywhere near its neighbours: three members, closed into their own triangle.

Inside that triangle, each member's end butts into the **side** of the next one, a little way along from the corner. Not into its end. Into its face. The three of them chase each other round the triangle, which is why it is called a pinwheel, and the mathematical vertex in the middle of that arrangement has nothing in it at all.

Three consequences, and they are the argument for the whole method:

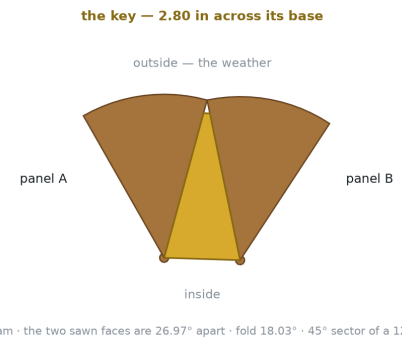
No hub. There is nothing to buy, machine or get wrong.

Flat sawn face on flat sawn face. A split wedge already has two flat faces at a known angle. The joint presses one against another. Nothing is mitred to meet at a point.

The corner is empty, so the corner cannot be crowded. The failure mode of a hubbed dome — five struts whose ends all want the same cubic inch — does not exist here.

What is actually at a seam

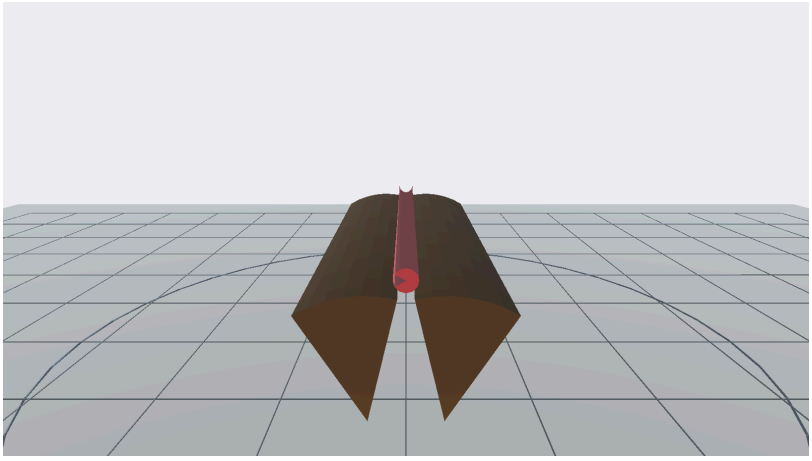
Where two panels meet there are **two** members, one belonging to each, with the key between them.



A seam in cross-section, at the angle the solver puts it: two sectors, the key between them, drawn to the solved dihedral.

That picture is a real cut. A plane through the seam, perpendicular to its own run, intersected with the solver's own geometry — not a diagram of the idea. Two 45.0-degree sectors, points inward, and the 26.971 degrees of sector angle they cannot close between them.

The key fills that. It is 2.798 inches across its base on an A seam and 2.345 on a B, and it is not a manufactured part



The gasket does the shaving: the part of the joint that is not made of wood.

: it is the truncated point of another raw sector, cut from the same log at the same angle.

Why put the variation there? Because the fold between two panels is not one angle. Across this dome it runs from 18.029 to 22.459 degrees. That spread is small in degrees and large in workshop terms: cut into the members, it would be a saw setup per seam and sticks that are no longer interchangeable. Moved into the key, the wood stays generic and the fussy part is small, cheap and replaceable.

The price of a panel being a thing

40 panels times three members is 120. The shell has only 65 edges.

So 55 members exist twice over, once for each panel meeting there. That is nearly twice the timber a shared-strut frame would use.

It is not lumber somebody forgot to remove. It is the price of a panel being an **object** rather than a position in an assembly — and what you buy with it is everything in Chapter 24: a panel you can build flat, check flat, carry, sheathe and lift as a finished thing, on a bench, before anything is standing.

Two trees instead of one and a half, against forty panels you can make indoors in winter. That has been an easy trade every time I have had to make it.

The pinwheel, measured

The pinwheel is not free. Each member lies with its bark face on its own edge line, so its centre sits half a member's width inside that line, and that offset is what opens the gap at the vertex no stick reaches. The bearing area where one member's end lands on the next one's face is the joint's real capacity; the vertex gap is what has to be trimmed clean so that panels do not fight each other at a corner. Both come out of the same solve, and both are what the jig in Chapter 24 has to hold.

Where the error goes

Chapter 1 claimed the frame absorbs error. This is the mechanism.

A member that is a bit long, a bit short, or at a bit of an angle changes the width of the gap at its seam. The gap already has a key in it. A key is a wedge, and a wedge accommodates a range of gaps by sitting a little deeper or a little shallower.

So the error does not accumulate around the triangle and it does not propagate into the neighbours. It lands in a joint that was always going to be filled, and it changes how far in the filling goes.

This is also why the head end of every member is left long and cut in place — Chapter 24 again. There is exactly one end of one stick where error is allowed to leave the building as offcut, and the method spends it deliberately.

What the butt cut actually is

> **WE GOT THIS WRONG**
>
> One correction, because this project published the wrong version of it and some of that is still out there.
>
> The butt is a **compound** cut. It has a bevel and a mitre.
>
> The bevel is constant at 22.5 degrees — half the sector angle — because it comes from how the log was split rather than from where the member sits in the dome. The mitre takes 3 values across all 120 members.

The wrong version is still on screen in one of the films that come with this book: the *Eight Cuts to a House* film's "pinwheel, measured" screen says every end in the frame is a square crosscut. Its other figures – each member sitting half its width inside its edge line, no wood reaching the vertex – stand. Its claim about the ends does not.

So the honest claim is not "no compound angles". It is **three saw settings for a hundred and twenty members**, which is a better claim anyway, and true.

What the pinwheel removes is the shared vertex. It does not remove the mitre.

Six ways to join the same sticks. This book uses the second.

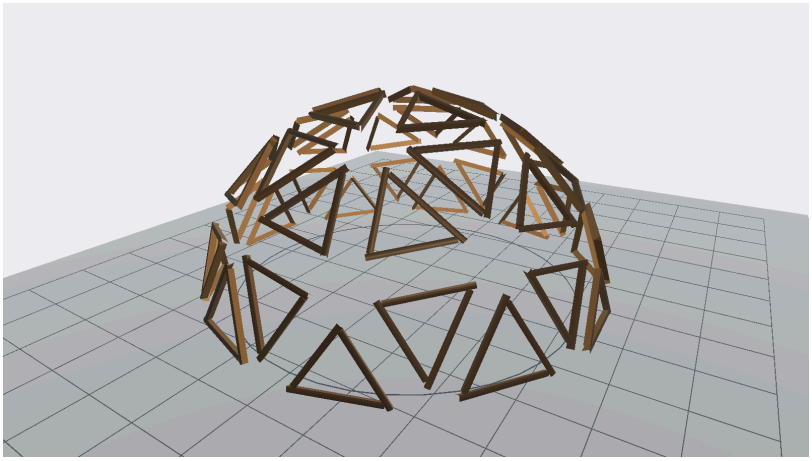
CHAPTER 21

Forty Frames, Lifted Whole

Each triangle is finished before it meets its neighbours -- which makes a panel a module you can swap, and the air a place nobody works

The shell of this dome is not assembled stick by stick in the air. It is 40 separate triangular frames, each one finished on the ground and lifted into place whole: three members each, closed into their own triangle, none of them shared with a neighbour. 40 frames of three is 120 members.

That changes where the precise work happens. Each triangle is built flat, on a bench, at waist height, where a square end and a screw are easy (Chapter 24). What happens up a ladder is bolting finished frames to each other. Nobody is holding three sticks at an angle in the wind. There is not one connector between two triangles that has to be bought, machined or waited for – every joint between panels is a lap between two flat sides.



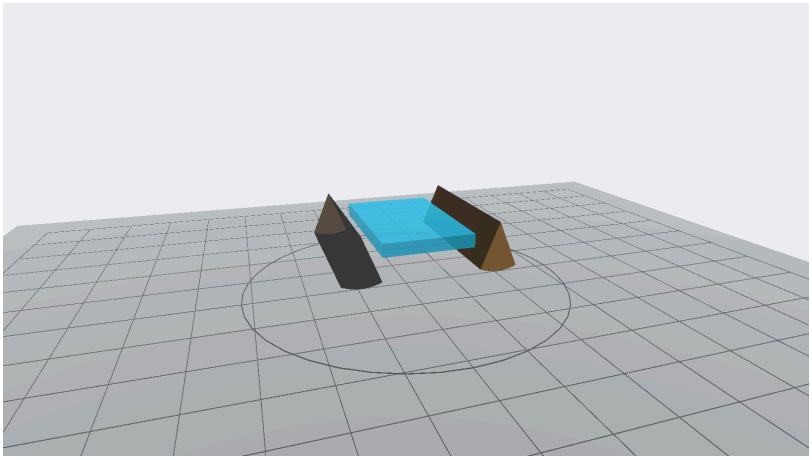
Forty frames, built flat, lifted whole.

A panel that cannot fall through

Here is the property that turns a frame into a chassis.

A wedge gets wider toward its point. So the opening in the middle of a closed triangle is not a plain hole; its sides slope, because each member's faces lean. Which way they lean depends on which way the members point. Turn them so the opening is wider on one side than the other, and a panel dropped in from that side **seats on the taper and is physically stopped**. It is a keystone.

Which means a panel is not cladding nailed onto a frame. It is a module in a socket. You want an air conditioner: it goes in a panel. You want a window, a vent, a solar collector, a hatch: each is a panel that drops into the same seat. This is the frame-level reason for the argument of Chapter 3.



The opening tapers; the panel seats.

Four rotations, four buildings

Stiffness was one reason to choose which way the wedge faces (Chapter 15). Turning it changes the building in other ways too, measured off the same cross-section:

- **Point in** (this book's reference): the panel is caught from the inside, so it goes in from inside and cannot fall out; and the services run on the warm side.
- **Point out**: the panel is caught from outside, and the channel's services sit toward the weather.
- **Turned sideways**, either way: the opening's sides go parallel. No lip, so panels need fixings, but the sawn faces come much closer to flat, which is what a sheet of sheathing wants to land on.

Same stick, same wood. The choice is which building you are making.

What holds it together

The first frame of this project was held together with folded steel V brackets: a strip bent to the corner angle, seated inside the vee between two struts, with long legs screwed down each one. A V bracket does not care what section it is holding, which is exactly why the salvage dome of Chapter 5 could use whatever the chainsaw made.

The wedge frame does not need it, because every member is the same shape as every other one, so the joint gets to be simpler:

- **Inside a triangle**, the members are glued on their mating faces and screwed in from the far side of the neighbour, so the fastening pulls the joint closed rather than hanging off it.
- **Between triangles**, panels bolt to each other edge to edge, and the bolt passes through the seam key that was going in anyway (Chapter 20).

No brackets to order, no hubs to weld, no single part the whole build waits on.

CHAPTER 22

Nine Processes, Whatever Size You Build

Nine operations turn standing timber into a shell. Not nine categories with sub-steps hiding inside them -- nine setups

The last chapters counted parts. This one counts *verbs*, and the verb list is the shorter of the two.

There are 9 things you do to turn standing timber into a shell. Nine. Not nine categories with sub-steps hiding inside them — nine operations, each one a distinct setup with its own tool, its own bench and its own way of going wrong.

You can learn them in an afternoon of reading and be bad at all of them by the end of the first week, which is exactly the right number of things to be bad at.

Here they are, in the order they happen:

> *fell, rip, crosscut, fold, drill, screw, raise, sheathe, glass*

That is the whole shop.

The tool's own construction order, played end to end.

Everything in this book that is not thinking is one of those nine.

Why verbs matter more than parts

Parts are things you buy. Processes are things you **get better at**.

A list this short is a list you can actually attack. Nine places to put your attention is a manageable number of places. A list you cannot hold in your head is not a list, it is a fog — and most building methods are a fog, which is why people who are perfectly capable of building things do not build houses.

Fix one process and every dome you ever build gets cheaper. That is not true of fixing one part.

The nine, one at a time

1. Fell. Drop the tree, buck it into sections the length of your longest member plus waste. One chainsaw. The failure is felling into something you did not intend; the fix is the same as it has always been and is not in this book.

2. Rip. Rip each section radially into 8 sectors with the chainsaw, every cut through the heart of the log. The failure is a rip that wanders off the radius on a section with a knot in it; the fix is to buck around the knot and accept a shorter stick.

3. Crosscut. Cut each stick to length — 70.182 or 62.681 inches — with the compound butt at one end and nothing yet at the other. This is the one cut that can be batched, and Chapter 24 is why. The failure is cutting both ends; the head end is not yours to cut yet.

4. Fold. Assemble three sticks into a panel on the jig, pull the butts up tight, and flush-cut the three heads in place. "Fold" because the triangle closes around itself. Forty times. The failure is doing it in the wrong order and trapping the last stick, which the jig's stage nine is specifically about.

5. Drill. Pilot and set the threaded inserts in the wedge faces — the ones Chapter 3 argues for. Done on the bench while the panel is flat, which is the only time it is easy.

6. Screw. Join panel to panel, key into seam, hardware into insert. The failure is treating this as structural; it is not. The shell holds its shape because it is a shell.

7. Raise. Ten panels on the base ring, thirty above them, five around the apex. Chapter 27 has the sequence. The failure is bracing too little on the first course, which is the only course that needs it.

8. Sheathe. Close the triangles, then the breather, then the cap. The one watertight layer of Chapter 28.

9. Glass. The openings — and the reason it is last is that an opening is a panel you did not close, so it costs nothing to leave the decision until the shell is up and you can see where the light actually goes.

What is not on the list

No milling. No planing. No jointing, thicknessing, moulding or sanding. No welding, no machining, no hub fabrication, no casting.

Every one of those is a process that some other building method needs and this one does not, and every one of them is a machine, a shed to keep the machine in, and a skill.

The list is short because of the decision in Chapter 1. A frame that is allowed to be lumpy does not need the processes whose only purpose is precision.

Why the list does not grow with the house

Here is the part that matters for anybody thinking about more than one building.

Make the dome twice as wide and the parts list grows: more timber, more sheathing, bigger everything. The *process* list does not change at all. It is still fell, rip, crosscut, fold, drill, screw, raise, sheathe, glass.

Build a second dome and it does not change. Build ten and it does not change.

That is what makes this a manufacturing argument rather than a carpentry one. Every hour you spend getting better at one of nine operations is an hour invested in every dome you will ever build, including the ones you have not thought of yet.

7 of the 9 happen on a bench rather than on a ladder, which is the other half of why this is a one-person method.

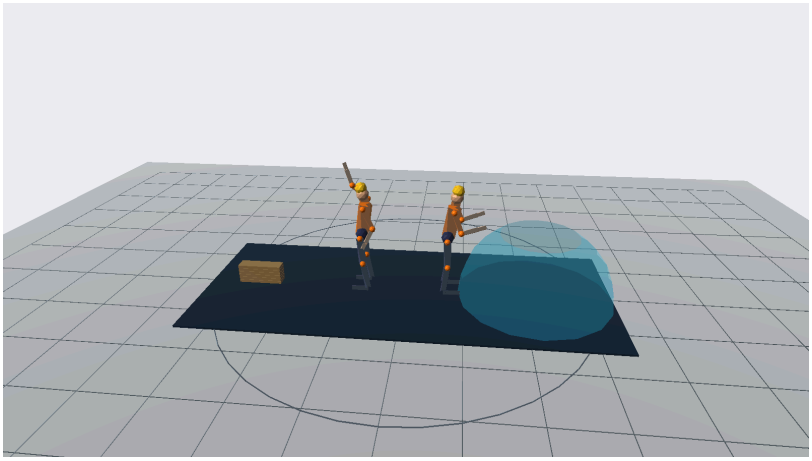
And it is why Chapter 45 can talk about a network of buildings without the argument falling apart: the thing that would normally not scale — the skill — is nine items long.

The honest limit

Two things this does not say.

It does not say the nine are equally hard. Fold is most of the work and rip is most of the danger, and a list that gives them one line each is flattening something real.

And it does not say nine is a natural number. Count differently and you get seven, or eleven. What is not arbitrary is the *order of magnitude*: this is a method with nine-ish operations, and the alternative is a method with fifty.



Inside one station: the manufacturing view of the same nine.

That difference is the argument, and it survives any reasonable recount.

CHAPTER 23

What the Work Costs the Body

Follow two people through every motion of a factory-built dome: the parts that look like effort are cheap, and holding still is the bill

Chapter 22 counted the work as operations. This chapter counts it as food.

The study it comes from is not this book's dome. It is a factory line building a larger product dome, followed part by part through fifteen stations, with two people at each one:

product	Iris-25, home, radius 4.39 m, 4V, Studio
stations	15
crew	2 x 82 kg, 1.75 m, age 34
resting metabolic rate	84.7 W = 1749 kcal/day

A line does not reduce the work; one crew doing every station does exactly the same labour. What a line buys is less **waiting**: split the work across stations and a dome comes off the end every cycle instead of every build. And the line runs at the speed of its slowest station – there, the frame. Speeding up any other station buys nothing. The lessons below hold for any build where people place parts, including one person building this book's dome in a field.

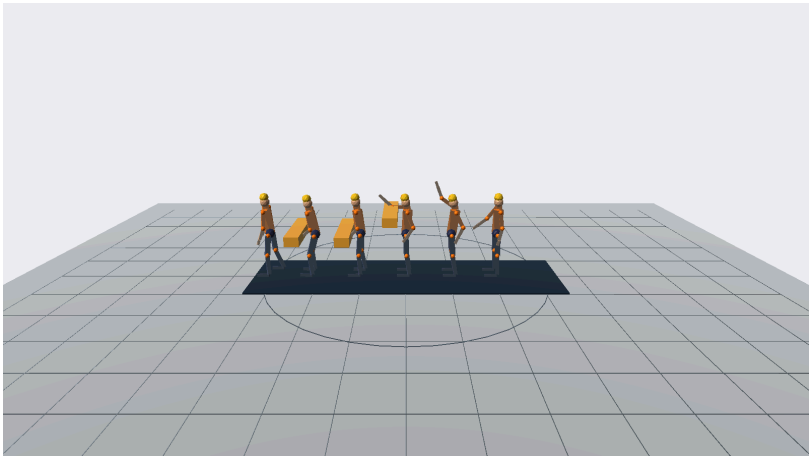
Six motions

Whatever the part, the body does the same six things with it. It **walks** to the stockpile empty. It squats, grips and **lifts**. It **carries** the load to where it goes. It **positions** it. It **fastens** it. It **recovers**, straightening up.

Four results from following those motions through the whole building:

- **The walk is cheap per trip and not cheap in total.** Which is why where the stockpile sits is a decision, not a detail left to whoever unloads the truck.
- **Carrying is not linear in the load.** The published equation for walking with a load has its load term squared: twice the weight costs more than twice the energy. That one fact is the argument for carts, conveyors and a closer stockpile.
- **Height changes the price.** Above 1.75 metres the arms are above the heart, the posture costs more and the crew tires faster. How much of a shell falls into that band is decided by its geometry – a design decision disguised as a shape.
- **Fastening is almost the whole bill.** It raises nothing, does no work against gravity, and still consumes most of what the crew burns, because it is minutes per part of holding a posture, gripping a tool and resisting its torque. The body pays for holding still.

And the rest is not slacking. Industrial engineering has added a recovery allowance to every task time for a century, with more for overhead work, because a schedule written without one is a schedule that will not be met.



Walk, lift, carry, position, fasten, recover.

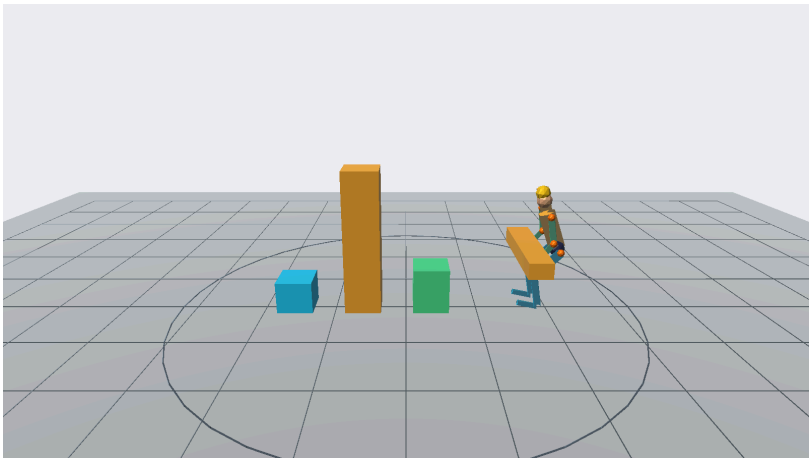
The body is the heaviest thing you lift

To cost a movement you have to know what is being moved, and the biggest thing moved is the worker. The model uses the standard anthropometric tables of biomechanics, built for a 82 kg person: each body segment a fixed share of the total mass. The trunk alone is about half of it. So in every placement the body rises and falls along with the part, and outweighs it several times over. Split by limb, across the whole building:

```
mechanical lifting work by limb group:
  legs           36.7 kJ   14.5 %
  trunk         164.0 kJ   65.0 %
  arms           51.6 kJ   20.4 %
```

The arms get the attention because they are what you watch, but the back does the work – and the back is where the injuries are. That is the honest case for a lower stockpile, a taller bench, and parts presented at waist height.

Above 23 kilograms, manual-handling guidance says a lift takes two people, and the model splits every such part between them. Work above the shoulders costs more per second and needs more recovery for the same output, and whenever the geometry pushes work up there, the cost lands on a body rather than a spreadsheet.



The trunk does most of the lifting.

Exact, and modelled

One part of this is exact: raising a part is its mass, times gravity, times the height it rises. No estimate anywhere in it.

Turning that into food is a model, and it says so. A muscle holding a panel steady does no mechanical work and still burns fuel, so there is no route from joules of lifting to kilocalories without published figures for how hard each task is. The model names every one it takes on authority:

```
external constants this model takes on authority:
  met_definition             1 kcal/kg/h    Metabolic equivalent of task, standard definition
  met_lift                   5.5 METs       Compendium of Physical Activities: lifting items continuously
  met_position               4 METs         Compendium: carrying and placing building materials
  met_fasten                 3.5 METs       Compendium: installing or repairing, general
  met_fasten_overhead        4.2 METs       Compendium: work above shoulder height
  met_recover                2.5 METs       Compendium: standing, light activity
  pandolf_terrain_factor     1 dimensionless Pandolf, Givoni and Goldman (1977) load-carriage equation
  concentric_efficiency      0.25 fraction  Standard exercise-physiology value for positive muscular work
  rmr_mifflin                kcal/day       Mifflin-St Jeor (1990) resting metabolic rate
  pfd_allowance              0.15 fraction  Industrial engineering: personal, fatigue and delay allowance
  pfd_overhead_extra         0.1 fraction   Ergonomics practice: added recovery for above-shoulder work
  met_pause                  1.5 METs       Compendium: standing at rest
  sustainable_shift_watts    350 W          Occupational physiology: sustainable 8-hour working rate
```

That is why two very different efficiencies are both true:

```
mechanical work done    0.363 MJ = 87 kcal
  as a share of food    0.2 %
  ceiling at 25 % muscle  25 %  (everything else is posture, grip and stabilising)
```

During a lift itself, a good share of the fuel becomes height, close to what muscle can manage. Across the whole building, mechanical work is a fraction of a percent of the food, because almost none of a working day is spent lifting.

The ledger

Every motion of every part, totalled:

```
elements                1322
material in the dome    16,751 kg
lifted by one worker    15,100 kg
time per worker         191.05 h = 23.88 shifts
energy per worker       54,870 kcal
energy for the crew     109,741 kcal
energy per shift        2,298 kcal
mean working rate       334 W = 3.50 METs
  sustainable bound     350 W
recovery allowance      17.5 % of the total (33.50 h)

food energy by motion:
  fasten       49,366 kcal   90.0 %
  pause         4,120 kcal    7.5 %
  carry           358 kcal    0.7 %
  walk_out        338 kcal    0.6 %
  lift            326 kcal    0.6 %
  position        272 kcal    0.5 %
  recover          90 kcal    0.2 %
```

Station by station, the energy follows the part count and the time, not the tonnage: a station placing many light pieces slowly costs its crew more than one placing a few heavy ones quickly. This is the table to read when deciding where a jig, a lift assist or an extra pair of hands would change someone’s day:

by station:				
frame	266 parts	3,925 kg	36.48 h	10,531 kcal
fiberglass	161 parts	1,691 kg	32.33 h	9,407 kcal
osb	160 parts	2,806 kg	22.49 h	6,550 kcal
sheetrock	160 parts	2,347 kg	21.90 h	6,357 kcal
shingles	160 parts	1,032 kg	18.96 h	5,514 kcal
insulation	160 parts	637 kg	14.58 h	4,220 kcal
wrap	160 parts	410 kg	12.55 h	3,638 kcal
solar	33 parts	404 kg	7.15 h	2,094 kcal
floor	22 parts	1,258 kg	6.07 h	1,615 kcal
interior	8 parts	767 kg	4.81 h	1,280 kcal
fixtures	7 parts	428 kg	4.10 h	1,090 kcal
water	11 parts	113 kg	3.06 h	813 kcal
column	3 parts	375 kg	2.42 h	660 kcal
power	8 parts	72 kg	2.42 h	642 kcal
hatch	3 parts	487 kg	1.73 h	459 kcal

And the total, in food, because until now it was on no drawing of the building:

```
the crew's total, in food:
       1,372  slices of bread at 80 kcal
       1,045  bananas at 105 kcal
          44  days of a 2,500 kcal diet
```

The parts that look like effort turned out to be cheap. The part that looks like nothing – holding a position while fastening – is almost the whole bill. So design the posture, not just the part.

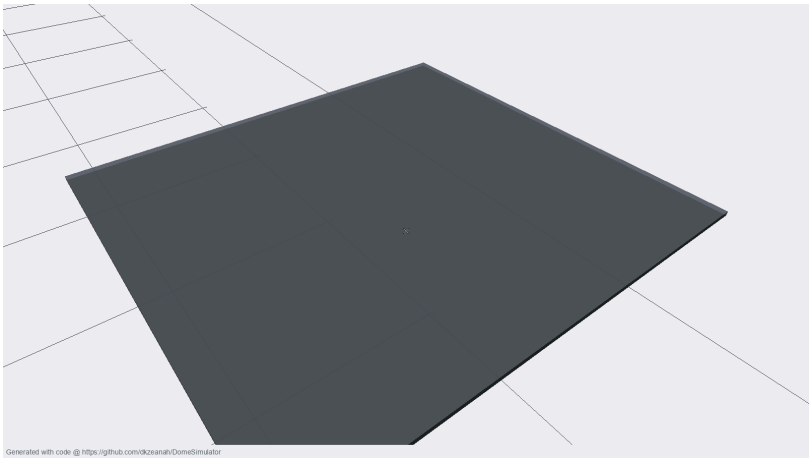
CHAPTER 24

One Flat Board, Forty Identical Panels

The only thing in the whole build that has to be true is the bench, and everything else is measured from it

The only thing in this entire build that has to be true is a sheet of plywood on two sawhorses.

Everything else is measured from it, and nothing on it is adjusted between panels — so panel forty is panel one.



One flat sheet on sawhorses, and nothing else.

THE 40-HOUR CLOCK

5.6 hours of panels on the jig – this book's estimate. The clock goes from 20.9 to **26.5 of 40**, leaving 13.5 hours of the week.

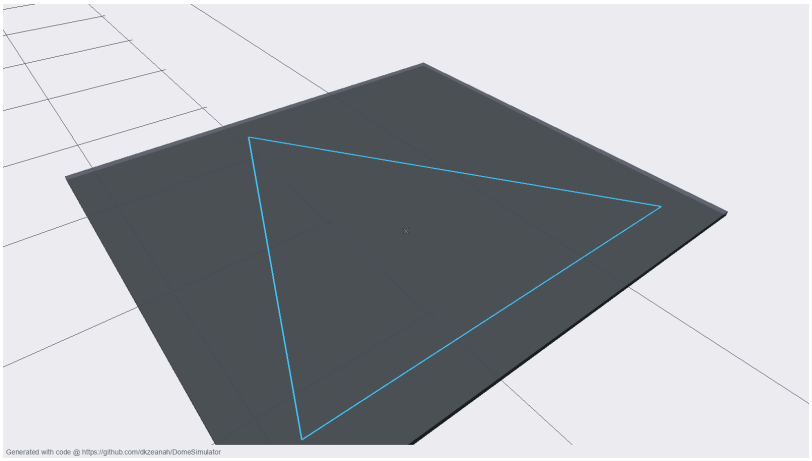
Building the jig

1. The bench

One sheet, flat, level enough that a marble does not run. Flatness is the whole specification: every angle below is measured from this surface, so the panel cannot wind or twist as long as the board does not.

Check it with a straightedge on both diagonals. If it rocks, shim the horses, not the sheet.

2. The triangle, struck once



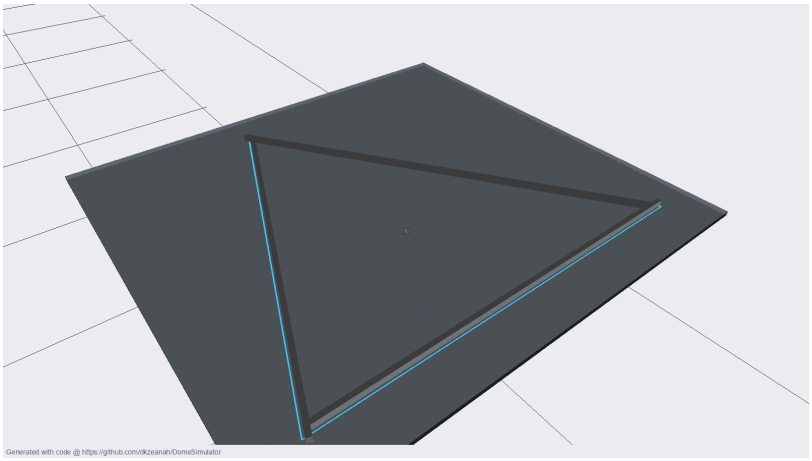
The panel's triangle, struck once from the solution and never measured again.

Snap the panel's mathematical triangle onto the board — the 72.000 and 63.670 inch sides of Chapter 16.

Strike it **one time**, from the geodesic solution, and never measure it again. Forty panels laid out from three lines struck once is a different proposition from forty panels laid out from forty tape measurements, and the difference is the entire reason this works.

The tool prints this triangle full size as a drawing you can lay on the sheet. Chapter 79 says how.

3. Axis rails



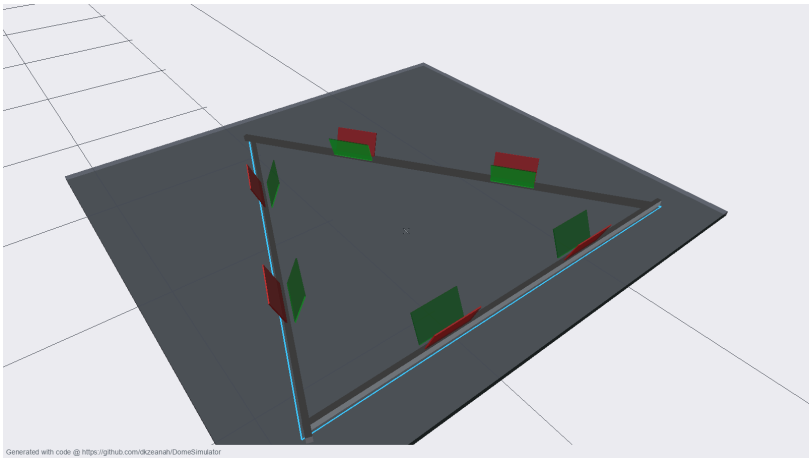
Three cleats, one per member, each parallel to that member's point-axis.

Three low cleats, one per member, each set parallel to that member's point axis and just to the seam side of it.

These fix **position across the panel**. Push the stick against its rail and the sector's point line sits exactly where the solution says.

The rail references the sawn point, never the bark. A lumpy log does not move the joint.

4. Red and green fences



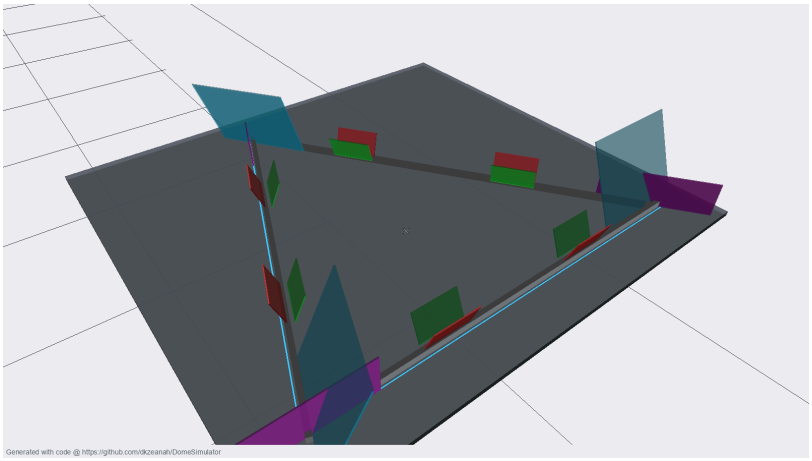
Red to red, green to green, or the stick does not seat.

Two short cradle plates per member, standing against the two sawn radial faces of the wedge.

These fix **rotation about the stick's own axis**, and they are the reason you cannot get a member in the wrong way round. A sector dropped in rotated ninety or a hundred and eighty degrees still *looks* fine. These plates make it not fit.

Mark them. Red side to red plate, green side to green plate. It feels childish for about two panels and then it saves you one.

5. The two cut planes



The magenta plate is the flush-cut fence; the cyan is the face the next butt lands on.

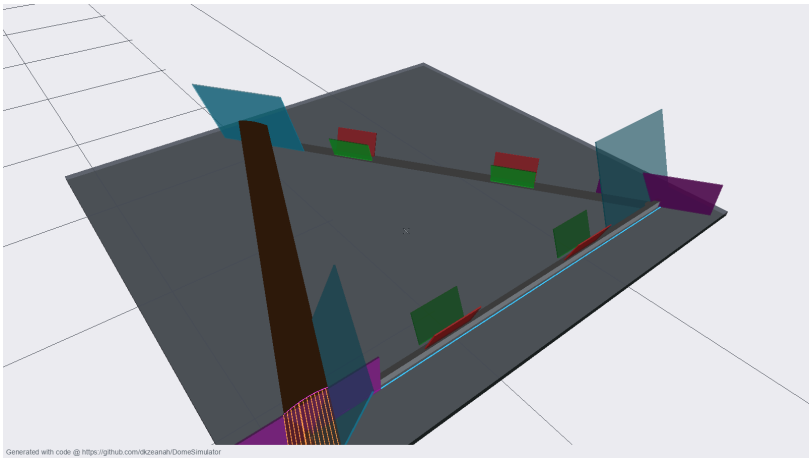
At each head end, a plate standing in the plane of the triangle's edge. That is the flush-cut fence: a saw run flat against it cuts the head to exactly the right plane without anybody setting a bevel or a mitre.

At each butt end, a plate in the plane of the receiving face the next stick's butt will land on.

That is the fixture. A sheet, three cleats, six cradle plates, six guide plates. Nothing on it moves again.

Making one panel

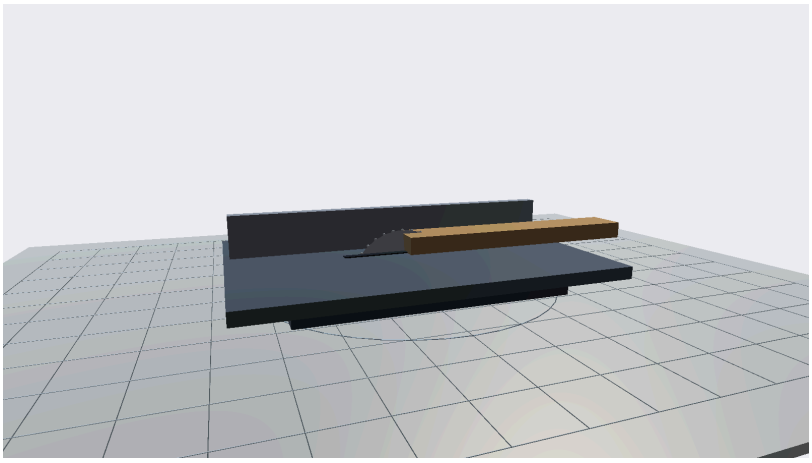
6. The butt cut, off the jig



The one cut made before assembly, and the only one that can be batched.

One stick, still full length at both ends, laid up to the cyan plate. Cut the butt.

This is the **only** cut made before assembly, and every stick of the same family takes the same one — so it runs through a saw bench in a batch rather than being fitted one at a time. Sixty A butts, then sixty B butts.



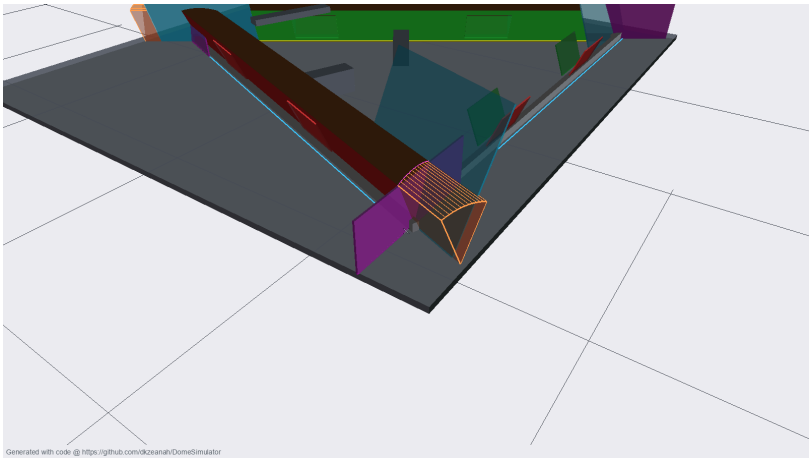
Setting the blade tilt.

7. First stick down

Butt now cut, dropped against its rail and its two fences. The head end is still long and hangs off the board.

Two ends, two completely different operations. The butt arrives finished. The head arrives long on purpose and you do not touch it yet.

8. Second stick, butt onto side

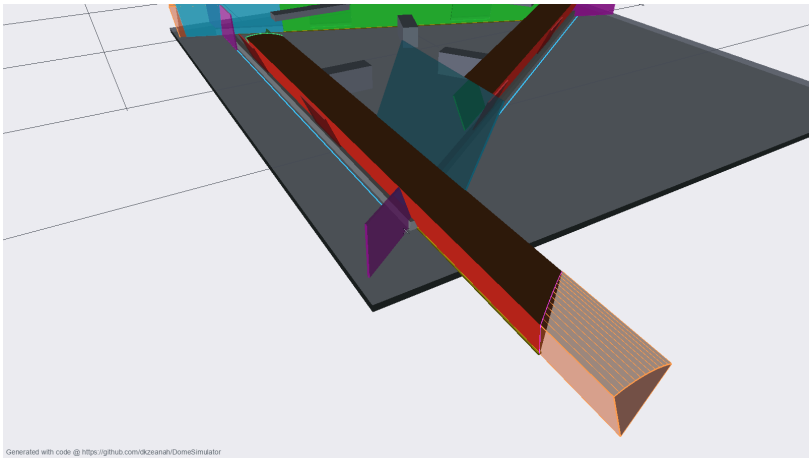


End to side: a flat sawn face on a flat sawn face.

The second stick lands so the first stick's butt bears on its **radial face** — on the side of it, not on its end.

Nothing is mitred to meet at a point. A flat sawn face is pressed onto a flat sawn face, which is what a split wedge already has without any further machining.

9. The last stick slides in



The last stick is captured at both ends, which is why the head is left long.

The third stick is captured at **both** ends once the other two are down, so it cannot drop in from above. It comes in along its own axis, sliding home into the pocket.

This is the one awkward move in the whole method, and it is why the head end is left long: the extra length is what gives the last stick somewhere to come from.

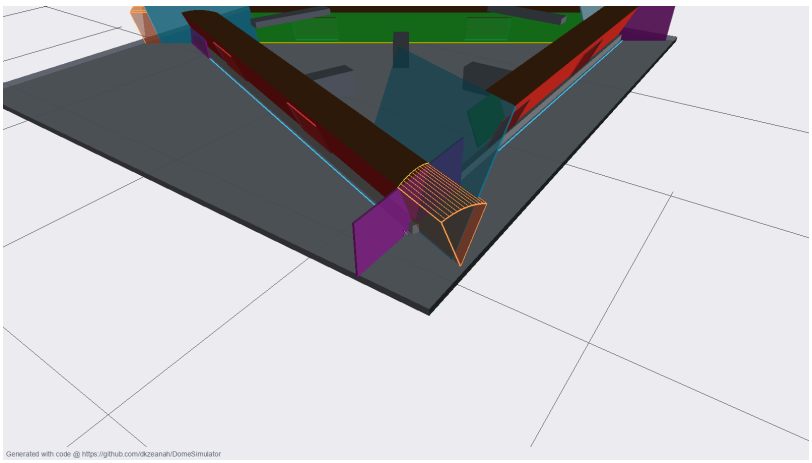
If you have trapped it, you put the sticks down in the wrong order. Take one out and start again; you will do this once.

10. Pull it up tight

Three butts bearing on three sides. The loop is closed, the frame holds its own shape, and the jig is now only keeping it flat while you finish.

All three heads still hang off long.

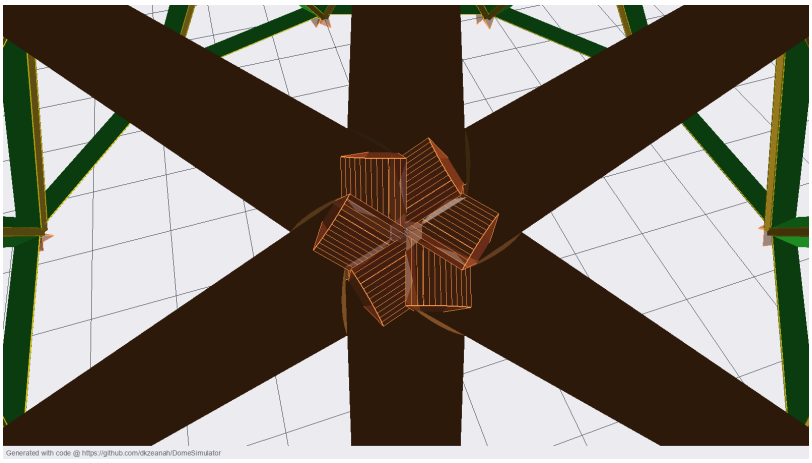
11. Flush-cut the three heads



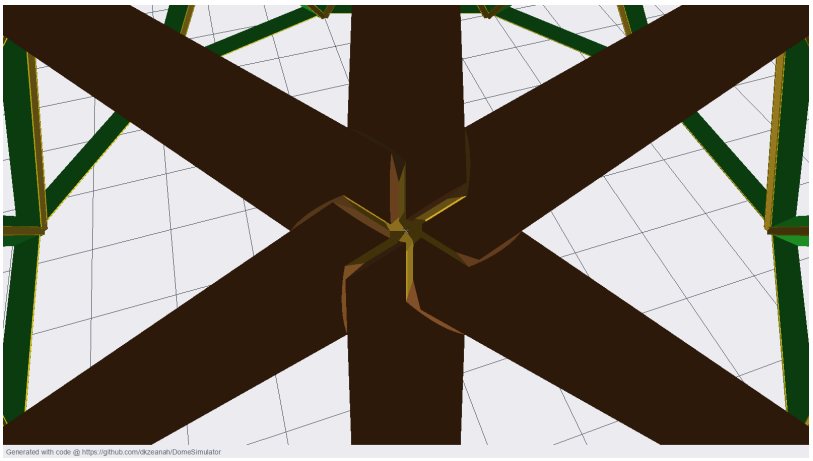
The head is never measured. It is cut in place.

Saw each head back to its magenta fence. The stock hanging past the board falls off.

This is the whole trick. The head is never measured. It is cut in place, to the plane the neighbouring member has already established — so error in stock length, or in the butt angle, or in how round the log was, leaves as offcut instead of accumulating around the triangle.



Uncut, every head trespasses across the vertex.



And after.

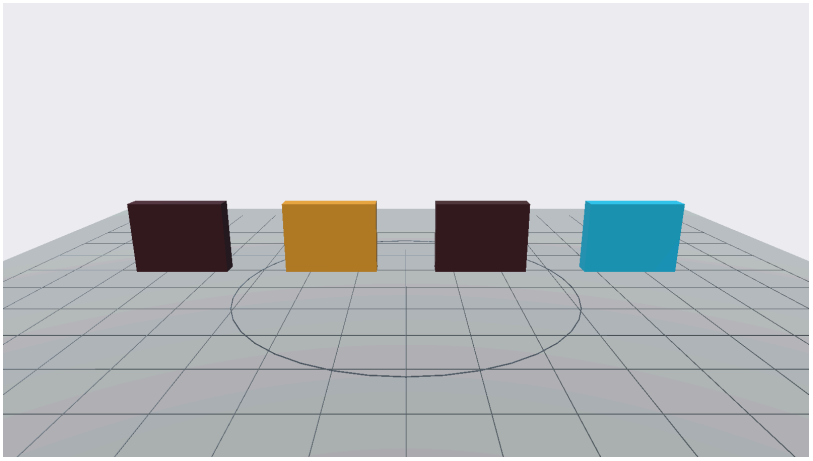
Those two pictures are the same corner of the same dome. In the first, every head still carries its stock and they trespass straight across the geodesic vertex into the neighbouring panel. In the second they end inside their own triangle and nothing crosses.

12. Panel off the jig

A finished triangular frame, its three corners clean inside the mathematical triangle. The jig is empty and unchanged.

Drill the inserts now, while it is flat — Chapter 3 — and sheathe it now if you are sheathing on the bench, which you should be.

Then do it 40 times, in two families: 10 equilateral and 30 isosceles.



Why the compound cut takes two saws.

What the jig is really buying

Repeatability, obviously. But more than that: it moves the hard work indoors and off the ladder.

Every operation above happens at waist height, on a bench, with both feet on the ground, in whatever weather. The part of dome building that everybody pictures — the awkward bit up a ladder holding something heavy at an angle — happens once, in Chapter 27, and takes three days out of a fortnight.

And it is worth being exact about what the fixture enforces, because it is only three things: where each member sits across the panel, which way up it is turned, and the plane its head is cut to. Everything else is free. Counted, the way the films count it:

```
40 triangles, 120 members, 120 corner joints -- all off one flat board
the fixture comes in 12 steps and holds three things:
  a RAIL per member, set to the sawn point-axis, not the bark
  a RED and a GREEN plate that make a wrongly-turned wedge physically refuse to seat
  a FENCE at each head end, which is the plane the saw runs on
each member arrives with ONE cut already made:
  the butt, cut to the angle its neighbour presents, batched
and one deliberate error: the head is left 6 in too long
three butts are pulled up, the loop closes, and only then is
  each head sawn against its fence -- unmeasured, in place
the offcut is 60 ft per dome = 8.3 % of raw length
and 40 panels need only 4 physically different fixtures
```

Every stick has some error in it. Cut both ends to a measurement and those errors add up around the triangle until the last corner will not close. Cut one end in place, against the neighbour, and the error has nowhere to accumulate: it leaves in the piece that falls on the floor. The offcut is what the accuracy costs, and it is cheap.

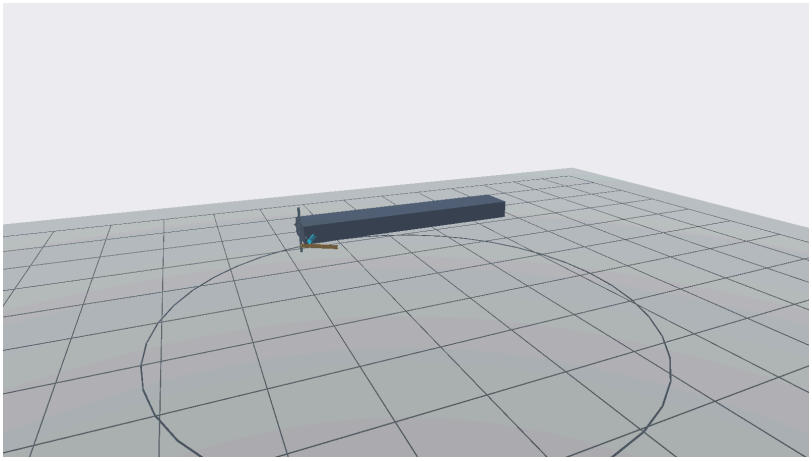
CHAPTER 25

If Your Sticks Are Rectangles

The same hubless dome from milled stock: two angles on every end, two machines, six saw settings, and a sled for the angles no saw reaches

Everything up to here cuts split log sectors. But the hubless frame – forty closed triangles bolted edge to edge – works with ordinary milled stock too, and plenty of people will start with a stack of two-by-fours rather than a stand of pine. This chapter is that version. Its joint is harder than the wedge's, and seeing why is the best argument for the wedge.

Every end of a rectangular strut needs two angles at once. The saw is **swung** to a mitre, so the strut meets its neighbour in the plane of its own triangle, and the blade is **tilted** to a bevel, so its face lies flat against the triangle next door. They happen on the same pass – that is what *compound* means – and getting one right while the other is wrong gives you a part that looks correct and fits nothing.



Two angles, one pass, every end.

Every setting the dome needs

The audit assumes:

```
stock 3.5 x 1.5 in, 10 in blade
mitre saw reaches 50 deg, table saw tilts to 45 deg
```

and the whole dome reduces to **6 distinct setups**:

```
6 distinct setups covering 240 strut ends:
  SET-1  mitre  55.569  bevel 11.229  x60
      table saw      blade tilt      11.229 deg
      mitre saw      table swing     55.569 deg   <- NOT REACHABLE
      crosscut sled  fence to blade  34.431 deg
      blade height for the rip: 1.5293 in (square would be 1.5)
      sled travel across the strut: 4.243 in
  SET-2  mitre  62.215  bevel 11.229  x60
      table saw      blade tilt      11.229 deg
      mitre saw      table swing     62.215 deg   <- NOT REACHABLE
      crosscut sled  fence to blade  27.785 deg
      blade height for the rip: 1.5293 in (square would be 1.5)
      sled travel across the strut: 3.956 in
  SET-3  mitre  60.000  bevel  9.015  x50
      table saw      blade tilt       9.015 deg
      mitre saw      table swing     60.000 deg   <- NOT REACHABLE
      crosscut sled  fence to blade  30.000 deg
      blade height for the rip: 1.5188 in (square would be 1.5)
      sled travel across the strut: 4.041 in
  SET-4  mitre  62.215  bevel  9.015  x50
      table saw      blade tilt       9.015 deg
      mitre saw      table swing     62.215 deg   <- NOT REACHABLE
      crosscut sled  fence to blade  27.785 deg
      blade height for the rip: 1.5188 in (square would be 1.5)
      sled travel across the strut: 3.956 in
  SET-5  mitre  60.000  bevel  0.000  x10
      table saw      blade tilt       0.000 deg
      mitre saw      table swing     60.000 deg   <- NOT REACHABLE
      crosscut sled  fence to blade  30.000 deg
      blade height for the rip: 1.5000 in (square would be 1.5)
      sled travel across the strut: 4.041 in
  SET-6  mitre  62.215  bevel  0.000  x10
      table saw      blade tilt       0.000 deg
      mitre saw      table swing     62.215 deg   <- NOT REACHABLE
      crosscut sled  fence to blade  27.785 deg
      blade height for the rip: 1.5000 in (square would be 1.5)
      sled travel across the strut: 3.956 in
```

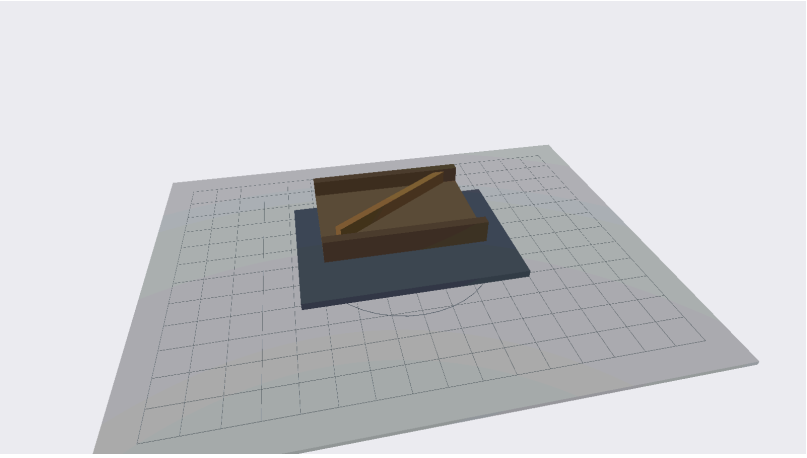
Read the right-hand column. **Not one of the mitres this dome asks for is on a common mitre saw's scale** – they all lie past its 50-degree stop. The geometry is easy; the tool is the problem.

The way through is that a mitre and its complement are the same cut approached from the other face. Swing to the complement instead – the sled column above – and turn the workpiece a quarter turn in a sled. Same joint, reachable setting. And each machine can only make the cut the other one cannot. The table saw rips the bevel down the strut's whole length, tilted to at most 45 degrees, which no mitre saw can do; the ends are crosscut on a mitre saw where its swing reaches, and on a sled fenced to the complement where it does not. Two triangle shapes means two assembly jigs, and the six settings cover every end of all 120 struts.

The sequence

1. Rip every stick to one width Every angle after this is measured from the edge of the board. If the boards are not all the same width, nothing downstream can be right, and no amount of care at the mitre saw will rescue it. Rip the whole pile in one session, with the fence untouched. **2. Mark the mating edge** Only one long edge of each strut gets the bevel: the one that will lie against the neighbouring triangle. Mark it on every stick before any of them go near the blade. This is the single most common way to produce a pile of mirror-image scrap. **3. Set the blade tilt** Set the tilt to the bevel for that strut class. Do not trust the saw's scale: rip two offcuts, put the cut faces together, and measure the pair. The pair reads twice the error, so anything you cannot see in the single is obvious in the double. **4. Rip the bevel, full length, one pass** Featherboard before the blade, push stick past it, and keep the marked edge against the fence for every stick without exception. This cut decides how the triangle meets its neighbour. **5. Build the sled before you touch the ends** The mitres this dome wants are past the stop on a common mitre saw, so the ends are cut on a crosscut sled whose fence is built to the complement and referenced off the blade instead of off square. Build it, prove it with the five-cut method, and only then cut a part you care about. **6. Cut the relief lap into the sled fence** The strut sits against the sled fence with one end projecting past it. The blade has to finish that cut flush, so the fence needs a lap cut away at the blade path -- deep enough for the blade to pass the full width of the strut, no deeper. Cut it once, with the sled in the position it will always be used in. **7. Set the stop block, not the tape** Length comes from a stop block clamped to the sled, never from a pencil line. Two hundred and forty ends measured individually will drift; two hundred and forty ends against one

block will not. **8. Cut the first end** Bevelled edge down and against the fence, marked face up, strut butted to the stop, end projecting over the lap. One pass, all the way through, and let the blade stop before lifting. **9. Turn the strut, do not flip it** The second end is not a mirror of the first. Rotate the strut end for end about its long axis in the sled -- a quarter turn, the way the sled was built for -- so the bevel stays on the same side of the dome. Flipping it face over face puts the bevel on the wrong side and the strut will not close its triangle. **10. Batch by setting, never by triangle** Cut every end that shares a setting before you change anything. Changing a setting is where error enters, so change it as few times as the job allows -- six times, not two hundred and forty. **11. Dry-assemble one triangle before cutting the rest** Three struts, no fasteners, laid on a flat floor. If it closes with no gap at any corner and lies flat, the settings are right. If it does not, you have lost three sticks instead of the whole pile.



The crosscut sled, fenced to the complement.

Proving it

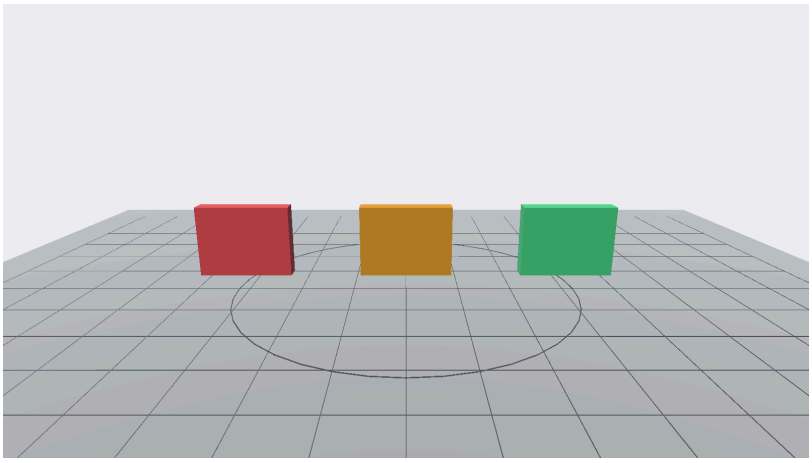
Two checks, because a saw's own scale is not to be trusted.

The tilt. Rip two offcuts at the setting, put the cut faces together and measure the pair. The pair doubles the error, so anything invisible on one is obvious on two:

```
bevel verification, by pairing two offcuts:
  bevel 11.229 -> the pair must read 157.542 deg
  bevel 11.229 -> the pair must read 157.542 deg
  bevel  9.015 -> the pair must read 161.970 deg
  bevel  9.015 -> the pair must read 161.970 deg
  bevel  0.000 -> the pair must read 180.000 deg
  bevel  0.000 -> the pair must read 180.000 deg
```

The sled fence. The five-cut method trims one board on all four sides and then once more, and the last strip's taper is four times the fence's error – so an error far too small to see becomes one a caliper reads:

```
five-cut check on the sled fence:
  0.001 in across a 20 in strip -> fence off by 0.0007 deg = 0.00004 in across a strut
  0.005 in across a 20 in strip -> fence off by 0.0036 deg = 0.00022 in across a strut
  0.020 in across a 20 in strip -> fence off by 0.0143 deg = 0.00088 in across a strut
```



An invisible error, made measurable.

The five ways it goes wrong

Every one of them looks correct until assembly:

- **Bevel on the wrong edge.** The strut looks correct and mates the wrong way. Marking the edge before ripping is the only reliable prevention. - **Complement confusion.** Setting the sled to the mitre instead of its complement gives a cut that is wrong by exactly the amount that looks plausible. - **Flipped instead of turned.** Flipping face over face between ends mirrors the part. It will look right on the bench and refuse to close a triangle. - **Stop block creep.** A clamped block walks under repeated impact. Check it against a reference stick every twenty cuts. - **Blade not through.** A tilted blade needs more height to clear the same thickness. Set the height after the tilt, never before.

So the whole sequence is: two machines, one sled, 6 settings, and every end of every strut batched by setting rather than by triangle. Six setting changes instead of a change for every end – which is the same lesson as the wedge, arrived at the hard way. The wedge simply never asks for most of these cuts: its bevel is the split, and its head is cut in place (Chapter 24).

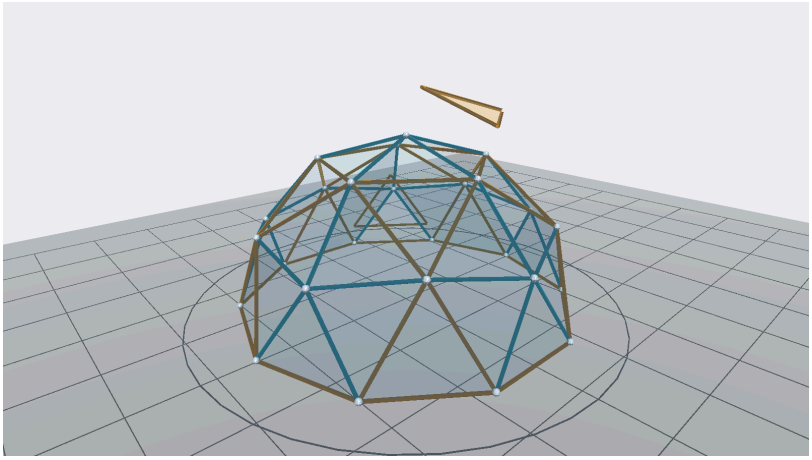
CHAPTER 26

Doors, Windows, and What Not to Cut

An opening is a panel you did not close -- which is why it costs nothing, and why cutting one afterwards costs everything

An opening is a panel you did not close.

That is the whole rule, and every expensive mistake people make putting holes in domes comes from not believing it.



Take whole panels. Cutting a strut means replacing a load path.

> **NOT ON THE CLOCK**
>
> Doors and windows are not on the 40-hour clock. The clock counts the frame, from the first log on the splitting stand to the last bolt in the shell – Chapter 27 adds it up. They come after the shell stands, and they are the easiest work in the building to do slowly.

Why a whole panel

Chapter 6 said a triangulated shell carries load because the triangles cannot change shape. Take one triangle out and the ones around it close the gap between themselves — the load goes round, the way it goes round a missing tile rather than through it.

Cut a **strut**, and you have not made a hole. You have removed a member from three triangles at once and given the shell nowhere for that load to go. Now you need a header, and posts either side, and something to carry what the header carries, and you are framing a wall in a building that did not have walls.

So:

Take whole panels. Never cut a member.

A panel left out is free. A panel cut into is a structural alteration.

Where the door goes

Ten panels stand on the base ring, so the ground floor has ten places a doorway can be, and they are all the same.

That is the first surprise for anybody used to rectangular buildings: you do not choose where the door goes on a plan, you choose *which panel*, and there are ten.

Take two adjacent first-course panels for a comfortable door. One is enough for a doorway you duck through; two gives you a proper opening with room either side.

Frame the perimeter, do not frame the hole. The three members bounding each missing panel are already there and already doing their job. The door frame hangs *inside* them, on the same threaded inserts of Chapter 3, and carries nothing but the door.

The threshold is the base ring. It is already level, already continuous, and already the thing the floor meets. Do not build a sill on top of it.

The sequence

1. **Decide after the shell is up.** This is the one decision in the whole build that gets better for being left late. Stand inside, see where the light actually comes from and where you actually walk, then pick the panel.

2. **Leave the panel out, or take it out.** If you are still raising, simply do not fit it. If the shell is closed, undo its keys and lift it out — the whole argument for panels being objects, collected.

3. **Fit the jamb inside the triangle.** Three sides, following the members, bolted to inserts rather than screwed into end grain.

4. **Hang a door in a rectangle inside that triangle.** Doors are rectangular; triangles are not. The leftover triangle above the head is either a fixed light or infill — not a vent: the dome breathes through its seams, with no opening in the cap (Chapter 34) — and a fixed light is the best answer because it puts daylight above head height where it does the most good.

5. **Flash it under the cap.** The cap of Chapter 28 runs over the whole shell and the door is a hole in the shell, so the cap has to be returned into the opening and terminated on the frame. This is the one place in the building where the single watertight layer is interrupted, and it is where a dome leaks if it leaks.

Windows, and why they go high

The wall leans in. That changes everything about glazing.

A window low in a dome looks at the ground a few feet away and lets in very little light, because the aperture is tilted toward the floor. A window high up looks at the sky, which is the bright thing, and it is why every dome photograph you have ever liked has the glass up near the crown.

So: **put glazing above head height** unless you need a view of something specific.

The other consequence of the lean is that every opening in a dome is partly a roof opening. Water arriving at a window in a wall runs down the wall; water arriving at a window in a leaning surface runs *onto the glass and sits at the bottom edge*. Detail every window as you would a rooflight — head flashing over, sill drained outward, and the cap returned properly — not as you would a window in a house.

The panel you do not take

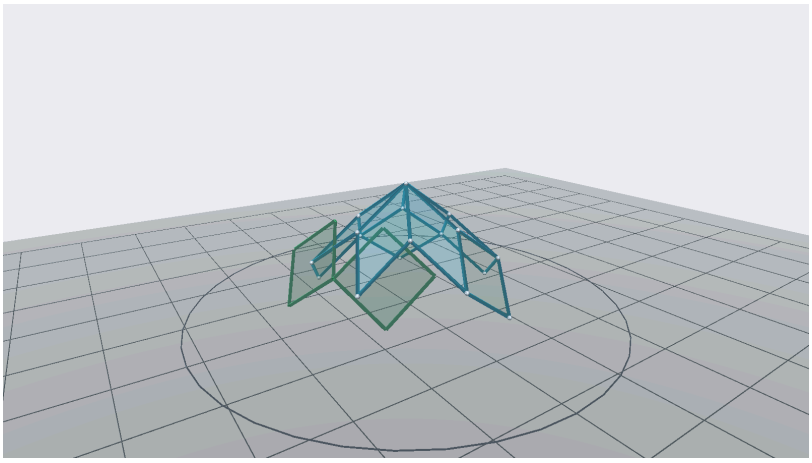
One warning, from the geometry rather than from experience.

The first course is ten panels and they are the ones standing on the ring. Taking two adjacent ones is comfortable. Taking three in a row is the point where the ring between them has nothing above it for a third of its length, and the base ring is the one part of this building that is genuinely carrying in bending rather than in compression.

Three is probably fine. This book does not know that it is fine, because nothing here has been load-tested, and a building with a third of its ground floor open is a different structure from the one that was solved.

If you want a wide opening, take two panels and put the third somewhere else.

Doing it on a zome



Take out a whole panel and no strut needs cutting.

The same answer, on a shape built the opposite way.

A zome's panels are bigger — one parallelogram is a lot of opening — and the rule is identical: take the whole panel, frame inside the struts, cut nothing. That the rule survives a completely different geometry is a good sign that it is about triangulated shells in general rather than about this dome.

What this chapter is not

It is not a schedule of door and window sizes, because those come from what you can buy where you are and from a panel whose dimensions are 52.516 inches across at the narrowest — Chapter 11.

And it is not permission. An opening in a load-carrying shell is exactly the kind of thing an engineer exists for, and the rule at the top of this chapter is what keeps you out of their office rather than a substitute for it.

PART 5

The Skin

“A frame that is out by an inch stands for fifty years. A skin that leaks for one winter takes the frame with it.”

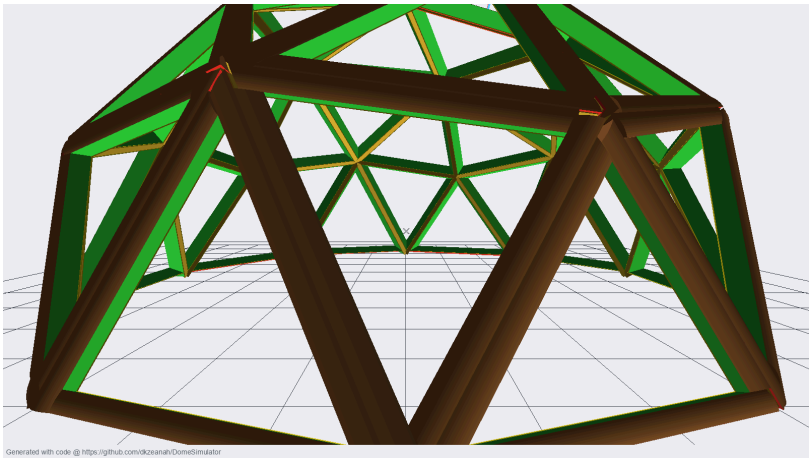
Raising the shell, the single watertight layer over it, and the hats that go on afterwards -- the part of the building this book says matters most.

CHAPTER 27

Raising It

Twenty panels round the ring, fifteen above them, and five around the apex -- in that order, and braced until the second course closes

Forty panels are stacked against a fence. This is the shortest stretch on the clock and the only part that looks like building a dome – spread over as many days as your helpers and your weather allow, because the bracing waits for the second course to close.



The ten panels that stand on the ring.

THE 40-HOUR CLOCK

3.5 hours of raising – this book's estimate. The clock goes from 26.5 to **30.0 of 40**, leaving 10.0 hours of the week.

Then the hours nobody plans for: **7.5 hours** for the stick that splits off the radius, the panel that goes up twice and the drill nobody can find. The clock stops at **37.5 of 40**.

stage	hours	clock	basis
splitting the logs into wedges	16.0	16.0	measured
cutting both ends and drilling	4.9	20.9	estimate
bolting members into panels on the jig	5.6	26.5	estimate
lifting the panels into a shell	3.5	30.0	estimate
the allowance for getting it wrong	7.5	37.5	allowance
the week		40	

Before anything goes up

Set the base ring out first. A regular decagon, 10 sides, each one an A chord — so each side is 72.000 inches and the perimeter is sixty feet. Chapter 38 is what it sits on.

Set it out from the centre, every time. Drive a pin where the middle of the floor will be and swing the base radius from it to mark all 10 corners. Do not step round the ring measuring side to side: every small error then adds to the next one.

Get this right and the rest is assembly. Get it wrong and every course above it inherits the error, because the base is the one place in this building where error does accumulate.

Check it by diagonal, not by angle. On a regular decagon the long diagonals, corner to opposite corner, are all the same length – so measure every one of them, and any difference tells you exactly where the setting out drifted.

Sort the panels into two piles. 10 equilateral, 30 isosceles. Mixing them up costs an hour on the second course.

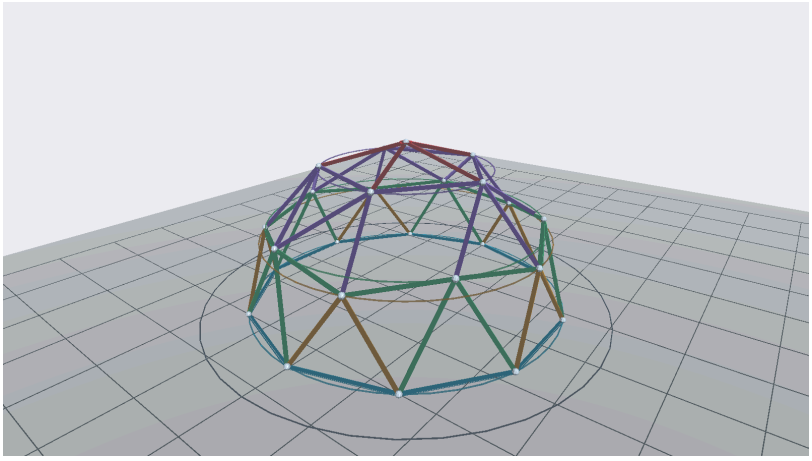
Have more bracing than you think. You will need it for about six hours total and you will need all of it at once.

Four rings and a crown

The corners of the shell sit at fixed heights, and those heights are your check numbers during the raise. Write them on the drawing before you start:

ring	corners	height	across
floor ring	10	0.00 ft	19.42 ft
belt, low corners	5	4.34 ft	17.37 ft
belt, high corners	5	5.10 ft	16.52 ft
pentagon ring	5	8.26 ft	10.21 ft
apex	1	9.71 ft	--

The two belt rows are one zigzag ring that rises and falls between two heights (Chapter 33). Every ring is a level circle you can measure across.



Twenty-six corners, in courses, at heights you can measure.

The first course

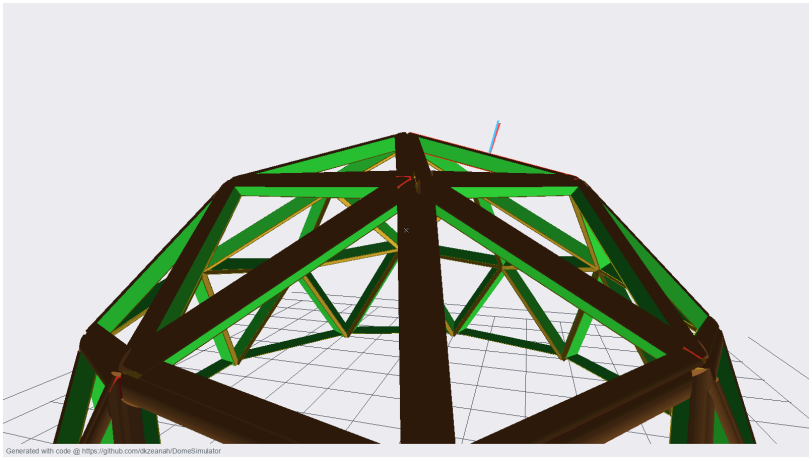
20 panels: 10 standing on the ring with two corners on it, and 10 inverted between them with one corner on it. They lean inward from the moment they go up, which is the first surprise: a dome is not a wall that gradually curves, it is leaning in from the ground.

Stand two adjacent panels, key the seam between them, and brace the pair externally. Then work round. Each new panel keys to the one before, and the run of them wants to fall inward until the ring closes.

The ring closing is the event. Until the tenth panel keys to the first, you have a curved wall held up by props. After it, you have a structure.

Do not take the bracing out yet.

The second course



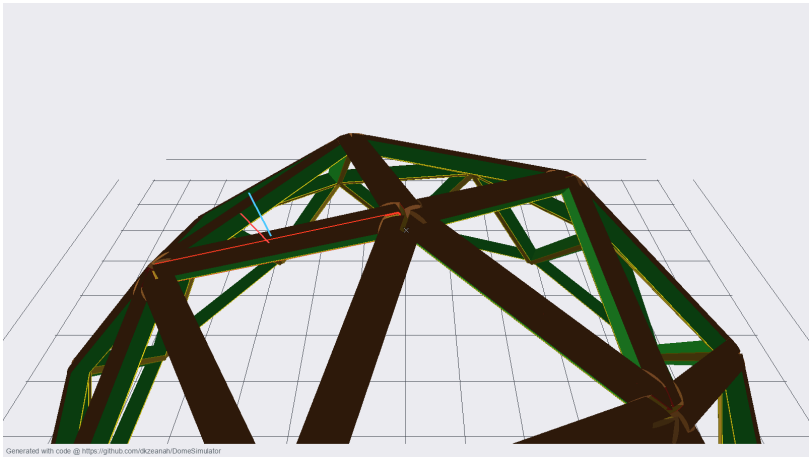
The band above: where the dome starts holding its own shape.

15 panels, leaning further in, each one keying to the course below and to its neighbours either side, up to the pentagon ring.

This is where the dome starts holding its own shape — and it is where the bracing can come out, once the course is closed all the way round. Not before. A half-closed second course is the least stable the building will ever be, because you have added weight and height without adding the ring that resolves it.

Work from opposite sides toward a closure point rather than round in one direction. It keeps the load symmetric and it means the last panel of the course is a panel rather than a wedge-shaped gap you have to persuade.

The apex



Closing the top.

5 panels around one vertex.

This is the last thing to go in and the only part of the assembly that has to be *held* rather than propped. Two people, or one person and a strap from below.

If you have built the mast of Chapter 72, the apex ring is already up there and the last five panels close around it. If not, the apex is five keys and a cap.

What goes wrong

The last panel of a course does not fit. Almost always the ring is out, not the panel. Check the base diagonals before you start cutting anything. The crown is where every accumulated millimetre finally arrives, so it reports the error rather than causing it: if the cap will not close, go back down and measure the ring beneath, which is nearly always slightly out of round or out of level.

Counting the courses wrong. An earlier draft of this chapter said ten panels in the first course, thirty in the second and five round the apex: forty-five panels, on a dome that has 40. The courses above are counted from the solved shell.

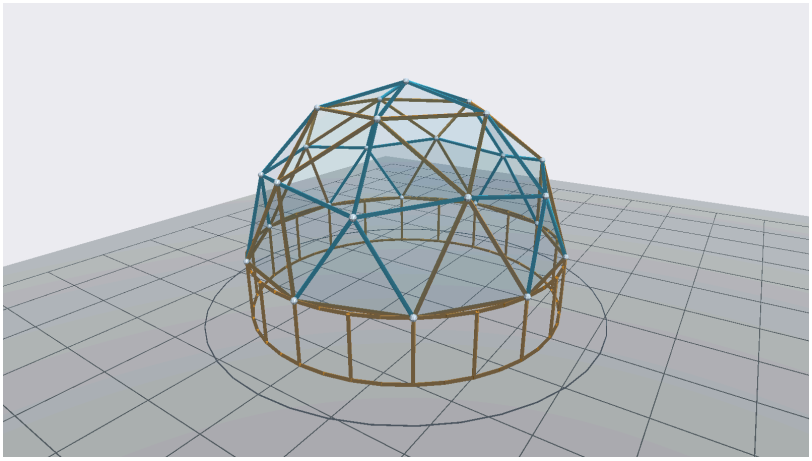
A seam gaps at the outside. The key is not driven far enough. It is a wedge; drive it.

A panel will not key to its neighbour. One of them is in upside down. This happens with the isosceles ones, whose two short sides look alike. Mark the long edge of every BAB panel on the jig, in the shop, before it ever comes outside.

The riser wall

A hemisphere has no headroom at its edge: the shell meets the floor at the rim, which makes the outer ring of floor nearly useless. Stand the whole dome on a short vertical wall and every point of the shell rises by that amount, so the perimeter becomes furniture height, or door height.

The floor area does not change. The usable area does, a great deal, and it costs one ring of studs. Decide on it before you set out, because it changes the foundation (Chapter 38).



The riser lifts every point of the shell by its own height.

Then stop

The frame is up. It is weather-tight in the sense that a bird cage is weather-tight, and it will stand like this indefinitely.

Do not stop here for a winter. That is the failure Chapter 3 describes, and this is the moment it happens: the frame is standing, it looks finished, it is intoxicating, and the skin is the expensive part.

The skin is the part that matters. It is the next chapter, and it is the reason all of this stands up in thirty years' time instead of in three.

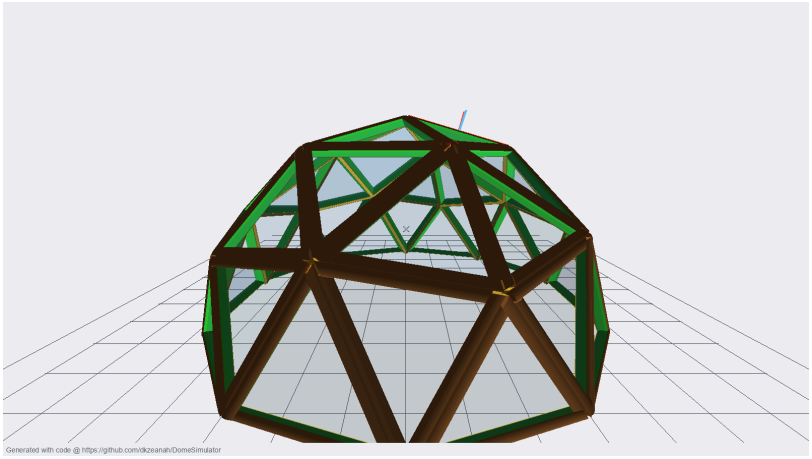
CHAPTER 28

One Watertight Layer, and Only One

Two impermeable skins with insulation between them is a trap, and this project shipped that design before it noticed

Put a waterproof layer on the outside of a building and you have kept the rain out.

Put a second one on the inside of the same wall and you have built a bag.



The frame with its skin on.

> **NOT ON THE CLOCK**
>
> The cap is not on the 40-hour clock. The clock counts the frame, from the first log on the splitting stand to the last bolt in the shell – Chapter 27 adds it up. It is the part that has to be right, and it gets its own days, not the end of somebody's week.

The design this replaces

The obvious shell is a waterproof membrane *under* the panels and a rain-slick cap *over* them, with the insulation between the two. Two skins, belt and braces.

It is a vapour trap. Water gets into every wall — from the weather, from the air inside, from somebody's breath — and the only question a wall design answers is where it goes next. A wall with an impermeable layer on both sides has no answer. Moisture arrives, cannot leave in either direction, and condenses on whichever surface is colder that night.

So: **one watertight layer, on the outside, like a shower cap** — and under it, a route for vapour to leave. The rest of this chapter is that route.

The stack, outside in

The cap. One continuous rain-slick layer over the entire shell. Welded seams, no fasteners through it, strapped down to ground anchors at the base vertices. Its only job is liquid water and it is very good at that job because it is doing nothing else.

A vented gap. Three quarters of an inch between the cap and what is under it. This is the part people skip and it is the part that makes the rest work: any moisture that gets past the cap, or arrives from inside, has somewhere to go and something moving through it to take it away.

The breather. A vapour-permeable sheet at around ten US perms. Not a poly sheet. It stops bulk water and lets vapour through, which is the exact opposite of what a waterproof membrane does and is why the distinction matters more than the price difference.

The quilted layers. As many as you have. Chapter 30.

The panels. Wood. Structural, not weatherproof, and not asked to be.

The frame. Dry, because everything above it is doing its job.

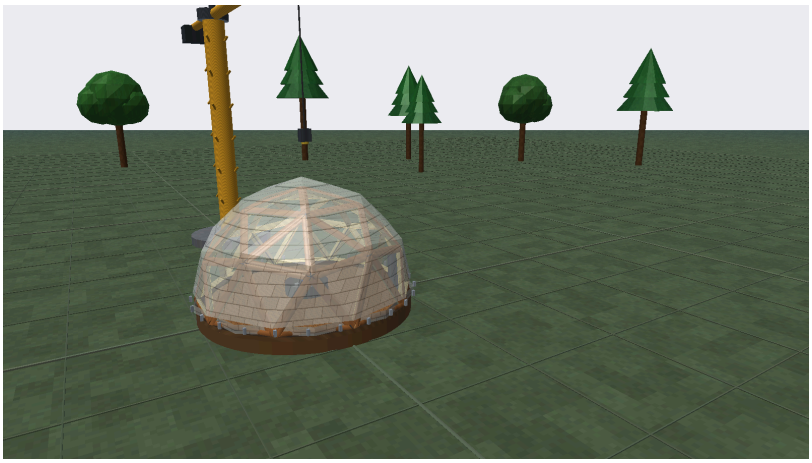
Which way the vapour goes

Warm air holds more water than cold air. Inside your building the air is warm and full of you, your cooking and your breathing. Outside it is cold.

So vapour moves outward through the wall, and it will condense at whatever point in the wall is below its dew point. The design question is: **can it get out before it gets cold?**

With a breather and a vented gap, yes — vapour passes through the permeable layer into moving air and leaves. With a second impermeable layer, no. It stops at the membrane, at a temperature partway through the insulation, and condenses there, inside the insulation, where you will not see it for years.

That is the whole argument, and it is why the cap is the only waterproof thing in the building.



Four ways to skin the same frame, and what each one costs.

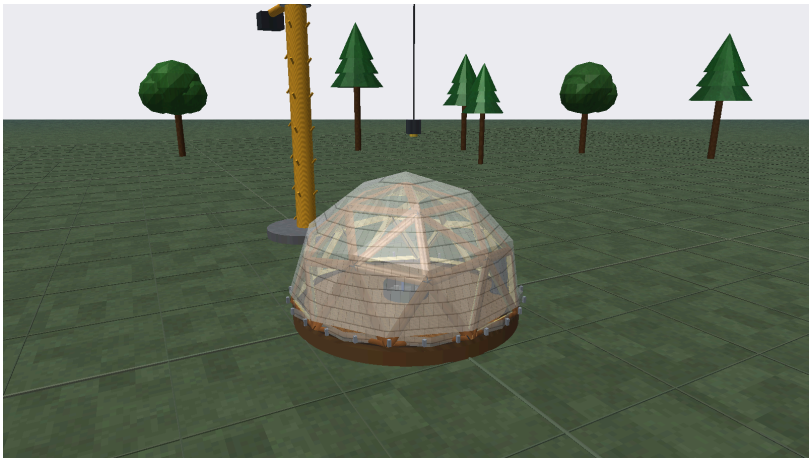
What it costs

A bare cap — no quilted layers at all — is $3,265 on this dome. With layers it climbs slowly; the shell in the standard article is $3,265.

The alternative is a laminated hull: glass fibre in resin over the whole envelope. Genuinely waterproof, sixty years, and $21,513 for the same dome against $12,036 — a real premium for a real difference.

Chapter 57 prices four laminates and the cap side by side. The short version: the hull is better at being waterproof and worse at being changed.

It is not exotic. It is a wood-cored composite skin – a sheet core, chopped strand mat, stitched biaxial cloth, laminating resin, and gelcoat on the weather face – which is how small boats have been built for sixty years, from products you can order this afternoon. In the stem-cell product of Chapter 62 it latches to the pad with over-centre catches, the kind that seal a preserving jar, and lifts off whole from a ring laminated into its rim. So it can come off. What it cannot do is change: a hull is one stiff, finished shape, so the quilted layers of Chapter 30 have no room to grow under it, and when it fails it fails as a repair on a bonded laminate rather than as a component you unstrap and replace.



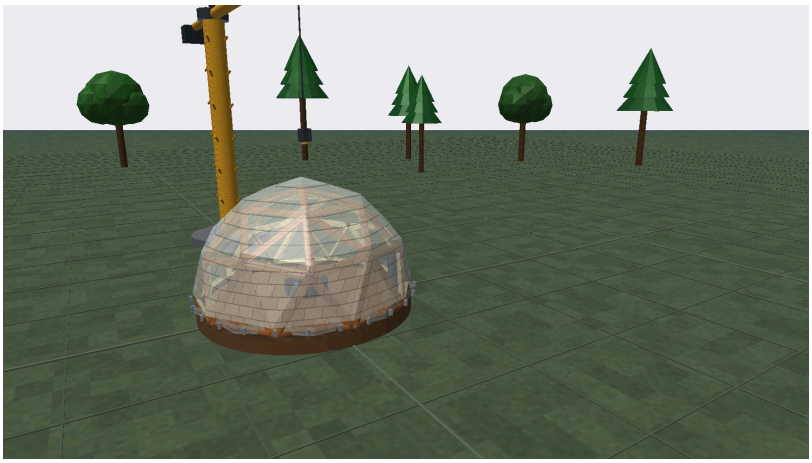
The shell is a boat hull.

The three things a cap has to get right

It has to be one piece, or welded. A cap made of overlapping panels is just a roof with more seams, which is the problem it exists to remove.

It has to be strapped, not fastened. Every fastener through a waperproof layer is a hole in a waterproof layer. Straps over the outside to anchors at the base put the load into the ground without putting anything through the membrane.

It has to be bigger than it needs to be. Chapter 3: a cap cut to fit a bare shell will not go over a shell wearing two quilted layers.



A thing we found out the hard way: no panel fits a four-foot sheet.

If you skin it in panels anyway

Some builders will sheathe the frame panel by panel and waterproof the panels, and the rules for that are old ones. Start at the rim and work up, so each panel laps over the one below it like a shingle and sheds water downhill. Every corner where several panels meet is a leak point: flash or tape each one before the next course covers it. And give the base a real drip edge that throws water clear of whatever the dome stands on. A dome almost never fails structurally; it fails at its seams.

For a small dome the panels can be the whole building. Cut the 40 triangles as solid sheet, pair the congruent ones into parallelograms and strip-pack them: on one 10 ft x 5 ft sheet the largest dome that fits is 62 inches across, with 21 square feet of floor. A shelter, a kennel, a seed store. Not a house, and it does not pretend to be.

One shell, any skin

The bones do not care what is painted on them. The same panels and the same 120 sticks wear a baseball, a basketball or a disco ball with a facet on every face. And be honest about what a “wall module” is: plywood, insulation and sheet metal, in a frame that holds them. The value is not in the materials, which anybody can buy. It is in the socket they drop into (Chapter 21).

The part that is not solved

Ultraviolet.

A rain-slick outer layer sits in the sun for its whole life, and that is what eventually finishes it — not water. The cap is a consumable on a decade-ish timescale, and the design’s answer is that it is a consumable **by design**: unstrap it, pull it off, put the next one on, in a day, with the building occupied throughout.

That is a genuinely different answer from a roof, and it is the right one for a component whose failure mode is known and gradual. It is still a cost, it is still a thing you will do four or five times in the life of the frame, and this book would rather say so than let you find out in year twelve.

CHAPTER 29

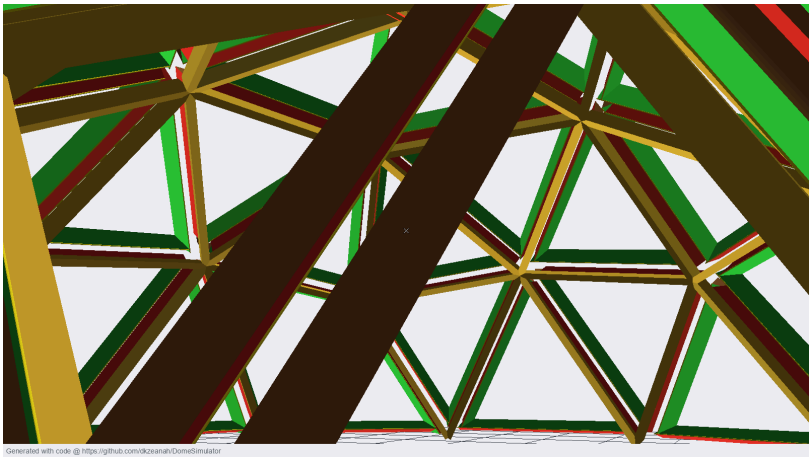
The Seam Does Four Jobs

Gutter, vent, dehumidifier and air barrier -- all of it in the space two sawn faces were always going to leave

The channel is already there.

Two sawn faces meeting at a dihedral angle cannot close flush — Chapter 20 — so every one of this dome's 55 seams carries a void running its whole length. 309 feet of it, threaded through the structure, touching every panel, arriving at the apex and at the base ring.

You are going to fill it with something. The question is what.



Every seam in the dome, and the channel in each one.

Four jobs, one space

Rain in at the ridge. A slot along the outer cap takes water where it already runs — the seam is the low line between two panels, which is where a roof sends water anyway. Under the slot is a gutter with suction inlets, and under that a drain to the tank. The building's guttering is inside its structure rather than bolted onto its edge. *(Chapter 33 corrects this: rain belongs in a gutter at the rim, outside the frame.)*

Air along the wood faces. A perforated liner washes air down both sawn faces of every member. That is the surface most at risk in a timber frame and the hardest to reach, and it is now the surface with moving air on it permanently.

A desiccant leg. A three-way gate routes the airflow either along the faces, or through a silica cartridge, or isolates the water path from the air path entirely. In wet weather when moving air will not dry anything, you pass it through the desiccant instead.

And a plate that condenses on purpose — the part that needs its own section, below.

The reason all four fit is that they want the same thing: a continuous, serviceable void that runs everywhere and reaches the ground. Nobody would build that void for any one of these jobs. It was free.

The air barrier

Here is the part that changes how you think about leaks.

Run the fan so the interior sits slightly **above** outside pressure, and the net flow at every gap in the envelope is outward.

A gap in an inward-leaking building admits weather. The same gap in an outward-flowing one does not — not because it is sealed, but because the air is going the wrong way for anything to ride in on. Wind-driven rain at a seam meets air coming out of it.

The numbers are undramatic, which is the good news:

```
interior volume        1,916 cu ft
at 0.55 air changes an hour   18 cu ft a minute
held above outside     4 Pa
spread over            309 ft of seam
```

4 pascals is far below what anybody feels on a door, and 18 cubic feet a minute is a bathroom extractor. Two fans — 2 — so the barrier can be reversed, because there are days when you want to draw rather than push.

What it costs honestly: it runs continuously, and it only works while it runs. A positive-pressure envelope with the fan off is an ordinary envelope with holes in it. That is an argument for sizing the array in Chapter 82 so this load never has to be shed, and it is an argument against treating the barrier as a substitute for the cap of Chapter 28. It is a second line, not the first.

Condensing on purpose

Thermoelectric plates — Peltier modules — on the cold side of the channel, dropping moisture out of the air as liquid straight into the gutter the rain already uses.

The appeal is obvious. The plate sits exactly where the moisture is highest, it is doing dehumidification whether or not you want the water, and the water it makes needs no new plumbing because the drain is right there.

Now the number.

one inch of rain on this roof	185 gal
the plates, running a day	0.21 gal (651 W)
the plates, running a year	76 gal
which is the same as	0.4 inches of rain
energy per litre	4.98 kWh

> **THE NUMBER THAT DOESN'T HELP**
>
> **A whole year of condensing is worth less than half an inch of rain.** These figures are solved from the owner's own module in Chapter 37; an earlier draft assumed a better yield than the module gives.
>
> At 4.98 kilowatt hours a litre, thermoelectric condensing is several times worse per watt than a compressor, and a compressor is already not how anybody sensible makes drinking water on a roof.

So the honest position, and it is the one this book takes:

The plates are a dehumidifier that happens to yield liquid. They are not a water supply. The water supply is the roof, and the roof is 185 gallons an inch.

Which leaves them worth fitting for one reason only: there are conditions — warm, humid, still, no rain for weeks — where the channel will not dry itself and the desiccant is saturated, and in those conditions a plate is the only thing in the design that can actively remove water from the timber. Run them on daylight surplus, treat the drips as a bonus, and never size a tank on them.

What the module costs

outer cap with intake slot	309.2 ft	$3.40	$1,051
vented liner, perforated	309.2 ft	$2.60	$804
water gutter with suction inlets	309.2 ft	$2.10	$649
drain line to the tank	309.2 ft	$1.15	$356
three-way gate, one per segment	55.0 each	$46.00	$2,530
silica cartridge drawer	55.0 each	$22.00	$1,210
service cover	55.0 each	$9.00	$495
thermoelectric condensing plate	10.0 each	$20.00	$200
ram fan, continuous duty	2.0 each	$185.00	$370
Seam module, whole dome			**$7,665**

$7,665 against a dome at $12,036 — a little over half the price of the building it is fitted to.

That is a lot, and it is why this is a chapter about a system rather than a line in the standard article. The parts of it scale differently, too: the cap, liner, gutter and drain are per foot of seam and grow with the dome, while the gates, cartridges and plates are per seam and grow more slowly.

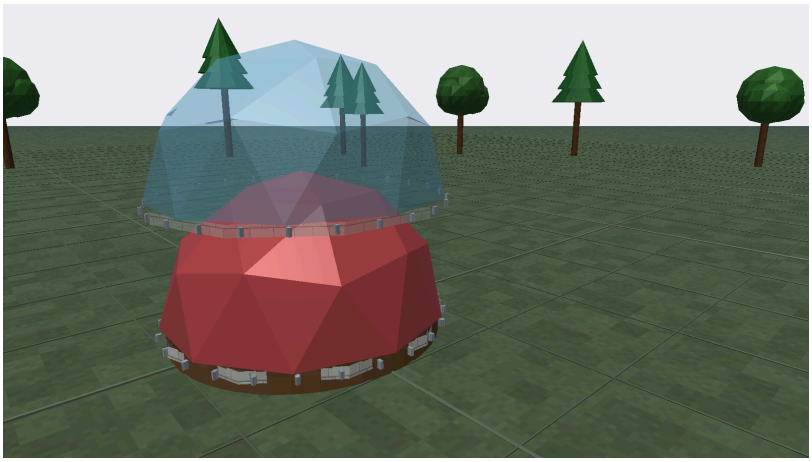
If you fit one part of it, fit the gutter. Water off the roof is the single best return in this chapter, it needs no power, and it is the only piece that pays for itself in a season.

CHAPTER 30

A Dome Is a Head, and It Wears Hats

Bare skin, then layers, then one rain-slick shell over the lot -- the way you dress for the Arctic, and for the same reasons

I learned how to dress a building standing on the outside of a ship in the Arctic Circle, on forward lookout, at night.



A quilt, and a bigger cap for each one.

What you wear out there

Nobody wears one warm coat. You wear layers — several of them, each thin, each doing one job — and on top of all of it goes a pumpkin suit: an immersion suit, bright orange, entirely waterproof, and utterly useless as insulation on its own.

The suit does not keep you warm. The layers keep you warm. The suit keeps the layers *dry*, and a dry layer is worth several wet ones.

The other thing about that arrangement is that it is adjustable in the middle of a watch. Too cold, add a layer. Too warm, lose one. You never take off the suit.

That is the whole shell design in this book, and I did not arrive at it by reading about vapour barriers.

The dome as a head

A dome is a head. It is round, it is the top of something, and it wears hats.

It can wear a bare skin — nothing at all, a frame with panels on it.

It can wear a toboggan: a quilted layer, then another, stacked, each one adding warmth.

And over all of it, always, one shower cap. Waterproof, thin, useless as insulation, and the reason every layer under it is dry.

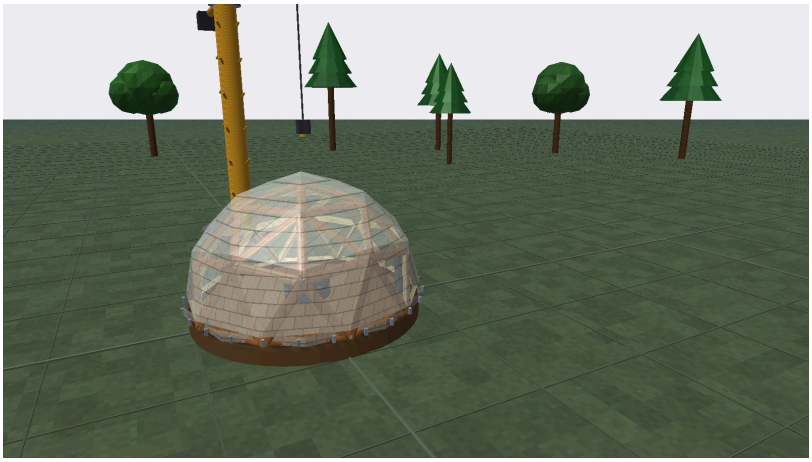
People understand this instantly in a way they do not understand “vented rain screen with a vapour-permeable weather-resistive barrier”, which is the same idea in a language that makes people stop listening.

Each hat is a size up

This is the part that is arithmetic rather than analogy.

A layer goes *over* the layers already there, so it is bigger than the one under it. Not a lot bigger — the shell is only a few inches thicker each time — but the areas compound, and by the seventh layer the outermost is meaningfully larger than the innermost.

So “one quilt costs fifty dollars” is a sentence about the first quilt. The model works out each layer’s own area, and the cap that goes over all of them is sized for the stack you intend to end up with rather than the stack you have.



The envelope gaining a layer a winter.

It gets warmer every winter

Here is the model’s ladder, with its one assumption stated:

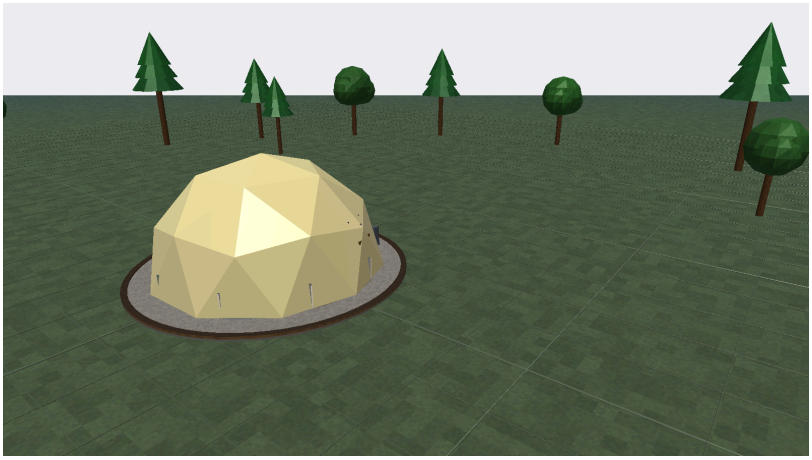
```
REMOVABLE SHELL / upgrade the insulation over time
Model assumption: +R-1.6 per quilted layer
0 added layers -> R-1.5
1 added layers -> R-3.1
2 added layers -> R-4.7
3 added layers -> R-6.3
4 added layers -> R-7.9
5 added layers -> R-9.5
6 added layers -> R-11.1
7 added layers -> R-12.7
Actual assemblies need tested moisture and fire performance.
The outer shell supplies the weather and wind barrier.
Insulation improves when you add suitable layers; age alone adds none.
```

The improvement comes from adding a layer, not from the house getting older; the outer shell keeps its separate job of keeping weather and wind out.

Here is what the layering model buys that a conventional wall does not.

A conventional wall is insulated once, when it is built, to whatever you could afford that year. Changing it later means opening the wall.

This one is insulated from the outside, in increments, forever. Each autumn you can add a layer — an evening’s work with a sewing machine and a box of old clothes — and the building is warmer than it was last winter.



What each quilted layer is actually worth.

The cap goes back over the top and nothing else about the building changes.

A house that improves as you live in it is a different relationship from a house that is as good as it will ever be on the day you move in.

And the quilts are made of clothes

The insulating layers in this design are quilted from recycled textiles — clothing that would otherwise be waste.

That is partly an economic argument and it is mostly a participation one. It is the only part of this entire method that somebody can do at a kitchen table with no land, no saw and no site. A layer is a real object with a number on it that goes on a named building, and it takes an evening.

The arithmetic is in the model: how many garments, how many pounds, what R-value the stack reaches, and what a layer is worth to the person who sews it for somebody else's dome. That last one has to be a real number or this is not a network, it is a request for free labour.

What actually closes a triangle

A wedge is not a rectangle, and the wall uses that. The member's inner face stands proud of where a panel sits, so every bay in the frame already has a lip in it – not machined, just left there by the shape of a split log (Chapter 21). So the wall goes together from the inside out:

1. **An inner panel** drops onto the lip from inside.
2. **The cavity** behind it is as deep as the member, 6.0 inches.
3. **An outer panel** drops into the same bay from outside and compression-fits against the frame.
4. **The shell** lands over everything, and nothing goes through it.

Less skin to start with

Before any hat, the shape has already paid. Compare a dome with a box of the same floor – worked here for the films' 20-foot dome, slightly larger than this book's reference:

```
same floor, both shapes: 314 sq ft
box: side = sqrt(314) = 17.7 ft, walls 8 ft, 0.50 pitch roof
box skin  = walls + roof + gables = 997 sq ft
dome skin = 40 flat triangles     = 583 sq ft
difference = 414 sq ft = 42% less to build, seal, paint, heat
volume enclosed: dome 2,093 cu ft vs box 3,208 cu ft
wind drag (published): dome Cd 0.42 vs box Cd 1.05 = 60% less load
```

Less skin means less to lose heat through in winter and less to gain it through in summer, and it is true before a single layer goes on. Run the arithmetic a heat-loss engineer would, with the same wall in both buildings and a middling American climate – the climate and the power price are borrowed, and named:

```
wall build-up: R-24.4 -> U = 1/R = 0.0410
heat loss rate = skin x U:
dome: 583 sq ft x 0.0410 = 24 BTU/hr/F
box:  997 sq ft x 0.0410 = 41 BTU/hr/F
x 4200 heating degree days and 1200 cooling
dome: 756 kWh/yr = $129 at 17 cents
box:  1,292 kWh/yr = $220
```

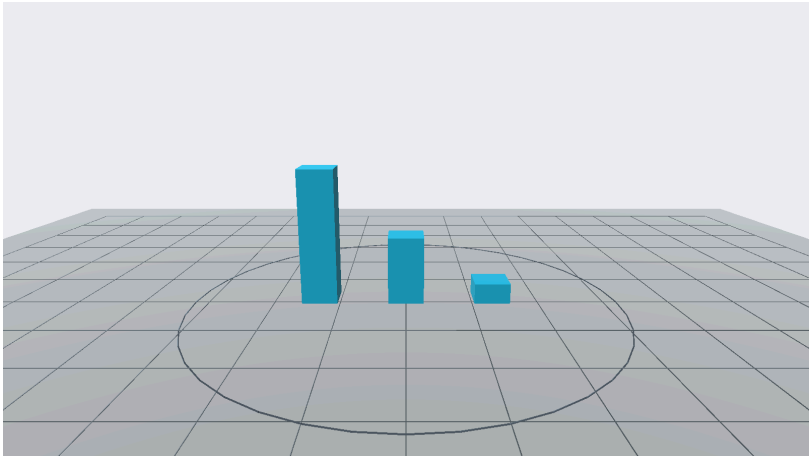
The dome wins by exactly its missing skin, every year. One more multiplier is available on the outside face: the white radiative paints developed in the last few years reflect almost all of the sun and shed heat to the sky even in daylight. That is a published research result rather than something this model computes, so it is left out of the figures above.

The wall as a filter

A dome holds a lot of air behind a lot of wall, and that ratio does something useful. Push fresh air in through the shell itself, even hard, and it crosses the wall so slowly that nobody inside feels a draught:

```
dome radius 60 in: shell 14.6 m2, air 262 cu ft
     6 ACH ->    26 cfm, through the wall at 0.17 ft/min
    20 ACH ->    87 cfm, through the wall at 0.56 ft/min
    40 ACH ->   175 cfm, through the wall at 1.11 ft/min
```

A draught you can feel starts at many times those speeds. So a breathable shell could be a filter face for the whole building, with the flow spread over every square foot instead of rushing past one vent. Chapter 29 and the seam chapters are what the air does once it is inside the frame.



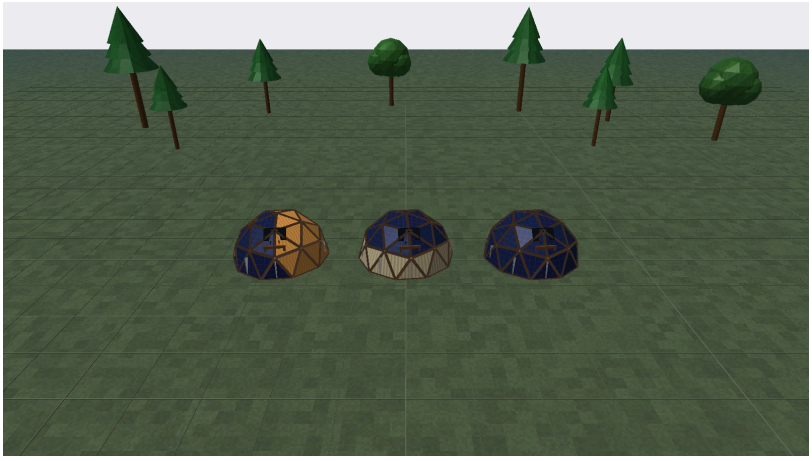
The shell so large the air barely crawls through it.

How much shell can carry solar?

The shell is also the dome's largest surface facing the sky. Three ways to use it, on the same dome:

```
SAME DOME: 20 ft across
Panel areas are measured from the model's faces.
Assumed panel density: 18.0 W/sq ft
one side: 292 sq ft -> 5.2 kW
top half, skirt below: 292 sq ft -> 5.2 kW
whole shell: 583 sq ft -> 10.5 kW
No panels = zero installed solar capacity.
Selected layouts: 5.2 to 10.5 kW nameplate.
Face area sets capacity, not guaranteed energy yield.
Sun angle, shade, climate and losses set actual kWh.
The useful limit is the available shell area.
```

Cells on one side give a rotating pad (Chapter 39) a clear purpose: turn the useful face to the sun. Cells on the top half with a metal skirt below make the skirt a rain catchment. Cells everywhere give the most capacity and the least reason to turn. In every case, face area sets the nameplate, not the year's energy.



One side, top half, whole shell.

Where the analogy stops

Two places, and both matter.

A person moves and a building does not. Your layers work partly because you are generating heat inside them and moving air through them. A building generates less heat per unit of surface than you do and does not move at all, so it relies on the vented gap of Chapter 28 to do what your movement does.

You take your suit off indoors. The dome does not. The cap is on it in July, in the sun, which is the ultraviolet problem at the end of the last chapter and the reason the cap is a consumable.

Neither of those breaks the model. They are the two places where a builder who reasons from the analogy alone will get it wrong, so they are written down.

PART 6

The Seam at Work

“Send air toward the wood only when it is drier than the water you want to remove -- unless it is going to a cold metal surface built to catch and drain it.”

What fits in the channel and what does not, the rule that decides which way its air moves, what each level of the dome does, where the desiccant belongs, and the metal that makes it work.

CHAPTER 31

What Fits in the Channel

Water and wire, yes. A round duct, no -- and the channel does not need one, because it already is one

Chapter 29 fitted the seam out as gutter, vent and dehumidifier and priced it. It did not say how much room there is inside it, and every idea anybody has for the channel – a water line, a circuit, a duct, a desiccant – runs into that number first.

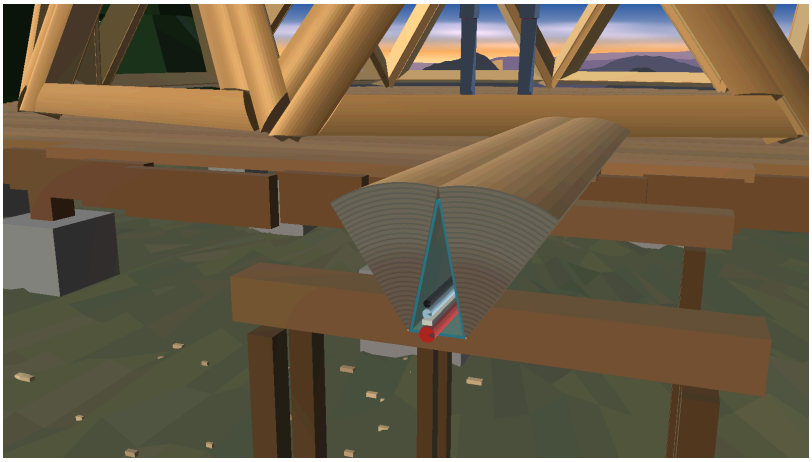
First, what the key is. It fills the V between the two members of a seam, and it is what the seam's bolts pull the two panels against. Make it solid and it is a spline. Make it hollow and it is a pipe running the length of every seam – 309 feet of it, touching every panel, reaching the crown and the ground. The room inside that pipe is the whole budget for everything below.

So here it is, from the solver's own seams, for the reference build.

The V between two members is 22.5 degrees at the tighter of the two seam types and 27.0 at the wider, and as deep as the log's radius. The key that fills it is printed with 3 millimetre walls, and it stops short of the ridge where the V is too narrow to be any use. What is inside those walls is the budget:

inside the key, tighter seam	5.3 sq in
inside the key, wider seam	6.5 sq in
largest round thing, tighter	1.69 in
largest round thing, wider	1.97 in

Everything in this chapter has to fit in that.



A water line, a circuit, the plates' pair and the drain, in the tighter of the two seams.

The sizes this chapter assumed

The pipe and tubing sizes are nominal, from their standards. The cable sizes are estimates, because every brand's sheath is different. The rule for how full a channel may be is borrowed from electrical conduit, which allows 40 percent of the area for a mixed bundle so the last thing can still be pulled through. The channel is not a listed raceway; the rule is used because it is a sensible allowance, not because it applies.

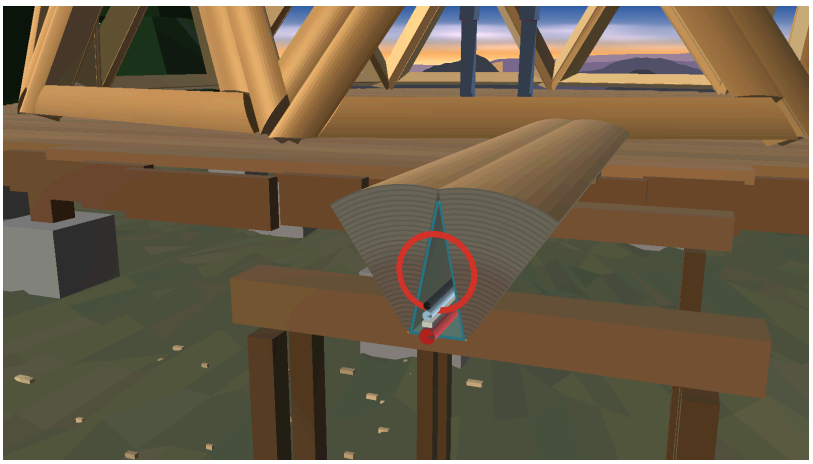
1/2 in PEX water line	nominal
3/4 in PEX water line	nominal
14/2 NM-B cable, 15 A circuit	estimate
12/2 NM-B cable, 20 A circuit	estimate
12 V pair to the Peltier plates, 14 AWG	estimate
condensate drain, 1/4 in ID vinyl	nominal
3 in round air duct	nominal
4 in round air duct	nominal
fill fraction	0.4 of area: rule borrowed from conduit fill (NEC Chapter 9, Table 1)
key wall in	0.11811 in: assumption
printer bed in	9.84252 in: assumption
filament g per cm3	1.27 g/cm3: nominal
filament usd per kg	20 USD/kg: assumption
print mm3 per s	15 mm3/s: assumption
quiet air fpm	400 ft/min: rule of thumb

What fits, item by item

item	size	22.5 deg seam	27.0 deg seam
1/2 in PEX water line	0.62 in round	fits	fits
3/4 in PEX water line	0.88 in round	fits	fits
14/2 NM-B cable, 15 A circuit	0.40 in x 0.20	fits	fits
12/2 NM-B cable, 20 A circuit	0.46 in x 0.22	fits	fits
12 V pair to the Peltier plates, 14 AWG	0.30 in round	fits	fits
condensate drain, 1/4 in ID vinyl	0.38 in round	fits	fits
3 in round air duct	3.00 in round	no	no
4 in round air duct	4.00 in round	no	no

Water and wire fit, with room to spare. A half-inch PEX line, a fifteen-amp circuit, the twelve-volt pair to the condensing plates and the condensate drain go into either seam together, filling 11 percent of the tighter one and 9 percent of the wider one, against the 40 allowed.

Round ducts do not. The smallest ordinary duct is three inches round. The largest round thing either seam passes is 1.97 inches. If the plan was to run ducts through the seams, the plan does not work as drawn, and a bigger log does not rescue it.



The smallest ordinary duct, against the V it would have to go in.

The channel is the duct

It does not need a duct in it, because it already is one.

With the water, wire and drain inside, the free space in one tighter seam still moves 13 cubic feet of air a minute at a quiet 400 feet a minute. The whole dome needs 18 for fresh air. So **2 seams carry the building's ventilation**, and every other seam is free for drying the frame, which is the job of Chapter 32.

If you want a duct anyway: the key becomes a spacer

There are reasons to want a real duct: a heat-recovery ventilator with round spigots, a stove exchanger, a fan you already own. Then the key stops being a filler and becomes a spacer.

Push every panel outward along its own face and every seam opens. The two seam types have different folds, so one push opens them by different amounts, and the push is set by whichever needs it most:

to fit a 3 in duct	panels out 7.0 in, dome +6%
a 4 in duct beside the bundle	panels out 16.4 in, dome +14%
seams open at the ridge	6.4 / 5.1 in

A spacer is not free. Moving every panel out makes the whole dome bigger: more wood, more skin, a bigger pad. And every opened ridge needs a cap over it. All of that, to carry air the plain channel was already carrying.

As built, and with the seams opened for a four-inch duct.

Or start with a bigger log

The other lever is the log. The room in the channel grows with the square of the log's radius:

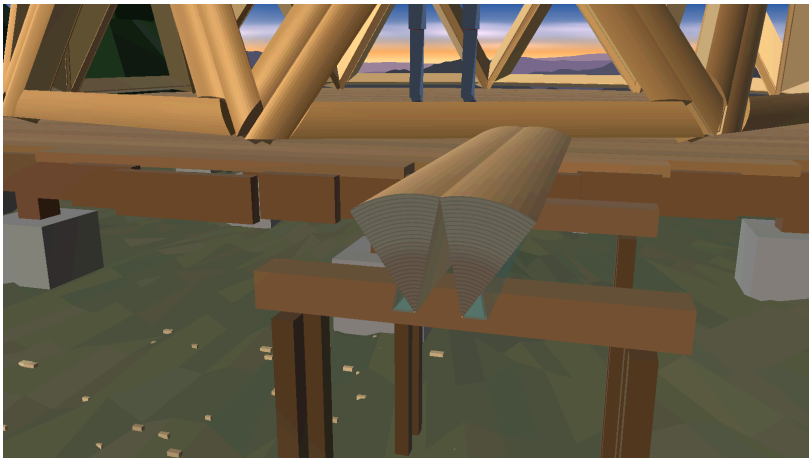
10 in log	3.5 sq in inside, 1.37 in round
12 in log	5.3 sq in inside, 1.69 in round
15 in log	8.8 sq in inside, 2.17 in round
18 in log	13.1 sq in inside, 2.65 in round

Even the biggest of those still falls short of passing a three-inch duct. A bigger log buys more room for water and wire; it does not buy a round duct.

Printing the key in halves

The key is printed, and it is printed in halves. Split it down the middle of the seam and each member carries its own half, screwed to its sawn face before the panel is built. When two panels meet, the two halves close into one channel. Nothing has to be threaded into a finished seam.

1. **Two profiles.** The dome has two seam types, so it has two key profiles: 2 files, not 55.
2. **Print after the frame has dried** (Chapter 17), to the V the dried wood actually has.
3. **Print in pieces** no longer than the printer's bed, assumed at 250 millimetres: 8 pieces to a member, butted end to end.
4. **Bolts cross near the ridge**, where the V is narrowest. Route services through the wide belly below them.
5. **Print only the seams that carry something.** Every seam in the dome is 110 half-keys, 820 pieces, 121 kilograms of PETG, about $2,425 and 1,768 printer hours. One seam is 2.2 kilograms, $44 and 32 hours. Cut solid keys from the offcuts for the rest.



Each member carries its own half of the key.

Where the channels meet

The frame has no hubs, but the services need them. Channels meet at 6 five-way and 10 six-way vertices, and at 10 more on the rim, where they drop to the pad.

As built, the Vs close to a point at each vertex, so each junction is a printed **rosette** fixed to the inside of the vertex: a small box where a pipe turns and a wire is joined, somewhere a hand can reach. With the four-inch spacer the vertex opens into a hole about 11 inches across, and the rosette becomes a node the channels plug into.

A printed rosette inside a five-way and a six-way vertex.

Two rules go with every rosette. **No pressure fitting inside a seam**: run the PEX in one continuous length and make every joint in a rosette you can open, because a hidden leak inside a wooden seam is the one failure this whole system exists to prevent. And **whether a cable may run in this chase is your inspector's decision**, not this book's.

CHAPTER 32

Compare Dew Points, Not Humidity

Three questions decide every move the seam's air makes, and none of them is about relative humidity

The seam channel can move air in almost any direction you like. Outside air in, inside air out, round in a loop, through a desiccant or past it, over a cold plate or a warm one. That is a lot of choices, and the way a system like this goes wrong is by making them on a feeling: it is a damp day, so shut everything.

This chapter makes them on one number.

Two airs, and the only number that says which one is drier.

The one rule

Never compare relative humidity. Compare dew points.

Relative humidity says how full the air is compared to how much it could hold **at its own temperature**. Cold air holds very little, so cold air reads as damp while carrying almost nothing. The dew point is the temperature at which the water in the air starts coming out as liquid, and it depends only on how much water there actually is. The lower the dew point, the drier the air, whatever the thermometer says.

Take a winter day:

outside	minus 5 C at 60% dew point minus 11.6 C 1.6 g/kg
inside	21 C at 45% dew point 8.6 C 6.9 g/kg

The outside air sounds damp. It carries 1.6 grams of water in every kilogram; the room’s air carries 6.9. Bring it in, warm it up, and it dries the dome. A rule that looked at humidity would have kept it out.

The dew point comes from temperature and relative humidity by the Magnus formula, which is what every weather station uses. A cheap sensor gives both inputs.

The three questions

The controller asks three questions, in this order.

1. Which air is actually drier? If the outside dew point is below the inside one, outside air has drying power. If it is above, outside air is a source of water, however cool it feels.

2. Will this air condense on that surface? Water comes out of air on any surface colder than the air's dew point. So a condensing plate must be colder than the dew point, and every piece of wood must be warmer than it, with a margin. This book keeps wood 3 degrees above the dew point of any air that touches it.

3. Which way should the air move? The answer to the first two decides it: toward the structure only with air drier than the water you are trying to remove – unless you are deliberately sending wetter air to a cold metal surface built to collect and drain it (Chapter 36).

The modes

That gives the one channel 7 ways to move air:

CLOSED	rain: exposed intakes shut
INTAKE	outside -> seam -> inside
EXHAUST	inside -> seam -> outside
CONDENSE	inside -> cold seam -> drain -> outside
FRAME_LOOP	seam -> desiccant -> seam
PURGE	outside -> seam -> outside
DRY_SUPPLY	dehumidified air in, slight positive pressure

Two of them never touch the living space at all. They are the ones that make this more than a ventilation system.

Eleven weathers, one controller

To test it, here is the table the owner of this design drew up – the weathers that matter and what the seam should do in each – plus one more: muggy outside with a wet frame. The temperatures and humidities are this book's estimates of typical conditions, not measurements from a site. Wet weathers carry a wood moisture reading over 19 percent.

The controller sees only the numbers, and works only from dew points:

weather (estimated)	out	in	dew out/in	mode
Cold + dry outside	-5 C 60%	21 C 45%	-12 / 9	INTAKE
Cold outside + humid inside	-5 C 80%	21 C 68%	-8 / 15	EXHAUST
Cold outside, harvesting water	-2 C 80%	21 C 68%	-5 / 15	CONDENSE
Mild + dry outside	16 C 40%	21 C 45%	2 / 9	INTAKE
Hot + dry outside	33 C 18%	21 C 50%	6 / 10	INTAKE
Hot + humid outside	32 C 70%	21 C 50%	26 / 10	DRY_SUPPLY
Hot + humid, cooling on	35 C 60%	24 C 50%	26 / 13	DRY_SUPPLY
Rain	14 C 97%	21 C 45%	14 / 9	CLOSED
After rain, dry outside	18 C 45%	21 C 45%	6 / 9	PURGE
Sun on a wet skin	24 C 55%	21 C 45%	14 / 9	PURGE
Muggy outside, wet frame	36 C 60%	24 C 55%	27 / 14	FRAME_LOOP

Every one of the 11 gets the answer the owner's table gives, and the book's own checks fail if one ever does not.

The loop and the purge

The two modes worth the extra plumbing are these.

The frame loop takes the seam's own air, dries it in the desiccant and sends it back round the seam. Nothing conditioned is thrown outside and nothing muggy is pulled in. On a hot, humid day with a wet frame – the one weather where neither the inside nor the outside air can help – it is the only thing in the design that dries the wood.

The purge takes dry outside air in at one port, through a wet seam, and straight back out at another. The wet air never enters the room. It is how a frame recovers after rain, and on a sunny day it vents the vapour the sun drives out of a wet skin before that vapour gets driven inward.

Both need something the plain barrier fan does not: two outside ports per circuit, and a gate that can route a seam to outside, to the room or to the drawer. That is the price of separating **drying the structure** from **ventilating the house**. It is worth paying. They are different jobs, and a building that does both with one airflow does neither well.

The controller, as code

The whole decision fits on one page. This is the function the book's tables come from, trimmed of its comments:

```
def decide(s):
    dp_in, dp_out = dew_point(s.t_in, s.rh_in), dew_point(s.t_out, s.rh_out)
    dp_seam = dew_point(s.t_seam, s.rh_seam)
    margin = C["dp_margin_k"]
    wet_frame = s.wood_mc >= C["mc_wet"] or s.rh_seam >= C["seam_rh_wet"]
    if s.rain:
        return "FRAME_LOOP" if wet_frame else "CLOSED"
    if wet_frame:
        return "PURGE" if dp_out < dp_seam - margin else "FRAME_LOOP"
    if s.rh_in > C["rh_high"]:
        if dp_out < dp_in - margin:
            if s.harvest and s.t_out <= dp_in - C["plate_below_dp_k"]:
                return "CONDENSE"
            return "EXHAUST"
        return "DRY_SUPPLY"
    if dp_out < dp_in - margin:
        return "INTAKE"
    if dp_out > dp_in + margin:
        return "DRY_SUPPLY"
    return "IDLE"
```

The sensors it needs are ordinary ones: temperature and humidity inside, outside and in the seam; a wood moisture pin; a rain switch. The margin stops the controller flapping between two modes when the two airs are nearly the same. Every value it compares against is declared, with its kind:

magnus_a	17.62 -	standard
magnus_b_c	243.12 C	standard
air_density	1.2 kg/m3	standard
air_viscosity	1.81e-05 Pa s	standard
gravity	9.81 m/s2	standard
sieve_uptake	0.2 kg/kg	nominal
sieve_bulk_density	650 kg/m3	nominal
bead_diameter_mm	2.5 mm	nominal
bed_voidage	0.37 -	standard
sieve_regen_c	200 C	nominal
silica_regen_c	120 C	nominal
sieve_heat_kj_per_kg	3700 kJ/kg	estimate
petg_soft_c	80 C	nominal
seam_air_max_c	60 C	assumption
pine_dry_density	420 kg/m3	estimate
mc_wet	0.19 kg/kg	standard
mc_target	0.12 kg/kg	estimate
dp_margin_k	1 K	assumption
wood_margin_k	3 K	assumption
plate_below_dp_k	6 K	assumption
plate_approach_k	2 K	assumption
frost_floor_c	1 C	assumption
rh_high	0.6 -	assumption
seam_rh_wet	0.85 -	assumption
fan_static_pa	250 Pa	estimate
drawer_face_m	0.3 m	assumption
drawer_depth_m	0.05 m	assumption
stove_exchanger_kw	2 kW	estimate
indoor_t_c	21 C	assumption

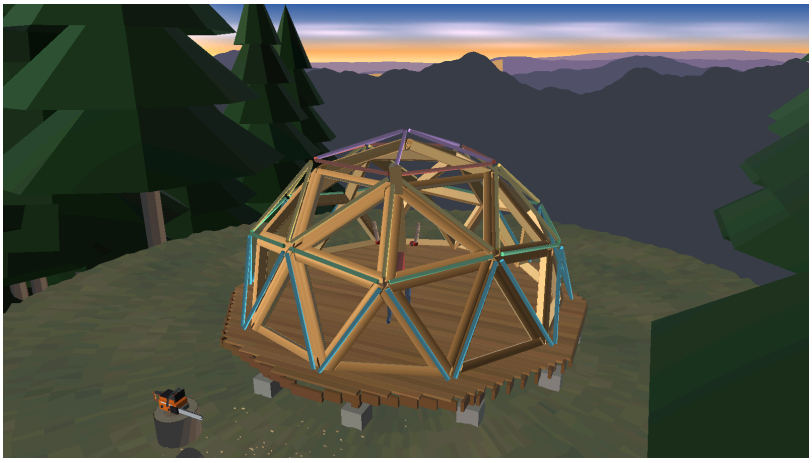
CHAPTER 33

Four Levels, and the Ring That Cannot Drain

The solver's dome stands on five heights, its seams fall into five bands, and the slope of each one decides its job

The owner of this design describes the dome in levels. The first ring is the floor. Above it is the band of triangles that stands on the floor. Above that, the band that meets the pentagon round the crown. At the top, the cap. And the question that goes with it: should each level of seam do a different job?

It should, and the solver says which.



The seams, coloured by the band they run in.

The levels, counted from the solve

The dome's 26 corners stand at five heights, not four: 10 on the rim; a belt of 5 and 5 corners that zigzags between two slightly different heights; 5 round the crown; and the apex. The zigzag is what makes it five rather than four, and it matters, as you will see.

The 55 seams run between those heights, and they fall into five bands. What decides each band's job is its slope:

band	seams	feet	slope	
lower band, rim to belt	20	113	57 deg	drains
the zigzag belt	10	53	8 deg	drains
upper band, belt to pentagon	15	87	37 deg	drains
the pentagon ring	5	30	0 deg	dead level
the cap, to the apex	5	27	16 deg	drains

Read down the last column. Four bands slope, so water in them runs downhill to the rim. One does not.

The ring that cannot drain

The 5 seams round the crown – the pentagon ring – are dead level. Every one of them runs between two corners at exactly the same height. Water that gets into them does not go anywhere. It stands in a wooden seam at the top of the building until it evaporates, and it evaporates into the wood.

So the pentagon ring carries air, never water. Which suits it, because air wants to be there: warm, humid air rises to the crown. The pentagon ring is the natural **manifold** – a closed loop of channel at the top of the room, touching every seam above the belt – joining the cap's 5 seams so the exhaust shares itself among them. Where that exhaust goes is Chapter 34: down, never up through the cap.

The belt is the opposite surprise. Its 10 seams rise at only 8 degrees, which looks like another level ring. But because the belt zigzags, it has 5 low corners, and every one of its seams runs downhill to one of them. **Each low corner of the belt is a drain point**, and the rosette there (Chapter 31) is where the belt's water leaves.

The pentagon ring is dead level; the belt's five low corners drain it.

Condense low, drain at the rim

The owner's first instinct was to put the condensing plates on the first ring, where the frame meets the floor. That instinct is right, and the reason is gravity.

Condensate falls. A plate at the rim drops its water straight into a drain at one of the 10 rim corners, with no seam for it to travel through on the way. A plate at the crown sends its water down the whole height of the dome inside the frame, which is the one place you least want water travelling.

The lower band's 20 seams, at 57 degrees, are the steepest in the building. They are the drains, and they are the path dry air takes up from the rim ports into the frame.

Rain does not come in through the ridge

> **WE GOT THIS WRONG**
>
> Chapter 29 described **a slot along the outer cap of every seam** taking rain in at the ridge, with a gutter under it. That was wrong, and this book corrects it here rather than quietly changing the earlier chapter.
>
> Three things are wrong with it. The slot is an opening into the structure along the entire length of every seam, on the face of the building the weather hits. Water taken in at the crown has to travel the whole height of the frame to get out, and on the pentagon ring it cannot get out at all. And every foot of seam that carries liquid water is a foot of wood with a leak waiting next to it.

Rain belongs outside. Let it run off the cap – the one watertight layer of Chapter 28 – to a gutter round the rim, outside the frame, and from there to the tank. The roof still delivers 185 gallons an inch; it just never enters a seam to do it. The only liquid water inside the channels is condensate, and it is made at the rim, where it has no distance to go.

And the rim gutter is a water plant. The cap's brim throws rain off a ring wider than the dome, and collecting what falls on that ring is the whole design:

```
hat rim radius = 8.9 ft; add an 18 in brim -> 10.4 ft
catchment = pi x 10.4^2 = 343 sq ft of plan area
x 40 in of rain a year x 0.6233 gal per sq ft inch
x 0.85 runoff
= 7,259 gallons a year, 20 a day, into a tank by the door
no pump, one downpipe -- the roof shape does the collecting
```

No pump and one downpipe: the roof shape does the collecting. (Those figures use the films' declared rainfall and cap brim, not a measurement of any one site.)

Warm air rises, and the fan still does the work

Warm air rising is free, so use its direction: take fresh air in low at the rim ports and let the room carry it up to the crown, where it is taken out of the room – into the channels, under the cap, not through it (Chapter 34).

But do not count on its strength going up. With the inside ten degrees warmer than the outside, the height of this dome produces 1.2 pascals of pull between the rim and the apex. The barrier fan of Chapter 29 holds 4. The chimney effect helps the fan along; it does not replace it, and on a summer afternoon it runs the wrong way. Sent *down* the seams instead, through a cold shell, the same physics carries the exhaust on its own in winter – Chapter 34 works the numbers.

A tube around the bottom

A dome has an unhelpful habit: warm air, and anything it carries, rises to the apex and stays there. The answer that came out of a badly ventilated workshop dome is a tube right round the base, on a blower, so the whole perimeter is one duct instead of the building having one extract point on one wall.

Run it either way. Push air in at the ring and it leaves through the upper shell and the pentagon ring, carrying fumes out through the whole surface. Reverse it and outside air is drawn in through the shell, warming against the structure on its way: a breathing wall, or dynamic insulation. Same hardware, one switch.

The flows are computed (Chapter 30); the wall is not. Direction decides it, because pushing warm, wet air outward through a cold wall condenses it inside the wall – exactly the failure Chapter 32 exists to prevent – and a sealed skin cannot breathe at all, so any permeable band has to be designed in.

The layout

Put together, every level has one job:

- **the rim** – the ports in and out, the condensing plates, the drains, the two desiccant drawers of Chapter 35;
- **the lower band** – air up, water down;
- **the belt** – its low corners drain it;
- **the upper band** – air up toward the crown;
- **the pentagon ring** – the exhaust manifold, air only;
- **the cap** – unbroken. A grille at the crown, under it, takes the room's air into the channels, and every port in the building is at the rim, facing the ground (Chapter 34).

Sensors inside, outside and in the seam read temperature and humidity; a pin in the wood reads its moisture; a switch on the cap says whether it is raining. Chapter 32 is what they feed.

CHAPTER 34

A Cap With No Holes Still Breathes

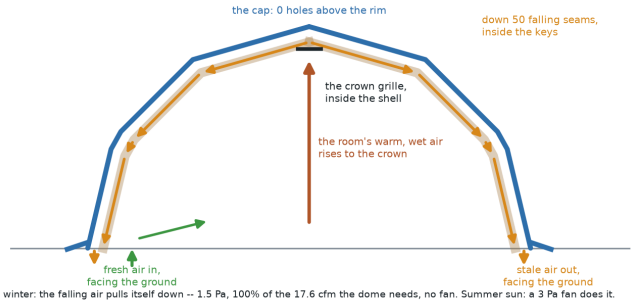
Take the room's air in at the crown, inside the shell; send it down the seam channels; let it out at the rim, facing the ground. The cap is never pierced, and the dome still breathes both ways

The cap of Chapter 28 is one watertight layer over the whole frame, and that is its whole job: nothing gets through it. But a dome that nothing gets through is a sealed jar with people in it. They breathe, cook and wash, and every one of those puts water into the air. With nowhere to go, that water waits for the coldest surface in the building and condenses on it – and in a timber dome the coldest surface is the frame.

So a watertight dome must still breathe. The question is how air gets in and out without making a way in for rain.

> **WE GOT THIS WRONG**
>
> The usual answer is a vent at the top. It is the wrong answer here: a hole in the roof, at the place the weather hits hardest, with the frame right under it. This book's own layout in Chapter 33 let the exhaust out through an apex vent in the cap, and that is corrected here.

The cap is never opened. The room's air is taken in at the crown, *inside* the shell, under the cap. It goes down the seam channels, inside their printed keys. It leaves at the bottom of the dome, through ports that face the ground. Fresh air comes in at the bottom the same way. Holes through the cap above the rim: **0**.



The cap is unbroken; the air goes down the seams and out at the bottom, facing the ground.

In at the crown, down the seams, out at the bottom

Why take it from the top. Warm air rises, and the warmest air in a room is also the wettest and the stalest: it is the breath and the steam, carried up. In a dome it all collects at the crown. A grille there, under the cap, takes in exactly the air that most needs to leave.

Why the seams. The channel inside every key is already a duct (Chapter 31), and it runs from the crown to the floor. The solver's own seams show the route: **50 of the dome's 55 seams fall all the way from the apex to the rim** without ever climbing. The other 5 are the pentagon ring, dead level, and they stay out of the falling path; their job is the one Chapter 33 gave them, a ring of air joining the five paths so the flow shares itself evenly. The shortest falling path is 15.9 feet, crown to rim, through two rosettes.

There is room for it. The dome needs 17.6 cubic feet of fresh air a minute. One empty channel carries 14.7 at a quiet speed, and the five cap seams that every falling path starts in carry 73.7 together – **4.2 times the need**. The exhaust seams carry no pipes or wires; the service seams of Chapter 31 are others.

Why the ports face the ground. Rain falls. An opening that faces down can only take in water that has to climb to reach it, and an opening at the bottom of the dome, under the brim of the cap, is out of the falling rain altogether. It is also the lowest point of the path, so anything that does get in drains straight back out. There are 10 rim corners to put ports at.

The air falls by itself in winter

Here is the part that makes the direction more than a convenience.

In winter the room's air is warm, so it rises: the room itself carries the air up to the crown grille. In the channel the exhaust gives its heat to the cold shell, gets heavier than the room's air, and **falls**. Two columns side by side – a warm one rising in the room, a cool one falling in the seams – make a loop that turns itself, a siphon of air. How hard it pulls depends on how cold it is outside:

weather	outside	siphon	carries fan for the rest	water in the keys
	C	Pa	of the need Pa	litres/day
hard winter	-15	2.13	100% 0.00	1.93
winter	-5	1.54	100% 0.00	0.24
cool	5	0.95	77% 0.42	0.00
mild	15	0.36	37% 1.02	0.00
summer sun	30	-1.66	0% 3.03	0.00

At -5 degrees outside the falling air pulls with 1.54 pascals and carries **100 per cent** of the need with no fan at all. In cool weather it still carries 77 per cent, and a fan adds the last 0.4 pascals. In summer sun the cap is hotter than the room, the channel air wants to rise, and the siphon runs backwards: then a fan does it all, against 3.0 pascals. The small in-line fan of Chapter 29 pushes 250. The worst case is a rounding error to it.

Chapter 33 said the chimney effect helps the fan and does not replace it. That is true of air going up. Sent down, through a cold shell, in the season a sealed house most needs to breathe, the falling air does the work itself.

The inputs this depends on, declared:

channel_share	0.5 -	assumption where the channel's air temperature sits between outside (0) and inside (1): the key lies inside the frame, about halfway through the wall's resistance to heat
summer_channel_c	35 C	assumption channel air on a sunny summer afternoon, the cap above it warmed by the sun
room_rh	0.45 -	assumption the room's relative humidity in winter
f_re	64 -	standard laminar friction, f x Re, for a round duct; a triangle's is lower (53), so 64 overstates the channel's resistance
k_junction	1.5 -	estimate loss at each vertex rosette where the air turns from one seam into the next
k_entry_exit	1.5 -	estimate entry at the crown grille plus exit at the outlet
paths	5 -	geometry the five cap seams leave the apex: all exhaust passes through them, so five parallel paths, each treated as one channel carrying a fifth of the flow the whole way (a conservative simplification: lower down the flow divides among more seams)

The water goes the same way as the air

When the channel is colder than the room air's dew point – 8.6 degrees for a room at the declared humidity – the exhaust drops water inside the key: 0.24 litres a day at -5 degrees outside, 1.93 at -15.

That water runs downhill, which is the way the air is already going, and it leaves by the same ports. **The air and its water go out of the same door.** This is the strongest reason the exhaust runs down and not up: sent up, the water it drops would run back against the air and pool wherever the channel levels off. Sent down, nothing pools. The one level ring in the building, the pentagon ring, is not on the falling path at all.

And the wet air never touches the wood. It runs inside the printed key, so the rule of Chapter 32 – keep every wood surface above the dew point of the air that touches it – is kept by construction: the air touching the wood is the seam's still air, not the room's breath.

Two ways in and out

The dome breathes both ways, and both doors are at the bottom.

Fresh air in through its own ports at the rim, facing the ground, and into the room low. The desiccant drawers and condensing plates of Chapter 35 sit on this intake, at the rim, where their water has no distance to fall.

Stale air out at the crown, down the seams, and out at the rim.

Fresh in low, stale out high: the room itself is the up-leg of the loop, and the air in it is always moving gently from the floor toward the crown. Run the two flows equal and the room is neither pressed nor pulled, so no air is shoved out through the frame and none sucked in through it. The controller of Chapter 32 decides when each runs and how hard.

Where it stops working

Ice. If the channel falls below freezing, the water in it freezes at the outlet and closes it. With the channel halfway between inside and outside, that happens below **-21 degrees** outside. Where winters go that cold, put a heat trace on the outlet stubs, or close the falling path and run the exhaust through the rim drawers on the coldest nights.

Snow. A port facing the ground is useless under a drift. Keep the ports above the snow line – on the deck, or on the riser wall of Chapter 27.

Insects and mice. Every port gets a screen. A downward opening keeps the rain out; it does not keep a wasp out.

Summer. The siphon runs backwards, and the fan carries the whole need. It is a small fan and a small number, but it is not nothing.

Cooking. The path is long and the flow is gentle. A range still wants its own hood, ducted out low and facing down, like everything else on this dome.

CHAPTER 35

The Desiccant Belongs in a Drawer

Molecular sieve holds a fifth of its weight in water -- and a seam packed with it passes almost no air and can never be dried in place

A desiccant is anything that pulls water vapour out of air and holds on to it. Two kinds matter here.

Silica gel is the beads in the packet in a new pair of shoes: porous glass that holds water in its pores. **Molecular sieve** is a zeolite – a mineral with a crystal lattice full of holes the exact size of a water molecule. The common grade, 13X, holds about a fifth of its own weight in water, keeps pulling vapour out of air long after silica gel has given up, and is what people mean when they talk about a solar-regenerated fridge or a dryer that works in the tropics.

Both do the same thing in the seam: they are the frame loop's dehumidifier (Chapter 32), the one way to dry the wood when neither the inside air nor the outside air is dry enough. The question is where to put them. The candidates are the obvious ones: packed along a seam, packed round a ring, arched over the top of the dome from one side of the floor to the other like a rainbow, or in one place.

The answer is one place, and two drawers. Here is why.

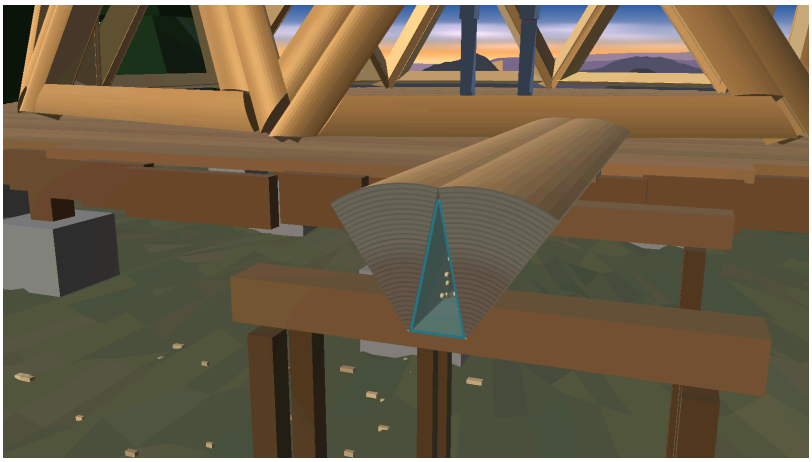
Why not in the seams

A seam packed with beads is a very long, very thin filter, and air does not like going through filters.

The resistance of a packed bed is well understood – it is the Ergun equation, which chemical engineers use for exactly this – and it rises steeply with both the speed of the air and the length of the bed. Take one seam's share of the dome's air, 13 cubic feet a minute, and push it through one seam packed with beads:

resistance of one packed seam	65,848 Pa
what a small in-line fan pushes	250 Pa
what that fan gets through	0.29 cu ft a minute

A packed seam does not slow the air down. It stops it. The rainbow, rim to apex to rim along the seams, is 32 feet of it, and worse again.



One seam packed with beads, and the pressure it would take to push the air through.

It would also hold a lot of beads to no purpose – packing every seam takes 233 kilograms of them – because a bed the air cannot get through is a bed whose water nobody can reach.

And why it cannot be dried where it sits

A desiccant fills up. To use it again you have to drive the water back out, and that takes heat:

```
13X molecular sieve gives its water back at    200 C
silica gel                                     120 C
the printed key goes soft at                   80 C
the hottest air this book sends through a seam 60 C
```

Beads in a seam could never be regenerated where they sit. You would have to bake the frame to do it, and the key would have melted long before the beads gave anything up.

Two drawers and a gate

So the desiccant lives in **drawers, at the rim**, where the channels come down to the pad and a person can reach them.

One drawer 30 centimetres square and 5 deep holds 2.9 kilograms of beads, which hold 0.58 kilograms of water between them, and it passes the whole dome's air at 20 pascals: a fan barely notices it. A short, wide bed is everything a long, thin one is not.

1. **Use two drawers.** One works while the other is out being dried. This is the arrangement every industrial compressed-air dryer uses, and for the same reason: the air never waits for the desiccant.
2. **Regenerate them out of the wall.** A drawer of 13X wants 0.60 kilowatt hours to dry out: about 18 minutes of a stove exchanger's 2 kilowatts, or an oven, or a black box in the sun on a hot day.
3. **Switch between drawer and bypass with a gate that fails open.** The gate is a damper with two positions: through the drawer, or round it. Build it so that when the power goes, a spring puts it on the bypass. Air that stops moving in a wooden seam is the failure; air that skips the desiccant for an afternoon is not.
4. **Put the gate where the channels meet**, in the rim rosettes, not halfway along a seam where you cannot reach it.

A drawer of beads and a bypass, switched by a gate that falls back open.

What the desiccant will not do

And the number that does not help.

> **THE NUMBER THAT DOESN'T HELP**
>
> A frame that has got properly wet – over 19 percent moisture – has 54 kilograms of water to lose to get back to 12. That is 93 drawer fills, and 56 kilowatt hours of heat to regenerate them.

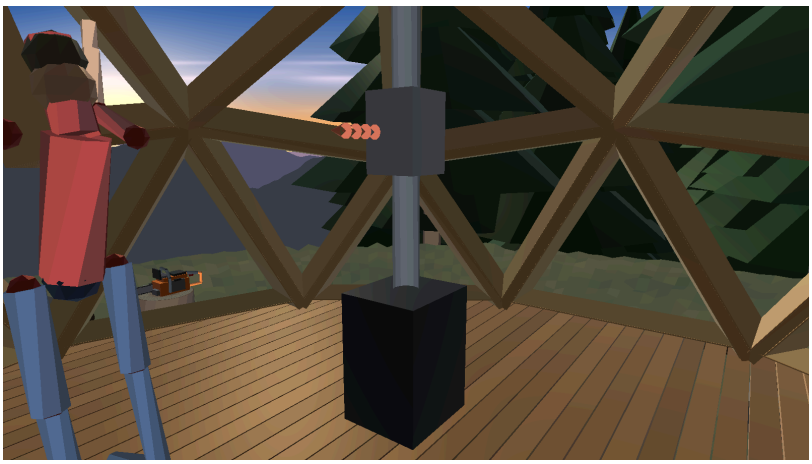
The desiccant does not dry a soaked frame. It is for the weather when no air is dry enough and the frame needs to stay where it is, not for getting it back after a leak. Drying a wet frame is the purge's job, with dry outside air, on the next dry day. The desiccant keeps it from getting wet in the meantime.

The stove's heat, never its air

A wood stove is a large, free source of heat in winter, and the owner's thought was to have it drive the seam's air: pulling it in or pushing it out depending on which way it draws.

Use its heat, never its air.

- **A sealed exchanger on the flue** – smoke on one side of a steel wall, clean air on the other – warms air that can dry the frame or regenerate a drawer. The smoke side never meets the seam's air.
- **Keep that air under 60 degrees** before it reaches a seam.
- **Give the stove its own outside air** for combustion, through its own duct. Building-science guidance says the same for any tight building: a stove that draws its air through the gaps in the envelope competes with every fan in the house.
- **Never connect the stove's draw or its flue to the seams.** A stove that can pull on the seam network is a stove that can back-draught smoke into the walls the first time a fan pulls harder than the chimney.



The flue sealed away; clean air warmed across a wall of steel.

CHAPTER 36

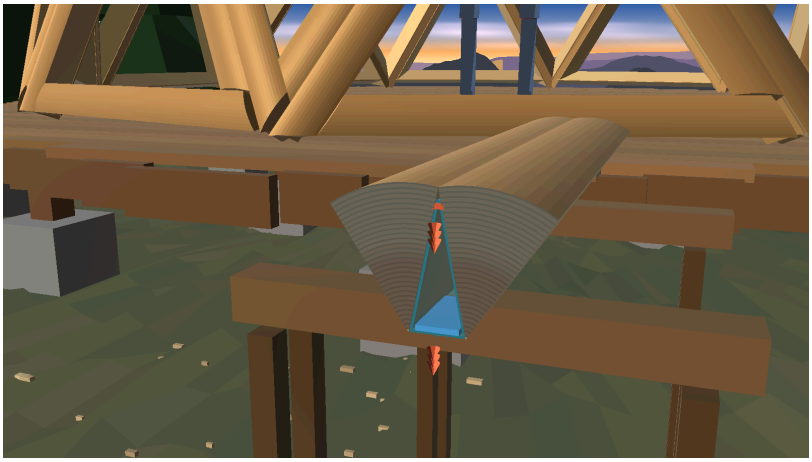
Metal in the Seam

A Peltier plate is a heat pump that runs both ways, and aluminium, copper and stainless make a battery the moment the seam is wet

The line that runs from the centre of the dome straight outward has a name: it is the **radial** direction. Through an ordinary wall it is the wall's thickness. Through a seam it runs from the V's open mouth on the room side up to the ridge on the weather side.

So the seam has two skins, the way a living cell has an inner and an outer membrane. The **outer skin** is the ridge, under the cap, touching the weather. The **inner skin** is the mouth of the V, touching the room. Put a thermoelectric plate on each and you can set the temperature of both independently.

That arrangement is the owner's, and it is a good one. This chapter is about what it takes to make it work: the plates, the metal channel that carries their cold, and the metals that must not touch.



The outer skin at the ridge, the inner skin at the room side, and a plate on each.

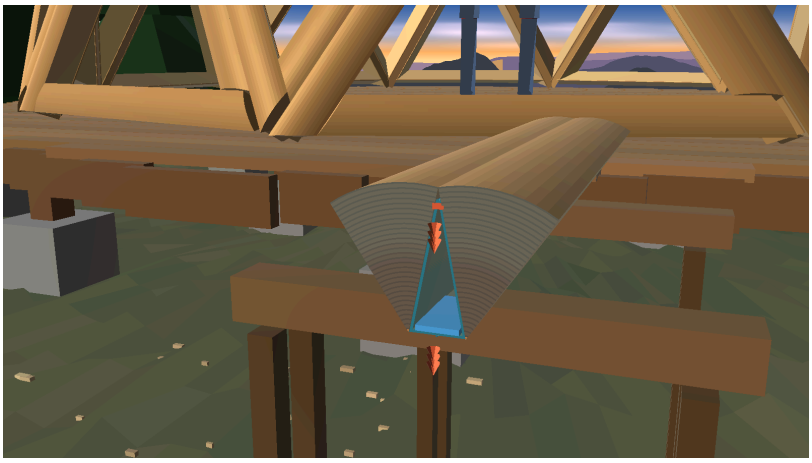
The plate runs both ways

A Peltier plate is not a cooler. It is a small solid-state heat pump: current through it moves heat from one face to the other, and reversing the current reverses the direction.

So the rule for both plates is simple: **throw the heat to the side you want warmer.**

- **In winter**, the cold face is in the channel and the heat goes into the room. The outer skin is already cold – often cold enough to condense on for free, with no power at all.
- **In summer**, the cold face is still in the channel, but the heat goes to the outer skin and out under the cap. Throwing it into the room would mean paying twice: once to make the cold, and again to the air conditioner to remove the heat you just put in the room.

Chapter 29 drew the plate throwing its heat into the room. That is right in winter and wrong in summer.



Winter: heat to the room. Summer: heat outside.

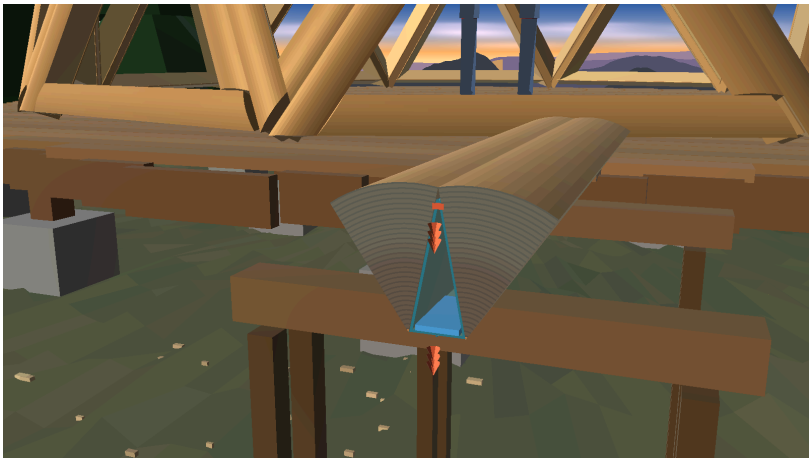
Fifteen to twenty degrees colder

The owner's question was this: if the framing wood is at the dew point, but the aluminium or copper channel is fifteen to twenty degrees below that, will the water go to the metal and leave the wood alone?

Take a room at 21 degrees and 55 percent. Its dew point is 11.6. The middle of the owner's range, 17.5 degrees under that, puts the plate at minus 5.9.

The plate freezes. Below zero, water comes out of the air as frost rather than liquid. Frost does not run to the drain; it builds up on the fins, blocks the air, and insulates the plate from the very air it is meant to dry.

So hold the plate 6 degrees under the air's dew point and never below 1: here, 5.6 degrees. That is cold enough to condense steadily and warm enough to drain.



A plate driven that far under the dew point freezes.

Plate first, wood second

And the real answer to the question: **a colder surface somewhere else does not protect wood that is already at the dew point.** Water comes out on every surface colder than the air touching it. Wood at the dew point, washed by raw room air, gets wet, however cold the plate across the channel is.

What protects the wood is **order**. Send the air over the plate first. It leaves carrying a dew point of 7.6 – about the plate's own temperature plus a couple of degrees – and then it reaches the wood. Wood at 11.6 is now comfortably above it; anything above 10.6 is safe with the margin this book keeps.

So in every mode that sends damp air along the frame, the plate is upstream of the wood. In the rim layout of Chapter 33 that happens by itself: the air comes in at the rim, over the plates, and then up the lower band.

Air over the plate first leaves the wood with room to spare.

The cold metal must never touch the wood. Aluminium carries heat 1,025 times better than the printed key's plastic. A metal channel laid against a sawn face chills that face toward its own temperature, and then the wood is the cold surface. Keep the metal inside the key, or behind a strip of closed-cell foam, never bare against the pine.

Which metals may touch

Put two different metals in contact, add water, and you have made a battery. The less noble metal dissolves into the more noble one. How fast depends on how far apart they are on the galvanic scale, and whether they stay wet. In a seam that condenses water on purpose, they stay wet.

The engineering rule, from military and aerospace practice, is that metals kept wet may touch only if their anodic index differs by 0.15 volts or less:

metal	against	difference	wet service
copper	stainless (passive 304/316)	0.15 V	yes
copper	aluminium	0.55 V	no
copper	galvanised steel	0.85 V	no
stainless (passive 304/316)	aluminium	0.40 V	no
stainless (passive 304/316)	galvanised steel	0.70 V	no
aluminium	galvanised steel	0.30 V	no

Three rules come out of that table.

1. **Copper and aluminium never touch**, and water never runs from copper onto aluminium: at 0.55 volts apart, dissolved copper plates out on the aluminium and pits straight through it. If the plates are copper and the liner is aluminium, put a plastic break between them and keep the copper downstream.
2. **Stainless bolts through aluminium need sleeves.** At 0.40 volts apart, a stainless bolt in a wet aluminium liner eats the liner round the hole. Nylon washers and sleeves isolate it.

3. **Copper and stainless may touch** (0.15 volts). If you are choosing one metal for the condensing channel and the bolts are stainless, that is the argument for copper. It also carries heat 1.9 times better than aluminium, which is the argument for it as a plate; it costs more and weighs more, which is the argument against.

Keep galvanised steel out of the seam entirely.

A liner grows against the wood

Metal moves with temperature much more than wood does along its grain. Over the longest seam, 72 inches, and a 50-degree swing between a winter night and a summer afternoon under the cap:

aluminium liner	2.1 mm
copper liner	1.6 mm
the pine beside it	0.4 mm

So fix a liner at one end only, and let the other end slide in a slot. A liner screwed down at both ends either buckles or pulls its screws, and the screw holes are where the water gets in.

The figures behind this chapter:

k aluminium	205 W/m K	standard
k copper	385 W/m K	standard
k stainless	16 W/m K	standard
k petg	0.2 W/m K	nominal
k pine	0.12 W/m K	standard
alpha aluminium	2.3e-05 1/K	standard
alpha copper	1.7e-05 1/K	standard
alpha pine	4e-06 1/K	standard
seam swing k	50 K	assumption
wet limit v	0.15 V	standard

CHAPTER 37

Ten Plates, Round the Clock

Ten twenty-dollar Peltier modules at the rim ports, run day and night on the sun: what they really move, what water they really make, and the array and bank it takes to feed them

The owner bought one thermoelectric module, a TEC1-12706, for $20. The listing calls it a 70-watt cooler with a 67-degree maximum temperature difference, running at 12 volts. The proposal: put 10 of them round the dome, one at each rim port, and run them day and night on the sun.

That asks three questions, and this chapter answers each one with numbers:

1. What does one module really do, mounted on the dome, on a humid day?
2. What do ten of them do for the room over a day – heat out, water out?
3. What array and battery does it take to run them all day and all night, and what does that cost?

The answer, up front: ten plates cool a little and dry a little, and the power system to run them round the clock costs about 81 times what the plates do. Run them on daylight that would otherwise go unused, and they are worth having. Run them around the clock, and they are not.

> **WE GOT THIS WRONG**
>
> A correction belongs here too. Chapter 29 counted six plates at 60 watts, making 0.35 litres of water per kilowatt hour. That yield was an assumption. Solved from the module's own listing below, the yield is 0.20 – each litre costs 4.98 kilowatt hours, not 2.86. The tables in Chapter 29 now carry the solved figure.

What one module really does

A Peltier module moves heat from its cold face to its hot face. Three things work against it, all inside the module itself: the electricity heats it as it passes through, heat leaks back from the hot face to the cold one, and the greater the temperature difference, the less heat it can move. Put in the listing's own maxima – 67 degrees, 15 volts, the internal resistance – and those three effects are fixed. The rest follows from them.

v_max	15 V listing TEC1-12706: rated 12 V, 15 V maximum
dt_max	67 K listing the largest temperature difference it can hold, at no load
r_module	2.25 ohm listing internal resistance 2.1-2.4 ohm: the middle
q_max_listed	70 W listing the cooling power on the listing; checked here, never used
i_listed	4.3 A listing working current at 12 V, 4-4.6 A: the middle
th_rating_c	27 C assumption hot-face temperature the maxima are rated at; makers use 25 to 50 C, and the low end gives the module the benefit of the doubt
module_usd	20 USD owner what the owner paid for one
modules	10 - owner one at each of the dome's ten rim ports
r_hot_sink	0.2 K/W assumption hot face to outside air: a finned aluminium sink with its fan, the kind sold with the module. A bare plate is several times worse
r_cold_sink	0.4 K/W assumption cold face to room air: a smaller finned plate with a slow fan, wet with condensate
fan_w	2.4 W nominal one 12 V 80 mm fan at 0.2 A; two per module, one each face

converter_eff	0.92 - assumption a buck converter from the 24 V bank to each module's 12 V
battery_round_trip	0.95 - nominal LiFePO4, energy in to energy out
battery_dod	0.9 - assumption the share of a LiFePO4 bank used each night
solar_window_h	8 h assumption hours a day the array is making more than the load; the other hours run on the bank
autonomy_days	1 days assumption one sunless day the bank must cover on top of the night, for 24/7 to mean 24/7
panel_w	400 W nominal one ordinary residential panel, about 1.0 x 1.9 m
rack_kwh	5.12 kWh nominal one 48 V server-rack LiFePO4 battery
people	2 - assumption the household the cabin is sized for
person_sensible_w	75 W standard ASHRAE: a person at rest, sensible heat
person_latent_w	45 W standard ASHRAE: a person lightly active, latent heat (their breath and skin)
household_extra_l_day	3 L/day estimate cooking, washing and a shower for two, beyond their breath; published household totals for two run about 5 to 8 litres a day all in
dehumidifier_l_per_kwh	1.8 L/kWh estimate a small compressor dehumidifier at the US Energy Star floor for its class
window_ac_btu_h	5000 BTU/h nominal the smallest window air conditioner sold
window_ac_ceer	11 BTU/Wh estimate its efficiency, about the US minimum
old_modules	6 - superseded wedge_book.systems until 2026-10-04: six modules at the wettest seams
old_watts_each	60 W superseded the draw it assumed per module
old_litres_per_kwh	0.35 L/kWh superseded the yield it assumed; quoted by The Seam Does Four Jobs and The Wedge Dome Explained
solar_usd_per_watt_diy	0.95 borrowed declared in seed_model / park_model
battery_usd_per_kwh	410 borrowed declared in seed_model / park_model
solar_derate	0.78 borrowed declared in seed_model / park_model
sun_hours_per_day	4.6 borrowed declared in seed_model / park_model

The first finding is the box: **the module's real maximum is 50 watts, not 70.** And that maximum is only reached with *no* temperature difference across the module – cold face as warm as the hot face, which is a module cooling nothing in particular.

Now mount it. The hot face goes onto a finned sink with a fan, outside the room, at 30 degrees. The cold face goes onto a smaller wet sink in the room's air, at 26 degrees and 60 percent. Both sinks resist the heat, so the hot face settles at 46.7 degrees and the cold face at 14.8. The module now draws 4.6 amps, inside the listing's 4 to 4.6. It moves **28 watts** of heat out of the room for 65 watts of electricity, counting its two fans.

The room's dew point that day is 17.6 degrees, so the cold face is 2.9 degrees under it. It condenses – but only 32 percent of the heat it takes is water. The rest just cools the air. That is 0.31 litres a day from one module.

Slower is better, until it stops condensing

A Peltier is far more efficient run gently. Most of what the electricity does at full drive is heat the module itself, so halving the voltage roughly halves the cooling but cuts the draw to well under half:

volts	amps	cold face cooling drawn	COP	water	ten modules
		C W each W each		L/day	kWh/day L/day
3	1.10	22.0 10.0 8.4	3.05	0.00	2.0 0.0
6	2.26	18.4 19.0 19.6	1.40	0.00	4.7 0.0
9	3.44	16.0 25.0 38.5	0.81	0.21	9.2 2.1
12	4.62	14.8 28.1 65.1	0.51	0.31	15.6 3.1

At 3 volts a module moves 3.05 watts of heat for every watt it draws; at 12, 0.51. So throttle it – with one catch, which the table makes plain. At 6 volts the cold face only reaches 18.4 degrees, *above* the dew point, and the water column goes to 0.0. A plate that does not get under the dew point is a cooler, not a dehumidifier. The most water for each kilowatt hour comes at 9 volts: 0.23 litres.

And here the module meets the dew-point controller of Chapter 32, which asks for the plate 6 degrees under the dew point: on this day, 11.6. **Mounted, at full drive, the module cannot get there.** It stops at 14.8, 3.1 degrees short, so a controller chasing that setpoint simply runs it flat out all day. A module this size, on sinks this size, cannot reach that setpoint in humid weather. The useful setting is the

opposite one: cap the drive at 9 volts. Ten then make 2.1 litres a day for 9.2 kilowatt hours, against 3.1 for 15.6 at full drive.

A day of ten

Run ten at 12 volts, all day and all night, in three kinds of weather. Against them, put what the room takes on: two people breathing and cooking and washing, and the fresh air of Chapter 34, which on a humid day carries water in with it.

weather	room	outside	cold face	cooling water	load: water heat
	C RH	C	C	W L/day	L/day W
humid summer day	26 60%	30	14.8	281 3.1	9.1 190
muggy night	24 65%	24	12.3	292 4.1	9.5 150
wet spring day	21 65%	16	8.8	305 4.1	6.2 150

On the humid day, ten plates take 281 watts of heat out of the room and 3.1 litres of water out of its air. The room takes on 9.1 litres: 3.2 from two people breathing, 2.9 on the fresh air, and the rest from cooking and washing. **The ring removes about 35 percent of the day's water.** It helps; it does not keep up.

The heat side is better than it looks. The people and the fresh air put in 190 watts, and the ring takes out 281. But the table leaves out the sun on the shell, which is the largest load of a summer afternoon. Against that, 281 watts is a fan's worth.

One rule from Chapter 36 matters more here than anywhere: **the hot faces go outside.** Ten modules throw 836 watts out of their hot faces – every watt they took from the room plus every watt they drew. Mount them with the hot faces inside, and the ring is a 651-watt heater that happens to drip.

Feeding it on the sun

Ten stations draw 651 watts, all day and all night. That is 15.6 kilowatt hours a day. The sun only outruns the load for about 8 hours, so 10.4 of those kilowatt hours have to come off a bank. The array has to make the day's use and refill the bank, from 4.6 peak sun hours. For twenty-four hours a day to mean it, the bank has to cover one sunless day as well as the night.

load, ten stations	651 W, all day and all night
energy a day	15.6 kWh
of it at night, off the bank	10.4 kWh
array	4,508 W $4,283 (12 panels of 400 W)
bank, a night + a dark day	28.9 kWh $11,864 (6 racks of 5.12 kWh)
ten modules	$200
total	$16,347

> **THE NUMBER THAT DOESN'T HELP**
>
> **About 4,508 watts of panels and a 28.9 kilowatt-hour bank: $16,347, for $200 of modules.** The array and bank are rated at the prices declared everywhere else in this book. Nothing was chosen to make this look bad.

The comparison that settles it: a small compressor dehumidifier removes the same 3.1 litres for about 1.7 kilowatt hours; the ring takes 9 times that. And the smallest window air conditioner sold moves 1,465 watts of heat for 455 – 3.2 watts moved for every watt drawn, against the ring's 0.43.

What to do instead

The plates are not wrong. They are the only part of this design that can take water out of a seam that will not dry itself – warm, still, humid weeks, with the desiccant full and outside air no help. Keep them for that.

- **Run them on daylight surplus**, not off the bank. At 6 hours a day the ring uses 3.9 kilowatt hours, all of it power the array was making anyway, and drops 0.79 litres. That is the figure Chapter 29 now carries, and it costs no extra battery.
- **Cap the drive at 9 volts**, and let the controller switch the plates on and off rather than chase a setpoint they cannot reach. Past that, every extra watt mostly cools the air rather than drying it.
- **Put the plates where the air is wettest.** Wet air gives up a bigger share of a plate's heat as water. At the rim, the fresh air of a muggy day meets the plate before it meets the wood – Chapter 36's rule.
- **For the house, buy the compressor.** If the room itself needs drying or cooling for weeks at a time, a dehumidifier or the smallest air conditioner does it for a fraction of the power. The plates look after the frame; the compressor looks after the people.

PART 7

The Floor and the Ground

“Everything you own sits on the floor, and the ground is the one part of the building you cannot take with you.”

The second thing that has to be right, the five ways to build it, and the argument for standing on somebody else's ground.

CHAPTER 38

The Floor Is the Other Thing to Get Right

It carries everything you own, you touch it all day, and it is where the ground sends its water

You will spend more hours in contact with the floor than with any other part of this building, and it is the only part in permanent contact with the ground.

Both of those are reasons to spend money here rather than on the frame.

> **NOT ON THE CLOCK**
>
> The floor is not on the 40-hour clock. The clock counts the frame, from the first log on the splitting stand to the last bolt in the shell – Chapter 27 adds it up. Most people build it first, before the clock starts, so the frame has somewhere flat to go up on.

What “right” means

Dry. The ground is wet. It is wet under a slab, under a deck, and under gravel, and it will move moisture upward into anything that touches it unless something stops it. A capillary break — gravel, a membrane, an air gap — is not optional and it is cheap at the time and impossible afterwards.

Warm underfoot. A cold floor makes a warm room feel cold, because you are touching it and you are not touching the air. This is worth more comfort per dollar than the same money spent on the walls.

Flat and stiff. You notice a floor that flexes every single day.

Level, to a few millimetres. Not because the dome needs it — the dome would stand on a slope — but because everything you put in the dome needs it.

Five ways to do it

Priced on this footprint, taken off board by board rather than rated per square foot:

Bare ground to a treehouse platform.

compacted gravel base	$443
full concrete slab	$2,468
framed deck on piers, sealed	$2,916
concrete ring with a timber middle	-- see the table
framed deck, ply and epoxy over it	-- see the table

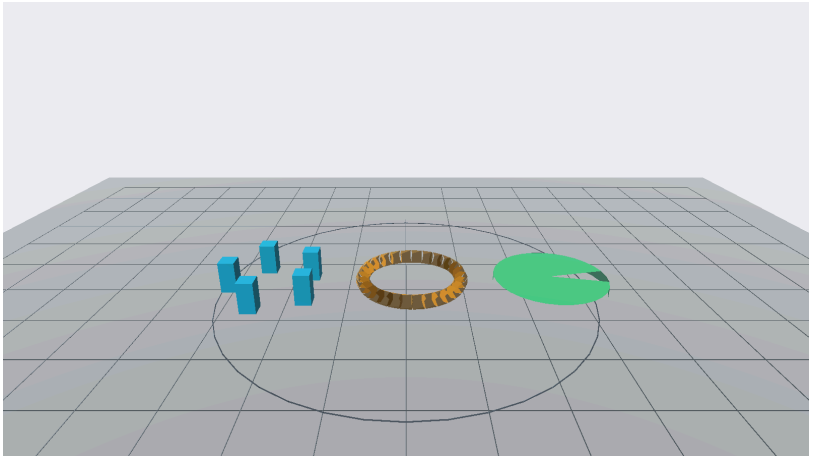
The staple is the framed deck on piers at $2,916: it is dry by construction, it is warm because there is air under it, it is repairable, and it does not require a concrete truck to reach your site.

The gravel base at $443 is not a floor. It is a prepared ground that a floor can go on. It is in the list because it is what a lot of people actually start with, and it is honest to say that it is a starting point rather than a cheap answer.

The thing that surprises people

Half a pour is not cheaper than a whole one.

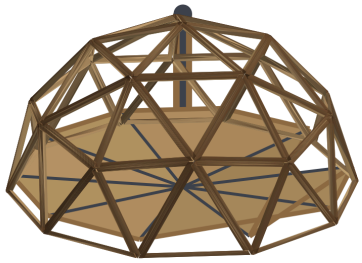
The concrete ring with a timber middle costs *more* than the full slab. The saving on concrete is smaller than the cost of the timber deck that replaces it, and the ring's formwork is fiddlier per cubic yard than a slab's.



Meeting the ground: piers, a ring beam, or a slab.

This is in the book because the intuition is so strong and so wrong, and because the model prints the number that does not flatter the argument.

The floor that clamps



The mast through the column, and the floor that clamps to it, sitting on the base ring where a floor goes.

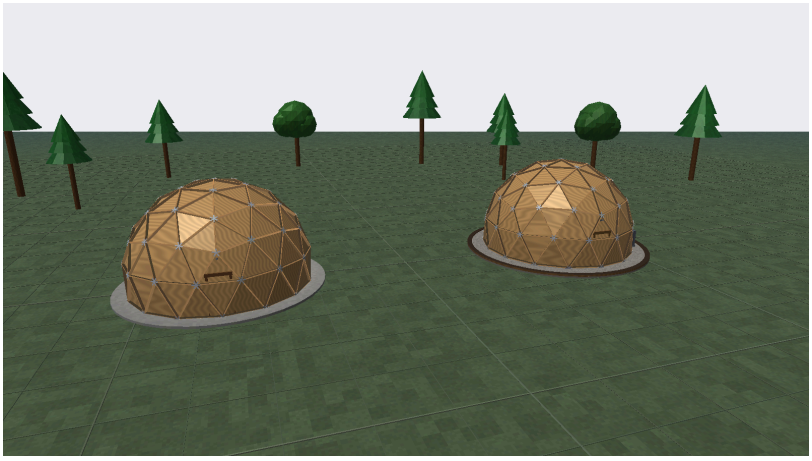
There are two ways to build a floor in a dome.

Framed into the base ring. Joists between the ring segments, decking over. Simple, strong, and permanent: it is now part of the structure and removing it is a demolition.

A hub, spokes and decking. A steel hub in the middle, radial spokes out to the base ring, timber over the top. It is a component. It comes out, it goes back, it can be replaced with a better one in ten years, and it can be left behind when the dome moves.

Chapter 3 argues for the second on general principles — the floor is one of the two layers that matters, so keep it changeable. It also turns out to be what makes Chapter 72 possible, which was not the reason for doing it.

The hub wants something in the middle to clamp to. In this design that is the utility column, which is already there, already centred, and already the one thing running floor to apex.



The part of a house you never take with you, priced across the catalogue.

Under-floor is storage

A deck on piers has a void under it. That void is dry, it is out of the way, and it is the cheapest cubic footage in the building.

Design for access to it from inside — a hatch, not a crawl from outside — and it is the difference between a small building that feels cramped and one that does not.

What this chapter will not tell you

Frost depth, soil bearing, seismic anchorage.

Those are local, they are the reason foundations are the part of a building most tightly bound by code, and this book does not know where you are. The five constructions above are priced take-offs of a design; what depth your piers go to is a question for somebody who has stood on your ground.

CHAPTER 39

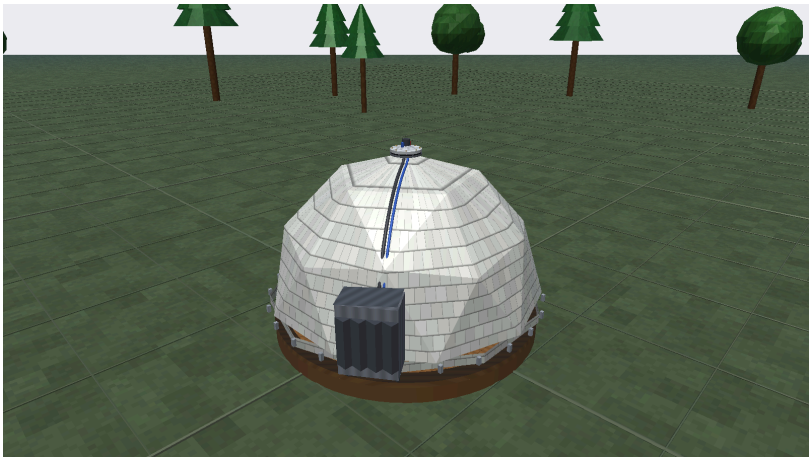
A Pad, Not a Plot

The ground is the one thing you cannot take with you, so stop buying it

The ground is the one part of this building you cannot take with you.

Everything else — 120 members, forty panels, a column, a cap, a floor — comes apart and moves. The ground does not, and it is usually the most expensive thing in the whole arrangement.

So stop buying it.



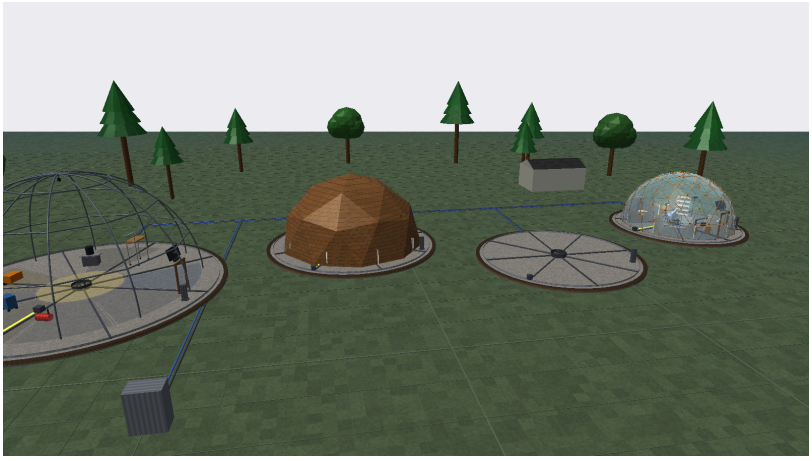
The landowner builds the pad. It stays when the dome leaves.

The split

The host owns the ground and everything permanently attached to it. The pad, the service port coming up through it, the water and power to that port, the access track, the under-floor storage if there is any.

The tenant owns the building. The frame, the panels, the shell, the column, the floor, the cap.

That line is not arbitrary. It falls exactly where “can this be carried away” falls, which means it never has to be argued about. Nothing straddles it.



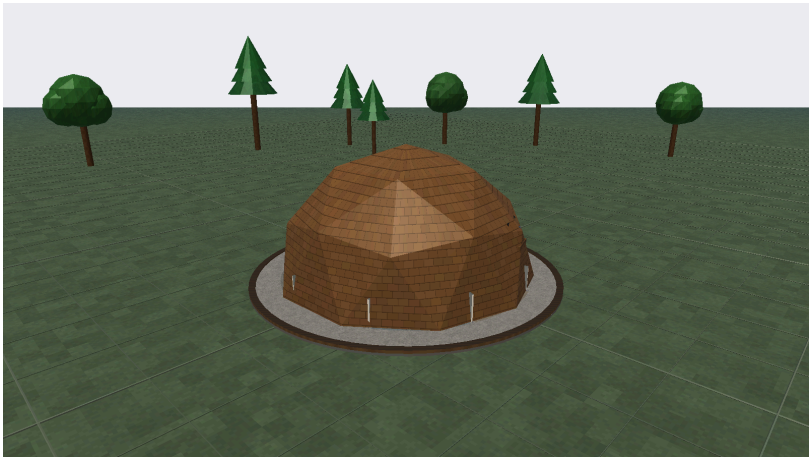
Two people, and what each of them owns.

Why it is a clean line

The usual version of this arrangement — renting land and putting something on it — is messy because the something is half-attached. Whose slab is it? Whose septic? Who pays when the deck rots?

Here the pad is a single object with a price. In this model it is $6,240, built once, by the host, to a specification the tenant can check before arriving. The dome is another object with a price, $12,036, and it arrives on a trailer.

When the arrangement ends, the tenant takes a building and the host keeps a platform. Both of those are worth something to their owner and neither is worth arguing about.



The line that never moves.

What the host is really selling

Not land. A **prepared, serviced, level place to put a building**, which is a much more specific thing and much more valuable than an acre of field.

Preparing that place is the expensive, skilled, permitting-heavy part of building anywhere. It involves earth, concrete, a trench, and somebody who knows where the frost line is. Doing it once and having it used for decades is a far better use of that effort than doing it once per owner.

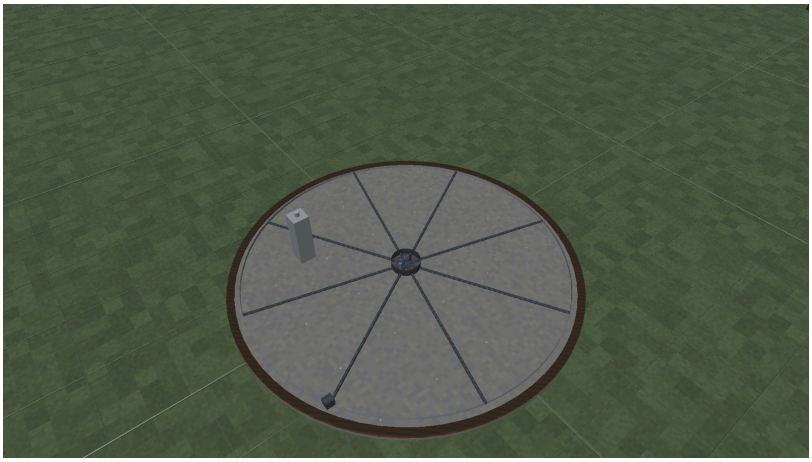
And what the tenant is buying

A foundation, without a mortgage.

The pad is cheaper than the foundation they would otherwise pour, because it is amortised over more than one occupancy. It is available now rather than after a land purchase. And — the part nobody expects to value until they have it — it is *leavable*.

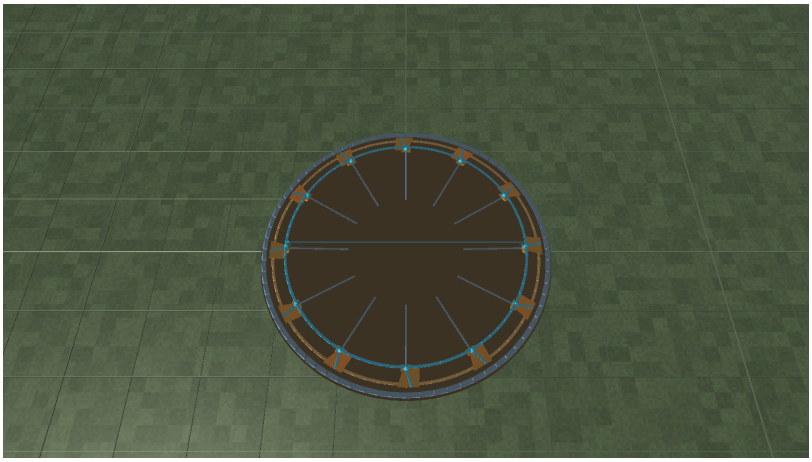
What a pad actually is

Graded ground. A deck – gravel, concrete or timber, whatever the host wants to pay for. A rim of wood round the edge that the dome latches to, which is the part that makes it a pad rather than a patio. Then power, water and drain coming up through the middle.



A deck, a rim, and the services up the middle.

The rim can be a fixed ring. Or it can be an **iris**: leaves on a track that open and close like a camera aperture, so one pad takes more than one size of dome without being rebuilt. That mechanism is a development concept, priced, and it belongs in the upgrade column, not the starter one.



One pad, more than one dome size.

Pad sizes are not invented. Every design the Dome Creator ships has a foundation diameter the tool computes from its own geometry; round each up to the next four feet and you have the pad catalogue:

```
the Dome Creator ships 12 designs, and each one has a
foundation diameter the tool computes from its own geometry.
round each up to the next 4 ft and you have the pad sizes:
      36 ft    1,018 sq ft   takes  1 of 12
      40 ft    1,257 sq ft   takes  4 of 12
      44 ft    1,521 sq ft   takes  7 of 12
      48 ft    1,810 sq ft   takes 10 of 12
      64 ft    3,217 sq ft   takes 11 of 12
      80 ft    5,027 sq ft   takes 12 of 12

but one pad does not have to be one size. the three domes
this system is for, by floor area and longest member:
   small personal    314.00 sq ft  ·  20.0 ft across  ·  6 ft member
   medium            822.49 sq ft  ·  32.4 ft across  ·  10 ft member
   large           1,184.39 sq ft  ·  38.8 ft across  ·  12 ft member
an iris rim -- leaves on a track, opening and closing like
a camera aperture -- spans 19 to 40 ft and takes all three.
the mechanism is $12,400 on top of the deck.

the treehouse canopy dome needs 35 ft of pad;
the whole trunk lodge - 20 ft needs 76 ft.
4 ft steps because decking and ring stock come in even lengths,
and the second pad a host builds wants the first one's cut list.
```

The pad is the floor

This is the move the whole idea turns on. The dome that arrives does not bring a foundation, because it does not need one. The deck it lands on is the deck it lives on, the services come up through the middle of it, and when the owner leaves, the pad is exactly as it was. No slab poured, no footing dug, no permit pulled for a structure that will be gone. Bring your carpet if you want one.

Dome rentals already exist, and most of them are glorified tents rented a weekend at a time. Own the glorified tent and rent the place to put it, and both sides have a reason to make it good.

Start with the cheapest working version

Before any iris or turning base, price the starting point: a small framed deck on blocks, a fixed rim, and four pads sharing one power panel and water manifold:

```
STARTER SCENARIO / declared unit prices
20 ft pad = 314 sq ft
deck on blocks, 314 sq ft: $4,398
share of one hub panel, 1 of 4: $600
spur from the hub: $450
permits: $1,200
site access + trunk service allowance: $6,500
TOTAL: $13,148
Author's all-in target: $10,000
Fixed rim; shared services; no iris or rotation.
Site allowance must cover the actual local works.
Start with a working floor and services; add features later.
```

That is above the target, and it is printed anyway. It shows exactly where the design work has to go – the site allowance, mostly – rather than pretending the difference is gone.

The pad turns

A pad on a rotating base can aim whatever stands on it. The panels are on the dome; the pad turns them. Worked for the films' medium, two-person dome:

```
every figure here is for a 10 ft radius dome -- the medium,
two-person size. a number a month means nothing without that.
the panels are on the dome, not the pad. the pad turns.

   one side               20 of 40 panels  ·    292 sq ft  ·    5.2 kW
   top half, skirt below  20 of 40 panels  ·    292 sq ft  ·    5.2 kW
   whole shell            40 of 40 panels  ·    583 sq ft  ·   10.5 kW

   one side               fixed   734  ·  tracking   918 kWh a month
   top half, skirt below  fixed   734  ·  tracking   918 kWh a month
   whole shell            fixed 1,468  ·  tracking 1,835 kWh a month

the skirt below the cells is the catchment -- the same metal that
keeps rain off the bottom ring collects it.
a tenant uses about 420 kWh a month.
and the one that does not flatter the tracking ring: cover the
whole shell and it is always facing the sun with something, so
turning the pad buys much less than it does on a half-clad dome.
the whole shell also makes 4.4 times what a tenant uses. that is surplus,
and the model refuses to credit surplus at retail: it is worth $0.045 sent back,
not the $0.14 it would have saved.
```

The host bought the ring. The tenant bought the panels.

The number that decides it

The pad is $6,240. The floor a dome brings with it is a few thousand more. Against a piece of land, which in most places is a five-figure number before anything is built on it and a thirty-year commitment after.

That ratio is the entire argument for the next chapter, which works it through from both sides and says what would have to be true for it to hold.

PART 8

What It Costs and What It Is Worth

“The tree is the same on every rung. What moves it up the ladder is how much of it you keep, and how many of the steps between the tree and the wall you are willing to take yourself.”

What a tree is worth on every rung, every hand between a stump and a stud, how much shelter an hour of your life buys, where a house's money goes, and the invoice line by line.

CHAPTER 40

A Twenty-Dollar Pine Is Not a Twenty-Dollar Pine

One tree priced on every rung: on the stump, burned, sawn, split, built in, and carried on a mortgage -- the wood never changes, the use does

What is a pine worth? Not one price. A ladder of them, one for every use the same tree can be put to – and the tree does not change from rung to rung. Only what happens to it does.

This chapter climbs the ladder for one representative pine: 60 feet of usable stem, 15 inches at the butt and 5.5 at the top – the same worked tree as Chapter 12.

What the ladder rests on

Measured, published and the author's own figures, kept apart, because the ladder is only as honest as its inputs:

```
THE AUTHOR'S  a representative pine: 60 ft of stem, 15 in to 5.5 in
THE AUTHOR'S  60% kept by a mill; 88% by splitting; 90 cu ft of wood a cord
THE AUTHOR'S  sawing $0.60/bd ft; a cord $275 picked up, $300 delivered
THE AUTHOR'S  $4,800 of framing for the shell; it uses 1.5 trees, 2 felled
PUBLISHED  pine $1.50/bd ft; stumpage $23.23/ton; a cord 128 cu ft stacked
CHECK  this log in the simulator: sawn to 2x4s 45%, split 88.4% before trimming
So the mill gets the benefit of the doubt, not the wedge
```

The stem, in board feet

The stem is 36.8 cubic feet of solid wood: 442 board feet, which is the fixed budget every method below starts from. Sold standing, at the published stumpage price, the whole tree brings about **$24**. That is the bottom rung, and it is roughly what the chapter title calls it.

Burned, it is heat

The first rung up is fire. A cord is 128 cubic feet stacked, but air fills the gaps and only about 90 of them are wood. So the stem makes 0.41 of a cord: **$113** picked up, about $123 delivered. Every rung above this one has to beat the value of the tree's heat.

Sawn, or split

Next, lumber. A mill keeps about 60 percent of the stem as boards, 265 board feet. Splitting keeps about 88 percent, 389 – the only wood a split loses is the width of the saw.

```
split against sawn          46.7% more usable wood
left behind, sawn           177 bd ft
left behind, split          53 bd ft (70% less)
trees sawn to match 2 split   2.93
```

Priced as wood at a sawmill's $1.50 a board foot, the sawn lumber is **$398** and the split wood **$584**: splitting is worth $186 more from the same tree. The mill gets the benefit of the doubt throughout – the simulator's own sawn recovery for this log is 45 percent, well under the 60 used here.

Built, it replaces framing

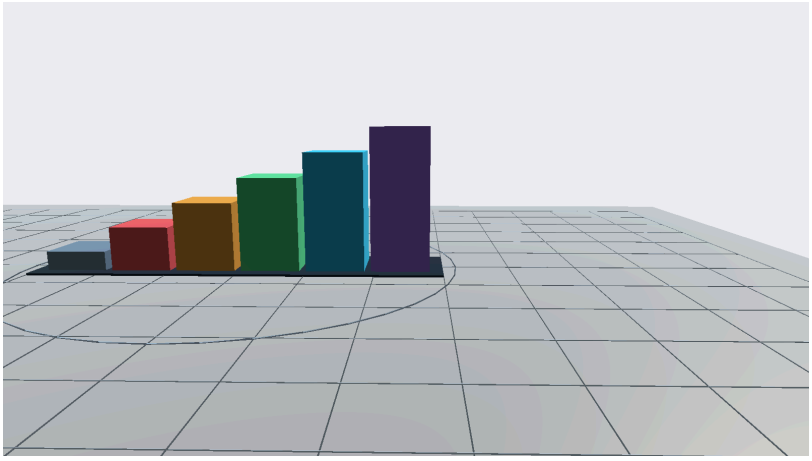
The next rung is not a price for wood at all. In the dome, these trees become a shell, and the framing that shell replaces is worth $4,800 at commercial rates. The frame uses 1.5 trees' worth of trunk, but 2 are felled, so each tree's share is **$2,400**.

Most framing is not paid for in cash; it rides on a mortgage. On 30 years at 6.5 percent, that framing becomes $10,922 of payments, and this tree's half is **$5,461**. Read that rung carefully: it is nominal dollars, paid monthly for decades, not money in your hand today. But it is money a household never sends to a lender.

The ladder

on the stump	$24
burned	$113
sawn, as lumber	$398
split, as lumber	$584
built into a shell	$2,400
that framing, financed	$5,461

As structure it is worth 21 times its firewood, 6 times its sawn lumber and 4.1 times its split stock. The wood did not get better on the way up. It got used.



One tree, every rung.

What the tree is worth to its owner

The honest figure is not the top rung. It is what the tree replaces, less what it costs the owner to make it do so: about 40 hours of the owner's own work for this tree, worth $1,000 at the author's wage, and $400 of cash. Net, **$1,000** – or $1,152 with the fortnight's actual receipts.

What the ladder does not say: the structure rung assumes you needed the house; the finance rung is spread over thirty years; and the author's own figures, though marked, are an author's. The generous numbers and the ones against them are both printed above.

The household as the whole chain

This is vertical integration at the scale of one household. A commercial house pays every hand between the stump and the wall: the logger, the hauler, the mill, the kiln, the yard, the framer, and very often the lender. Each takes a margin, and every margin is inside the price. This method does not make those businesses unnecessary for everyone. It lets one owner do the few steps that matter for one building – and keep the margins. The next chapter counts the hands.

CHAPTER 41

Fifteen Sets of Hands

Every machine and every middleman between a tree and a stud, counted -- and the same frame bought at the rack, with everything nobody counts

Almost none of a stud's shelf price is the tree.

There are two ways to get from a standing tree to a wall. The long chain: earn an income, pay tax on it, drive to a store, buy lumber somebody else has sawn, dried, graded and shipped, pay a contractor to put it up, and very often borrow to do all of that. The short chain: tree, saw, wall. This chapter counts what the long chain is made of, so the short one can be priced honestly against it.

Counting the machines out

Every conversion step between a log and a stud is a building full of machinery, with its own setup, labour and margin. Station by station, which of them the wedge path still needs:

```
what has to happen to a log before it can hold a roof up:
  BOTH  fell and buck            chainsaw
        identical in both, so it cancels
  SAWN  primary breakdown        SAWMILL
        wedge path splits radially with the same chainsaw
  SAWN  edging                   EDGER
        there is no waney edge to remove from a split face
  SAWN  drying and grading       kiln / grader
        wedges are used green; the shell is not a graded-lumber assembly
  SAWN  surfacing to dimension   PLANING MILL
        the sawn faces come off the split already flat and already exact
  BOTH  crosscut to length       crosscut saw
        one square cut, both paths
  BOTH  end joinery              compound saw / jig
        sawn path measures two compound ends; wedge path cuts one butt and flush-cuts the head in place
machines the wedge path never buys, feeds or houses: 3
  SAWMILL, EDGER, PLANING MILL
what is left on the wedge side is one chainsaw and a flat board
```

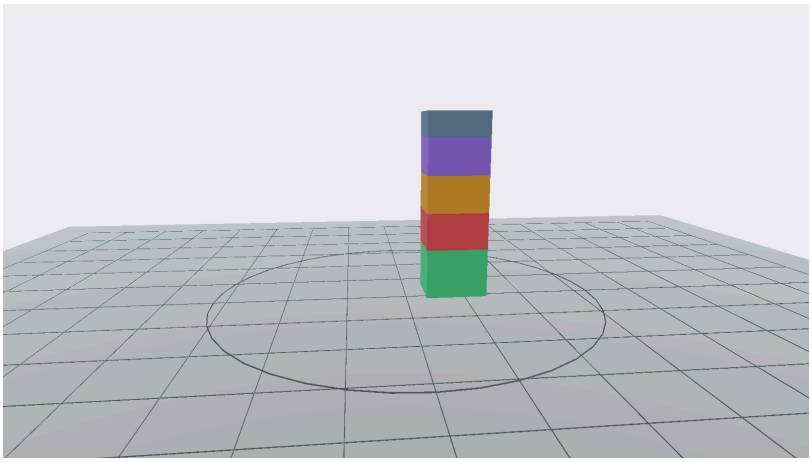
The wedge path drops the sawmill, the edger, the kiln and grader, and the planing mill. It keeps the chainsaw, the crosscut and the joinery.

What a stud has been through

And the hands. A stud on a rack has been sold and handled this many times on its way there, listed rather than rounded, because a round number is easy to say and hard to check:

```
a stud on a rack has been sold and handled this many times:
   1. timber broker        sells the standing timber
   2. logging contractor   fells and skids it
   3. log haulier          trucks it to the mill yard
   4. log scaler           measures and grades the log
   5. headrig sawyer       breaks the log down
   6. edger operator       squares the waney sides
   7. trimmer operator     squares the ends
   8. kiln operator        dries it to a moisture spec
   9. lumber grader        stamps a grade on it
  10. planing mill         surfaces four sides to dimension
  11. bundler              packs and wraps a unit
  12. mill sales           sells the unit
  13. freight broker       books the flatbed
  14. distributor          holds stock and breaks bulk
  15. retail yard          sells you one stick
then sales tax: $8.88 on the rack becomes $10.00 paid
every one of the 15 took a margin, and every margin is in that price
the wedge path is: fell the tree, rip the trunk, use the wood
```

Each of them is a real job done well. Each of them also takes a margin, and all of those margins are in the price on the rack.



Every hand between the stump and the rack.

What one wedge is worth

So what is a wedge worth, priced against the bought sticks it would take to replace it? Two ways, and how far apart they land:

```
a 2x4x16 untreated pine is $8.88 on the rack, $10.00 with tax
  = $5.00 per eight-foot section
one 1/8 wedge of a 12 in trunk has 14.14 in2 of section
a dressed 2x4 has 5.25 in2
  computed multiplier = 14.14 / 5.25 = 2.69 x
  the estimate on site was 2.5 x
  the multiplier actually used for planning was 2.0 x -- lower than both
RULE OF THUMB:
  2.0 x $5.00 = $10.00 per wedge
STRICT, counting section AND the length difference:
  2.69 x 0.75 x $5.00 = $10.10 per wedge
  the two disagree by 1.0 %
at 5 wedges an hour that is $50.00/hour gross, $45.00 after fuel
one 6-hour afternoon: 30 wedges, $300 of wood, $30 of fuel
```

The rule of thumb and the strict calculation agree to about a percent, and the planning figure is set below both.

The same frame, bought

Now buy the frame instead: enough studs to match the section the wedges provide, plus everything a shopping list never shows – the boards bought and rejected as bowed, twisted or wet, the ones lost on the way home, the truck, and the hours of driving, picking and queueing. The overheads are estimates, and they are all on the store-bought side, which is the side it flatters least to guess about:

```
to match the section 120 wedges provide you would need 2.69 dressed studs alongside each one
  120 x 6 ft x 2.69 = 1939 stud-feet
  / 16 ft = 121 sticks of actual wood needed
then the part nobody counts:
  10 % bought and rejected -- bowed, twisted, wet, split
  3 % lost to handling and weather on the way
  so you buy 139 sticks, 18 of which become firewood ($176)
lumber = 139 x $10.00 = $1388
4 truck loads x 40 mi = 160 mi at $0.55/mi = $88
  and 6.0 h of driving, picking, queuing, loading and unloading = $150 of time
BOUGHT: $1476 cash, $1626 with the time
CUT: 24 h of ripping, $120 of fuel and oil, no trips at all
```

The whole argument on one page

Wood, machines, hands, rate, value, cost and strength, each from the chapters that worked it out:

WOOD	88 % of the tree kept split, 45 % sawn -- 1.95x
MACHINE	3 fewer: SAWMILL, EDGER, PLANING MILL
HANDS	15 middle men removed between tree and wall
RATE	6 wedges/hour measured, 5 used for planning
VALUE	$10.00 of wood per wedge, $45/hour net of fuel
COST	$120 to cut this dome's frame against $1476 to buy it
STICK	226 % of a dressed stud in bending at 12 in trunk; under 9.1 in it would lose
EDGE	every interior edge carries two, 539 % of a stud's section
ANGLE	4.43 deg of dihedral variation lives in the key, not the timber
CUTS	one batched butt per member, one flush cut in place, 4 fixtures for 40 panels

None of this says split wood is better wood. It says the long chain has fifteen sets of hands in it, and the short one has yours.

CHAPTER 42

How Much Shelter Does an Hour Buy

Not how cheap the house is: how much house an hour of your life buys, how debt multiplies it, and the wage above which you should just buy

The usual question is how cheap a house is. The better one is how much shelter an hour of your life buys – because an hour is the thing you actually have a fixed number of, and money is only stored hours.

The figures here are a mix, and they are labelled: the author's own round numbers for one cycle of the shell, beside published ones.

```
THE AUTHOR'S  $100 a sq ft; framing 16% of it; 80 hours of shell work
THE AUTHOR'S  6.5% for 30 years; $25 an hour take-home; $800 cash
PUBLISHED  new manufactured home: $86.62 a sq ft (Census)
PUBLISHED  new site-built house: $216.94 a sq ft without land (NAHB)
PUBLISHED  framing 16.6% of building cost; median take-home $19.69 an hour (BLS)
So the author's figures sit on the cautious side of the published ones
```

What an hour produces

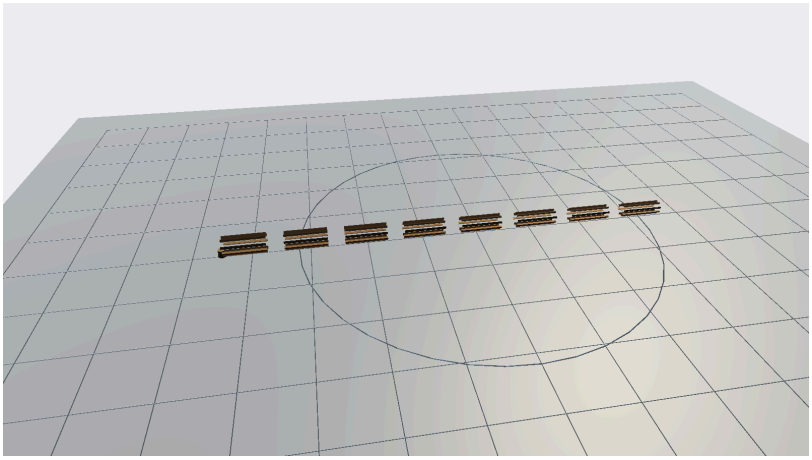
One cycle builds 300 square feet of shell in 80 hours: 3.75 square feet an hour. At $100 a square foot, that floor is worth $30,000 finished, but framing did not create all of it; plumbing, wiring, windows and finishes are still to come. So take only the framing, 16 percent: **$4,800**. Divided by the hours, that is **$60 an hour**.

Nobody hands that over as a wage. It is value you no longer have to buy. And it is more than the labour in the framing would earn: labour is 40 percent of that framing, $1,920, or $24 an hour. The dome way produces 2.5 times as much, because you supply the labour *and* replace the wood. Two other things help: a wedge arrives with its angle already in it (Chapter 15), and the jig solves the geometry once instead of for every panel (Chapter 24).

Debt makes it larger

Now suppose that framing would otherwise ride on a 30-year mortgage at 6.5 percent. $4,800 does not stay $4,800; it becomes $10,922 of payments. Over the same hours that is $137 an hour. That is not a wage and it is not today's money: it is payments that never have to be made, spread across decades. Today's value is still $60 an hour.

Take a whole $100,000 mortgage at the same rate. Over its term you pay **$127,544 of interest**, $227,544 in all. At $100 a square foot, the interest alone would buy 1,275 square feet. Interest builds no bedroom, no workshop, no wall. It is the price of having the house before you have the money. Building directly does the other thing: effort becomes equity, one hour at a time.



Interest builds no bedroom.

Count it in hours of your life

Money is stored labour, so price the frame in hours. At an example take-home of $25 an hour, buying that framing costs 192 hours of pay. Building it costs 80 hours of work plus 32 hours of pay for the $800 of cash it still needs: 112. **80 hours of life kept**, about 2 working weeks.

Out of the commodity chain

A conventional house also runs a long critical path – about 33 weeks from start to finish, by the Census count – carrying financing, weather, scheduling and overhead the whole way. And it rides on commodity lumber, whose record price in one recent spring added tens of thousands of dollars to an average new home (Chapter 43). A dome still needs fasteners, glazing and wiring from the same supply chain, but every member that comes from a nearby tree is one less thing exposed to that chain's shocks.

When to build, and when to buy

This is a rule, not an ideology. What building earns per hour is the money it avoids, less the cash it still costs, over the hours it takes: $4,000 over 80 hours, **$50 an hour**. If your take-home is under that, those hours are worth more building than working extra to buy the same frame. If it is well over, buying may be the better use of your time, and the model says so.

What the model does not promise: it is a model. The author's figures are round, the mortgage is illustrative, and the payments avoided are nominal dollars over decades. It is a way to see a decision clearly, not a guarantee.

The score

One cycle, on one sheet:

```
300 sq ft of shell in 80 hours: 3.75 sq ft an hour
$4,800 of commercial framing displaced
$60 an hour production value; 2.5x the framing labor alone
$10,922 of 30-year payments avoided: $136.53 an hour, not a wage
80 hours of life kept, against earning $25 an hour to buy it
Worth building while your take-home is under $50 an hour
One shell a year for 30 years: $144,000 of framing, 9,000 sq ft
One hour: $60 of shelter today, $137 of payments never made
```

Built once a year for 30 years, that is $144,000 of framing and 9,000 square feet of shell. More shelter from each hour of a finite life is the whole of the why.

CHAPTER 43

Where a House's Money Goes

The builders' own survey of a new house, the factory's split of a trailer, and where the dome's saving actually comes from

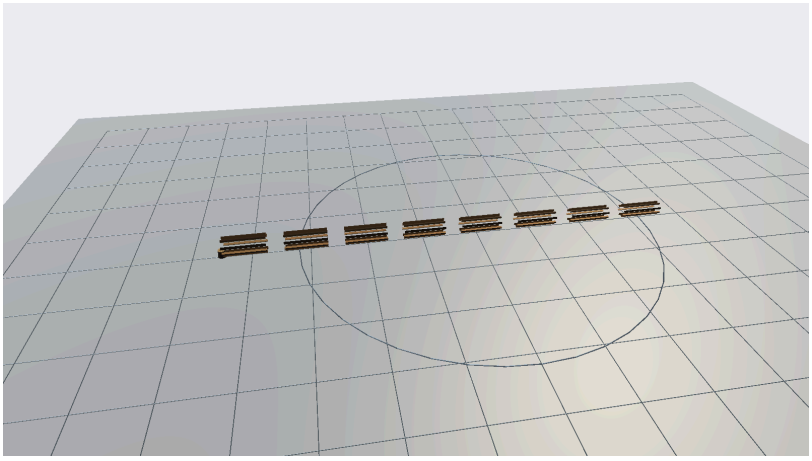
Before claiming a saving, look at what a house actually costs and who gets it. This chapter uses the builders' own survey of new houses, the Census, and the factory figures for manufactured homes, each one labelled with where it came from:

```
PUBLISHED  NAHB 2024 cost survey: who gets a house's price, and 37 line items of building it (41 builders; not representative)
PUBLISHED  Census: 7.6 months, start to finish; BLS wages, May 2025
PUBLISHED  Lumber peak: $1,500 per 1,000 bd ft; +$35,872 on an average home
PUBLISHED  Manufactured homes: $86.62 a sq ft; factory split from 2002
THE AUTHOR'S  $200,000 house, $100,000 trailer, 40% labor, 80 hours of shell work
DECIDED    the 12% federal bracket; state tax left out
ESTIMATED  dealer 25%, labor by item, run share 35%, curve +25%
So every estimate below is a guess that can be changed, and says so
```

Where a house's money goes

Of a $200,000 new house, in the shares the builders' survey found:

the land	$27,373
the building itself	$128,729
the builder's overhead and profit	$33,434
selling it: commission, marketing, loan	$10,464



The land, the building, and the people who sell it.

Where its time goes

A new house takes about seven and a half months from the first shovel to finished, by the Census count, and much longer when its owner builds it. One builder's week-by-week schedule shows where that time goes, and most of it is inside: the ground and foundation are a small share, framing a smaller one, and insulation, drywall, floors and trim the largest. The frame – the part this book is about – is one of the shortest stages of building a house.

Labor is not most of it

So is labour most of what you pay for? Not in a new house. Take the building's labour as the survey's share and it is $51,491: **25.7 percent of the price**. Lumber at its record high moves the number further toward materials, not away.

A manufactured home is built the other way. Of a $100,000 one, the dealer's share is $20,000, the factory's materials $45,969, its payroll $16,817 and its overhead and profit $17,214. Factory labour is only **16.8 percent** of the price. Building indoors on a line is what makes it cheap – the same lesson as Chapter 23.

Every part of the house

The same sum, stage by stage: what each part costs bought, against the dome way if every hour of the work is yours, and how many hours of take-home pay the bought version is:

```
Site work: buy $4,973 · dome $3,708 · 64 h of pay
Foundations: buy $6,801 · dome $1,816 · 253 h of pay
Framing: buy $10,788 · dome $845 · 505 h of pay
Exterior: buy $8,741 · dome $3,788 · 251 h of pay
Pipes, wires, HVAC: buy $12,511 · dome $3,759 · 444 h of pay
Interior: buy $15,714 · dome $7,420 · 421 h of pay
Yard, driveway: buy $4,212 · dome $1,771 · 124 h of pay
Other: buy $1,343 · dome $631 · 36 h of pay
All: buy $65,082, dome way $23,738, if every hour of the work is yours
```

The frame line is the one the harvest of Chapter 12 builds: two weeks of one person against buying it.

Where the saving comes from

Stack it up and the saving on the floor has four sources, and the two largest have nothing to do with the dome:

```
bought whole                            $65,082
the builder's and seller's share you keep $16,550
labour you do yourself                  $17,291
lumber from your own trees              $3,183
the shape itself                        $4,319
```

Most of the saving is building your own, which anyone who builds their own house keeps, whatever its shape. The dome's own contribution is real – less framing, less skin, a shorter foundation, shorter runs from the column – and it is a small line beside those two.

What this does not show

And the half of the sum that does not flatter the dome. **37 percent of what a house costs to build does not care about its shape at all**: fees, windows, cabinets, appliances, floors, the driveway. Only the frame's hours were measured; the rest of the labour is weeks more of somebody's time, and wiring, plumbing and gas usually need a licensed trade and an inspection. And a triangle is harder to finish inside than a flat wall.

CHAPTER 44

What It Costs, Line by Line

The invoice, the price guide, the pad, the stay and the factory -- every figure typed in declared before anything is claimed

What was typed in, before anything is claimed

Every price in this chapter is one of two kinds of number, and they are kept apart. Quantities are measured off the dome's solved geometry and cannot be argued with. Prices and rates are typed in by a person, each with a unit and a reason, and they are assumptions, not market research:

```
this film has two kinds of number in it.

MEASURED -- off the dome's own solved geometry:
   120 members, 664 ft of stock
   40 panels, 549.8 sq ft of surface
   277 sq ft of floor, 19.42 ft across
   none of these can be argued with.

DECLARED -- prices and rates, typed in by a person:
   140 inputs, each with a unit and a reason
   31 of them borrowed from suppliers' own lists
   these are assumptions. they are not market research.

change one and every figure after it moves,
```

Change one and every figure after it moves. That is the point of declaring them.

What the wood was never

Start with the frame, priced the way anybody can check it:

```
start with a board anybody can go and price.

   one 2x6x12 at Lowes  ·  $14
   rips and cuts into 4 struts of 2x3x6  ·  $4 each
   120 of those  ·  $420

that is the whole frame of this dome in bought sticks.

a wedge off a 12 inch trunk is bigger than a 2x3:
   one wedge  ·  1,018 cu in  ·  3.77 x a 2x3x6
   so 120 of them, bought as timber  ·  $2,465
   felled on site instead  ·  $692

and here is what splitting does to a tree:
   split into wedges  ·  88.4% of the trunk  ·  391 bf
   milled into 2x4s  ·  45.2% of it  ·  200 bf
   so a wedge takes  ·  1.95 x as much  ·  +191 bf a tree

squaring a round log throws away the round part.
we do not square it.

the pinwheel costs more stock, though:
   shared struts  ·  369 ft
```

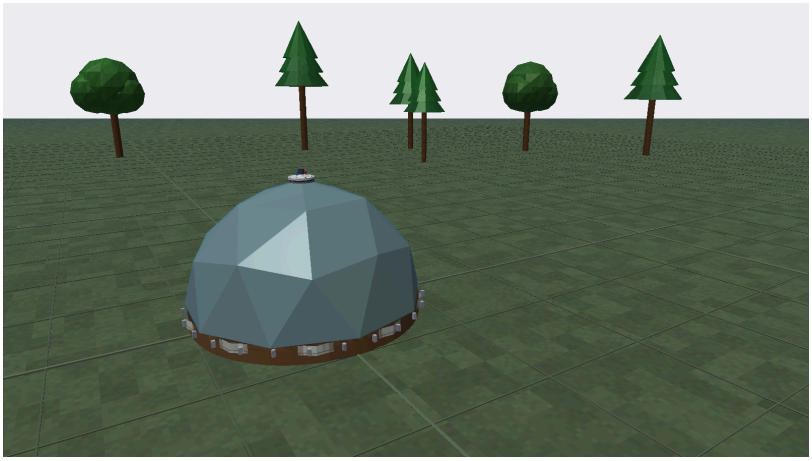
The wood was never the expensive part of a house. That is the whole argument of Chapter 40, in one shopping list.

The invoice

What one costs, all of it, with the maker's margin shown rather than hidden in a round number:

```
what it costs, all of it.

   Wedge-cut frame  ·  $692
   Shower-cap shell  ·  $3,265
   Utility column and seal cap  ·  $1,405
   Light, fan and cooling  ·  $485
   Utility panels  ·  $491
   Build labour  ·  $1,884
   ---------------------------------------------
   materials and labour  ·  $8,221
   shop overhead  ·  $1,480
   warranty reserve  ·  $329
   COST TO BUILD  ·  $10,030

   our profit, 20% marked up on cost  ·  $2,006
   WHAT YOU PAY  ·  $12,036

   per square foot of floor  ·  $43.45

and the ground is quoted beside it, not inside it:
   a platform to stand on  ·  $6,240
```

That is about $43.45 a square foot of floor, and it does not include the ground. The ground is a different bill, paid by a different person.



Ten thousand to build. Two thousand is ours.

Four ways to buy it

The same dome bought new at the till, from the cheapest honest version to a pristine one, for the films' 20-foot dome:

```
one dome, 314 sq ft of floor (20 ft across), priced four ways,
every part bought new at the till:
CHEAPEST: $5,116  ($16/sq ft)
BUDGET: $6,583  ($21/sq ft)
PRISTINE, HAT: $6,943  ($22/sq ft)
PRISTINE, FULL: $9,496  ($30/sq ft)
```

The ground's bill

The landowner builds the pad once and keeps it (Chapter 39). Itemised:

```
two people are paying here, and they are not the same person.

THE LANDOWNER builds the pad, once, and keeps it:
   framed deck on piers, sealed, 369 sq  ·  $2,916
   moisture barrier  ·  $152
   service port up through the deck  ·  $340
   water tank, 120 gal  ·  $162
   under-floor storage  ·  $420
   share of one hub panel, 1 of 4  ·  $600
   spur from the hub  ·  $450
   permits  ·  $1,200
   THE PAD  ·  $6,240

THE DOME BUYER pays none of that.

the deck stays when the dome is lifted off.
the next dome lands on the same port.
```

A bigger, gravelled pad for a park, built and run as a business – including the lines people leave out, the management and the tax:

```
a 48 ft gravel pad, built:
   compacted gravel base, 1810 sq ft     $2,031
   electrical pedestal                   $1,400
   water and drain                       $1,100
   site infrastructure, per pad          $6,500
   permits                               $1,200
   to build                             $12,231

lease            $667 a month, leased 80% of the year
utility margin    $10 a month -- paperwork money, not rent
management, tax     -$920 a year (insurance too)
net            $5,578 a year

and the cheap version, for a host starting with nothing:
a framed deck on precast blocks, and one power panel and water
manifold in the middle of 4 domes instead of a pedestal and a trench each.
   deck on blocks, 1257 sq ft           $17,593
   share of one hub panel, 1 of 4          $600
   spur from the hub                       $450
   permits                               $1,200
   built                                $19,843
services: $2,250 against $10,060 -- nothing dug, nothing poured.
the deck is still the cost of a pad. it always is.
```

And the deluxe one, with rotation, a column, solar and the iris, which is an optional future build rather than anything a first site needs:

```
DELUXE SCENARIO / declared unit prices
Same model's 48 ft concrete pad:
full concrete slab, 1810 sq ft: $11,315
electrical pedestal: $1,400
water and drain: $1,100
site infrastructure, per pad: $6,500
permits: $1,200
rotating base: $9,120
utility column: $2,600
solar, 15,339 W: $17,640
Add adjustable iris: $14,880
TOTAL WITH IRIS: $65,754
A larger pad with rotation, column and solar.
Optional future build; outside the first-site campaign scope.
```

The declared inputs behind the park figures:

```
60 numbers in this film were typed by a person.
they live in one table, with a unit and a reason each:
   a serviced pad rents for $450 a month, plus $0.12 per square foot
   power costs the host $0.14 a kWh; the host adds $0.02
   a hotel room is $140 a night, a short let $95
   a one-bedroom is $1,450 a month
   the sun delivers 4.6 peak hours a day here
   pads are leased 80% of the year
everything else on screen is measured off the Dome Creator:
which domes fit which pad, how much deck that is, how much
shell can carry panels, what the frame and the skin cost.
none of the typed numbers are market research. they are the
owner's working assumptions, printed where you can replace them.
```

What a stay costs

A year in one place, four ways, with power and water on every line so the compare is not rigged:

```
a year in one place, four ways. every line carries power
and water, because two of them include it in the nightly rate
and dropping it from the other two would rig the compare.

   hotel room            $4,256 / month    $51,072 for the year
   short let             $2,888 / month    $34,656 for the year
   apartment lease       $1,641 / month    $19,696 for the year
   own dome on a pad     $1,277 / month    $15,325 for the year

the dome line is the split-log homestead at $16,601 on a pad,
plus $1,800 to move it, and it carries the two
costs an ownership pitch usually hides: $600 a year of upkeep
and 6% on the money tied up in it.
at the end of the year it is worth $13,640, after a
15% secondhand haircut taken on day one.
```

The smallest version

At the other end of the scale: the 62-inch dome from one sheet of Chapter 28, cut by a laser in an afternoon. You do not stand up in it and you barely sit up in it. It is a hole you can afford, above ground. With a riser wall under it, the same sheet becomes somewhere to sit; with a taller one, somewhere to stand.

The factory’s view

The software that wrote this book also prices domes the way a factory would: a fifteen-station line in trailer-plant order, every element with a material cost and an install time. Its consumer price separates three things that a sticker merges – the bill of materials, the labour at a burdened wage with its overhead, and the margin a comparable product carries. A cheaper shell does not erase labour, and a fast build does not make glass cheap. And what that line costs the people on it is Chapter 23, in one sentence:

```
simulate every part placement, all six motions, crew of 2:
time on task = 191 hours each = 23.9 eight-hour shifts
food energy = 54,870 kcal per worker
mechanical work actually raising things = 87 kcal (0.16% of the fuel)
fastening alone = 49,366 kcal (90% of the fuel) -- and it lifts nothing
so a factory should not chase the lifting;
it should chase the screw gun
```

PART 9

The Network

“Own the building. Rent the ground. Move when you want to, and leave a platform behind rather than a house.”

The pad network from both sides of it: what a pad is, what a host earns and stops paying for, what the house that lands on it can become, the deluxe pad that turns, and who all of it is for.

CHAPTER 45

Why a Network Beats a Park

An RV park rents you a slot for a week. This rents you a foundation for a decade, and both sides come out ahead

An RV park rents you a slot for a week. This rents you a foundation for a decade.

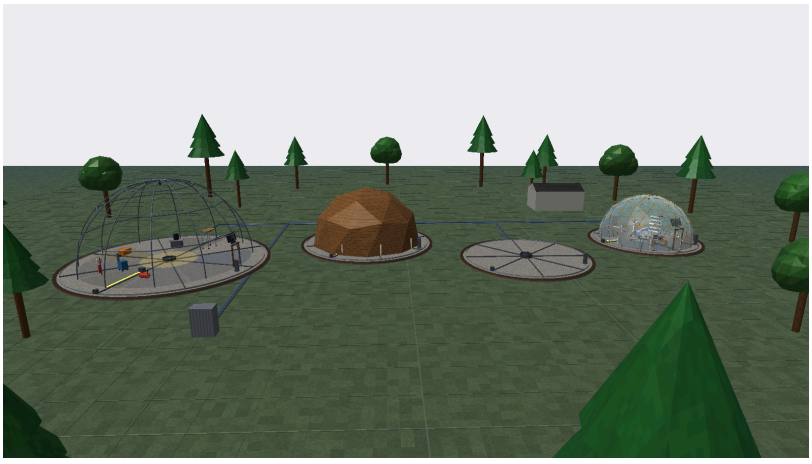
That is the whole difference, and everything below follows from it. The arrangement only works if it is better for the landowner *and* for the person who arrives with a building, so this chapter works it out twice.

First, why it is not a caravan site

An RV park sells **nights**. Its costs are turnover: booking, cleaning, hookup, disconnect, a pitch standing empty out of season, and a tenant who is leaving from the moment they arrive and behaves accordingly.

A pad sells **years**. It is built once, connected once, metered once, and occupied by somebody who owns the building standing on it. The revenue per square foot is lower. The cost per occupied month is very much lower. The difference is turnover, and turnover is most of what makes short-stay accommodation a job rather than an income.

The tenant's own building is also the deposit. Nobody abandons a dome.



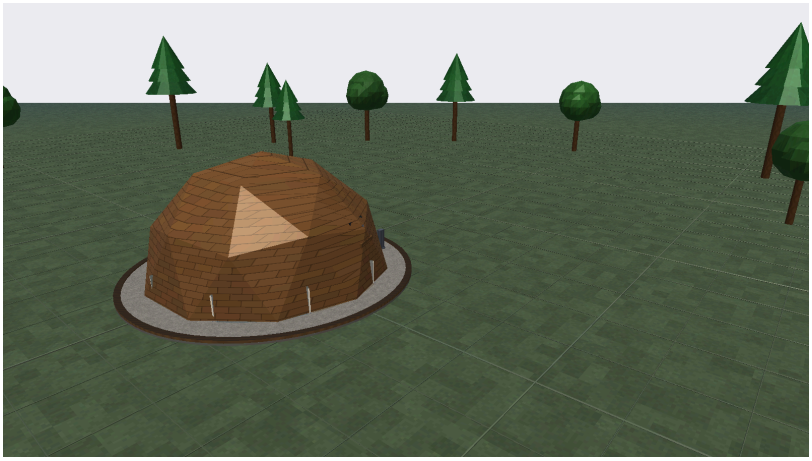
An RV park, for houses.

The tenant's side

Four ways to be housed, on this book's dome and on rates declared in the park model with the assumption written beside each one:

	per month	at the door	over 10 yr	owned after
hotel room	$4,200	$0	$504,000	$0
short let	$2,850	$0	$342,000	$0
apartment lease	$1,450	$3,625	$175,450	$0
own the dome, rent the pad	$528	$12,036	$75,365	$12,036

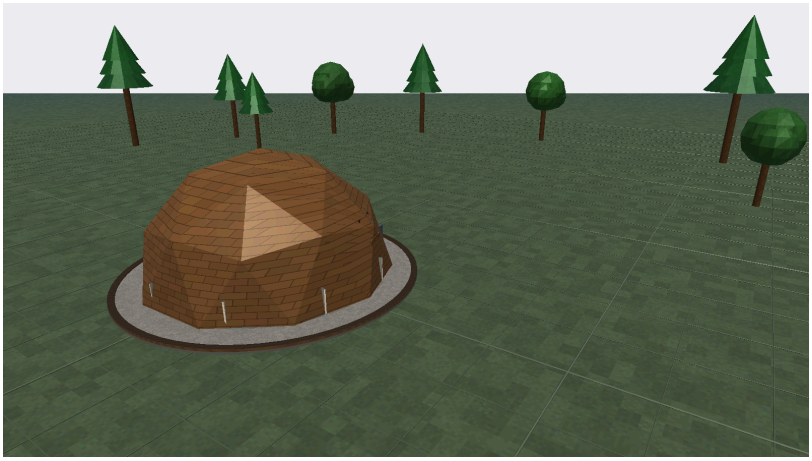
Two things in that table are worth slowing down for.



Four ways to have a roof, for a year.

The monthly figure includes the dome wearing out. $528 a month is the pad's rent plus the dome's own depreciation over a 30-year service life. It is not rent-and-ignore-the-asset; the building is being used up and the table says so.

The entry cost is the real objection. $12,036 at the door against an apartment's $3,625. That is the catch and it does not go away. What the table shows is how quickly it is repaid: owning overtakes renting after **11 months**, and after ten years the difference is about $100,085 — with a building still standing at the end of it that you own.



How long you have to stay.

What the tenant actually gets

No land purchase and no mortgage. The most expensive item in almost every housing decision is removed, not reduced.

A building that moves. Not "portable" in the sense of a caravan — this is a real timber shell — but demountable. It comes apart into panels, a column, a floor and a cap, and it goes onto a trailer. The thing you cannot take is the pad, and the pad is the cheap part.

Flexibility in how long. A season or a decade, decided as you go. There is no twelve-month lease to break because the thing tying you to the place is a rent on a platform, not a claim on a home.

Flexibility in where. This is the part that has no equivalent. If the work moves, or the climate turns, or the neighbours do — you take the house. In every other arrangement on that table, moving means losing either a deposit or a building.

A floor that is better than the pitch. An RV park gives you a slab and a hookup. A pad in this arrangement is a prepared, levelled, serviced platform sized for the building that is going on it, and the dome's own floor clamps to its own column.

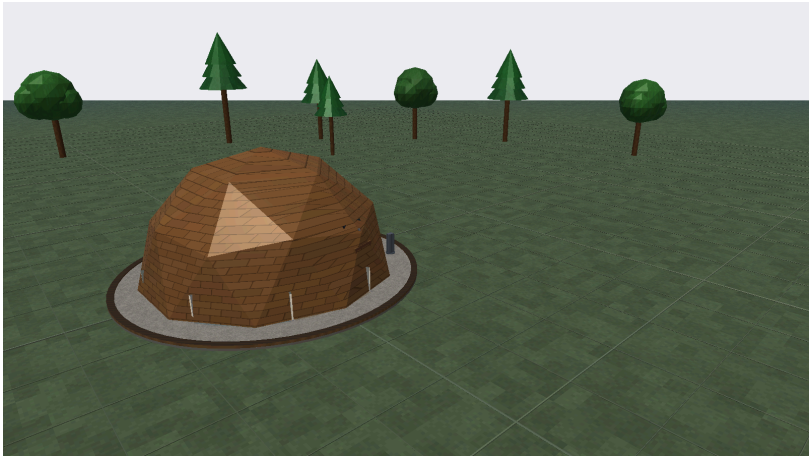
The host's side

Three things to do with a piece of ground:

	upfront	a year, net	payback
bare ground, unused	$0	$-380	never
one serviced pad	$14,280	$4,868	2.9 yr
furnish a unit and let it short	$26,316	$26,238	1.0 yr

> **THE NUMBER THAT DOESN'T HELP**
>
> And here is the number that does not flatter the argument, printed rather than buried: **furnishing a unit and letting it short earns the host more.** More than five times the annual net, and it pays back in about a year against a pad's 2.9 years.

So why would anybody build pads?



The host's side, against the obvious alternative.

Because those two lines are not the same kind of thing

The short-let line is given every night of the year full, which nobody achieves. Halve the occupancy and it halves.

It also requires the host to own a *building* — to build it, insure it, furnish it, clean it three times a month forever, hold a reserve against damage, renovate it every year, and answer the phone when the heating fails at eleven at night. That is not an investment. It is a small business, and the income in the table is the pay for running it.

The pad is an annuity. $14,280 once, $4,868 a year net, and the work is reading a meter and sending an invoice. The tenant owns the building, so the tenant fixes the building.

The honest framing is: **a short let earns more per pound of effort at the start and a pad earns more per hour of attention forever.** Which of those you want is a question about your life, not about your land, and this book is not going to pretend the money answers it.

What the host is really selling

Not land. A prepared, serviced, level place to put a building — which is the expensive, skilled, permit-heavy part of building anywhere.

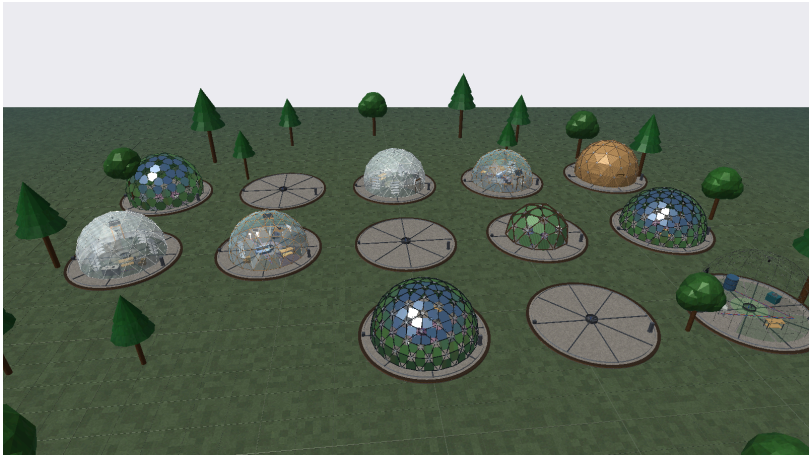
Doing that once and having it used for thirty years by a succession of owners is a far better use of the effort than doing it once per owner. The pad is also worth something while it is empty, which bare ground is not: it is an improvement with a rent attached to it.

Why a network and not a park

One host with one pad is a transaction. The thing that makes it a network is that the pads are **the same**.

A dome built to this book's geometry fits any pad built to this book's specification, which means a tenant can move between them without modifying anything, and a host can take any tenant without knowing in advance which dome is coming. Chapter 39's clean split — the host owns what is attached to the ground, the tenant owns what can be carried — is what makes that interchangeability possible, because nothing straddles the line and so nothing has to be negotiated.

That is the same argument as Chapter 22, applied to land instead of labour: a short, fixed interface is what lets a thing scale.



Why this has to be a network.

What would have to be true

This chapter is a design possibility, not a tested arrangement. Four things would have to hold, and none of them is proven:

Zoning would have to permit it. A demountable dwelling on a leased pad is not a category most jurisdictions have. Some will treat it as an RV, some as a permanent dwelling, some as neither. This is the single largest risk and it is entirely local.

The rates would have to survive contact. $494 a month for a serviced pad is a declared assumption against local RV and storage rates, not a price anybody has charged.

Enough domes would have to exist. A network of pads with nothing to stand on them is a field of concrete. The first hosts carry that risk, which is why the model's test build is three pads and not thirty.

Somebody would have to underwrite the first year. A host who builds a pad on the strength of a book deserves better than hope, and the park model carries a twelve-month manufacturer guarantee for exactly that reason.

State those four plainly and the argument is still a good one. Leave them out and it is a brochure.

CHAPTER 46

What a Pad Actually Is

A deck, a rim, three pipes up the middle and a meter at the edge -- and the line between the host's half and the tenant's that never moves

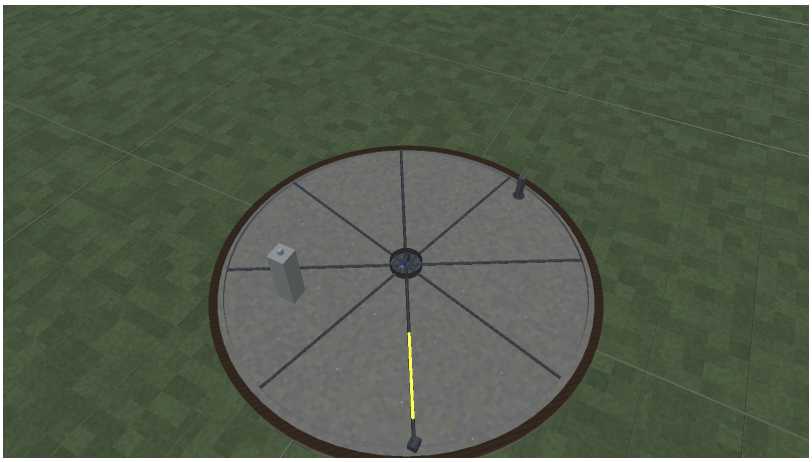
Chapter 45 argued for a network of pads. This chapter is about one of them: what it is made of, who owns which part, and why the part the house stands on is the part the house never has to bring.

Start with graded ground. On it goes a deck – gravel, concrete or timber, whichever the host wants to pay for. Round the edge of the deck runs a rim, and the rim is the part that makes this a pad rather than a patio: it is what a dome latches to. Then power, water and drain arrive, in the middle. At the edge, a metered pedestal, which is how the host sells the power and water on. If the host wants to, a utility column in the middle with a sink, a shower head, a toilet and outlets on it.

That is all of it. Look at the list, because it is short:

- a deck;
- a rim;
- three pipes and a conduit;
- a pedestal with a meter in it;
- perhaps a column, perhaps a turning ring.

There is no kitchen on that list. No sofa, no carpet, no bathroom fittings somebody else chose. The host's asset is a piece of civil engineering, and the thing most likely to damage it is the weather.



Everything the host is responsible for.

The cheapest version that is still a serviced pad is not poured at all: a framed deck on precast blocks, with one power panel and water manifold in the middle of four pads instead of a pedestal and a trench for each. Nothing excavated. Chapter 47 prices it, line by line.

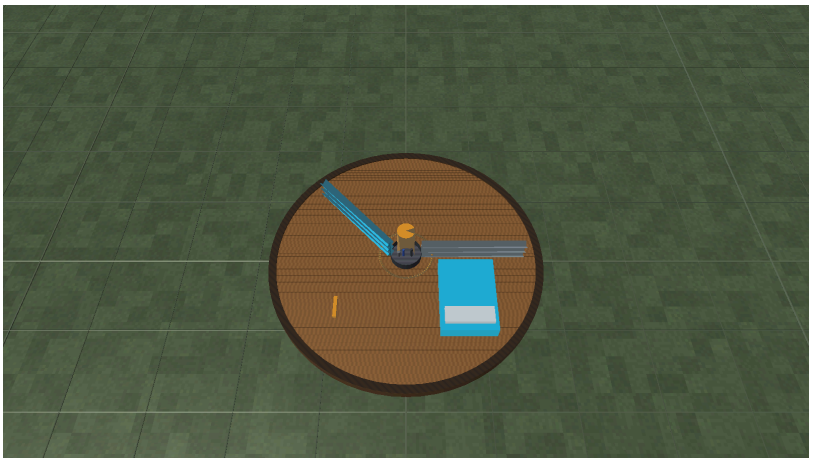
The reserved spot

Every builder knows to reserve the middle of a floor for the services, and this system makes that a rule rather than a habit. Power, water and drain come up **inside the footprint, in the centre**.

Centre, not edge, for a plain reason: the dome that lands here puts its floor on the pad. Services arriving at the edge would have to cross the doorway, or come up through a wall that is not there yet, or be moved every time a different house arrives. From the centre they go up the utility column, and from the column they go out along the shell.

That second leg is where the wedge dome earns something. Its seams are channels – Chapter 31 measured what fits in them – and those channels connect round the whole shell. Supply lines and wiring can run up from the centre and then out along the seams like veins, latched in place and still reachable when something has to change. Drainage stays down at the floor, where gravity wants it.

The result is a house whose rooms can be rearranged around a fixed service point instead of being redesigned every time the household changes. Chapter 48 is about what that buys.



The reserved spot: services up the middle.

Sizes are rounded up, not invented

A pad catalogue could be a marketing table. This one is arithmetic. Every design the Dome Creator ships has a foundation diameter the tool computes from its own geometry. Round each one up to the next 4 feet and you have the pad sizes:

```
the Dome Creator ships 12 designs, and each one has a
foundation diameter the tool computes from its own geometry.
round each up to the next 4 ft and you have the pad sizes:
      36 ft    1,018 sq ft   takes  1 of 12
      40 ft    1,257 sq ft   takes  4 of 12
      44 ft    1,521 sq ft   takes  7 of 12
      48 ft    1,810 sq ft   takes 10 of 12
      64 ft    3,217 sq ft   takes 11 of 12
      80 ft    5,027 sq ft   takes 12 of 12

but one pad does not have to be one size. the three domes
this system is for, by floor area and longest member:
   small personal    314.00 sq ft  ·  20.0 ft across  ·  6 ft member
   medium            822.49 sq ft  ·  32.4 ft across  ·  10 ft member
   large           1,184.39 sq ft  ·  38.8 ft across  ·  12 ft member
an iris rim -- leaves on a track, opening and closing like
a camera aperture -- spans 19 to 40 ft and takes all three.
the mechanism is $12,400 on top of the deck.

the treehouse canopy dome needs 35 ft of pad;
the whole trunk lodge - 20 ft needs 76 ft.
4 ft steps because decking and ring stock come in even lengths,
and the second pad a host builds wants the first one's cut list.
```

Four-foot steps, because decking and ring stock come in even lengths, and because the second pad a host builds wants the first one's cut list. Add a bigger dome to the tool and a bigger pad size appears on the next run; nobody edits a list.

But one pad does not have to be one size. The rim can be an **iris**: leaves on a track that open and close like a camera's aperture, so one pad takes a 19-foot dome or a 40-foot one without being rebuilt. The deck stays where it is; only the acceptor changes size. That spans all three of the domes this system is planned around – 20.0, 32.4 and 38.8 feet across.

It is not cheap. The mechanism is about $12,400 at full aperture. Against three fixed pads, one per size – $39,242 – one iris pad at $32,243 saves $6,999, which is the case for it on a site that expects all three. On a site that expects one size it is an upgrade, not a necessity, and the fixed rim is the right answer. The iris is also a development concept: nobody has built one, and it belongs in the upgrade column until somebody does.

The pad is the floor

This is the move the whole idea turns on.

The dome that arrives does not bring a foundation, because it does not need one. **The pad is the floor.** The deck it lands on is the deck it lives on, the services come up through the middle of it, and when the owner leaves, the pad is exactly as it was for the next one.

No slab poured. No footing dug. No permit pulled for a foundation under a building that will be somewhere else next year. Bring your carpet if you want it; some people want a shop floor in their house, and that works too. It is the dome owner's floor to finish, laid on the host's deck.

We already see domes for rent, most of them glorified tents let out a weekend at a time. I do not think their value is ever realised that way. Own the glorified tent, rent the place to put it, and both sides have a reason to make a long arrangement work.

What the ground is worth

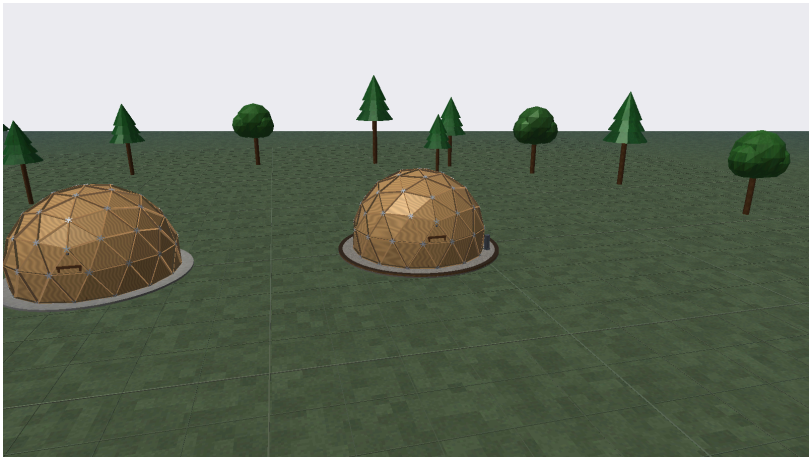
How much of a house is the part you can never take with you? That is a measurable question, so it was measured. Ask the Dome Creator for each of its 12 shipped designs twice – once as shipped, and once with its foundation set to bare ground – and the difference is exactly what a pad stands in for:

```
ask the Dome Creator for each design twice: once as shipped,
once with its foundation set to bare ground. the difference
is what the pad is standing in for.
   Treehouse Canopy Dome       63.3%    $17,251
   Timber Workshop             45.2%     $8,829
   Glass Studio Loft           25.3%     $6,232
   Continuous Steel Arc Hangar 23.7%    $23,796
   Concrete Monocoque Form     23.6%    $13,843
   ... and 7 more
8 of 12 designs stand on something that costs
real money: 8% to 63% of the build.
the other four sit on grass, and for those the pad replaces
nothing -- which is worth saying out loud.
the timber workshop is the one in the picture:
$19,514 on its own concrete slab, $10,685 on a pad.
```

For the designs that stand on something that costs real money, the ground is anything up to **63 percent** of the build. The design this book's tenant figures are run on, the split-log homestead, saves the least of them: $1,466, about 8.1 percent.

THE NUMBER THAT DOESN'T HELP

For a dome that would have sat on a lawn, a pad saves nothing. Several of the shipped designs stand on bare ground, and for those the pad replaces no foundation at all. What it sells them is services – power, water, a drain, a level deck – and that is a smaller argument than "the pad is your foundation." It is still an argument, but it is the one to make.



What the ground is worth, per design.

Below the line and above it

The films that go with this book keep the two people in different colours the whole way through, because they want opposite things. **Amber is the pad host**: the deck, the trench, the pedestal, the meter. **Cyan is the dome owner**: the house, and everything they ever do to it.

The line between them runs along the top of the deck, and the most important thing about it is that it never moves. Below the line is the host's money, and it is in the ground. Above the line is the owner's money, and it drives away with them.

Every other way people rent somewhere to live puts the landlord's money in the walls the tenant lives between, and that is where the conflict comes from, every time: the stained carpet, the deposit, the walk-through with somebody's clipboard. Here the host never owns a building somebody else is living in, and the tenant never pays rent on a thing that will never be theirs. Fewer things held in common means fewer things to argue about – and an easy goodbye, which is worth more than people think. If it is not working, the dome leaves and the pad is exactly as it was.

CHAPTER 47

What a Host Earns, Line by Line

The cheapest pad that works, the ten-thousand-dollar target it misses, the rents that would repay it, and the bill a host stops paying

Why would anybody host a dome?

Cash. Income from land, without also buying and maintaining the house somebody else lives in. Everybody knows the pains of landlords and everybody knows the pains of tenants, and this arrangement is an attempt to take the worst of both out at once: the host looks after a deck and a connection, and replacing somebody's kitchen, sofa or bedroom carpet stops being the host's business.

That is the pitch. This chapter is the arithmetic under it, worked with the same pad model as the films and with the lines a pitch usually leaves out left in.

The cheapest working pad

Before any rotating base or adjustable rim, the starting point: a small framed deck on precast blocks, a fixed rim, and one shared power panel and water manifold serving four pads, with a short spur to each.

```
STARTER SCENARIO / declared unit prices
20 ft pad = 314 sq ft
deck on blocks, 314 sq ft: $4,398
share of one hub panel, 1 of 4: $600
spur from the hub: $450
permits: $1,200
site access + trunk service allowance: $6,500
TOTAL: $13,148
Author's all-in target: $10,000
Fixed rim; shared services; no iris or rotation.
Site allowance must cover the actual local works.
Start with a working floor and services; add features later.
```

Note the site allowance. Roads, trunk services and drainage do not become free because four people share them, so a share of them is in the total whether or not anybody likes it.

> THE NUMBER THAT DOESN'T HELP
>
> **The starter pad is $13,148. The target was $10,000.** It misses by $3,148, and the model does not get to pretend otherwise. What the miss tells the designer is where the work has to go: the deck is most of a pad's cost, and it always is.

What ten thousand dollars would return

Suppose the design work does get a complete serviced pad – power, water and drain included – to $10,000. What would it earn at the rents a host might actually ask?

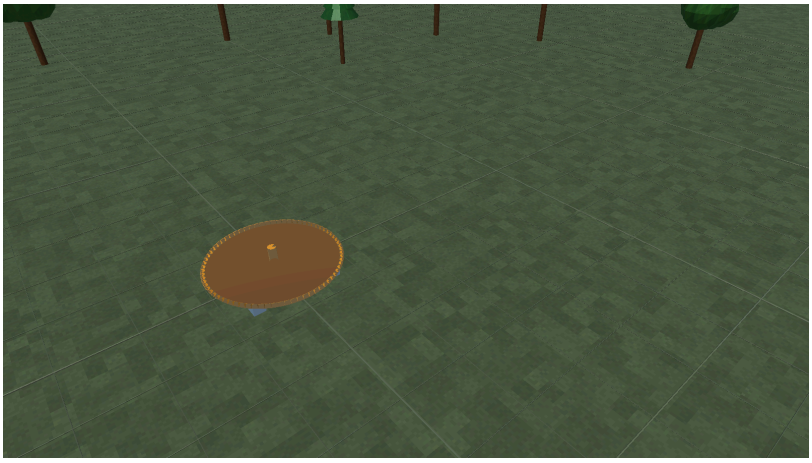
```
WHAT IF the complete pad costs $10,000?
Power, water and drain included in that TARGET.
Leased 80% of the year
Management + tax + insurance: $920/year
Utility resale profit assumed zero here.
$500/mo -> $3,880/yr net -> 30.9 months
$1,000/mo -> $8,680/yr net -> 13.8 months
$1,500/mo -> $13,480/yr net -> 8.9 months
Before financing, income tax and additional maintenance.
One-year payback is a scenario, not a promise.
```

At $1,500 a month it repays in 8.9 months. At $1,000, 13.8. At $500, it takes 30.9 months – more than two years.

Three things about that table, all of them caveats:

- **It is payback on net**, after management, tax and insurance of $920 a year, with the pad leased 80 percent of the year rather than all of it. Payback on gross is a brochure number.
- **The build cost is a target**, not a quote; the line above this one shows the model's current pad costing more.
- **It is before financing and income tax.** A one-year payback is a scenario, not a promise.

The goal under all of it is infrastructure that serves for 10 years and longer. A pad that repays in two and lasts ten is a good pad. A pad that repays in one and cracks in three is not.



The ten-thousand-dollar pad, at three rents.

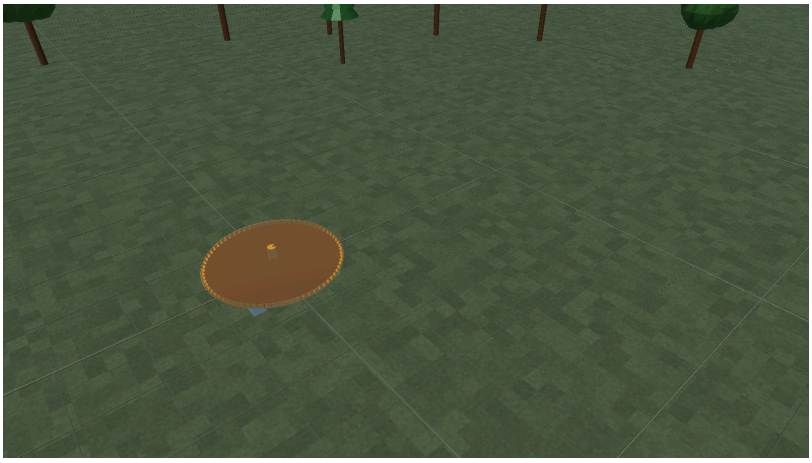
What the host maintains

The host's list is the one from Chapter 46: a deck, a rim, the connections, perhaps a meter, perhaps a column. Weather and ordinary wear still matter. The work does not go to zero.

But compare it honestly. The alternative a landowner actually weighs is not "nothing" – it is furnishing a place they already own and letting it out. Carry the same management, tax and insurance on both sides, then add what a furnished let needs on top:

```
two ways to earn the same money off the same piece of land.

pad host          $12,231 up front      $920 a year
short-let host    $30,231 up front    $6,441 a year

the short-let host's yearly bill is the same management, tax
and insurance, plus cleaning at $95 a turnover,
a 6% damage reserve, a 3% platform fee, and $2,200 of renovation.
that is 7.0 times the pad host's, every year, for ever.

and the figure that does not help: a loaded pad -- rotating,
plumbed, carrying an array -- costs $50,874 to build,
against $30,231 for a bare pad and a furnished let on it.
on up-front cost alone, the short let wins.
```

A pad host pays $920 a year. The furnished let pays $6,614: turnover cleaning, a damage reserve, a platform's fee and a renovation cycle, for ever. That gap is the recurring work a pad host is trying to stop doing.



The comparison a host actually makes.

That comparison assumes the rental building already exists. Buying or building one is a different starting point with a much bigger number on it, and the table does not include it.

The deluxe pad

Now the top of the range: a larger concrete deck, a rotating base, the plumbing column, an array of panels, and the adjustable iris on top, because it is another mechanism with another bill.

```
DELUXE SCENARIO / declared unit prices
Same model's 48 ft concrete pad:
full concrete slab, 1810 sq ft: $11,315
electrical pedestal: $1,400
water and drain: $1,100
site infrastructure, per pad: $6,500
permits: $1,200
rotating base: $9,120
utility column: $2,600
solar, 15,339 W: $17,640
Add adjustable iris: $14,880
TOTAL WITH IRIS: $65,754
A larger pad with rotation, column and solar.
Optional future build; outside the first-site campaign scope.
```

$65,754. That is more than a furnished let costs to set up, and Chapter 45 said so: on up-front cost alone, the short let wins. The deluxe pad is a different feature set at a different starting point, and a house that turns toward the sunrise is a future option – Chapter 49 – not the ticket into the system.

The ticket into the system is a floor and services. Everything else is a branch a host can grow into: floor heating, water storage, a deep freeze, a concrete shop floor, rotation, sun tracking, and eventually something smart enough to run all of it. Each of those can be its own development, and each can be sold to the host who wants it. None of them is needed to start.

CHAPTER 48

A House That Moves, and Grows

Start small, reallocate the rooms as life changes, add a layer when you can afford it, and take all of it with you when you go

Why would anybody put their own money into a dome, when they could rent?

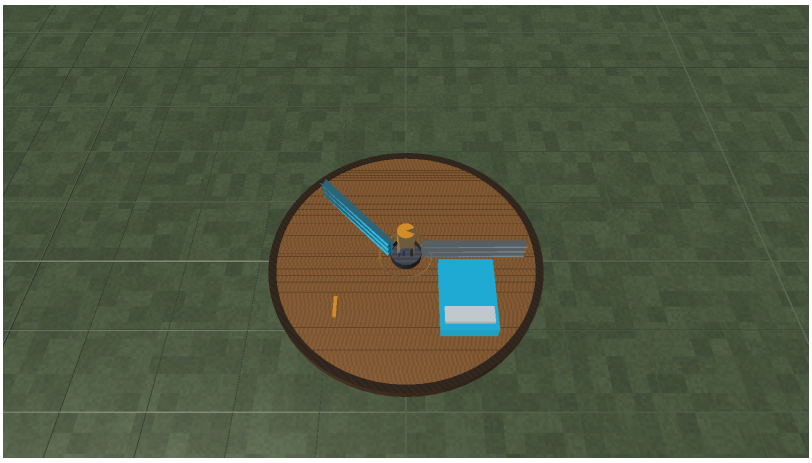
Because a small home you own can start simple and become more – more comfortable, larger, better equipped – as your life changes, and every dollar of that stays yours. The target for the most basic version of this system is a $5,000 dome that leans on the pad for the essentials. That is a design target, not a finished-home quote, and Chapter 47 has already shown what happens to targets when the arithmetic gets to them. But it is the right shape of target: the smallest building that is a home, on a pad that supplies what a small building cannot.

The durable money goes somewhere specific, and the rest of this chapter is where.

The rooms can change

I live in a trailer. Four bedrooms, a living room, a den, an office, two bathrooms, a kitchen and a dining area – every allotment a house is supposed to have. What changed was not the house. It was us. I was gone for six years and came back. Three people became four. Eventually that same house will be too much for one person, and none of its walls will have moved an inch.

A dome does not have that problem, because its walls are not holding anything up. The shell carries the load; the partitions inside are furniture with ambitions. Split a big bedroom to make an office. Three years later you retire, stop needing the office, and open the bedroom back up. The services stay where Chapter 46 put them – in the middle – and the rooms rearrange around that fixed point instead of around a plumber's quote.



The rooms move. The core stays.

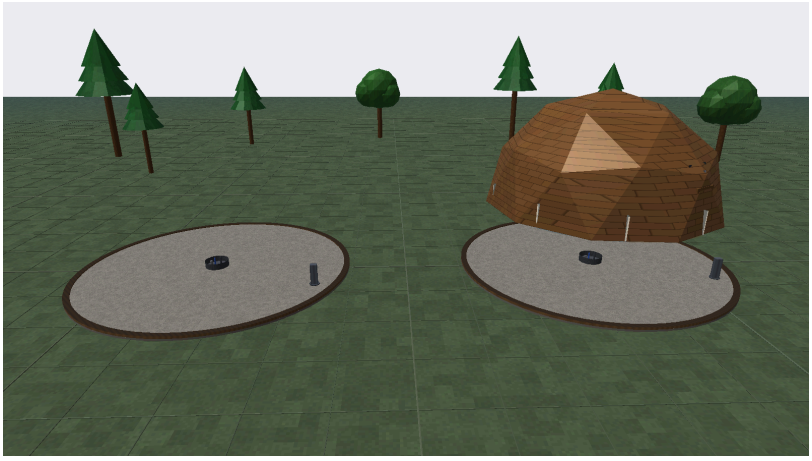
Semi-nomadic

This is the part a lease cannot do at any price.

Not a caravan you tow away on a whim. **A house that comes apart.** The working estimate is 2 days at the quick end and 14 for something large or heavily layered: disassemble, transport, reassemble, connect. That is a real burst of effort and nobody should pretend otherwise – but it is a burst, not a condition. You spend it when a move is worth it, and the rest of the time you are simply living in a house.

Off one pad, onto another. A different park, a different city, a different state. No lease broken, no deposit argued over, no walk-through, nothing left behind – and every improvement you ever made goes with you. Moving the dome between pads is $1,800 in the park model.

Chapter 45 found the stay length where owning starts to win: 11 months. Under that, rent; a short let is the right answer and this book is not going to pretend otherwise. Over it, the gap opens and keeps opening. That is the stay neither a hotel nor a twelve-month lease was ever designed for: **too long to be a guest, too uncertain to sign for a year.**



Lift, drive, set down, plug in.

> **THE NUMBER THAT DOESN'T HELP**
>
> **A dome is worth less the day after you buy it.** The park model takes a 15 percent secondhand haircut on day one, because you may build your dream home but only you value every choice in it at what you paid. After a year on a pad, the house it prices is worth $13,640. That loss is in every owning figure in this Part, which is why they can be trusted.

One hardware set, three sizes

Here is where the durable money should go. Take one design in the Dome Creator and move a single slider – the radius – holding everything else exactly where it was:

take the timber workshop and move one slider:	
the radius. every other menu is held where it was.	
19.7 ft · 295 sq ft · 165 struts · 61 hubs ·	105 panels
32.8 ft · 819 sq ft · 165 struts · 61 hubs ·	105 panels
52.5 ft · 2,096 sq ft · 165 struts · 61 hubs ·	105 panels
the counts do not move. the hub bill does not move either:	
$596 at 20 ft and $596 at 52 ft.	
what moves is the stick and the skin -- panels go from	
$1,161 to $8,253, because those are priced by area.	
7.1 times the floor off the same list of parts.	

From 20 feet across to 52, the counts do not move: 165 struts, 61 hubs, 105 panels. The hub bill does not move either – $596 at both ends. What grows is the stick and the skin, because those are priced by length and by area. That is 7.1 times the floor off the same list of parts.

For the wedge frame in this book, the plan is the same idea at three steps:

```
HARDWARE DESIGN BRIEF / size for two upgrades
6 ft longest member -> ~314 sq ft
10 ft longest member -> ~822 sq ft
12 ft longest member -> ~1,184 sq ft
These are the author's target sizes, not a load rating.
Keep qualified connectors; collect longer replacement members.
Larger frames also need more skin, insulation and pad area.
Beyond the rated span, change hardware or frequency.
One purchase should support a planned range of future homes.
```

Start with a six-foot member. Keep the connectors, collect the longer sticks over time, and the next house is a rebuild with parts you already own rather than a purchase.

That illustrates the opportunity. It does **not** prove that a connector sized for the small house can hold up the large one. The hardware has to be qualified for the whole range of sizes and loads first, and the skin, the insulation and the pad all have to grow with the frame. Past the rated span, change the hardware or the frequency.

A house that gets warmer every winter

I stood lookout outside the skin of a ship inside the Arctic Circle, for hours at a time, over months. There is one way to stay warm out there: layers on layers on layers, and then a wind-breaking, watertight suit over the whole lot. You look like the Michelin man. You also stay alive.

A dome with a removable shell can work the same way. Lift the shell, add a quilted layer – about 132 recycled t-shirts' worth – and put the shell back over it. Which means the R-value stops being a number somebody decided on the day the house was built:

```
one quilted layer is declared at R-1.6: half an inch of loft,
set under the published R-3.5 an inch for cotton fibre.
the heat loss is the same degree-day model the performance
film uses, run on this dome's own shell area.

   bare      ·   R-1.5  ·  $8,260 a year
   1 layer   ·   R-3.1  ·  $3,955 a year  ·  pays back in   6.9 months
   3 layers  ·   R-6.3  ·  $1,937 a year  ·  pays back in  45.0 months
   5 layers  ·   R-9.5  ·  $1,282 a year  ·  pays back in 114.5 months
   7 layers  ·  R-12.7  ·    $958 a year  ·  pays back in 215.4 months

the first layer pays for itself in 7 months; the seventh takes 18 years.
that is 1 over R, and it is why deciding the whole R-value
on day one is the wrong move.
seven layers is R-12.7 -- a good wall, not a code-beating one --
and it cost $17,404 spread over years.
this prices conduction only. it does not price the windbreak,
and the windbreak is what makes layers work.
```

The first layer pays for itself in months. The seventh takes years. That is one over R, and it is exactly why deciding the whole R-value on day one is the wrong move: add the layer that is worth adding this year, and the house gets better at heating and at cooling for as long as you own it.

Two honest limits. Seven layers is R-12.7 – a good wall, not a code-beating one. And the ladder prices conduction only. It does not price the windbreak, and the windbreak is what makes layers work; that is the shell's job, and Chapter 30 is about it. A real assembly also has to answer for moisture and for fire, which a stack of quilts on a model does not.

CHAPTER 49

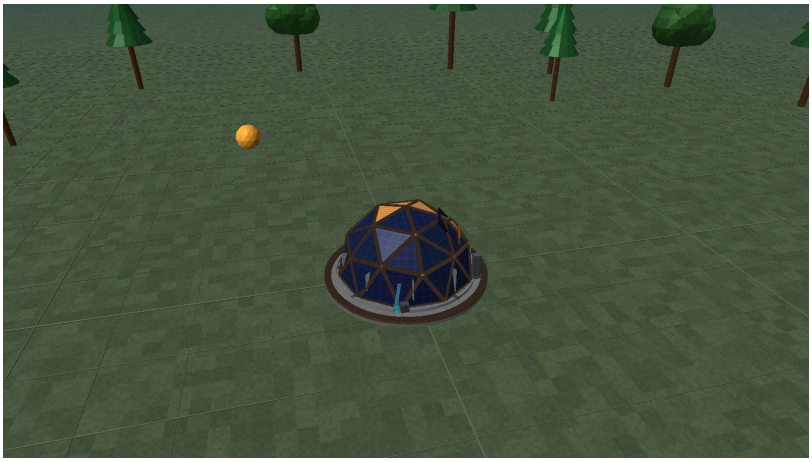
The Pad Turns, and the Site Shares

A rotating base, cells on the shell, a bathhouse so a house can carry no plumbing -- and the surplus power a tenant cannot use

The top of the range is a pad that turns.

A rotating foundation is the expensive option and, honestly, it is just very cool. Maybe today you want the window facing the street. Maybe somebody is coming over and you want it facing the private, fenced corner. Maybe you want the morning sun on your face, and maybe tomorrow you definitely do not. A house that turns can change which way it faces without a single solar panel on it.

Put an asymmetric array on it – cells on one side of the shell – and turning it is also aiming the array. The bearing, the drive and the rotating utility connections all need their own development, and the first site will not have one (Chapter 50 says why). This is future deluxe hardware. But it is worth knowing what it would be worth.



The deluxe version: the pad turns.

What tracking is worth

Every figure in this section is for a dome 10 feet in radius, the medium two-person size, because a number of kilowatt hours a month means nothing without the dome it came off. The panels are on the dome, not the pad; the pad only turns.

First, how much a shell can carry at all. Panel areas are measured off the model's own faces:

```
SAME DOME: 20 ft across
Panel areas are measured from the model's faces.
Assumed panel density: 18.0 W/sq ft
one side: 292 sq ft -> 5.2 kW
top half, skirt below: 292 sq ft -> 5.2 kW
whole shell: 583 sq ft -> 10.5 kW
No panels = zero installed solar capacity.
Selected layouts: 5.2 to 10.5 kW nameplate.
Face area sets capacity, not guaranteed energy yield.
Sun angle, shade, climate and losses set actual kWh.
The useful limit is the available shell area.
```

That is installed capacity, not a promise. Sun angle, shade, weather and losses decide what a curved face actually delivers. Now the energy, fixed and tracking:

```
every figure here is for a 10 ft radius dome -- the medium,
two-person size. a number a month means nothing without that.
the panels are on the dome, not the pad. the pad turns.

   one side               20 of 40 panels  ·    292 sq ft  ·    5.2 kW
   top half, skirt below  20 of 40 panels  ·    292 sq ft  ·    5.2 kW
   whole shell            40 of 40 panels  ·    583 sq ft  ·   10.5 kW

   one side               fixed   734  ·  tracking   918 kWh a month
   top half, skirt below  fixed   734  ·  tracking   918 kWh a month
   whole shell            fixed 1,468  ·  tracking 1,835 kWh a month

the skirt below the cells is the catchment -- the same metal that
keeps rain off the bottom ring collects it.
a tenant uses about 420 kWh a month.
and the one that does not flatter the tracking ring: cover the
whole shell and it is always facing the sun with something, so
turning the pad buys much less than it does on a half-clad dome.
the whole shell also makes 4.4 times what a tenant uses. that is surplus,
and the model refuses to credit surplus at retail: it is worth $0.045 sent back,
not the $0.14 it would have saved.
```

Single-axis tracking adds about 25 percent over a fixed array: on one side of this dome, 184 kilowatt hours a month. The skirt below the cells in the middle layout is also the rain catchment – the same metal that keeps water off the bottom ring collects it.

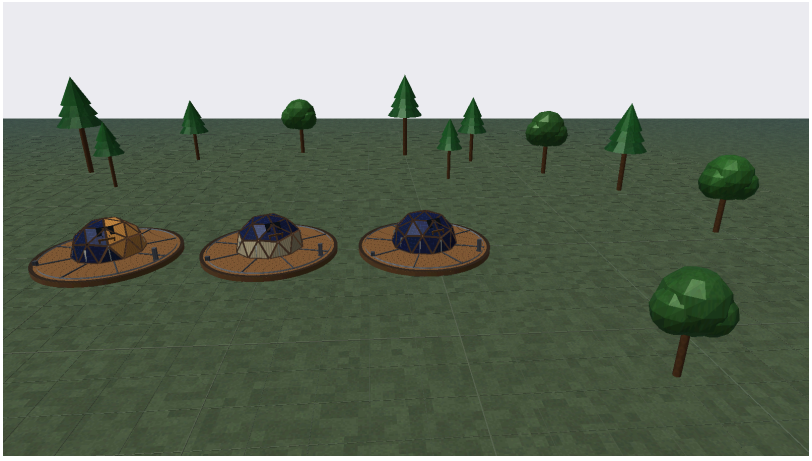
WE GOT THIS WRONG

The first version of this model was wrong about solar. It valued every kilowatt hour the shell made at the retail price, which quietly assumed the tenant was buying power they would never have bought. A tenant uses about 420 kilowatt hours a month; a fully clad shell makes 4.4 times that. So the model changed: power that displaces the tenant's own draw is worth retail, $0.14; everything past it is worth the export rate, $0.045. That took the pad's solar income down by well over half.

THE NUMBER THAT DOESN'T HELP

At export value, the turning ring takes 77 years to pay for itself. A year of tracking gain is worth $99 sent back to the grid, against a ring that costs $7,600 under a 40-foot pad. Even if every gained kilowatt hour were worth retail, it would take 25 years. Turn the house because you want it to face somewhere. Do not turn it for the money.

The real case for cladding a dome in cells is a park that can sell the surplus, not a tenant who cannot use it.



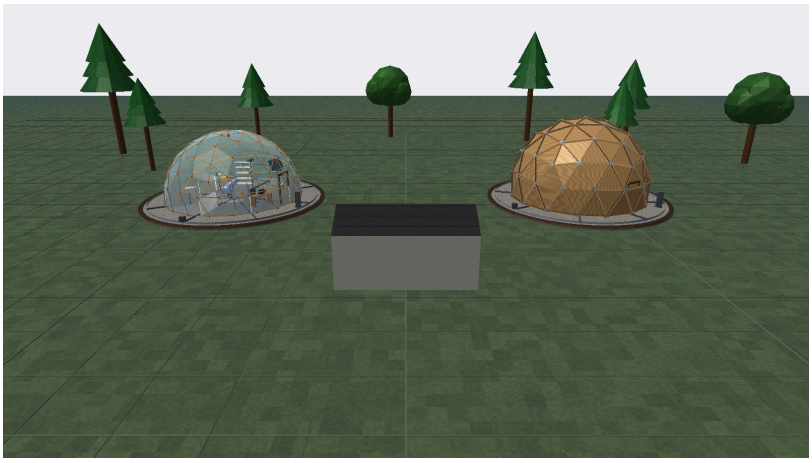
What tracking is worth, and what it is not.

What a site can share

Some of what a house needs can be bought once by the site instead of once per house, and that changes what a dome has to be.

A **bathhouse** on the site means a dome can skip plumbing entirely – and plumbing is the single most expensive and most regulated thing in a small dwelling. Or the host fits a utility column on the pad, toilet and sink and shower and drain, once, for every owner who ever parks there. Either way it is paid for by the party who stays and used by everyone who comes and goes.

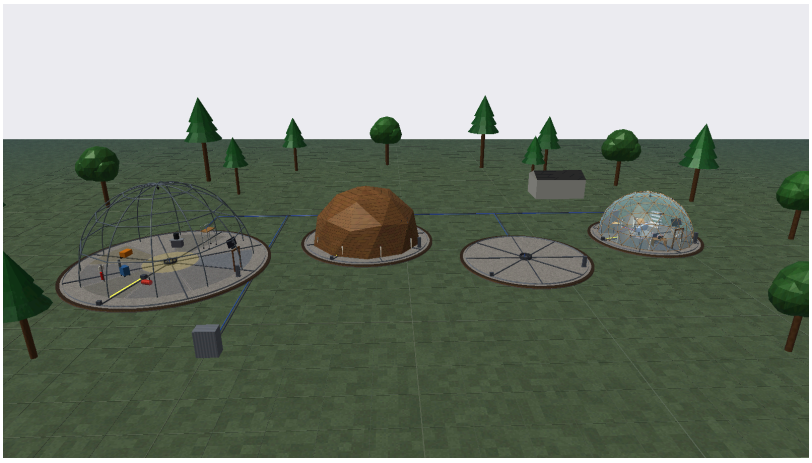
Note what a host is *not* obliged to do. Reselling power and water with a margin is one option, not the model; a host may simply pass the meter through. A more ambitious site might run a shared off-grid plant with central storage for the whole cluster. What hosts really compete on is the range of things a site offers – location, price, services, how well the arrangement works for the people living there – and the price follows from that.



A bathhouse, so a house need carry no plumbing.

Another use

A side thought, and only that. The same kit – a compact shell, a quilted layer or two, a reusable pad – would make an interesting climate-controlled storage building. It would need its own check of cost and performance before anybody sold it as one. But housing is not the only thing the parts are good for, and a site with three empty pads in a slow season might find out what else they are good for.



The same parts, storing something other than people.

CHAPTER 50

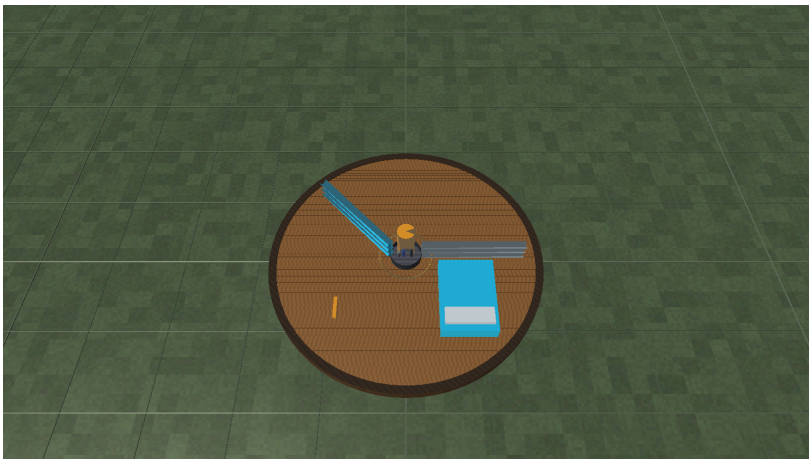
Who This Is For

A nineteen-year-old with nowhere steady to live, a single mother working two shifts, and a man on his seventh holiday watching a rental portfolio

When I was nineteen I could borrow five thousand dollars against my name for college. I blew that chance, joined the military, and came back at thirty-two for revenge. Constructive revenge, as it turns out, involving domes.

I did not fail because I was stupid. I failed because where I lived was not stable. I did not have the support I needed, and I could not keep my mind on what mattered, and it cost me more than I can easily add up.

So I ask a specific question when I design this, and it is not "what would a buyer want." It is: **what would a small home I owned have done for my eighteen-year-old self?** A floor that was mine, on a pad that supplied what a small house could not, that I could improve as I went and take with me when I left. I think it would have been revolutionary. That is the person I keep in mind.



What this would have meant to me.

Built with the little guy in mind

Rich people will find a way to have what they want. They always have. I want to make a difference for the little guy: single parents, people with very little money, families just starting, bachelors, anyone trying to get stable.

My mother was a poor single parent. She worked so much that I went to the before-school program *and* the after-school program. I wish she had had an economical home that worked for her, instead of one more thing eating her life. If there are people with the means to fund something like this, women especially, that is who I would love to reach. Help build the kind of stability my mother deserved.

I would like to become successful enough to give domes away. Maybe by then I am making dome furniture too. That is an ambition, not a plan. The plan is the first step: a home people can afford to start with and improve.

Three things at once

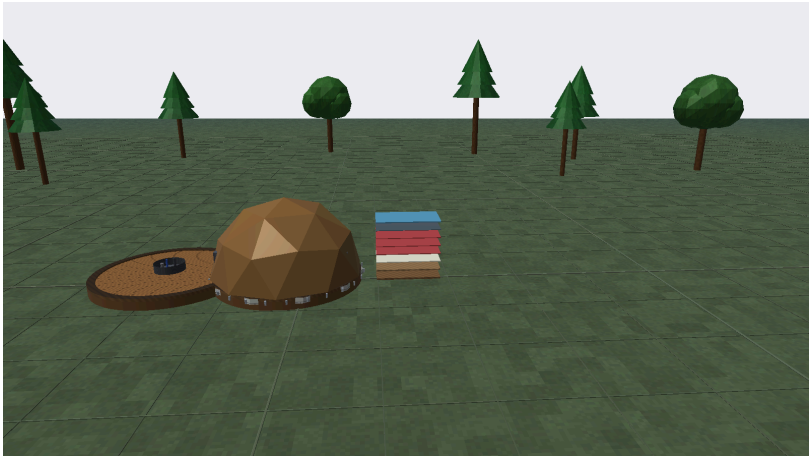
This was never a plan to sell domes. It is three things that need each other.

Quilters. A dome's insulation is a quilted layer of recycled clothing – about 132 t-shirts a layer, adding R-1.6 each, the only building insulation I know of that a person can make at a kitchen table. So the network wants a register of people who will sew them, paid and listed. Not a charity drive.

Pad hosts. People with land and no vocation as a landlord, who will keep a deck and a service connection and not a roof, a boiler or somebody's kitchen.

Dome owners. People who want a building, with trees of their own or without.

A dome with nobody to quilt for it is a cold dome. A dome with nowhere to stand is a kit in a garage. A pad with no dome on it is a deck. And one host with one pad is a favour; what makes it worth building is the second host, and the third, and the four hundredth – choice of place and price for the owner, a pool of residents for the host that is not limited to whoever is in town, and a resale market for the houses. A house you can sell is an asset. A house nobody will buy is a very expensive tent. I want to build demand for the real thing, not assume it is already there.



Each one needs the other two.

Small scale, small risk

The first site is a pilot, not a claim that the engineering is finished. The land and somewhere to work are already mine, so the money goes on the next practical steps, costed with the same pad model as everything else in this Part:

```
the goal is $100,000. here is what it is actually for.

   3 pads on land already held, 36 to 44 ft:
      36 ft gravel pad, 1,018 sq ft     $11,343
      40 ft gravel pad, 1,257 sq ft     $11,611
      44 ft gravel pad, 1,521 sq ft     $11,907
      built                            $34,860

   the remaining $65,140 is the part that is not concrete:
      a website, and a list of people who want to host or park
      one to three dome designs that suit this, built and measured
      a camera, because all of it is published as it is made

what the money does not buy: the tracking foundation. a pad that
turns is a nice thing to own and it is not what a first site needs,
so it is not on this list.

a whole 6-pad park costs $213,718 to build.
this raise is not that. it is the first three pads and the
drawings that let anyone else build the fourth.
```

$34,860 of a $100,000 goal is the 3 test pads. The other $65,140 is the part that is not concrete: modular design and the means to do it – woodworking and metalworking, CNC plasma, 3D printers, a router, casting – one to three dome designs that suit this, built and measured, the web app and its hosting, documentation, a camera, and real promotion, because I want this idea to spread like a cold, not sit in a repository.

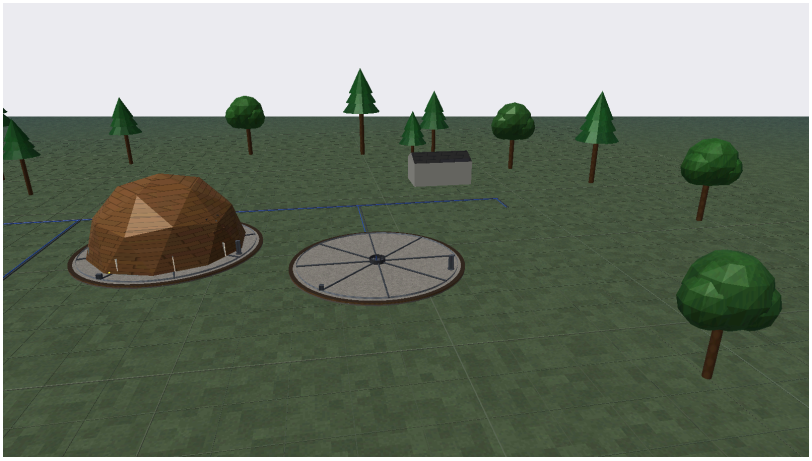
What it does not buy is the turning foundation of Chapter 49. A pad that follows the sun is a fine thing to own and I intend to build one, but on a first site it is whimsy, and I am not spending anybody else's money on it.

I see my own job becoming pad design and getting hosts started: a repeatable foundation and connection standard, like RV hookups are repeatable, with room for hosts to offer different things on top. The first job is to make the simple version work well enough that another host can build it from a clear plan.

For the people who help: founder status, a printed dome keychain or a window decal, a signed copy of this book – and, higher up, sitting down with me to imagine your own dome, build it digitally and make a plan for it, whether you are a future host, a future owner or somebody collecting possibilities for later. Those are the categories; a campaign sets the actual tiers.

THE NUMBER THAT DOESN'T HELP

A whole 6-pad park costs $213,718 to build. This raise is not that. It is the first 3 pads and the drawings that let anyone else build the fourth. And if it does not fund, my plan and my work are the same: I am building this whether it takes one year or ten. Support decides the speed, and how many people the drawings reach along the way.

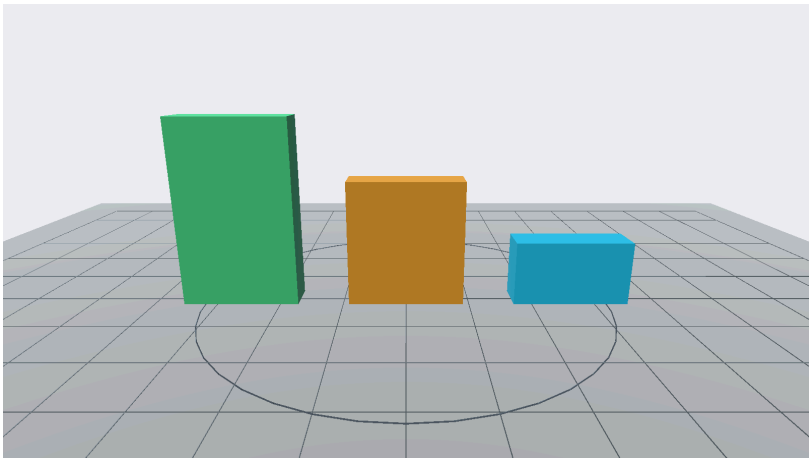


Small scale. Small risk.

The strategy, stated plainly

The Frankendome film put it in three phases. Phase one: build geodesic frames. Phase two: spread them across civilisation like a plague of structurally efficient triangles. Phase three: become filthy rich selling upgrades.

And to be completely transparent about the endgame: somewhere there is a man on his seventh holiday of the year, watching the projected yield on a portfolio of rental houses, and I would like that line to do something sudden and downward on account of this. That is the strategy. It is entirely transparent and only slightly relatable, and I have made my peace with it.



Three phases.

Bring your own home

A hotel sells you a night. A short let sells you a fortnight. A lease sells you a year and charges you to leave early. None of them sells the thing most people actually want, which is to live somewhere for a while without handing over the value of everything they do to the place while they are there.

Bring your own home. Pay the host for the ground and the services. Improve your house as much as you like, because it is your house, and take all of it with you when you go.

The source, the model's inputs and the experiments are open – actually open, not open as a marketing word. The drawings, the prices and the things that failed are all there to check, and every number in this Part came off them, including the ones that argue against it.

If this book was worth your time, hand it to one person who can carry the idea further than I can: somebody who could build it, host it, or needs it.

PART 10

Other Shapes

“A zome is swept, a hex dome is tiled, and a geodesic dome is subdivided. Three answers to one question about curvature.”

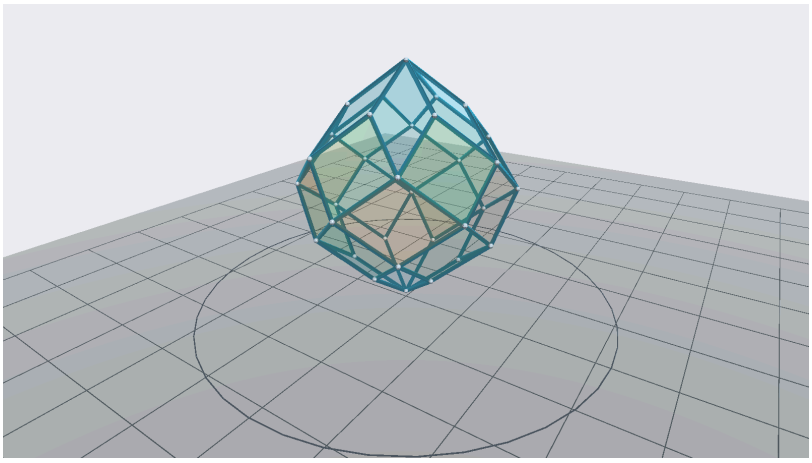
The two other round buildings this project solves, what each one is good at and what it costs against the dome in the rest of this book -- and the framed-panel method most bought dome plans use, solved again from this book's own geometry.

CHAPTER 51

A Zome Is Not a Piece of a Sphere

Swept from a star of directions instead of subdivided from a solid -- which is why every panel comes out flat, guaranteed

Everything so far has been subdivision: take a solid, cut its faces smaller, push the new points onto a sphere. A zome is made the other way and the difference decides everything about it.



A zome is not a piece of a sphere.

The sweep

Start with a **star of directions** — a set of vectors radiating from a point, evenly spread around a vertical axis.

Now take a stick. Point it along the first direction and walk to the end. Point it along the second and walk to the end of that. Keep going through all the directions and you come back to where you started, having traced the outline of a solid.

Every face of that solid is a **parallelogram**, because every face is made by two of the directions and nothing else. Two directions define a plane. A parallelogram made of two directions is flat, necessarily, always, without anybody checking.

That is a zome.

Every panel flat, guaranteed

This is the property worth the whole chapter.

In a geodesic dome, a panel is flat because a triangle is always flat. Fine — but it means every panel must *be* a triangle, and triangles tile a sphere with a lot of members.

In a zome, panels are four-sided and still guaranteed flat, because of how they were constructed rather than because of how many sides they have. You get big flat quadrilateral panels, which is a genuinely better thing to sheathe and to glaze than a triangle is.

And the star means **one strut length**. Make the directions equal in length and every stick in the building is the same stick. Not two lengths. 3 is not the number of anything. One.

What it costs

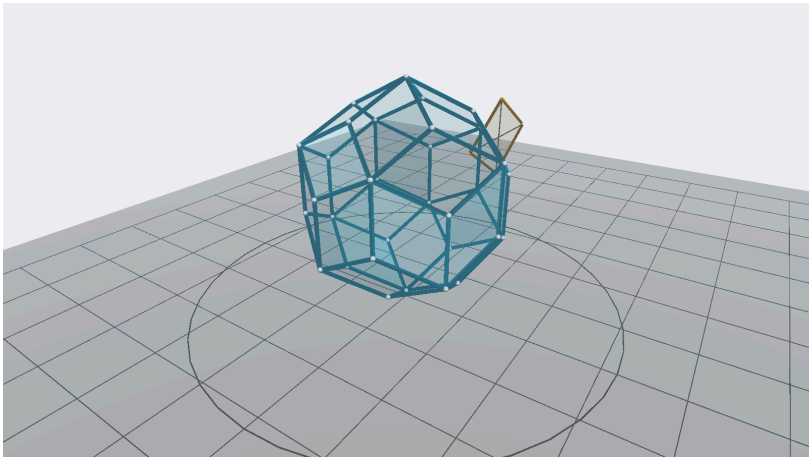
Three things, and they are real.

The hubs get particular – in a frame of single sticks. In a geodesic dome, the hub problem is what the pinwheel of Chapter 20 exists to remove. Build a zome as a frame of single sticks and the problem comes back: the sticks arrive at a hub in a pattern set by the star, at angles that are not all the same, and the joint has to hold that pattern. An earlier draft of this chapter stopped there and said zome hubs are where zome builders spend their time. That was wrong for the way most zomes are actually built: frame each panel on its own and the hub disappears, exactly as it does in the wedge dome. Chapter 53 shows how.

The top comes to a point. The sweep converges on the axis, so a zome finishes in a spike rather than a cap. That is either a feature or a detailing problem depending on your rainfall.

The shape is not a sphere. It is taller for its footprint, it has more surface per square foot of floor, and if you wanted a hemisphere you did not want a zome.

The famous one-panel zome



The famous one-panel zome.

Here is where the golden ratio comes back, and this time it really is the design.

Choose the star's directions using φ and the pitch to match, and every single panel in the zome becomes the **same rhombus**. Not two panel shapes, not five. One.

One strut length and one panel template, for a whole building. You can cut the entire skin from one jig and the entire frame from one saw setting.

That is the golden zome, and it is the single most elegant result in any of the geometries this project solves.

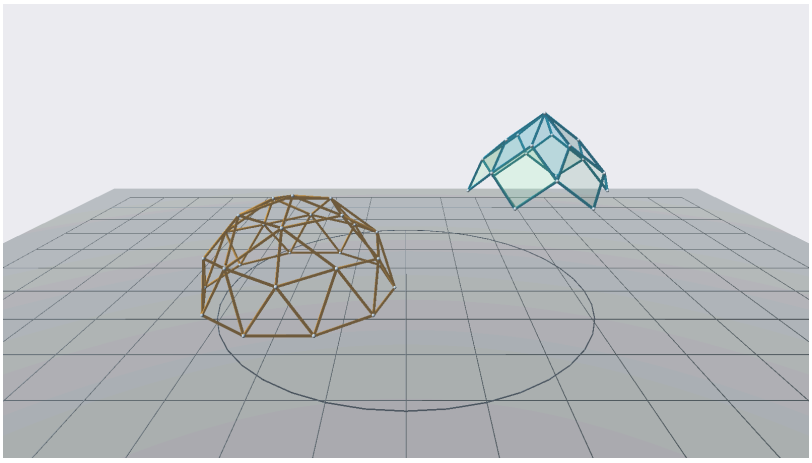
What the golden zome charges you

Two things.

It charges at the floor. The ordinary polar zome of Chapter 52 has its hubs on evenly spaced level rings, so a level floor line meets every strut it crosses at the same point along its length – one cut setting, repeated all the way round. The golden zome's 30 faces come from the icosahedron's axes, not from a star round a vertical one, and its rings of hubs are *not* evenly spaced. A level line through it meets struts at 2 different points, so the bottom row needs 2 cut settings instead of one. It is the exact mirror image of the polar zome's trade: one pays at the panel bench, the other at the floor.

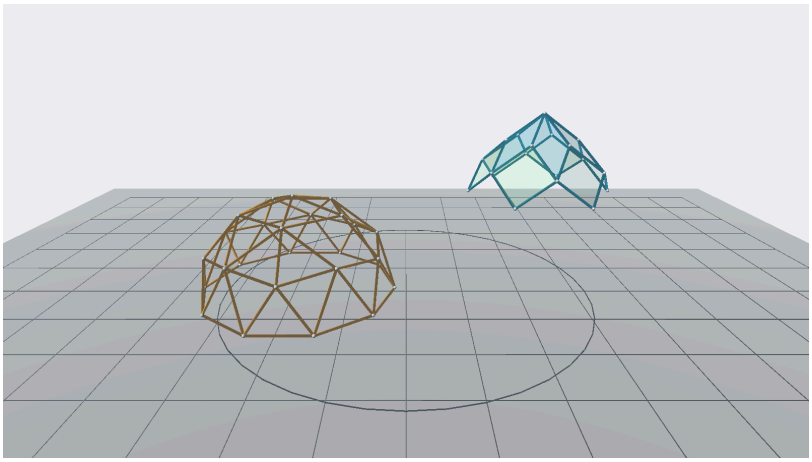
And it charges you the freedom to choose your proportions. Every panel comes out identical – diagonals in the ratio 1.6180, which is φ – because the star is pinned by φ. Change the height-to-width ratio you want and the property is gone; you are back to several panel shapes.

So the golden zome is beautiful and it is exactly one building. If its proportions suit your site, it is very hard to beat. If they do not, you are designing around a constraint rather than a feature, and it is worth finding that out before the jig is built.



A zome, drawn by the tool that solves it.

Zome against geodesic dome



Zome against geodesic dome.

strut lengths	zome 1	geodesic 2
panel shape	parallelogram	triangle
panel flatness	by construction	by being a triangle
hubs	particular	removed by the pinwheel
top	a point	a cap
surface per floor	more	less

Neither wins. They are answers to different questions, and the question a zome answers well is *"I want big flat panels and one stick length and I do not mind a hub."*

This book builds a geodesic dome because its question was *"I want to use a split log and avoid a hub entirely"*, and the wedge's flat sawn faces make the pinwheel possible in a way they would not in a zome's hub.

But if you have a bandsaw and a box of brackets and no chainsaw, the zome may be your building, and this chapter exists so you can find that out here rather than after the tree is down.

CHAPTER 52

Building a Zome, Ring by Ring

Two numbers -- a count and a pitch -- give the parts list, the level rings, the templates, the hub schedule and the one repeated floor cut

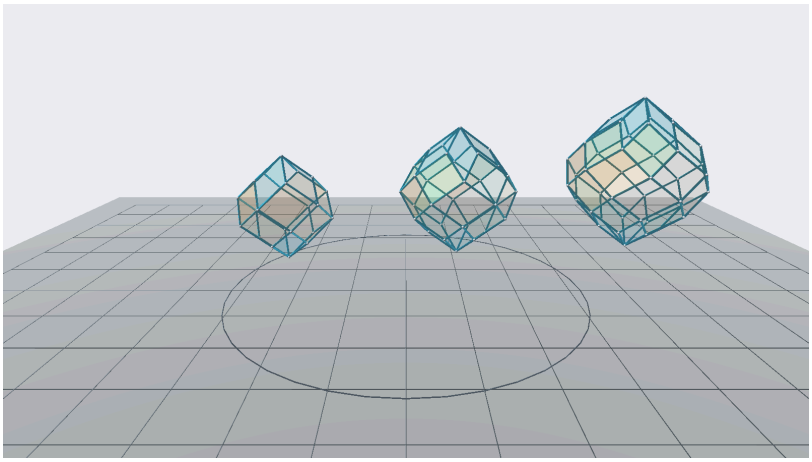
Chapter 51 said what a zome is: a shape swept along a star of directions, every panel flat, every strut the same length. This chapter is what a builder does with that. Everything in it falls out of two numbers you choose at the start – **how many directions, and how far they lean** – and the worked example is the zome film's own: 6 directions at 54 degrees, framed in 48-inch struts.

Counting the parts before you buy anything

Each pair of directions makes exactly two panels, one on each side of the whole closed shape. So with n directions the panel count is n times $n - 1$. Each panel has four edges and every edge is shared, so there are twice as many struts. Add the two points and you have the hubs:

panels n(n - 1)	30
struts 2n(n - 1)	60
hubs n(n - 1) + 2	32
V - E + F	2

That last line is the check. Euler's formula says a closed cage comes out to two, every time; if your count does not, you have miscounted, and the time to find that out is before the timber is ordered. A dome is the top of the closed shape: here, 12 panels on 30 struts.



Counting the parts before you buy anything.

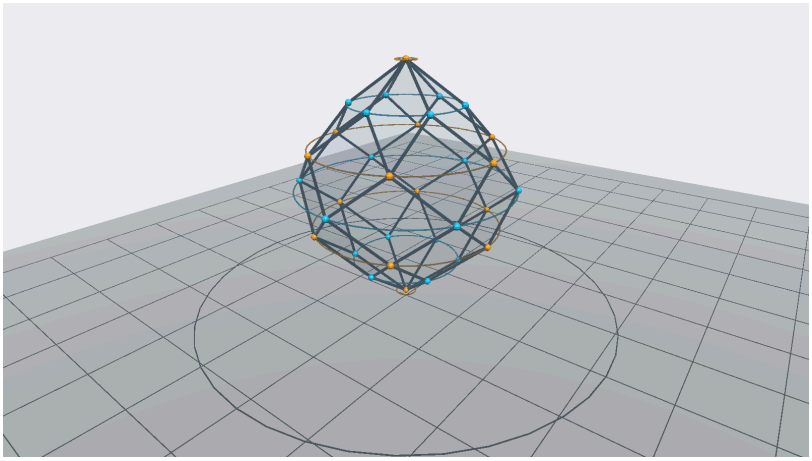
The top comes to a point

Walk along every direction in turn and you arrive at one particular corner: the far end of the whole sweep. Only one corner can be reached that way, so the roof closes on a single point, with a panel arriving from each direction. Add up the panel corners that meet there – 286.3 degrees here – and they fall 73.7 degrees short of a full turn. That shortfall is why it closes into a peak instead of lying flat: the same missing-angle bill Chapter 54 found the pentagons paying.

The hubs land on level rings

This is the property that makes a zome pleasant to build. A corner's height is just the number of directions you travelled to reach it, times the rise of one direction. Every corner reached in the same number of steps is at **exactly** the same height – so the hubs sit on perfectly level rings, 7 of them, evenly spaced. No ring is approximate and no tape is needed to find it.

A hexagon dome has nothing like this, and it is why a zome wall meets a floor and a lintel so easily.

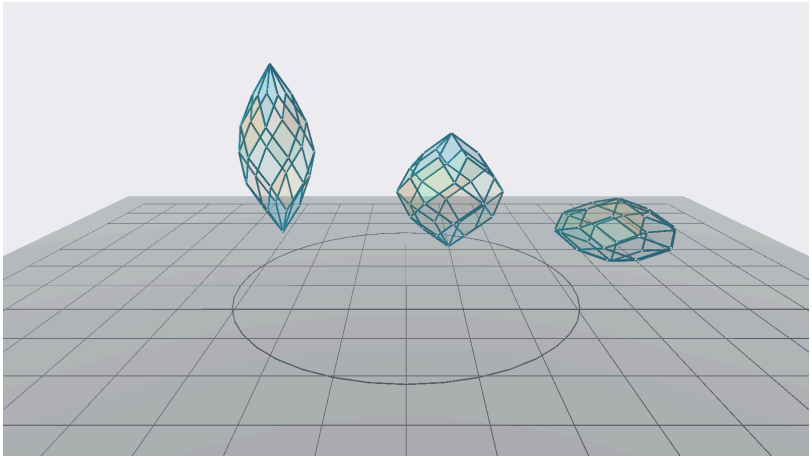


The hubs land on level rings.

How many templates, and what pitch buys

A panel's shape depends only on how far apart its two directions sit round the star. Neighbours give a long, thin rhombus; opposite directions give a fat near-square. A pair separated by three steps one way is separated by the same the other way, so the shapes pair up, and a zome with *n* directions needs only about half that many templates. Here: **3**. Cut one jig per shape and trace the rest.

Then the free knob. Lean the directions in steeply and the zome climbs into a spire; open them out and it settles into a saucer. On the same 48-inch struts, the film's 54 degrees stands 7.1 feet above its rim; at 34 degrees, 9.9; at 70, 4.1. **Nothing about the strut length or the part count changes** – only the panel angles do. You can shape the building to the site, the snow load or the headroom you want without adding a line to the cut list. Very few structural systems give you a knob like that for free.



Pitch is the free design knob.

The hubs

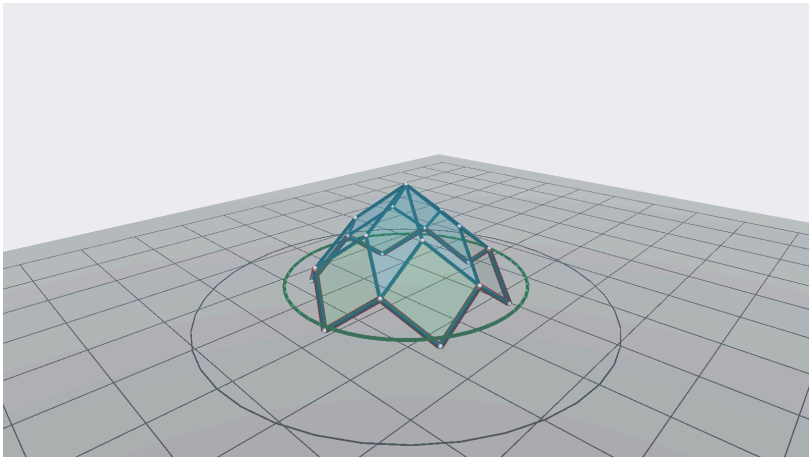
The saving on struts is paid back at the joints – in a zome framed from single sticks. The corners come in 4 kinds: some take three struts, some four, and the two points take one from every direction. Each kind has its own splay angles, so a hub that works at mid-height will not sit right near the top. If you build that way, make a full set of angle jigs before you fabricate any hub, and stamp each one with its type as it comes off the bench.

Or frame each panel on its own and the hubs disappear, which is what Chapter 53 does.

Meeting the floor

Not one strut in a zome runs level, so a level floor line always crosses struts in mid-span. On a polar zome that turns out not to matter. Because every hub in a ring is at the same height, the level line meets every strut it crosses at exactly the same place along its length: **1 cut setting**, repeated all the way round. Mark one, check it, and use it as the master for the rest.

(The golden zome of Chapter 51 needs 2. That is its bill.)



Meeting the floor: one repeated cut.

The cut list and the raise

What the shop receives:

1. **One strut length**, 48 inches centre to centre, cut as many times as there are struts in the roof – less the connector allowance measured on a real joint, never from a catalogue.
2. **A template for each panel shape** – 3 of them.
3. **A hub schedule**, if you are building on hubs: how many of each kind.

Everything on that list came from the two numbers chosen at the start. Change your mind about the pitch and the list regenerates; it does not have to be re-derived.

Then the raise, which the geometry hands you:

1. Set out the bottom ring on a level base and bolt it down.
2. Add **complete tiers**, never a single column of panels up one side. A zome is a shell, and it is not stable until each ring closes.
3. Prop the tier you are working on until the ring above it goes in.
4. Measure each completed ring's diameter in three directions before starting the next.
5. Leave the apex panels for last.

That is the whole method: a count and a pitch, swept out; every panel flat, every strut the same, the hubs on level rings, the roof closed on a point, and the bottom row one repeated cut.

CHAPTER 53

A Zome Without a Bevel

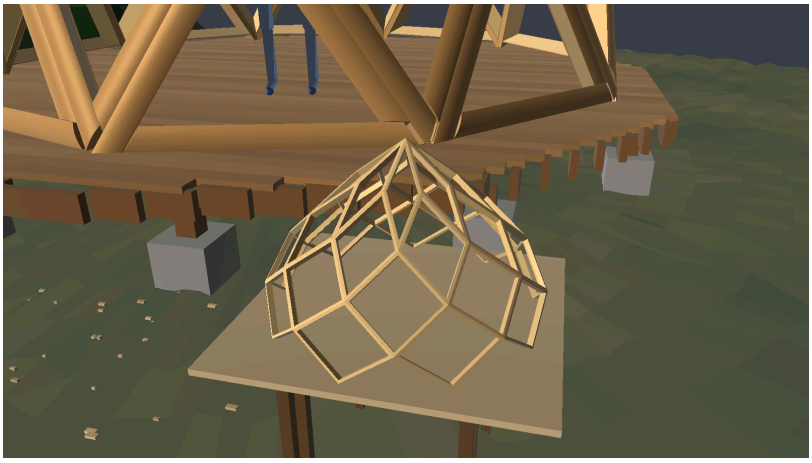
Four identical sticks to a panel, a miter at each end and no bevel anywhere -- and the gap on the outside that the method allows on purpose

Chapter 51 said a zome's hubs are where zome builders spend their time, because the sticks arrive at a hub in a pattern the joint has to hold. That is true of a zome built as a frame of single sticks. It stops being true the moment each panel is framed on its own, the way Chapter 56 frames a dome's triangles. Then the panels meet each other along their edges, and there is no hub at all.

It is called the *non-beveled* method, and builders who have framed zomes both ways overwhelmingly choose it. The figures are this book's, solved for its own zome:

base_ft	16 ft the reference 3V dome, base corner to base corner: a round size, and the largest that stands beside the wedge dome on the Cabin World hill
stock_w_in	1.5 in a 2x3's actual thickness: the stick lies wide side down, so this is its width on the jig
stock_d_in	2.5 in a 2x3's actual width, ripped in half along a bevel to make two sticks
kerf_in	0.125 in a common table-saw blade
base_board_in	5.5 in a 2x6 on edge under the lowest base corner: lifts the first panels off the deck's wet
zome_n	9 directions a nine-sided zome
zome_pitch_deg	54 deg the pitch the zome chapter uses, from the zome module
zome_side_m	0.9 m panel side: puts the rim about four metres across, a garden-studio size
zome_w_in	1.5 in a 2x4's thickness, flat on the bench
zome_d_in	3.5 in a 2x4's width, on edge through the wall

The zome is nine-sided – 36 rhombus panels in 4 rows of 9 – about 13.8 feet across its rim and 8.7 feet tall above it, every panel side 35.4 inches.

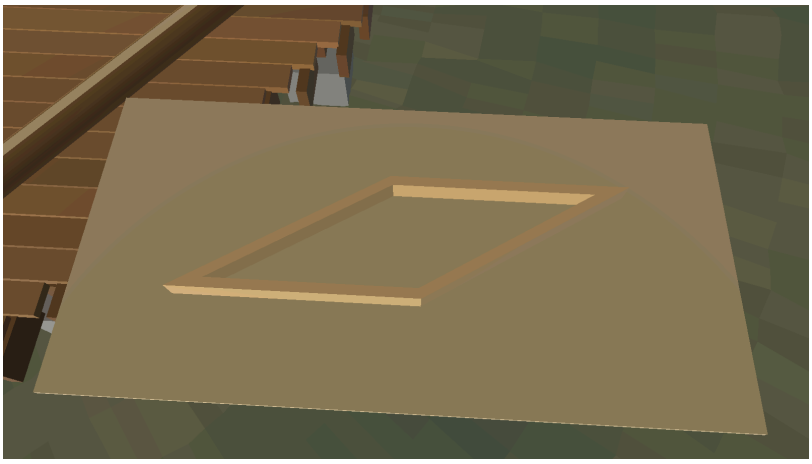


One stick, one miter, per row.

Four sticks in a pinwheel

Each panel is four sticks in a pinwheel, the arrangement of Chapter 20 once more: each stick runs to one corner and butts the next at the other. A rhombus's corners alternate sharp and blunt, and the sine of an angle is the sine of its supplement, so the shortening at every tail is the same – **all four sticks of a panel are one stick**. Its two ends are cut at the same angle and parallel, long point to short point.

Nothing is bevelled. The sticks stay square through their whole depth, so there is no compound cut, no tilted blade, and no jig with a bevel in it. Cut one end, butt two sticks together on the bench, mark the panel length across the joint, scribe the miter and cut to the mark; set a stop block and cut the rest of that row's sticks from full-length boards. Screw pairs into halves and halves into panels.



Both ends cut parallel; nothing bevelled.

Row by row

A zome's rows differ, so each row has its own stick:

row	panels	sharp corner	miter both ends	fold	outside gap, in	block bevel
1	9	32.1	57.9	24.2	1.50	12.1
2	9	62.7	27.3	19.3	1.19	9.6
3	9	89.0	1.0	27.8	1.73	13.9
4	9	74.4	15.6	27.8	1.73	13.9

One stick and one miter for every panel in a row; that is the whole cut list.

The gap, and what fills it

Square-edged panels meet only along their *inside* edges. Where two panels fold away from each other, that leaves a V-shaped gap on the outside whose width is twice the stick's depth times the tangent of half the fold. In a two-by-four on edge it opens from **1.19 to 1.73 inches** on this zome.

The gap is allowed on purpose. On a small zome, screw the inside edges together and let the roof bridge it – a zome's panels each slope one way, so roofing laps down the structure like shingles. On a large one, rip a **wedge block** to half the fold on each face and fix it between the panels: it fills the V and gives the joint something to bear on. The block's bevels are in the table.

Assemble with two screws per stick near the inside edge, not tight, so the panels can hinge. As the inside edges line up, the fold sets itself. Brace the top rows – they sag under their own weight until the ring closes – then go back and add the rest of the fasteners.

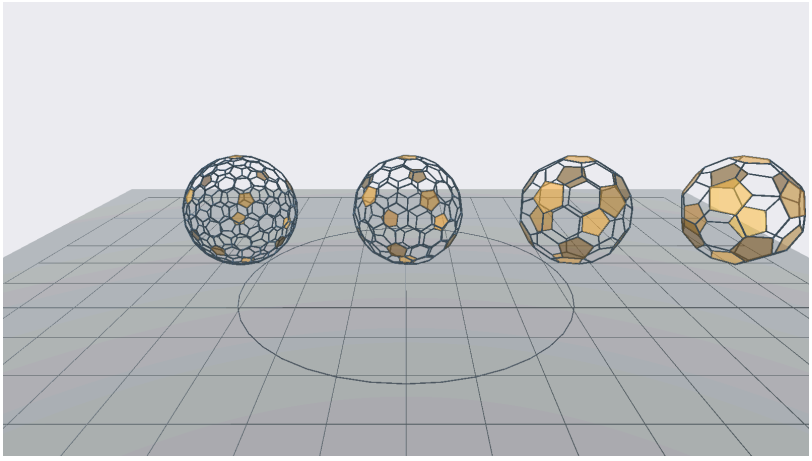
CHAPTER 54

Exactly Twelve Pentagons, Always

A sheet of hexagons will not curve. Curvature is bought with missing angle, and the price is always twelve pentagons

Everybody reaches for hexagons. Bees use them, basalt cools into them, a soap froth settles into them, and for one reason: a hexagon walls off more floor per foot of wall than a square or a triangle does, while still leaving no gap. So people keep asking for a dome made of hexagons, and this chapter answers that honestly – what you can have, what you cannot, and exactly what the difference costs in the shop.

There is a catch, and it is absolute.



Exactly twelve pentagons. Always.

A sheet of hexagons will not curve

Stand at any corner of a hexagon tiling and **three** panels meet there. Each brings 120 degrees, and three of them is 360: a full turn, flat, with nothing left over.

That is what makes hexagons perfect for a flat surface, and it is exactly what stops them curving. Take a paper disc, cut one wedge out of it, and pull the two cut edges together: the paper cannot lie flat any more, and it lifts into a cone. Nothing was stretched. The only thing that changed is that there is less than a full turn of paper round the centre. **That missing angle is the whole mechanism of curvature**, and it is the only currency a dome has to spend.

A regular hexagon corner has none to give. A sheet of them stays a sheet however many you add.

Where the angle comes from

A pentagon. A regular pentagon's corner is 108 degrees, so where three of them meet there are 36 degrees missing from the full turn. Fold that slack out and the surface pulls into a bowl. Every hexagon dome you have ever seen is doing this: **hexagons carry the field, and pentagons do all of the curving.**

And how many pentagons is not a design choice. Any closed convex cage is missing exactly 720 degrees in total, whatever its size – Descartes proved it – and Euler's formula gives the same answer another way, with the hexagons cancelling out of the arithmetic entirely. A closed cage of hexagons and pentagons, three faces at every corner, needs **exactly twelve pentagons**. Twenty hexagons or two

thousand, still twelve.

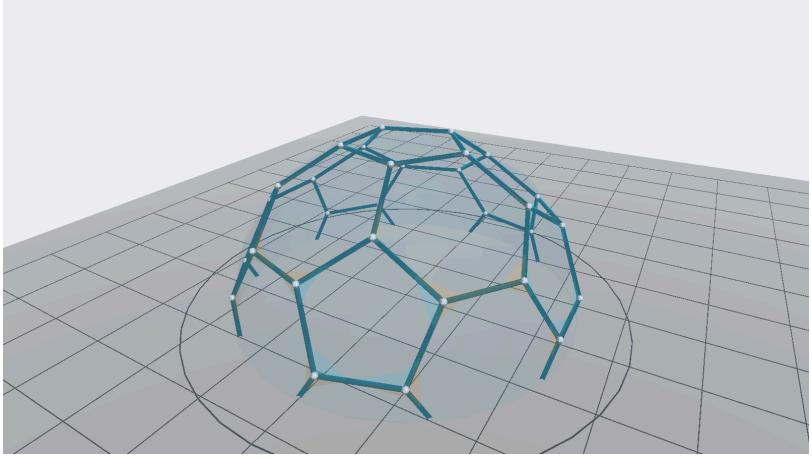
A football has twelve. A fullerene has twelve. A hexagonal dome has twelve, or it is not closed.

The football dome

The cage you want if one shape is the goal is the pattern on a football, the truncated icosahedron: 20 identical regular hexagons and twelve identical regular pentagons, 32 panels on 60 hubs. Every one of its 90 struts is the same length. No other hexagon cage on a sphere is this kind to a builder.

So the cut list is one line. At a dome 20 feet across, the strut is 48.4 inches hub centre to hub centre. Set a stop block, cut every piece, and check a sample against a master gauge rather than a tape. The geometry's number is centre to centre: subtract the connector allowance for your hub before you cut, and measure that allowance on a real assembled joint, never from a catalogue.

Both panel shapes are genuinely flat – their corners lie in a plane – so one full-size hexagon template and one pentagon template, made from a scrap sheet, trace the whole skin. Cut slightly oversize, fit to the raised frame, then trim. The frame is always a little less perfect than the arithmetic, and the panel is where you absorb it.

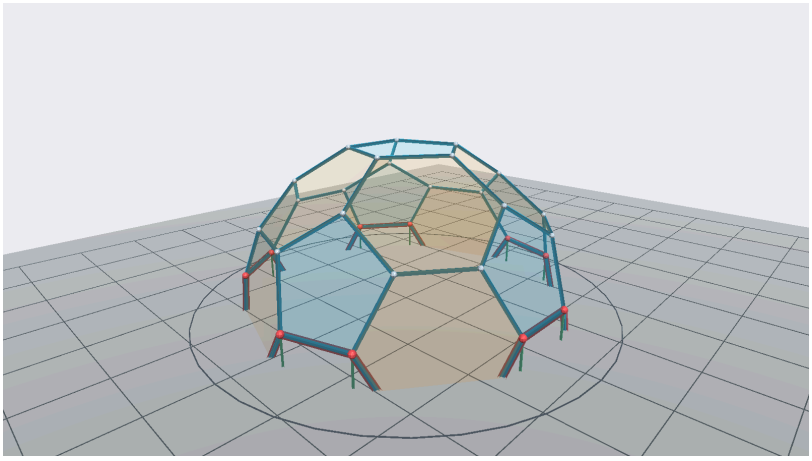


The football dome.

Where do you cut it off?

A triangulated dome can be cut along a flat ring of hubs. A hexagon cage cannot: no ring of its struts runs level, so wherever you stop, the bottom edge rises and falls. On the 20-foot football dome that zigzag is **41.2 inches deep** – 3.4 feet.

Two honest options. Build a level stem wall and let the zigzag sit on top of it, or cut the bottom row of panels along a chalked level line and flash the cut. Either way, decide before the first strut is cut.

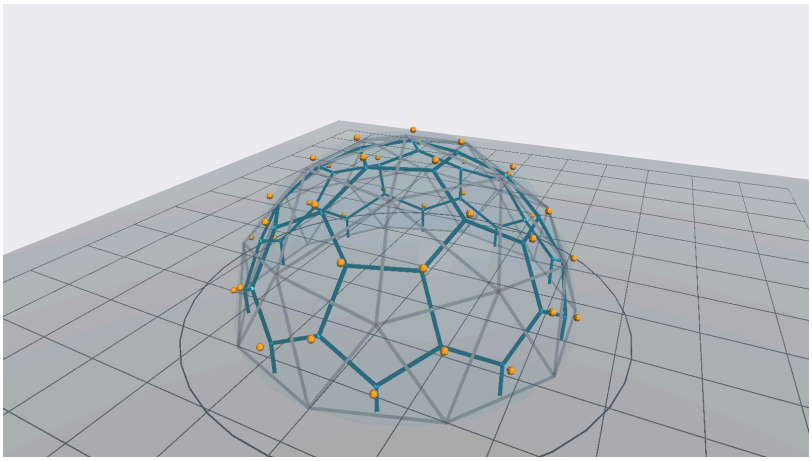


The rim comes out as a zigzag.

Then raise it rim to crown. Bolt the first course of hubs to the stem wall and work upward **one complete ring at a time, never up one side**, so error spreads evenly instead of piling up in one place. Measure each finished ring's diameter in three directions before starting the next. Leave the crown panel until last: it is where every accumulated millimetre shows up, and it is far easier to trim one panel than to unbolt a ring.

Then you want it bigger

The football is one fixed pattern. You can scale it, but you cannot make it finer, and past a certain size its panels get too big to lift and too flat to look round. For a finer cage, take a geodesic sphere of the kind Part 2 built, mark the centre of every triangle, and join the centres around each corner. That operation is the **dual**. Corners where six triangles met become hexagons; the twelve corners where only five met become the pentagons – the same twelve the icosahedron started with, which no amount of subdividing ever creates or destroys. The family is called the Goldberg cages.



Every triangle corner becomes one panel.

The price shows up at once. Lay a Goldberg hexagon beside a football hexagon: its sides are no longer all equal and its corners are no longer all 120 degrees. It is still a good panel, but it is a specific shape with a specific orientation, and the strut list now has two lines on it instead of one.

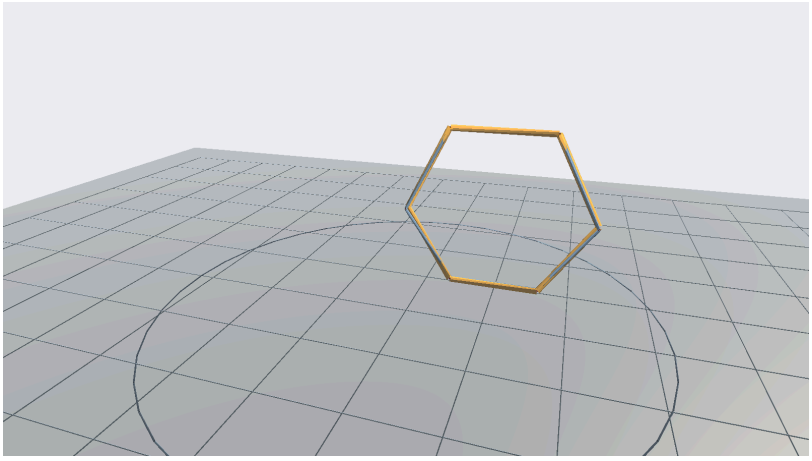
Every step up multiplies the shapes

Measured at the same 20-foot dome:

```
cage                      panels strut lengths hexagon shapes worst warp, in
Truncated icosahedron GP(1,1)     32             1              1           0.00
Goldberg 2V hexagon cage GP(2,0)     42             2              1           0.40
Goldberg 3V hexagon cage GP(3,0)     92             4              2           0.12
Goldberg 4V hexagon cage GP(4,0)    162             6              3           0.07
```

2 strut lengths, then 4, then 6; 1 hexagon shape, then 2, then 3. The dome gets rounder and its panels easier to lift, but every extra length is another jig, another label and another chance to reach for the wrong stick.

And the panels stop being flat. Three points always define a plane, so a triangle is flat for free. Six points on a sphere generally do not, so a subdivided hexagon panel is **warped**: its corners want to sit slightly out of any flat sheet you cut. The football measures exactly zero, and always will. The first Goldberg cage is the worst, at 0.40 of an inch at the worst corner; finer cages warp less, down to 0.07 at four-frequency, because smaller panels sit closer to their own plane.



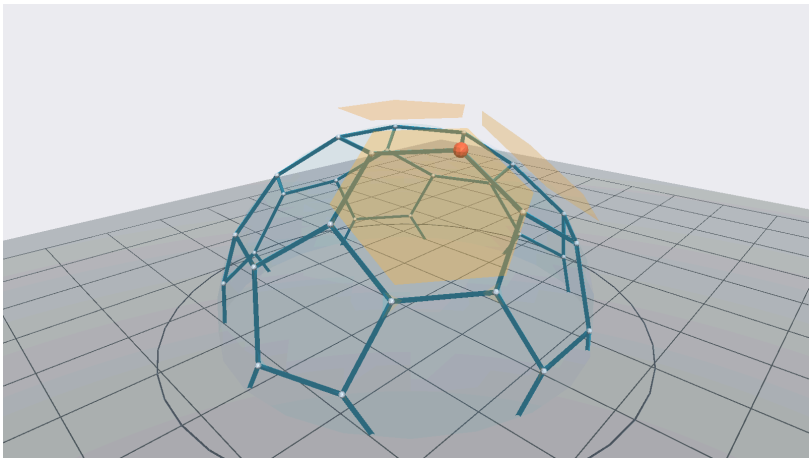
The panels stop being flat.

THE NUMBER THAT DOESN'T HELP

A flat sheet on a warped opening gaps at two corners, or springs. You can seal the gap, let a sheet take the twist and hold it with every fastener, fold the panel into the two triangles it is trying to be, or skin the dome with a membrane. All of them work if they are planned. None of them works if you find out with the panels on site. The triangulated dome does not have this bill at all.

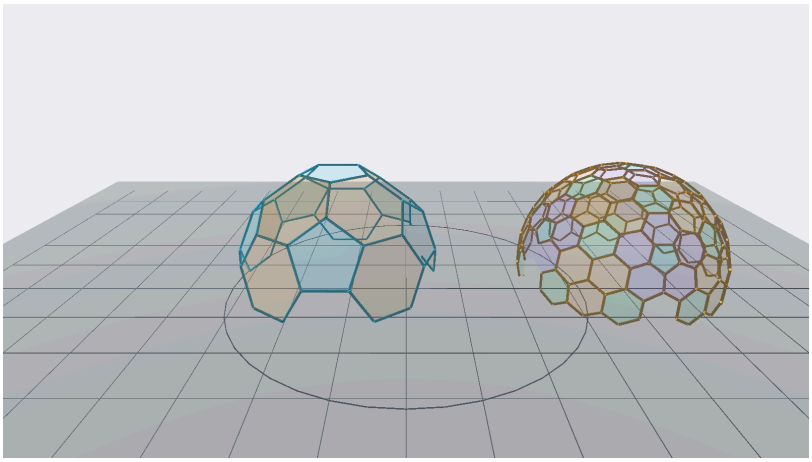
Three panels, one hub, three seams

Every corner of a hexagon cage is three-way, so every hub is a junction of three panel edges – and on the upper half of the dome those seams point at the sky. Water finds them. Shingle the skin from the rim upward so every lap sheds downhill, tape or flash each hub, and give it an eave that throws water clear of the wall rather than down it. Chapter 28 said it about the wedge dome and it is truer here: **a dome leaks at its seams long before it fails at its struts.**



Three seams meet at every hub.

Choosing between them



The two domes, side by side.

At the same radius, the football dome's half is 16 panels and 55 struts of a single length. The four-frequency cage is a far rounder shell that needs 6 lengths and 3 hexagon templates.

If the dome is small enough that a two-metre panel is liftable, take the football: one strut, two templates, flat panels, no warp. If you need a bigger or rounder shell, accept the subdivision – and then be ruthless about labelling, because a six-length dome punishes a mis-sorted stick far more than a one-length dome does. Colour-code the ends, and do it at the saw, not at the scaffold. Repeated work is cheap. Varied work is where the mistakes live.

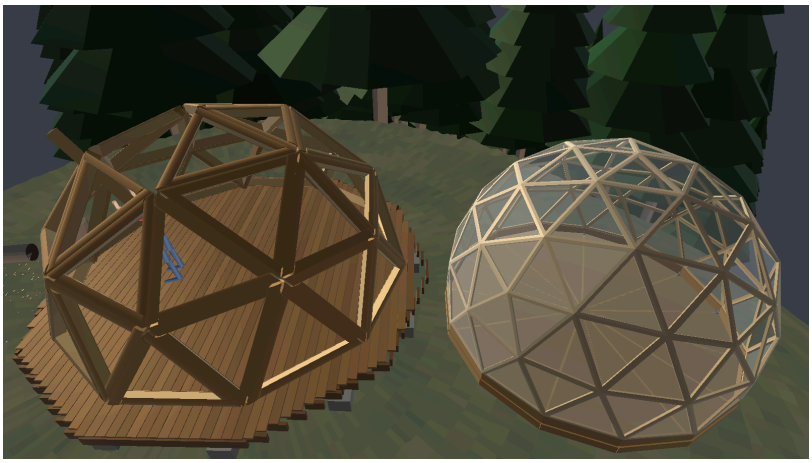
And the thing to carry away: curvature costs angle, and something has to pay. In a hex dome, twelve pentagons pay it. In a geodesic dome, the subdivision pays it – which is precisely why the wedge dome has two chord lengths instead of one. There is no tiling that curves for free; every round building in this book is a different answer to the same bill.

CHAPTER 55

Three Frequency, Five Eighths

The dome most bought plans build: a three-frequency sphere cut below its middle, a base that zig-zags, and two panel shapes

Most small timber domes built from bought plans are not split from logs and are not hub-and-strut frames. They are **framed panels**: every triangle of the dome is its own small frame of sawn sticks, built flat on a plywood jig, skinned in the shop, and screwed to its neighbours on site. It is a good method with a great many domes behind it, and anyone who builds the dome in this book should understand it, because it is the road most people take.



Two domes, framed two ways.

Why it is here. The three-frequency five-eighths dome of framed panels is a popular build, tried and true: a great many are standing. That makes it the fairest comparison for the wedge dome, and this chapter and the next set it beside it. Every length, angle and count is solved by this project's own geometry (`two_v_demo/panel_method.py`), for a dome of this book's own choosing, and drawn by this project's own software.

The reference dome, and the stock its sticks come from, are choices rather than measurements, and they are declared before anything is claimed:

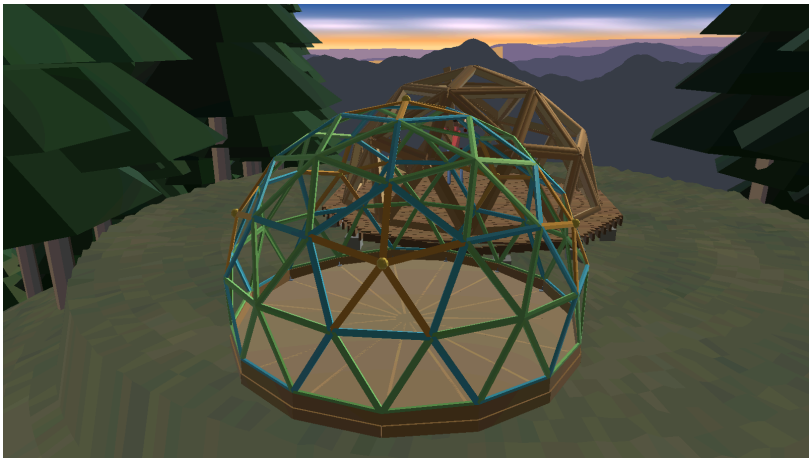
base_ft	16 ft the reference 3V dome, base corner to base corner: a round size, and the largest that stands beside the wedge dome on the Cabin World hill
stock_w_in	1.5 in a 2x3's actual thickness: the stick lies wide side down, so this is its width on the jig
stock_d_in	2.5 in a 2x3's actual width, ripped in half along a bevel to make two sticks
kerf_in	0.125 in a common table-saw blade
base_board_in	5.5 in a 2x6 on edge under the lowest base corner: lifts the first panels off the deck's wet
zome_n	9 directions a nine-sided zome
zome_pitch_deg	54 deg the pitch the zome chapter uses, from the zome module
zome_side_m	0.9 m panel side: puts the rim about four metres across, a garden-studio size
zome_w_in	1.5 in a 2x4's thickness, flat on the bench
zome_d_in	3.5 in a 2x4's width, on edge through the wall

Three lengths, two kinds of point

Chapter 8 made this book's dome by cutting each face of the icosahedron into four and pushing the new points out. Cut each face into *nine* instead – three to an edge – and push every point out to the sphere, and you have a **three-frequency** dome. It is rounder, its triangles are smaller, and it has only three strut lengths. At 16 feet across the base:

A	34.00 in	(864 mm)
B	39.36 in	(1,000 mm)
C	40.23 in	(1,022 mm)

Twelve points have five neighbours – the icosahedron's own corners – and every other point has six. Chapter 54 explained why it must be exactly twelve. The triangles round a five-way point make **pentagons**; all the others make **hexagons**. That is the whole vocabulary of the dome: three lengths, two kinds of point.



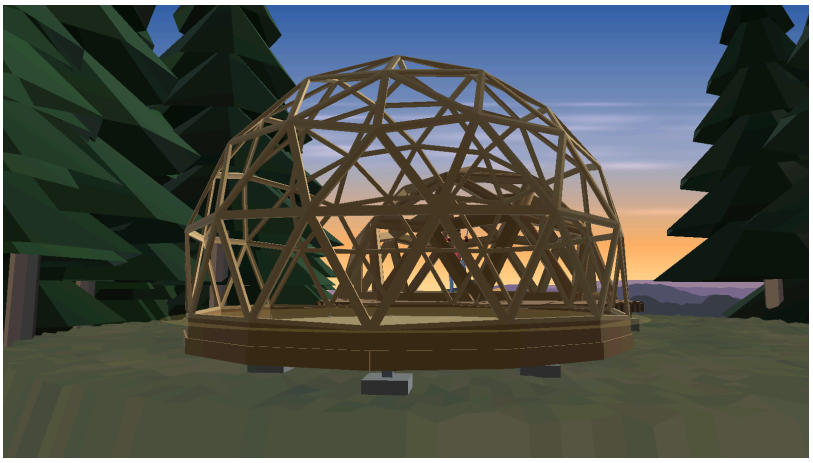
Three chords: A, B and C.

Why five eighths

A two-frequency sphere has a level ring of points at its middle, which is why this book's dome is a clean hemisphere. A three-frequency sphere does not. Its points sit at heights that never line up across the middle, so the cut goes lower, to the ring of points nearest five-eighths of the way down.

That cut keeps **105 triangles**: 30 that belong to pentagons and 75 that belong to hexagons. Its height is **0.594 of its width** – a little under the five-eighths in its name – which for the reference dome is 9.65 feet.

The extra wall below the middle is the point of the cut. Near the base it stands almost upright, so the floor near the edge is floor you can stand on. A hemisphere's wall leans in from the floor; this book's dome answers that with a riser wall under the shell (Chapter 27), and the five-eighths cut answers it with more of the sphere.

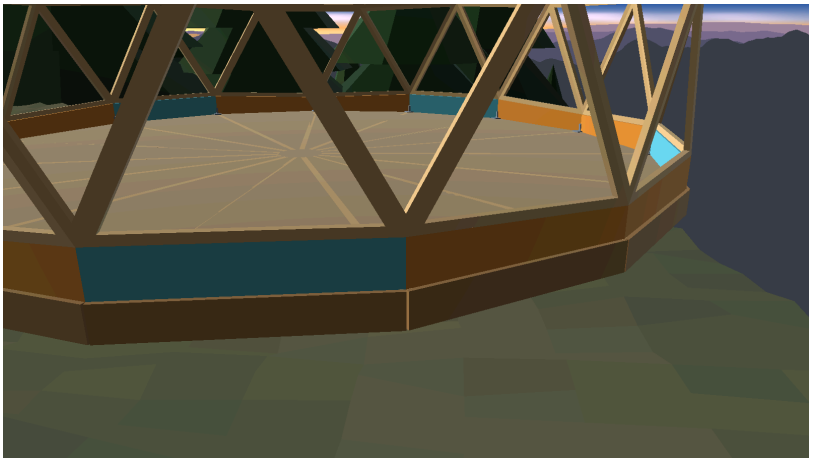


Cut below the middle.

A base that zig-zags

The price of the five-eighths cut is its base. The bottom edge is 15 struts long and its corners do not lie on one level: they alternate between two heights **40 mm apart** – about 1.56 inches. So the base sections, the boards the bottom panels sit on, are not all alike. **10 must be tapered**, low at one end and high at the other, and **5 sit level**.

Build that ring first. Set the base sections out on the deck, fasten them to each other, pull the diagonals until they agree, and fasten them down. Every panel above lands on this ring, and every error in it is passed up the dome.



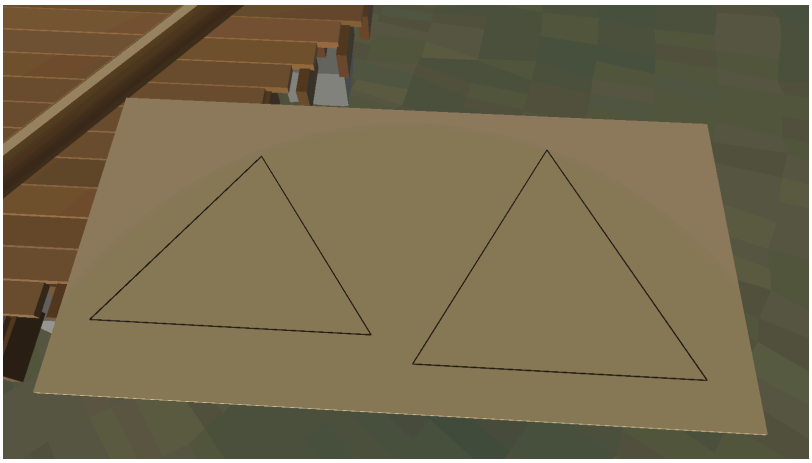
Ten tapered, five level.

Two panels, over and over

Every panel in the dome is one of two triangles, and the jig is drawn for each of them full size:

	pentagon	hexagon
panels	30	75
sides (letters)	B-A-A	B-C-C
base, in	39.36	39.36
left side, in	34.00	40.23
right side, in	34.00	40.23
apex angle, deg	70.73	58.58
base angles, deg	54.63	60.71
height on jig, in	27.73	35.08

The outlines are the *outside faces* of the panels – the face the weather sees. Nothing may stick out past them and nothing may fall short of them, which is what the jig of the next chapter enforces.



A pentagon panel and a hexagon panel.

A floor the shape of the base

A dome on a deck wants the deck to copy its base. A **radial deck** does: one joist from the centre to every base corner – 15 of them – and a rim board along every base edge, 39.9 inches long point to long point, each end cut **12 degrees** off square. That angle is not looked up anywhere: it is half of the 360 degrees round the deck divided by the number of sides, 180 over 15.

Under the joists, the girders follow the dome's own symmetry. A three-frequency dome has five-fold symmetry about its top, so the girders make a ring of 5, each end cut 180 over 5, 36 degrees, with a post under every joint where two girders meet. Mark the centre on the ground, lay the girders out on it to place the posts, set the post tops to a level line, then roll in the joists. The deck is 195 square feet. Chapter 38 is about why the floor is the other thing to get right, and Chapter 39 about where it stands.

Joists to the corners, girders on the five-fold.

CHAPTER 56

A Panel on a Jig

Rip, draw, rail, pinwheel, skin, raise: the framed-panel method start to finish, and the bevel no single saw setting gets right

This is the framed-panel method from first cut to anchored shell, for the three-frequency dome of Chapter 55: a popular, tried-and-true way to build, shown here for comparison. The figures are this book's, solved for its own reference dome; the pictures are this project's own renderings.

The idea is to move the work indoors. Each triangle is built flat, on a bench, at a comfortable height, out of the weather. It can be finished and skinned before it ever leaves the shop. On site, the dome is mostly a matter of screwing finished frames together.

Finish, then rip in half

The sticks are sawn lumber: a 2x3, ripped in half down its length, gives two sticks 1.50 inches wide and 1.19 inches deep. A 2x6 ripped into two 2x3s first is often straighter than any 2x3 on the rack, and straightness matters more here than species or grade.

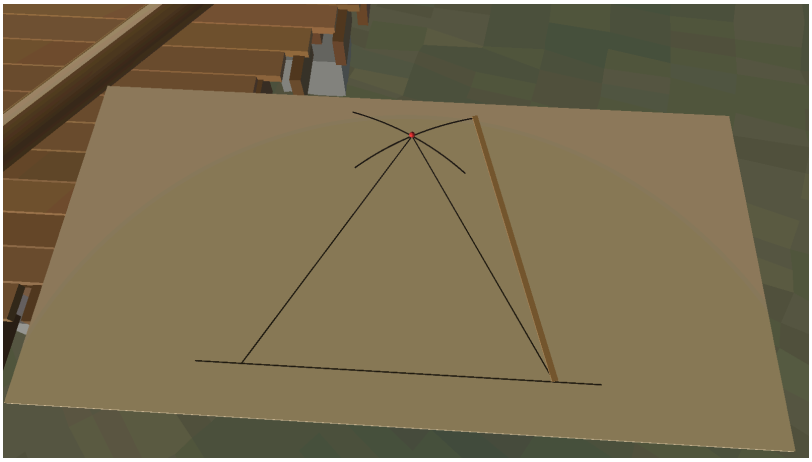
Finish the boards before you rip them. The face the saw makes is hidden in the joint, so finishing whole boards is half the work of finishing sticks.

Then rip on the bevel, through the centre. Tilt the table-saw blade to the bevel and set the fence so the cut runs through the middle of the board's end. A cut through the centre of a rectangle leaves two pieces that are copies of each other, turned end for end – the same symmetry that makes a split log give identical wedges in Chapter 13. Turn the offcut over and send it back through for a hair more, and the two halves come out exactly alike. A stick a little smaller than planned is fine; sticks that differ from each other are not. The bevel itself must be right, so set it with a digital angle gauge, not the saw's scale.

Draw the jig with two arcs

The jig is a sheet of 3/4-inch plywood – thick enough not to warp – with the panel drawn on it full size. Snap a straight line for the base and mark the base length on it. Make a **trammel**: a thin stick with a nail through each end. Set it to one side length and swing an arc from one end of the base; set it to the other side length and swing an arc from the other end. Where the arcs cross is the apex. Measure all three sides again before joining the points. For the hexagon panel the apex lands 35.08 inches above the base line; for the pentagon panel, 27.73. Cut the sheet down to an inch or two outside the lines.

It is the same jig, in spirit, as Chapter 24: geometry solved once on a flat board and then copied by every panel.



Where the arcs cross is the apex.

Rails, and the line

Lay a stick along one side, flat face down and bevel out, with its outer edge *splitting* the pencil line. Clamp it. Butt a straight rail against its inside edge and screw the rail down hard. Unclamp, and do the other two sides. Now any stick pushed against a rail lands exactly on its line. The jig does not care what section the sticks are, so long as every stick is the same.

Sticks that turn like a pinwheel

The three sticks do not meet at a point. Each runs all the way to one corner and butts into the inside face of the next, so seen from above the corners turn like a pinwheel. It is the pinwheel of Chapter 20, met again in a different dome: nothing meets at the corner, so no corner needs a hub.

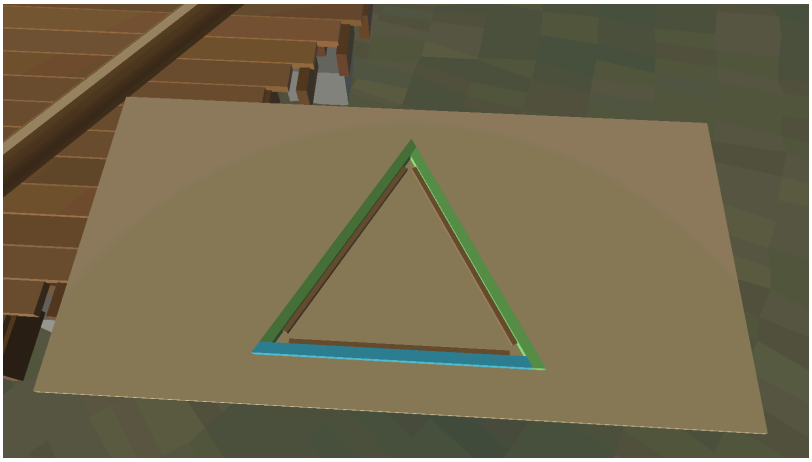
A pinwheel has a consequence that surprises people: the two equal sides of a panel do **not** take the same stick, because one butts and the other is butted at the apex. A stick of width *w* that butts at a corner of angle *phi* stops *w* / sin *phi* short of the corner and is cut 90 - *phi* off square. Every stick in the dome:

panel	side	each panel	outer edge, in	tail miter	bevel
pent-BAA	B	30	37.52	35.4	6.5
pent-BAA	A	30	32.42	19.3	7.2
pent-BAA	A	30	32.17	35.4	7.2
hexa-BCC	B	75	37.64	29.3	6.5
hexa-BCC	C	75	38.47	31.4	6.8
hexa-BCC	C	75	38.51	29.3	6.8

That is **6 different sticks**, 315 in all, about 972 feet of them.

Cut each stick's first end, lay it in the jig against its rail, and mark where the next stick's inside face crosses it. That second end is a compound cut – the tail miter and the bevel together. A miter saw that bevels both ways makes it in one pass; a handsaw, cutting flush along the next stick after the panel is screwed, works too and is pleasantly slow. Clamp a stop block for the repeats, and check a stick in the jig every few cuts in case the block has crept.

Half panels – at a door, or where the base needs one – are drawn on the same jig, but their middle stick runs across the triangle rather than along an edge, so it takes a different bevel or none at all. Check it before you cut it.

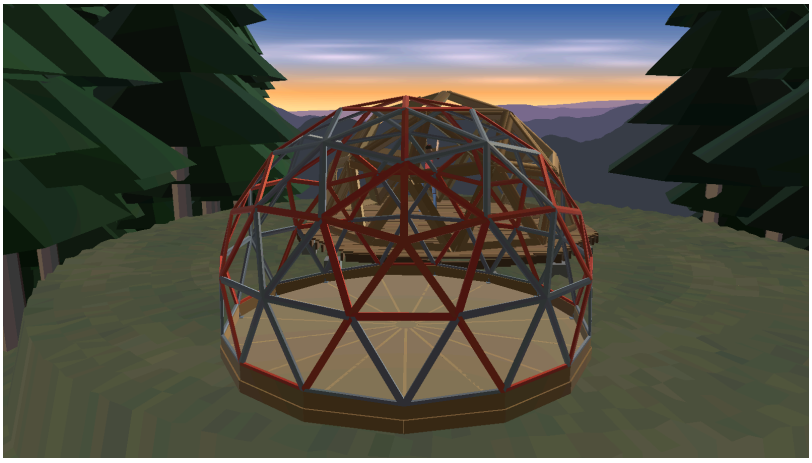


Three sticks, six kinds in the dome.

One bevel cannot be exact

> **THE NUMBER THAT DOESN'T HELP**
>
> Here is the number that does not flatter this method. The panels do not all turn by the same angle where they meet. Across a B edge alone the fold runs from 11.4 to 14.5 degrees, and over the whole dome the exact stick bevel runs from **5.7 to 7.2 degrees**. A shop rips every stick at one setting – 6.5 degrees, halfway – so the worst joint is **0.8 degrees** off.

That is too small to see and not too small to leak. Lay a bead of sealant or a strip of gasket between the frames as they go together; that is what takes up the difference.

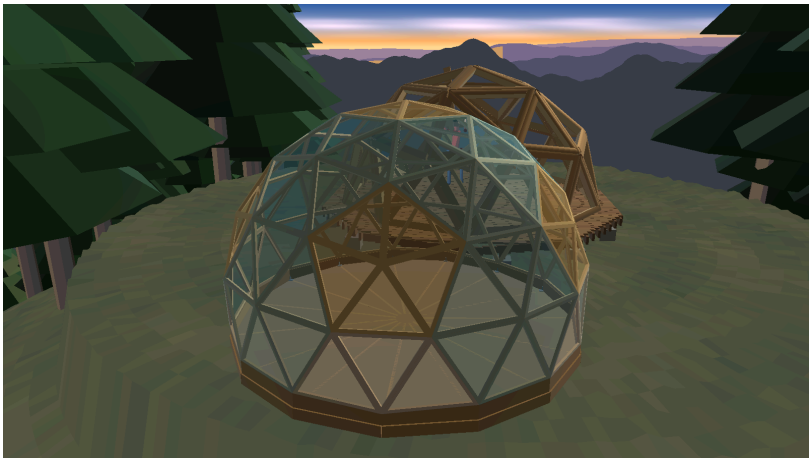


Red where one setting is furthest off.

Skin it in sections, raise it loose

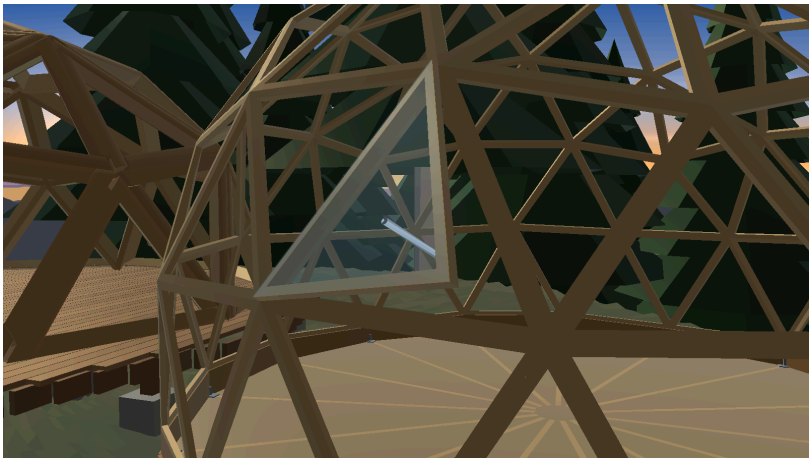
Sections. Before raising, screw panels into whole pentagons and hexagons on the shop floor and cover each group with one sheet. At this cut 6 pentagons and 10 hexagons come out whole, and 15 panels are skinned one at a time. Pull the film tight and staple it on the bevelled edges, where the next section hides the staples. A woven poly film stretches less than a thin greenhouse film and forgives a cold day; a thin film tears where a woven one does not.

This is a different trade from Chapter 28, which argues for one continuous watertight layer over the whole frame. Sections are repairable – a torn one is one section to replace – but every seam between sections is a joint to tape and to keep.



One sheet per pentagon or hexagon.

Windows. Build one extra frame for each window. Leave the skin off that panel in the dome, cover the spare frame, and hinge it along its top edge. A small gas strut of the kind kitchen cabinets use holds it open. Weather-strip the face it closes on, so rain running down the dome cannot creep under it, and tape the hinge seam once the dome is up.



A spare frame on two hinges.

Raising. Assemble from the base ring up, with a helper and plenty of clamps. Put only a few screws in each joint until the last panel closes the dome: until then the shell flexes, and that flex is what lets the last panels in. Lay the sealant or gasket between frames as they meet. Do not raise a loose shell in wind.

Taping. Tape the open seams between sections from the bottom up, each strip lapping over the one below, so water always runs onto tape and never under it. The surfaces must be clean and dry or the tape is decoration.

Anchoring. A dome is light for its size, and wind lifts. On a deck, angle brackets tie every base section to the frame below. On gravel or earth, steel stakes driven at a slant and strapped to the base sections do the same job. Anchor before the skin is finished, not after the first storm.

A doorway that sticks out

A five-eighths dome's own wall gives room for a door, but a door set into triangles is a door with a slanted head. The better answer is a short projecting entry – an arch doorway – built from the same kind of panels: a row of roof panels and a row of wall panels standing out from the dome, with two posts and a header inside them carrying an ordinary rectangular door. The same piece built twice, back to back, is a covered passage joining two domes into one building, which is the network of Chapter 45 at the scale of a single site.

Its panels are not triangles of the sphere, so they are built on **templates** rather than a trammel jig: each panel's outline is cut full size from a sheet, the sticks are fitted inside it, and the template is then turned over to build the panel's mirror twin for the other side. One stick on each roof panel – the one that lands on the dome – is the only thing allowed past the template's line, because its bevel is what seats the entry against the existing frame. That bevel is sharp; it is easier to rip with the board standing on edge, the saw set to the complement of the angle. The wall panels are drawn the same way, then shortened by the depth of the roof's edge stick so they fit snugly under it. Put the roof panels up first, then the walls, then the posts and header.

This book does not solve the entry's geometry – its angles depend on where it meets the dome and how far it projects – so it gives no figures for it.

Against the wedge dome

	2V wedge dome	3V panel dome
across the base, ft	19.42	16.00
height, ft	9.71	9.65
floor, sq ft	277	195
triangles	40	105
sticks	120	315
distinct sticks	2	6
base sections that taper	0	10

The panel dome is the better-trodden road: many people have built one from bought plans, and the method forgives a beginner. It asks for a table saw, a miter saw, bought boards, and a lot of identical cuts. The wedge dome asks for trees and a chainsaw, and gives back fewer, bigger pieces that carry their angles in their own shape. Neither is wrong. Choose by what you have.

PART 11

The Catalogue

“Twelve finished buildings, drawn by the tool that builds them, with the price of every choice in each one.”

The whole substitution space -- frame, section, panel, cladding, foundation, finish and fit-out -- the two arguments that decide most of it, the twelve designs one at a time, and what every choice in the tool costs.

CHAPTER 57

Every Other Way of Doing It

Eight frame materials, eight sections, sixteen panels, nine claddings and twelve foundations -- priced on this dome, not in the abstract

This book recommends one way of doing each thing. You are owed the others.

Not a paragraph saying alternatives exist — the alternatives, named, with what each one costs and weighs **on this building**, and a mark against the one the reference build uses.

How to read these tables

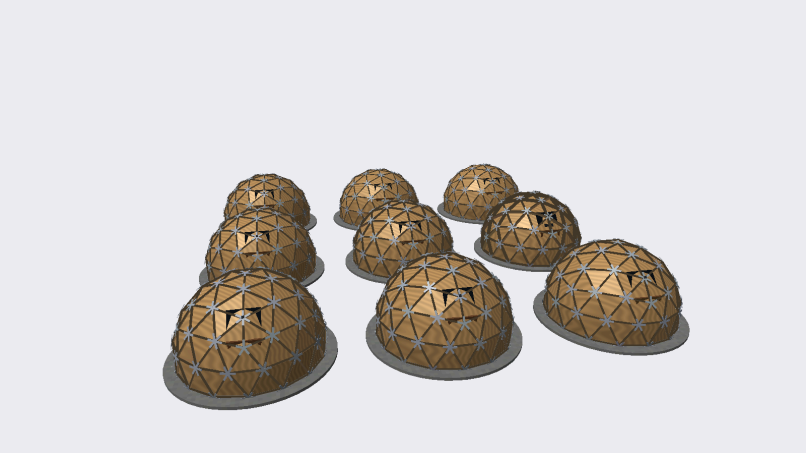
A list of materials with prices per square metre is not a decision. A decision is *"on a 19.42-foot dome with 550 square feet of shell, cedar shakes are this much and this heavy, and asphalt is that"* — and that is arithmetic this project can do, because it costs domes for a living.

Every option comes from a tool that already models it. Nothing is invented for the book. The full tables are in the reference section at the back; these pages are the shape of the choice.

The frame

The same frame, priced and weighed eight ways.

Eight materials, the same 120 members. The reference build is the whole tree trunk, and it is the cheapest by a long way — which is the entire economic argument of this method in one row.



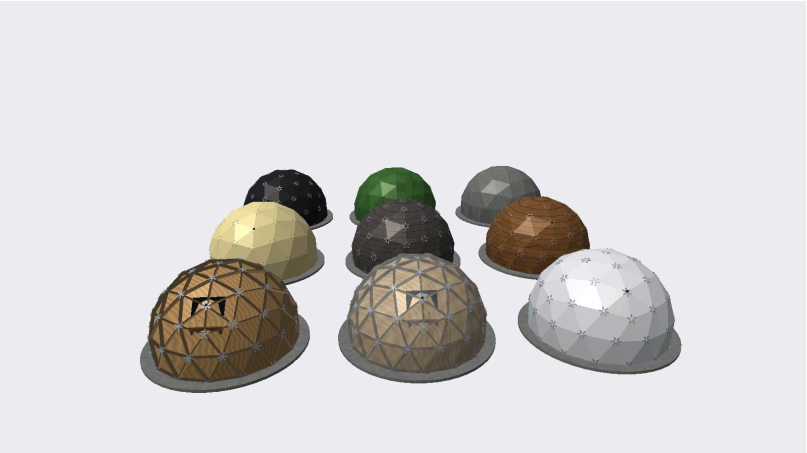
Round tube to quarter wedge, all at the same width.

Eight cross sections at the same nominal width. Read the wedge row carefully: the Creator's quarter wedge and this book's member are both split log, sized differently — one is a quarter of a log scaled to the strut's width, the other an eighth of a 12.0-inch one. Same family, different stick.

The envelope

Everything the tool can put between the struts.

Sixteen panel types, from canvas to solar, over 550 square feet. The span from cheapest to dearest is more than a hundred to one, which is why this is the decision that actually sets what your dome costs.



What goes over the sheathing.

Nine claddings. A green roof is twenty-four times the weight of house wrap, and the frame has to carry whichever one you pick.

What is deliberately missing

There is no straw-bale row and no earthbag row.

Not because they are bad — because this project cannot cost one, and a table with a plausible number in it that came from nowhere is worse than a table with a gap.

The same applies to anything else you find missing. If it is not here, it is because nothing in this repository models it, and the honest thing is to say so rather than to estimate.

The four you cannot change later

Everything above can be decided after the dome is standing. Four things cannot, and they are worth a page of their own in the back:

- which way the wedge's point faces
- how many sectors the log splits into
- how big the tree was
- which way the triangle pinwheels

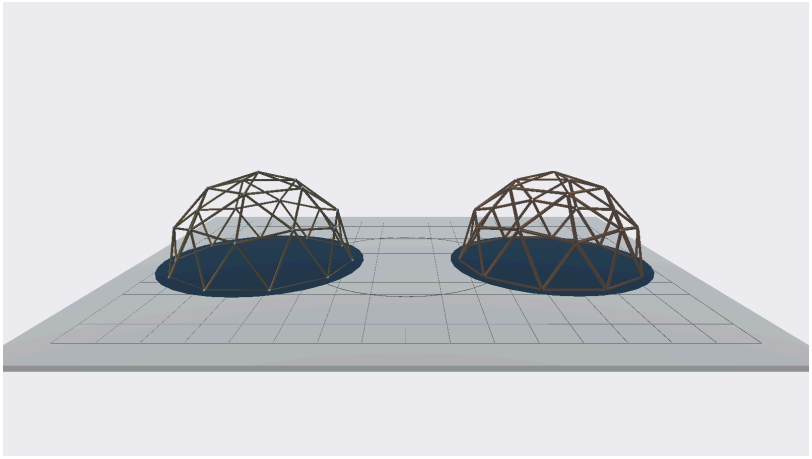
Each of those changes the frame itself, and each has a figure in this book showing what it does to the seam. Chapter 15 is where they are argued; the tables at the back are where they are priced.

CHAPTER 58

Hubs, or No Hubs

The trade this whole method is an answer to, counted across twelve designs rather than argued

This is the decision every dome builder makes, usually without noticing they have made it, and this entire method is one answer to it.



Hubs, or no hubs.

The two ways

Hubbed. Every edge of the geodesic mesh is one strut. Where five or six edges meet, their ends all arrive at the same point, and a connector holds them there.

- Fewest members: 65 struts rather than 120.
- Least timber, by a long way.
- But the hub is now the whole engineering problem. It has to hold five or six sticks at five or six different angles, in three dimensions, and it has to do it 26 times.

Hubless. Every panel is a closed triangle of its own three members, and panels meet panels rather than sticks meeting sticks.

- Nearly twice the timber — 120 members for 65 edges.
- But there is no connector to design, buy, machine or get wrong.
- And a panel is a *thing*: it can be built flat, checked flat, sheathed and lifted as a finished unit.

The Dome Creator ships one of each, both two-frequency, so the trade can be read straight off two real models:

```
both are 2V domes from this world:
Whole Trunk Lodge - 20 ft  --  Hub & Strut
  struts 65,  hubs 26,  bolts 0
Split-Log Homestead  --  Hubless Doubled
  struts 120,  hubs 0,  bolts 130
the hubless frame carries 1.8x the sticks
and exactly 0 hub connectors
every triangle brings its own three boards, so each shared
seam ends up two boards thick and bolts to its neighbour
and because its struts are split logs, the model reports 30 trees to harvest at 4 wedges a log
```

More sticks, zero hubs, and nothing on the critical path you cannot cut yourself.

Why the hub is worse than it looks

The strut count makes hubbed framing look obviously cheaper, and on a materials list it is.

The hub's cost is not on the materials list. It is:

- **Design.** Angles that are specific to the frequency, the vertex type and the strut section.
- **Fabrication.** Either bought, which is expensive per unit, or made, which needs a machine most people do not have.
- **Tolerance.** The one place in a triangulated frame where error *does* accumulate — five sticks whose ends all want the same cubic inch, and if they are each a little long the hub cannot close.
- **Time at height.** A hubbed dome goes up stick by stick, up a ladder, with somebody holding things.

Chapter 1 argued that a dome forgives error. That is true of the shell and untrue of the hub, and the hub is the place a hubbed dome's precision requirement lives.

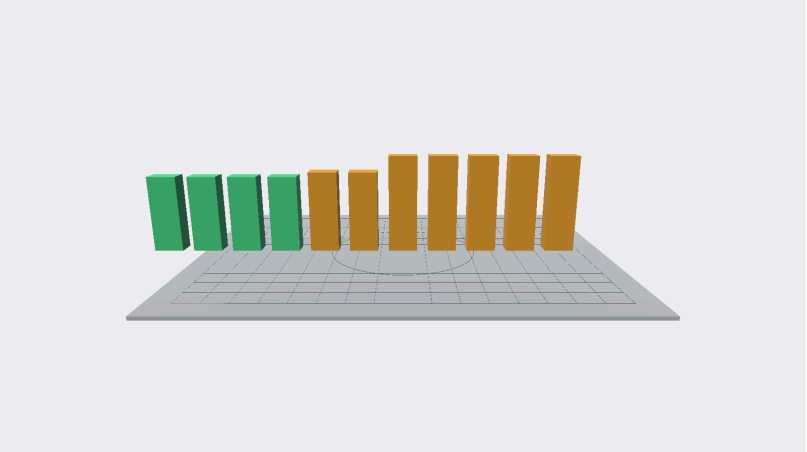
What the duplicated timber buys

Everything in Chapter 24.

The extra members are what make a panel a closed object. A closed object can be made on a bench at waist height, in a shed, in February, by one person, and checked before it goes anywhere. Then the site work is lifting 40 finished things into place instead of assembling 65 sticks in the air.

Timber is the cheapest thing in this method — Chapter 59 prices it against seven alternatives and it wins by a factor of several. Buying more of the cheapest input to remove the most expensive process is a good trade, and it is more or less the only trade this book makes.

The cheapest wall is the one you never build



Envelope per floor, measured.

One more argument from the same family, and it is about the shape rather than the joint.

A dome encloses a given floor area with less envelope than a box does. Less envelope is less material, less cladding, less insulation, less surface losing heat, and less area for water to find a way through.

That saving is permanent and it compounds with everything else: a smaller envelope means the skin of Chapter 2 — which is the part this book says matters most and costs most — is smaller too.

It is measured rather than asserted, across the whole catalogue – the skin you must build and seal, per square foot of floor:

```
skin you must build and seal, per square foot of floor:
Split-Log Homestead          2260 /  1217 = 1.86
Whole Trunk Lodge - 20 ft    6092 /  3281 = 1.86
Continuous Steel Arc Hanga   4018 /  2164 = 1.86
Treehouse Canopy Dome        1108 /   597 = 1.86
...
Rebar Garden Dome            2333 /   991 = 2.35
Concrete Monocoque Form      2776 /  1179 = 2.35
Woodland Hex Mirror          2333 /   991 = 2.35
best: Split-Log Homestead at 1.86
worst: Woodland Hex Mirror at 2.35
the spread is frequency and where the dome gets cut,
not the material bolted to it
```

Every one of those is a dome, and they still differ. The spread is frequency and where the shell gets cut, not the material bolted to it, and it is worth knowing before you choose a frequency, because that number follows the building for its whole life.

When to take the hub anyway

Be fair to the other side.

If you are working in steel or aluminium, the hub is a bracket and a bolt and none of the above applies — the machine that makes it is a drill press. If you are buying a kit, somebody else has solved the hub and you are paying for it once. If the dome is very large, the timber duplication stops being cheap.

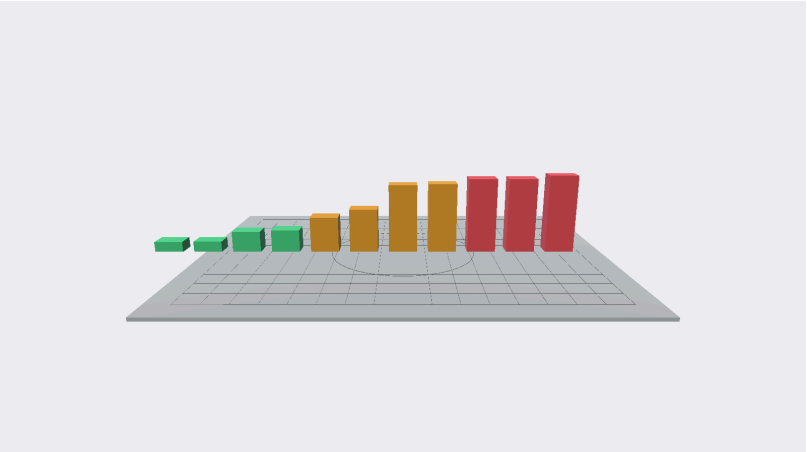
The hubless argument is strongest exactly where this book is: **timber, split rather than sawn, one person, no shop.** Outside that, it is a real trade with two defensible sides.

CHAPTER 59

Twelve Buildings, One Tool

Every finish, floor division and fit-out the Dome Creator will put on a shell, priced against each other

The previous chapter priced the frame and the envelope. This one is everything else the Dome Creator will put on a shell — and the point of it is that all twelve of these are the same program, so the comparison between them is real rather than rhetorical.



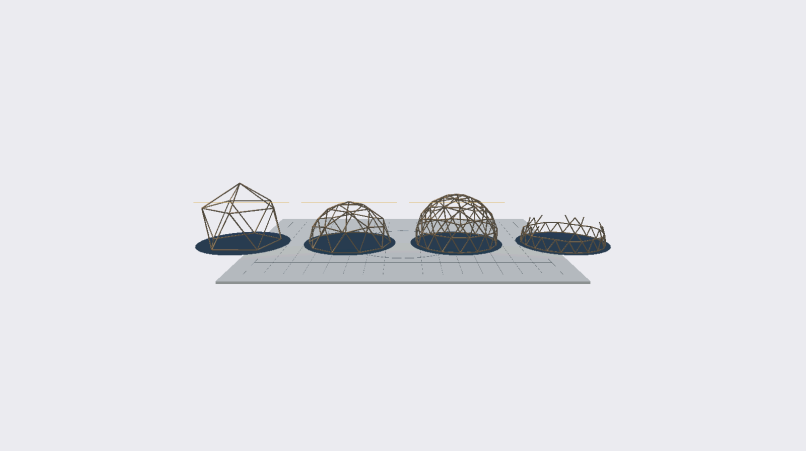
Every design in the catalogue, priced.

What the catalogue contains

Twelve finished buildings, from a timber workshop to a concrete monocoque, a glass studio loft, a rebar garden dome, a treehouse canopy and a steel arc hangar.

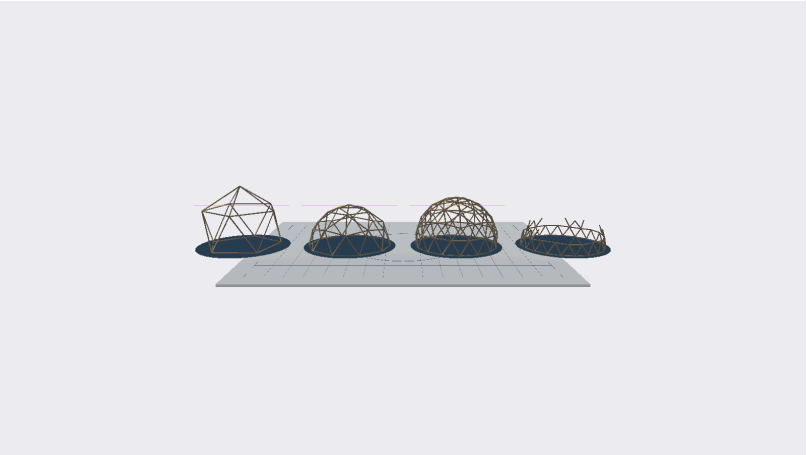
They are not illustrations of an idea. Each one is a configuration of the same four dials — frequency, frame material, strut section, panel type — plus cladding, foundation and finish, and each one comes with its own bill of materials.

Set the dials differently and you get a thirteenth.



Frequency, and where the dome gets cut.

The catalogue is a set of saved positions, not a product range.



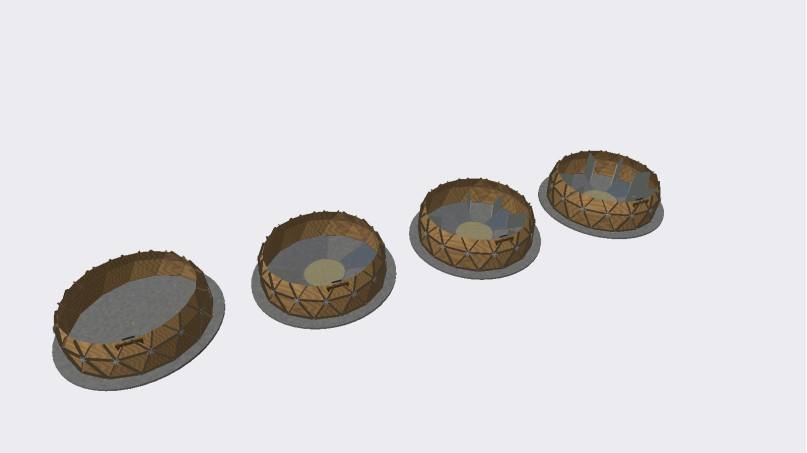
What frequency actually costs.

The decisions nobody prices

Three of them, and they change how a dome *lives* rather than how it stands.

Sixteen finishes.

Finish. Sixteen of them over the same shell. Cheap to change, and the only decision in this entire book that is purely aesthetic — which is worth saying, because almost every other choice here has a number attached and this one honestly does not.

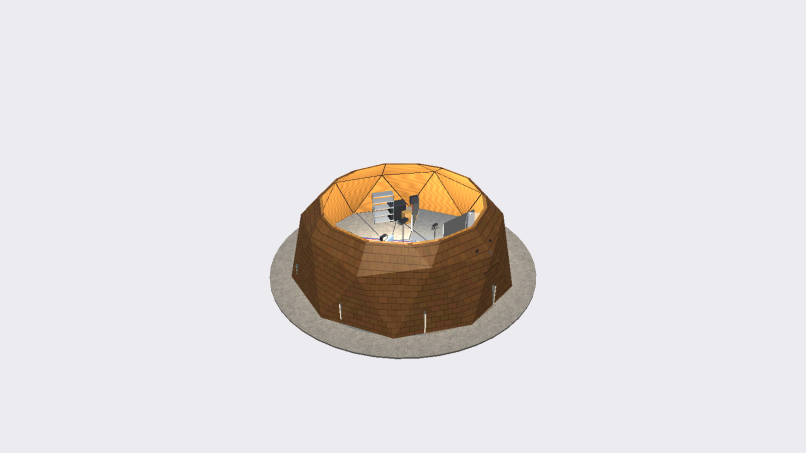


Four ways to divide a round floor.

Floor division. This is the question everybody asks second, right after “does it leak”, and it is the one round buildings are genuinely worse at.

A round floor does not take rectangular rooms. The four divisions the tool offers — radial, a central core, a chord wall, and open — are four answers to that, and the honest summary is that a dome wants **one big room and a periphery**, and fighting that produces awkward rooms with curved outside walls.

The periphery is also where headroom runs out. Chapter 11’s 277 square feet of floor is not 277 square feet of usable room, and a knee wall is the cheapest fix there is.



The part nobody films.

Fit-out. Benches, storage, services, the things that make it a workshop rather than a shell. It is most of the cost of a finished building and none of the glamour, which is why it is in the tool and not in the photographs.

How to use this chapter

Do not read it. Open the tool.

The tables at the back have every figure in them, but a table of sixteen panel types is a worse instrument than sixteen domes side by side with the prices on them. Chapter 81 says how to start it, and the one thing worth doing first is building the dome you were going to build before you read this book, pricing it, and then building this one.

That comparison is yours rather than mine, and it is the only one that will change your mind.

CHAPTER 60

The Twelve, One at a Time

Every design the Dome Creator ships, rebuilt from its preset and read off its own bill of materials -- from a seven-thousand-dollar garden dome to a hundred-thousand-dollar hangar

The Dome Creator is one of the tools that came with this book: a walkable site where you click the ground and an avatar walks there, click a panel and swap it – glass, shingle, solar, plastic sheeting – and change the frequency, the radius, the frame material or the foundation, while a bill of materials updates as you go: cut lists, panel weights, costs, even the number of trees to harvest. It reads the same geometry module as every other tool in the project, so a strut length in one cannot disagree with the same strut in another.

It ships with 12 finished designs. This chapter is all of them, each one rebuilt from the preset the tool loads and every figure read off that model. Nothing here is a drawing of a dome. It is the dome.

The catalogue, counted

Start with the whole catalogue in one table, because the spread is the point:

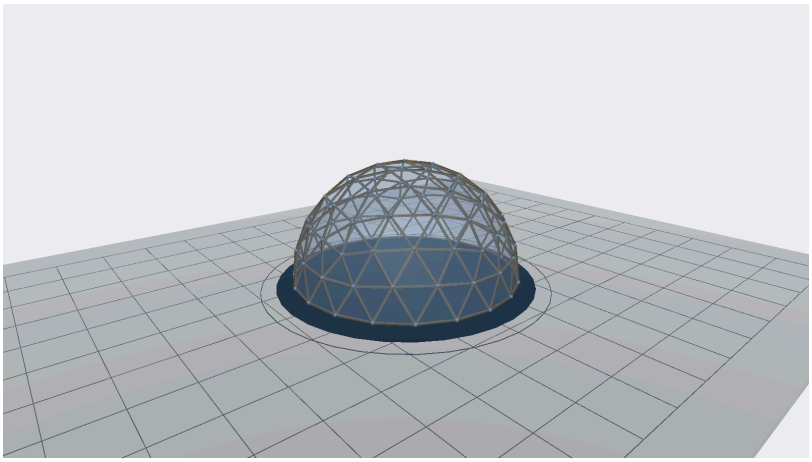
12 designs, each rebuilt and measured whole:		
Timber Workshop 3V 165 struts 105 panels 12 items	819 sqft	$ 19,514
Glass Studio Loft 4V 250 struts 160 panels 11 items	845 sqft	$ 24,670
Split-Log Homestead 2V 120 struts 40 panels 12 items	1217 sqft	$ 18,067
Whole Trunk Lodge - 20 ft 2V 65 struts 40 panels 9 items	3281 sqft	$ 22,747
Grow Dome 3V 165 struts 105 panels 11 items	819 sqft	$ 11,489
Hex Cell Pavilion 3V 165 struts 105 panels 5 items	1179 sqft	$ 61,355
Continuous Steel Arc Hang 2V 65 struts 40 panels 6 items	2164 sqft	$100,454
Rebar Garden Dome 3V 165 struts 105 panels 5 items	991 sqft	$ 7,002
Concrete Monocoque Form 3V 165 struts 105 panels 6 items	1179 sqft	$ 58,661
Woodland Hex Mirror 3V 165 struts 105 panels 4 items	991 sqft	$ 67,346
Woodland Square Mirror 4V 250 struts 160 panels 4 items	1023 sqft	$ 50,968
Treehouse Canopy Dome 2V 65 struts 40 panels 5 items	597 sqft	$ 27,243

The same engine makes a 597-square-foot canopy dome and a 3,281-square-foot lodge, and the only things that changed between them are a frequency, a radius and a material list. The equipment count is the part most catalogues cannot show at all: every bench and tank in these buildings has a weight, a price and a power draw in the model, and the wiring was routed to reach it.

The timber four

Timber Workshop – 3V, 819 sq ft, $19,514. The one the tool opens on, and the one every comparison in the next chapter starts from. Dimensional lumber on metal brackets, plywood between the struts, a concrete slab under it, and a working shop on the floor: an assembly bay, wood and metal benches, storage and an office corner. If you want to know what a dome costs as a *building* rather than as an idea, this is the honest starting point.

Glass Studio Loft – 4V, 845 sq ft, $24,670. Raise the frequency to four and the shell gets smoother, rounder and far more complicated: 250 struts instead of 165, in more length classes. Every panel is glass on a wood deck. It shows what smoothness costs – not in dollars first, but in part variety, which is the thing that actually slows a build down.



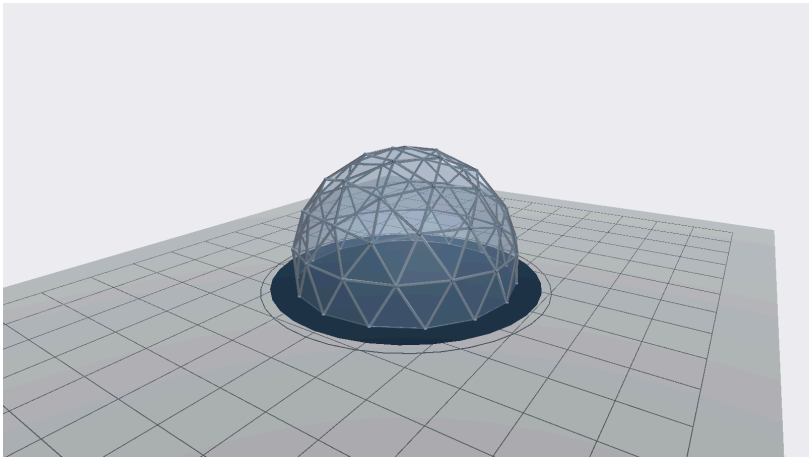
The glass studio loft.

Split-Log Homestead – 2V, 1,217 sq ft, $18,067. The one this book argues for: hubless doubled framing from quarter wedges split straight out of logs, cedar shakes over plywood. Not one hub connector in the building. Every triangle brings its own three members and bolts to its neighbours, and the model reports the trees to harvest, four wedges to a log.

Whole Trunk Lodge - 20 ft – 2V, 3,281 sq ft, $22,747. The tree used whole. The radius is set so the longest member lands just under twenty feet, which is what two people can actually move and lift. 65 round trunks with canvas between them, gravel underneath: the largest floor in the catalogue by a wide margin, framed from the fewest pieces, and at $7 a square foot the cheapest floor too.

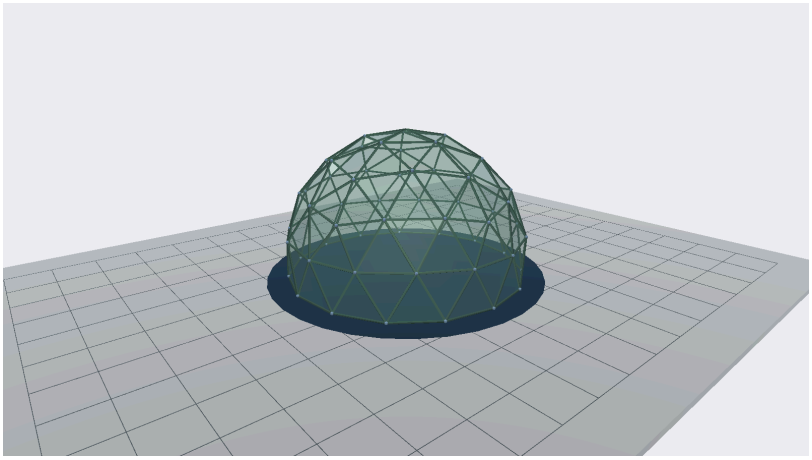
The other eight

Grow Dome – 3V, 819 sq ft, $11,489. The workshop's exact skeleton, in aluminium and polycarbonate, with grow racks and water storage inside. The geometry does not change at all; only the material list does – which is exactly the argument of this whole Part. One frame, many buildings.



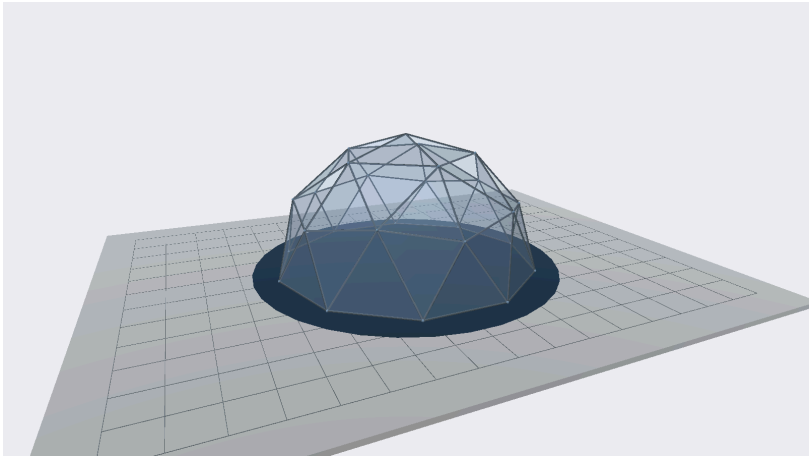
The grow dome.

Hex Cell Pavilion – 3V, 1,179 sq ft, $61,355. The geometry grouped into hexagonal cells on a structural steel frame with composite tiles, fitted out as a presentation space. Heavier and dearer than the timber designs, and it looks like nothing else on the site.



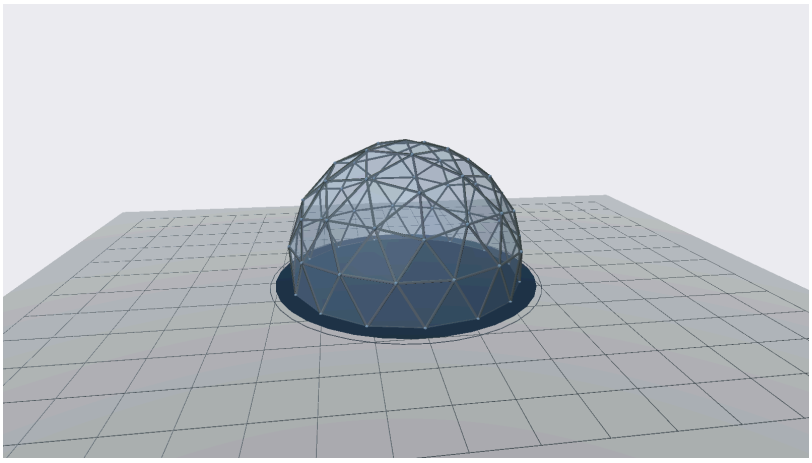
The hex cell pavilion.

Continuous Steel Arc Hangar – 2V, 2,164 sq ft, $100,454. Short straight struts swapped for continuous curved steel ribs and rings, with fabrication and site-power equipment inside, open between the ribs. This is what the same geometry looks like when a steel shop builds it instead of a carpenter – and the dearest thing in the catalogue.



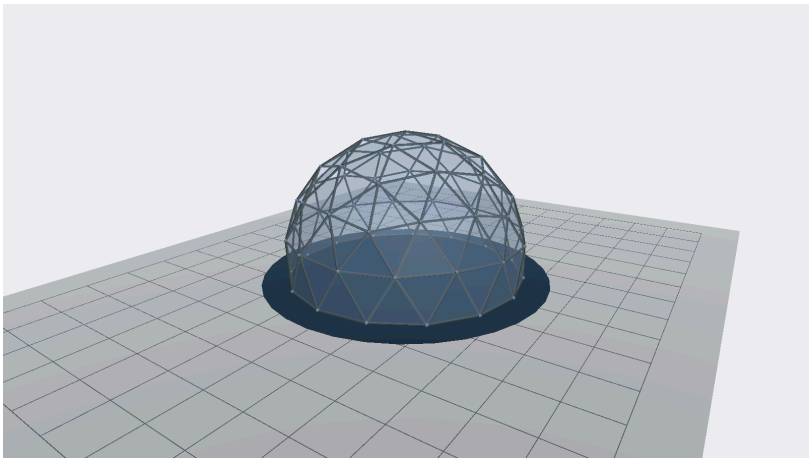
The continuous steel arc hangar.

Rebar Garden Dome – 3V, 991 sq ft, $7,002. Rebar is the cheapest structural steel there is, and it bends, so the tool draws it as a dense lattice of meridians and rings rather than sticks between hubs. The cheapest building in the catalogue, and the one closest to what people actually put up in a back yard on a weekend.



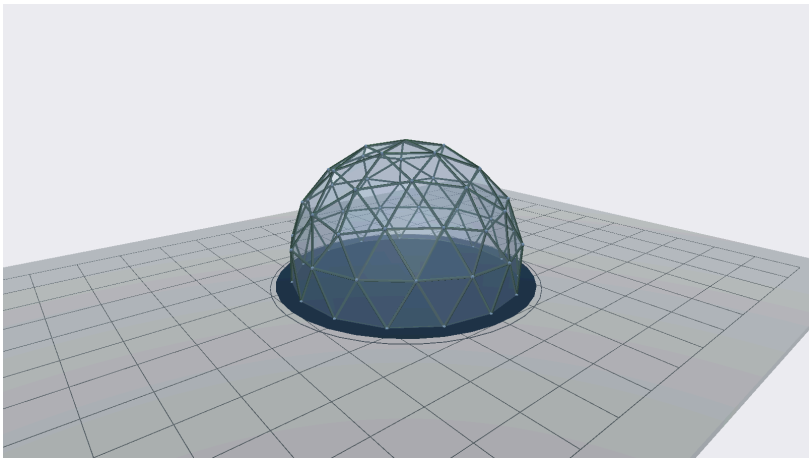
The rebar garden dome.

Concrete Monocoque Form – 3V, 1,179 sq ft, $58,661. Here the frame is not the building. It is the formwork: rebar and shuttering carry a poured concrete shell, with shoring, scaffold, a mixer and a rebar bender on the floor. The dome is doing what concrete does best – standing in pure compression – and at 37.9 tonnes it is by some distance the heaviest thing here.



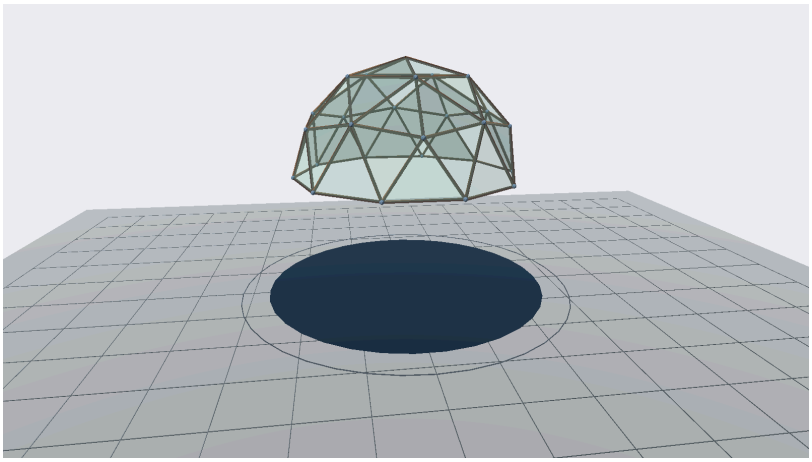
The concrete monocoque form.

Woodland Hex Mirror and **Woodland Square Mirror** – $67,346 and $50,968. They exist because the renderer can do it: mirror tiles that genuinely reflect the sky, the ground and the trunks around them, a pavilion that disappears into a wood. The hexagonal one is the same 3V shell as the workshop; the square one is 4V on a steel frame, with more and smaller panels and a finer reflection. Side by side they show exactly what frequency does to a faceted surface: it stops looking like a machine and starts looking like a curve.



The woodland hex mirror.

Treehouse Canopy Dome – 2V, 597 sq ft, $27,243. The smallest: a two-frequency dome on an elevated timber platform with braces and ladder access, up in the canopy. Proof that the same 65-stick frame works when the ground it stands on is not the ground.



The treehouse canopy dome.

Check it yourself

That is the whole catalogue: 12 designs, several frequencies, six framing systems and one geometry engine under all of them, from $7,002 to $100,454 – 14 times apart, on the same shape.

Do not take this chapter's word for any of it. Open the Dome Creator, press the preset button until you reach the one you want, and read its bill of materials. It will say what this chapter says, because this chapter read it from there.

CHAPTER 61

What Every Choice Costs

Which menu is the expensive one, where the hours go, what the fit-out adds, how many domes the menus make -- and what no model prices

Chapter 60 walked the catalogue one building at a time. This chapter turns it sideways: hold one design still, move one menu at a time, and find out which choices actually cost money, where the hours go, what the furniture adds, and how many different domes the menus can make between them.

What is declared, and what is measured

Before any dollar figure, where the dollars come from:

```
nothing below is measured -- these are declared inputs:
frame materials: 8 entries, density kg/m3 and price per kg
   timber (SPF)  480 kg/m3  $2.00/kg
panels: 16 types, $1.50 to $210.00 per m2
   cheapest Plastic Sheeting, dearest Hex Mirror
cladding layers: 9, foundations: 7 ($85/m2 for a slab)
partition walls $45/m2, wire $1.40/m, PEX $2.10/m
labour hours come from the mesh builder's construction record,
which is an estimate per work step, not a stopwatch
```

The densities and unit prices live in one file in the project. They are declared inputs, and they are the only numbers in this chapter typed by a person. Every quantity they are multiplied by – areas, lengths, counts, weights – is measured off the model that drew the building. Change a price in that file and every figure below moves with it; nobody edits a sentence.

Which menu is the expensive one

Start from the Timber Workshop at $19,514 and price every setting of one menu at a time:

```
start from Timber Workshop at $19,514,
change one menu at a time, price every setting:
Frame material   $ 18,563 (Whole Tree Tru) -> $  93,185 (Structural Ste)
Panel type       $ 16,290 (Open) -> $  53,899 (Hex Mirror)
Foundation       $ 10,685 (Bare Ground) -> $  30,420 (Treehouse Plat)
Cladding layer   $ 19,514 (None) -> $  36,527 (Poured Concret)
Strut shape      $ 17,013 (Round Tube) -> $  19,514 (Dimensional Lu)
Frequency        $ 17,312 (1V) -> $  19,755 (4V)
and 3 menus cost nothing at all: frame colour, panel colour, wedge curve
```

The covering and the frame material swing the price by tens of thousands. The geometry barely moves it. Some menus cost nothing at all. That is the catalogue's own verdict on itself, and it is good news: **the skin decides the price**, and the skin is the decision you can defer, stage or upgrade later. The frames across the catalogue are close cousins; the cladding – plywood, glass, canvas, polycarbonate, composite tile, poured concrete – is not.

Which menu is the expensive one.

Six ways to frame one shell

The same workshop shell, framed every way the tool knows:

```
one timber workshop shell, framed six ways:
Hub & Strut            165 struts   61 hubs fitted    0 bolts  $ 15,519
Hubless Doubled        315 struts    0 hubs fitted  330 bolts  $ 17,837
Hex Cell               165 struts   61 hubs fitted    0 bolts  $ 15,519
Continuous Steel Arcs  165 struts    0 hubs fitted    0 bolts  $ 15,519
Rebar Lattice          165 struts    0 hubs fitted    0 bolts  $ 15,519
Concrete Formwork      165 struts   61 hubs fitted    0 bolts  $ 15,519
and one disagreement inside the tool, said out loud:
Continuous Steel Arcs  bills $596 for 61 hub connectors its own assembly never fits
Rebar Lattice          bills $596 for 61 hub connectors its own assembly never fits
```

WE GOT THIS WRONG

The tool argues with itself here, and it is said out loud rather than averaged away. Two of the six framing styles report hub connectors in their bill of materials that their own assembly never fits. The picture and the price disagree. The picture is the one to believe, because the picture *is* the assembly – so read those two totals as high by their hub line.

Where the hours go

The tool records an estimated time for every step of every build, and grouped by phase they say something useful:

```
Timber Workshop: 348 work steps, in build order:
Site               2 steps    14.0 h
Frame            165 steps    19.8 h
Joins             61 steps    12.2 h
Openings           1 steps     2.0 h
Sheathing        105 steps    42.0 h
Fit-out           13 steps     5.1 h
Commissioning      1 steps     2.0 h
total 97.1 hours of work
two people, eight hour days: 6.1 days
```

The frame – the part everybody pictures when they imagine building a dome – is not the expensive phase in hours. The sheathing is. That is a model and not a stopwatch, and it is a different building from this book's forty-hour frame: an estimate attached to each work step. But it is the same estimate applied to every design, so comparing two designs is fair even where the absolute number is soft.

The dome is the cheap part of the dome

Every design ships furnished, and some with framed walls and plumbing:

```
12 of 12 designs ship furnished, 4 with framed walls
Split-Log Homestead, as the tool ships it:
   12 pieces of equipment, $2,595, 388 kg, 1,050 W connected
   7 framed partition walls, 22 m2, $1,000
   4 plumbed fixtures, 84 m of PEX and 42 m of drain, $549
   0 circuits -- and that is not an oversight:
   conduit runs start at a battery bank or a charge controller,
   and no design in this catalogue ships with either
add a battery bank, a controller and two outlets:
   3 conduit runs, 22 m of wire, $30
fit-out and services: $4,144 of $18,067, or 23% of the building
```

THE NUMBER THAT DOESN'T HELP

The shape saves you on envelope. It does not save you on anything that plugs in. On a design whose shell is split logs and plywood, the equipment, the walls and the plumbing are a serious share of the building. And the electrical line reads zero not because domes need no wiring, but because no preset ships with a battery or a charge controller for the conduit to start from. Add them, and the model routes the runs without being asked twice.

Size is a number you type

Frequency is the one menu that changes how many pieces there are. Same radius, same material, same foundation, and only the number of times each face is divided:

```
same 5 m radius, four frequencies, all built and measured:
1V:  25 struts in 1 length class ,  15 panels, 11 hubs
2V:  65 struts in 2 length classes,  40 panels, 26 hubs
3V: 165 struts in 5 length classes, 105 panels, 61 hubs
4V: 250 struts in 6 length classes, 160 panels, 91 hubs
now the part nobody mentions -- where the dome gets cut:
even frequencies (2V, 4V) have a ring of hubs exactly on the equator,
so height = radius exactly: 5.00 m, 5.00 m
odd frequencies (1V, 3V) have no such ring; the cut lands above it,
so they stand taller than a hemisphere: 7.24 m, 5.88 m
```

Frequency buys a smoother shell and charges in part variety, which is what actually slows a build down.

The tool's sliders look continuous. They are not: each one moves in fixed steps written into the menu code, so they can be counted like the drop-downs can, and **not one of them changes the part count**:

```
the sliders are not continuous -- the tool moves them in steps:
Radius                 2 to 15  step 0.5     27 settings   m
Strut width       0.02 to 0.35  step 0.005   67 settings   m
Recess depth      0.05 to 0.95  step 0.05    19 settings   fraction of strut depth
Foundation size       1 to 1.6  step 0.05    13 settings   x radius
```

Which is the whole economic argument for this shape, proved on the catalogue:

```
part counts depend on frequency alone, never on size:
Timber Workshop            r = 5.0 m -> 165 struts, 105 panels
Grow Dome                  r = 5.0 m -> 165 struts, 105 panels
Hex Cell Pavilion          r = 6.0 m -> 165 struts, 105 panels
Rebar Garden Dome          r = 5.5 m -> 165 struts, 105 panels
same three numbers, different buildings
smallest in the catalogue: Treehouse Canopy Dome, 28 ft across, 597 sq ft
largest: Whole Trunk Lodge - 20 ft, 65 ft across, 3281 sq ft
that is 5.5x the floor from the same list of operations
```

Part counts depend on frequency alone, never on size. The largest dome in the catalogue has several times the floor of the smallest, off an identical list of operations. Load a preset, drag the radius slider, and watch the counts stay exactly where they were.

How many domes that is

Multiply the menus:

```
the shell menus, multiplied:
Frequency              4
Frame style            6
Hub style              2
Strut shape            8
Frame material         8
Frame colour           8
Panel type             16
Panel colour           8
Cladding layer x3      9 ^ 3 = 729
Foundation             7
Wedge curve            2
product = 32,105,299,968 shells
but some settings do nothing in some combinations:
hub style changes nothing on hubless doubled, which has no hub to change
the wedge curve bends 1 of 8 strut shapes (quarter wedge)
count only the ones that differ: 16,554,295,296 shells
then the floor: 4,398,046,511,104 partition and room layouts
and the dials, at the steps the tool moves them in: 446,823 settings
```

Then the honest deduction, because some settings do nothing in some combinations – a hub style changes nothing on a frame with no hubs – and count only the shells that genuinely differ: **16,554,295,296**. At three seconds each, showing every one would take 1,574 years of film.

How many domes that is.

So here are eight the dice picked instead, drawn at random from the menus, built and priced, with the seed fixed so anyone can draw the same eight:

#	freq	frame style frame material	panels		bill
1	3V	Hubless Doubled Structural Steel	Hex Composite	$	48,735
2	1V	Continuous Steel Arcs Aluminum	Square Mirror	$	48,448
3	4V	Hub & Strut Rebar Steel	Square Mirror	$	55,810
4	2V	Hub & Strut Structural Steel	Plywood	$	63,654
5	3V	Continuous Steel Arcs Timber (SPF)	Plastic Sheeting	$	14,177
6	4V	Hex Cell Structural Steel	Square Mirror	$	56,226
7	4V	Hub & Strut Whole Tree Trunk	Precast Concrete	$	46,247
8	3V	Continuous Steel Arcs Timber (SPF)	Solar Panel	$	42,893

Some of them are sensible. At least one is a thing nobody would ever build, which is the honest shape of a space this large.

Eight the dice picked.

Ten builds, one ledger

Last, ten of the designs carried all the way through: the Dome Creator’s bill of materials, the hours in its build record, and the assembly-line economics that turn both into a planning price.

```
Creator BOM  +  Creator build events  +  Assembly Line economics:
Timber Workshop            BOM $  19,514     96 h  direct-sale $   31,991
Glass Studio Loft          BOM $  24,670    193 h  direct-sale $   53,584
Split-Log Homestead        BOM $  18,067     96 h  direct-sale $   35,130
Whole Trunk Lodge - 20 ft  BOM $  22,747    104 h  direct-sale $   42,663
Grow Dome                  BOM $  11,489     96 h  direct-sale $   21,466
Hex Cell Pavilion          BOM $  61,355     99 h  direct-sale $   84,898
Continuous Steel Arc Hang  BOM $ 100,454     49 h  direct-sale $  130,125
Rebar Garden Dome          BOM $   7,002     43 h  direct-sale $   11,926
Concrete Monocoque Form    BOM $  58,661    114 h  direct-sale $  109,164
Treehouse Canopy Dome      BOM $  27,243     60 h  direct-sale $   45,272
lowest modeled direct-sale price: Rebar Garden Dome at $11,926
least labor: Rebar Garden Dome at 43 hours
```

The lowest price and the least labour can be different winners, which is why a product decision cannot be made from one number.

Ten builds, one production ledger.

THE NUMBER THAT DOESN'T HELP

None of these numbers is a bid. A dome does not make the difficult parts of housing disappear. Curved walls complicate ordinary cabinets and openings. An unfamiliar system can slow permits, insurance and appraisal. Freight and site work can dominate a remote job. Every occupied structure still needs local structural engineering and code review. Those costs are not folded quietly into the totals above. They are outside them, and this box is here to say so.

PART 12

The Stem Cell

“Buy the plumbing once. The shell is the thing that grows, and the core is the thing that moves.”

The product line: one undifferentiated body, the utility core that is thirty per cent of it and outlives it, how to build one, the fifteen buildings it becomes, the ten reasons for the shape, and the standard article with its invoice and its asks.

CHAPTER 62

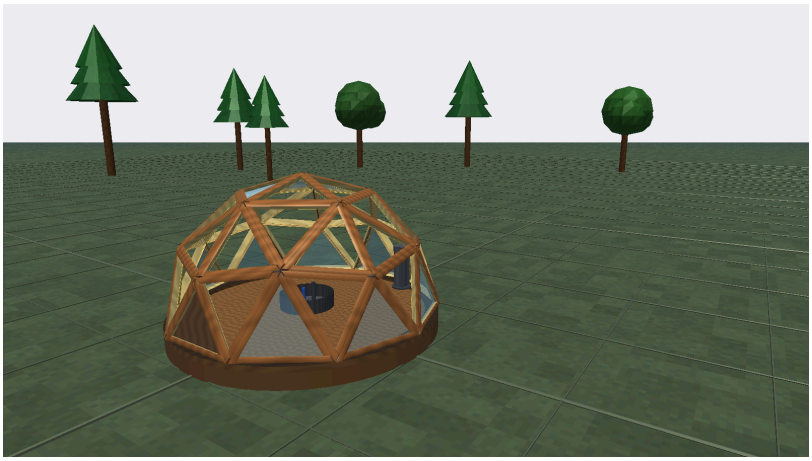
Why We Call It a Stem Cell

One body, undifferentiated, that becomes fifteen different buildings depending on what you put in it

A stem cell is a cell that has not decided what to be yet.

It has the full machinery — a nucleus, a membrane, everything a working cell needs — and no specialisation. Put it in one place and it becomes liver. Put it somewhere else and it becomes bone. The body is the same body; the environment does the differentiating.

That is the product line in this book, and the word is not decoration.



One body, many things it can become.

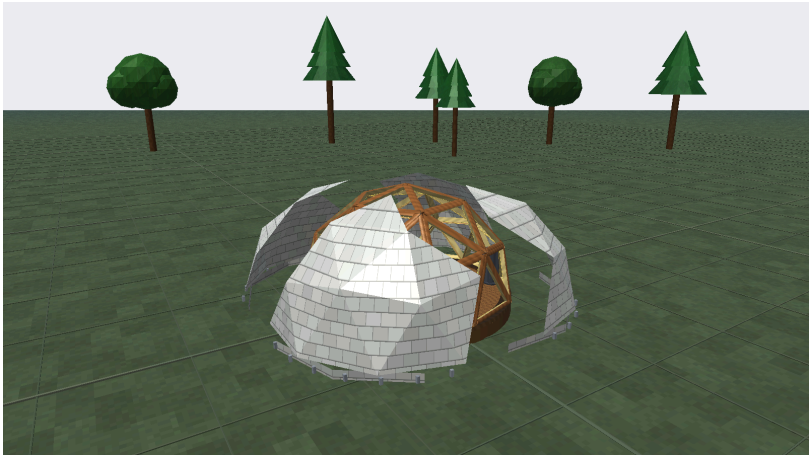
One body, undifferentiated

The standard article is a dome: 120 members, 40 panels, a shell, a floor, and a utility column running from the floor port to the apex. It lists at $12,036.

It is not a house. It is not a workshop. It is not a studio, a gym, a sauna or a guest room. It is the **body** of all of those, and what it becomes depends on what you put in it and what you plug into the column.

There are 15 of them in the catalogue — Chapter 65 — and every one is this same body. The frame line of the quote does not change between them. Not by a dollar.

The socket at the top



The roof comes apart too.

What makes differentiation possible is that there is exactly **one** place where things connect.

The utility column is the only object in the building that touches the ground, the wall and the sky. It carries 5 services in 14 documented parts, and it makes exactly one penetration in the weather surface — the apex sleeve, with a gasketed seal cap over it that is meant to stay shut for years and come off in ten minutes.

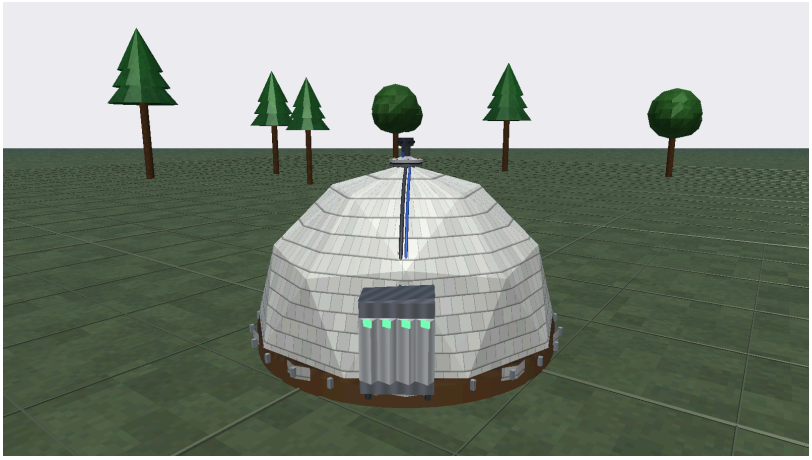
Everything a fit-out needs comes off that column. Power at a sub-panel inside it. Water at a four-port manifold. Drain down a stack inside the same chase. Air, eventually, through the seam channel of Chapter 20.

Nothing is plumbed into a wall. The wall is a shell that resolves loads and keeps weather out, and it has no services in it to disturb. That is what makes a fit-out a thing you do rather than a thing you commission.

And a skin that snaps on

The outside works the same way. Services that want to be outside the footprint — a water heater, a cooling unit, a shower — hang off the shell on threaded inserts rather than cutting a new hole. The design calls them polyps and the name is the same joke as the stem cell: an organism growing an appendage rather than a builder cutting an opening.

Every one of them is reversible. Take it off and there is a panel with four inserts in it.



Everything else snaps onto the outside.

Why this is a different kind of product

A normal building is specialised on the day it is designed. A workshop has a workshop's floor, a workshop's power, a workshop's door. Turning it into a guest room is a renovation.

This one is specialised on the day it is **fitted**, and un-specialised by taking the fit-out out again. The most expensive parts — the frame, the shell, the column — never learn what the building is for.

That is the argument, and the next chapter is the economic consequence of it, which is the part that actually decides whether any of this is worth doing.

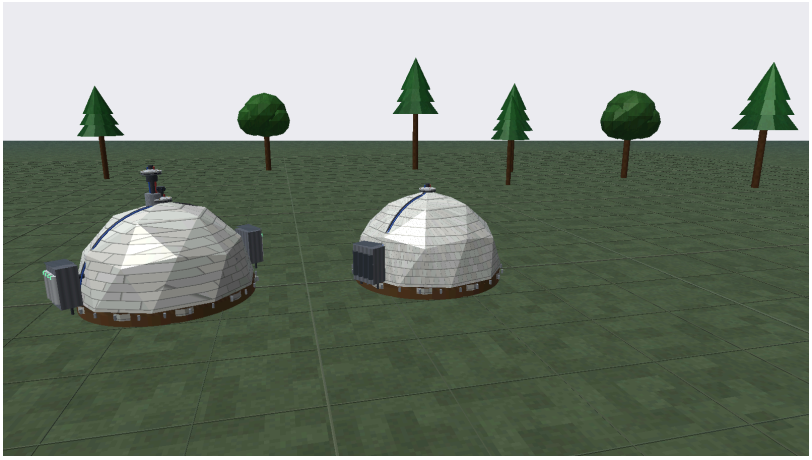
CHAPTER 63

Sink the Money Into the Part That Moves

The utility core is thirty per cent of the dome and the only part that transfers -- so it is the part worth overbuilding

Here is the theory, in one sentence.

Put the money in the hardware, because the hardware transfers and the shell does not.



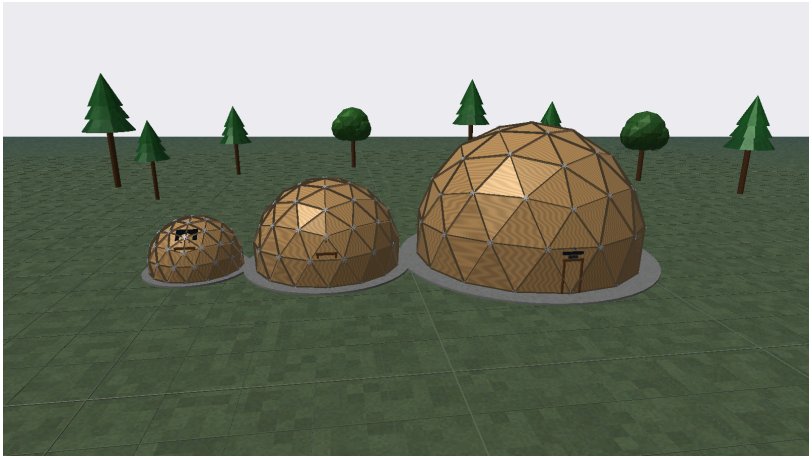
What buying it once is worth.

The two halves of a dome

Every building in this book is two things wearing one roof.

The shell — 120 members, 40 panels, the cap, the floor. It is shaped by *this* dome: its members are cut to this diameter, its panels are this geometry, its cap is this size. Move to a different dome and none of it comes with you, because none of it fits.

The core — the utility column and everything in it. A chase, a sub-panel, a manifold, a stack, a seal cap. 14 parts carrying 5 services. It is shaped by *what a small building needs*,



One hardware set, any size of house.

which is the same in every small building.

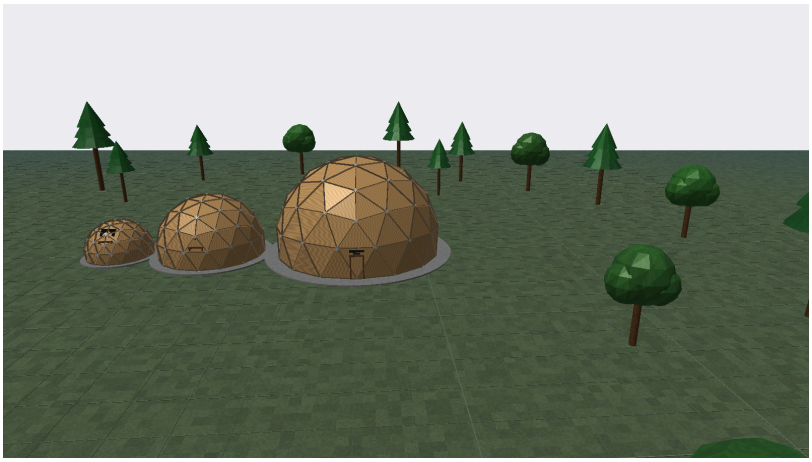
The shell is specific. The core is generic. And the core is **30 per cent of what the dome costs to build** — $1,890 of it.

What generic means for money

A shell is consumed by the building it is part of. When that building's life ends, so does the shell's.

A core is not. It was never shaped by this dome, so it fits the next one. Its declared service life is 25 years — not because a pipe wears out in 25 years but because at some point it is outgrown rather than worn out — and moving it to the next dome costs $260.

Two hundred and sixty dollars to move 30 per cent of a building.



What does not move when the house does.

That is the whole theory. The expensive, fiddly, skilled, permit-adjacent third of the project gets bought **once** and then amortised across every dome you ever put it in.

Why this changes what you should overbuild

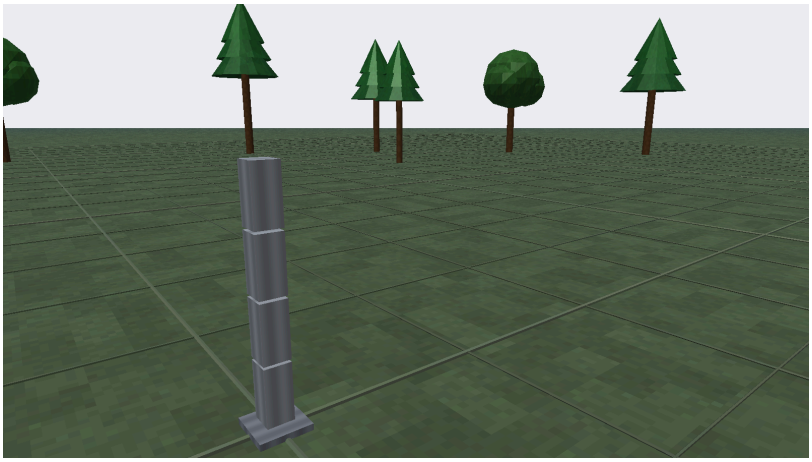
Ordinary building advice says spend where it shows and economise where it does not.

This says something different: **spend where it transfers.**

Every dollar in the shell is spent on this dome. Every dollar in the core is spent on this dome *and the next one and the one after*. So the core is the place where overbuilding is not overbuilding — it is prepayment.

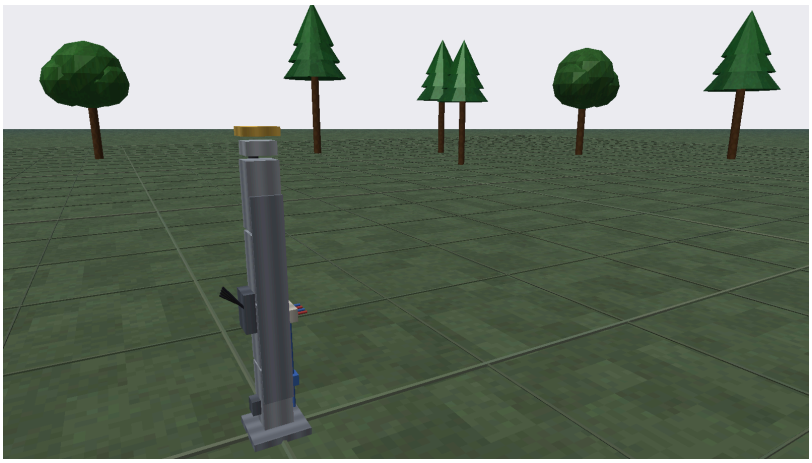
Concretely, this is why the design specifies:

- a sub-panel with a **spare breaker**, so the first snap-in module does not need a panel change;
- a four-port manifold with **capped stubs**, in a dome that ships with no plumbing at all, because the alternative is opening the chase later;
- a feeder that **unplugs** — a cord and a recessed inlet rather than hard wiring, which is what makes the dome moveable without an electrician;
- a floor-port tie-in that the **host** makes once and nobody disturbs when a dome is swapped.



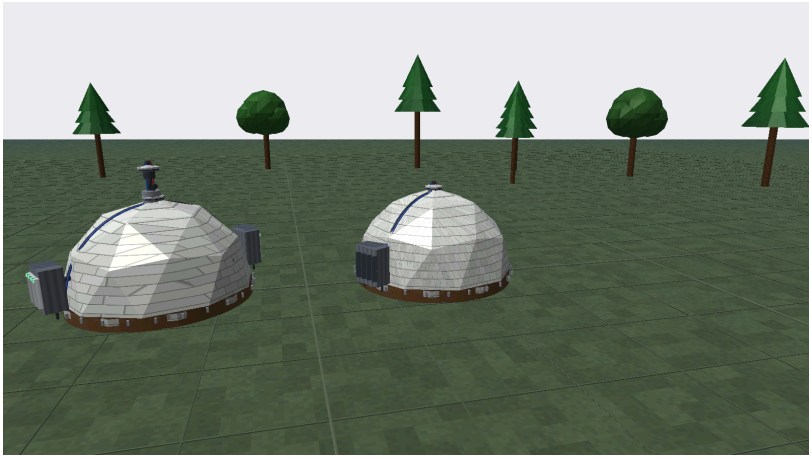
Build the chase on the bench.

Every one of those is a small extra cost now against a large avoided cost later, and each one is only worth it because the thing carrying it survives the building it was bought for.



Close it, prove the cap, stand it up.

The same argument, one level up



This only works as a system.

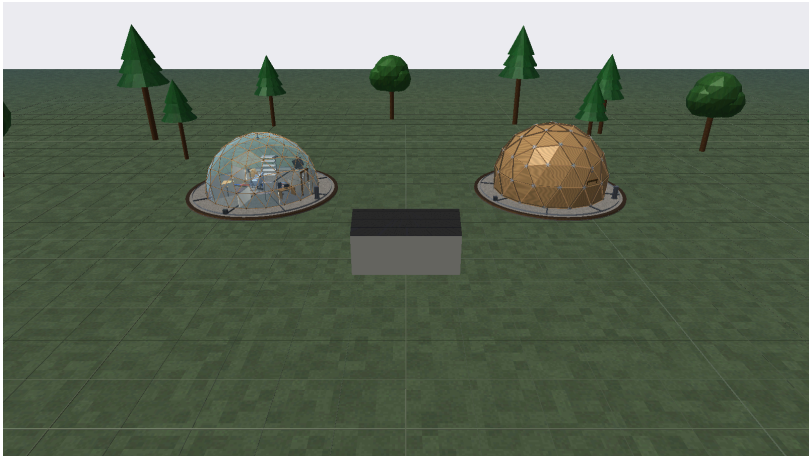
The core is to a dome what the pad of Chapter 39 is to a site and what the jig of Chapter 24 is to a panel.

In all three cases, something is built once, carefully, and then used many times without being rebuilt:

the jig	built once, makes 40 panels
the core	bought once, serves several domes
the pad	built once, hosts a succession of buildings

And in all three cases the thing being amortised is not material — it is **precision and setup**. The jig holds the angles so you do not have to. The core holds the services so an electrician does not have to come back. The pad holds the level and the connection so a foundation does not have to be poured again.

That is what “this only works as a system” means. Any one of them alone is merely a nice idea. Together they move almost all of the difficulty out of the building and into three objects that are built once.

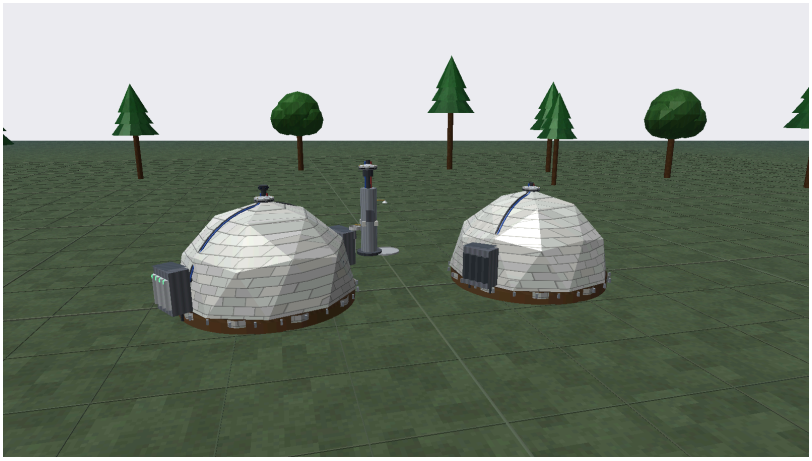


What a site can share, so the dome does not have to.

The honest objections

It raises the entry price. A dome with a proper core costs more than a dome with a light socket and a hose through the wall. The transfer argument only pays if there is a second dome, and for somebody who will only ever build one, this chapter is wrong and they should spend the money on the skin.

Transfer is not free. $260 is a declared assumption, not an invoice. It covers disconnection, handling and reconnection on a pad that already has the port. It does not cover a drive across a country.



The core moves to the next dome.

Generic is a claim, not a fact. The core is generic across the 15 structures in this catalogue, which were all designed around it. Whether it is generic across somebody else's building is unproven, and the honest answer is that it is generic across this system and that is all anybody has tested.

The sentence to keep

If you are building one dome: put your money in the skin and the floor, as Chapter 2 says, and keep the core simple.

If you are building more than one: **put it in the core.** It is the only third of the project you will not have to buy again.

CHAPTER 64

Build One Utility Core

Fourteen steps in five stages: drain first, water proved before anything electrical exists, power last into a core already known to be dry

This is not the argument for the core – Chapter 63 made that. This is the work.

A utility core is the one part of this dome that is identical in every dome, at every size and with every fit-out, because the interface it plugs into never changes. So it is the first thing worth learning to build, and the first thing worth building a jig for.

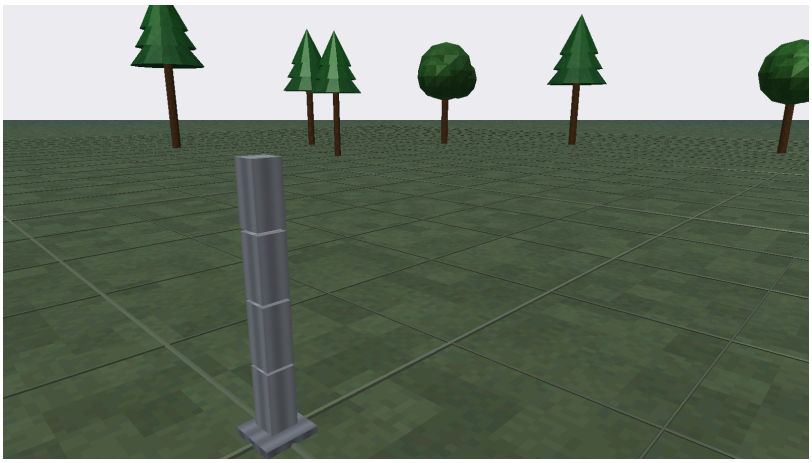
14 steps in 5 stages, about 7.2 hours once you have done a few. Your first will take closer to 17.2, and anybody who tells you otherwise has not built one.

The whole core is built upside down on a bench and stood up at the end. Working at waist height on a flange you can walk round is the difference between an afternoon's job and a day of kneeling.

What has to be on the bench

Cordless driver	everything is screwed, nothing is nailed
Step drill	clean holes through the housing for the grommets
PEX crimp tool and go/no-go gauge	the gauge is not optional -- an uncrimped ring passes a pressure test and fails in a year
Tubing cutter	a square cut on PEX and on the 2 in stack
Deburring tool	a burr in the stack is where the blockage starts
Torque screwdriver	terminals in the sub-panel are torque-specified and a loose terminal is how a panel burns
Multimeter	continuity and polarity before the feeder goes live
Pressure test pump and gauge	for the water test at the end of stage three
Spirit level	the chase is plumb or the seal cap does not seat

9 tools, and two of them are the ones people leave in the van: the **go/no-go gauge** for the crimps, because a bad crimp passes a pressure test and fails a year later, and the **torque screwdriver** for the panel, because a terminal that feels tight is not a terminal that is tight. Everything else you probably own.



What has to be on the bench.

The chase

Before any service goes in, the box it goes in: the base flange set square on the bench, the insulated chase sections stacked and pinned and checked for plumb – an error at the bottom is an inch at the apex – and every hole a pipe or cable will pass through drilled and grommeted now, with the sections apart, never later with a hole saw inside a finished core.

```
1. Set the base flange on the bench   20 min   Spirit level, Cordless driver
   check: Flange square to the bench and the bolt pattern matching the pad's service port template.
2. Stack and pin the chase sections   35 min   Spirit level, Cordless driver
   check: Plumb within 1/8 in over the full height, dry.
3. Drill and grommet every penetration   40 min   Step drill, Deburring tool
   check: Every hole grommeted. No bare edge anywhere a cable or a pipe will move against.
   stage total: 95 min
```

Drain first, because it is the one you cannot re-route

```
4. Cut, dry-fit and mark the stack   30 min   Tubing cutter, Deburring tool
   check: Every joint marked. Trap at the foot, and the outlet pointing at where the floor port will be.
5. Prime and cement the stack   25 min   no tool
   check: No joint moved after setting. Stack rigid when the top is twisted by hand.
6. Fit the floor-port boot   15 min   Cordless driver
   check: Boot clamped, square, and the clamp screw reachable from inside the finished core.
   stage total: 70 min
```

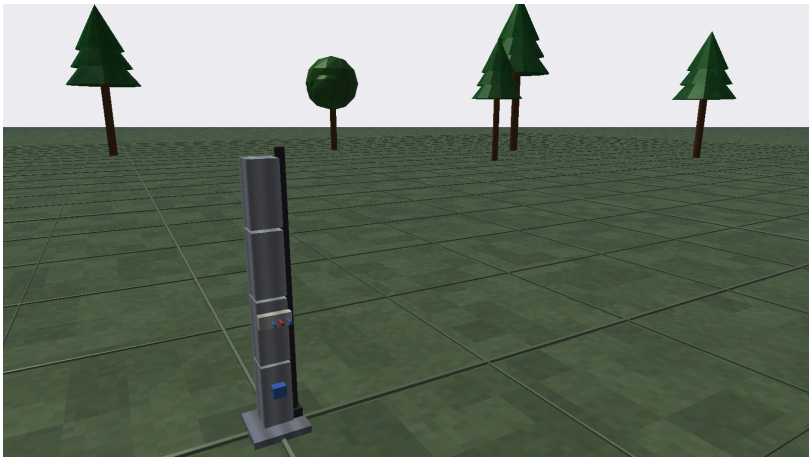
A two-inch stack the full height, with the trap at its foot. Dry-fit the whole run and mark every joint across the socket before the cement comes out, because solvent cement gives you about five seconds and no second chance at alignment. Work bottom to top, so gravity is not pulling a fresh joint apart. Then the floor-port boot – the joint the host makes once, and nobody touches again for the life of the pad.

Water, then prove it before anything electrical exists

```
7. Mount the manifold and the isolating valve   25 min   Cordless driver
   check: Valve handle has clearance to swing fully with the cover on.
8. Run and crimp the riser and the tails   45 min   PEX crimp tool and go/no-go gauge, Tubing cutter
   check: Every crimp passes the gauge. Tails capped, not left open.
9. Pressure test, and do not skip it   35 min   Pressure test pump and gauge
   check: No measurable drop over 30 minutes. A drop means find it now, not after stage four.
   stage total: 105 min
```

Manifold at chest height, isolating valve at knee height where somebody who has never seen the building will look for it, both on the opposite face from where the panel is going. Four capped tails off the manifold, because a dome with no plumbing still ships with them – the alternative is opening the chase later.

Then the test: **80 psi for 30 minutes, with a gauge on it.** This is the last moment a leak costs you nothing. After this the panel goes in, and a leak costs the panel.



Water, proved before anything electrical exists.

Power last, into a core already known to be dry

```
10. Mount the load centre and the inlet   30 min   Cordless driver, Spirit level
    check: Panel door opens fully with the cover on. Inlet reachable from outside the dome's skirt.
11. Land the feeder   30 min   Torque screwdriver, Multimeter
    check: Every lug at its specified torque. Continuity and polarity checked before anything is energised.
12. Fit breakers and run the branches   50 min   Cordless driver, Multimeter
    check: Every branch identified at the panel with a written legend, not from memory.
    stage total: 110 min
```

Load centre at chest height on the dry face, inlet at the base. Feeder to the main lugs, strain relief at both ends, every lug torqued to the figure printed inside the panel door – with a torque screwdriver, not by feel. A loose terminal is how a panel burns, and it is the failure nobody sees coming.

Four breakers: the outlet ring, the lighting, the window unit, and one spare, so the first module somebody snaps on does not need this door opened again. Label them in writing, on the door, now – not from memory in a year.

> **NOT ON THE CLOCK**
>
> At 7.2 hours practised and 17.2 the first time, the core is not on the 40-hour clock either. It is built on a bench, on its own schedule, and it goes onto the pad as one object.

Close it, prove the cap, stand it up

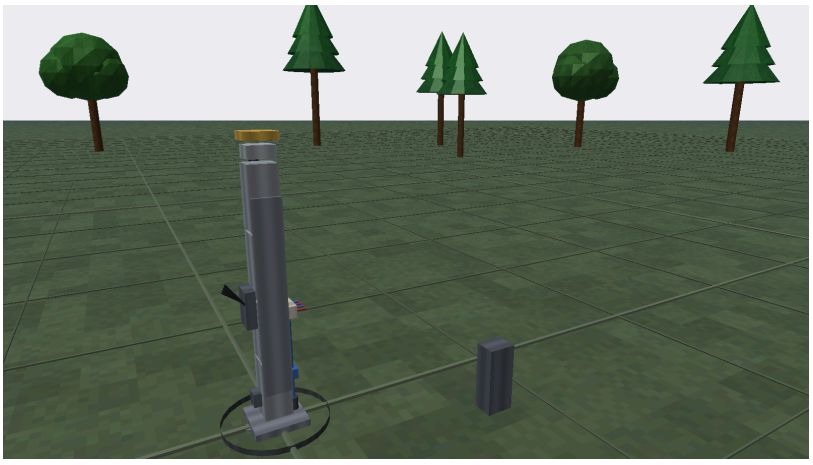
```
13. Fit the apex sleeve and the seal cap   25 min   Cordless driver
    check: Cap seats and releases by hand, no tool, gasket evenly compressed all round.
14. Hang the cover panel and stand it up   25 min   Cordless driver
    check: Every joint made in the last thirteen steps still reachable with the cover off and nothing behind anything else.
    stage total: 50 min
```

Open and close the cap five times on the bench. If it is stiff there, it will be impossible at the top of a dome in the rain.

Everything you buy for one

structure · 12 in square insulated chase secti	·	4 sections
structure · base flange and gasket	·	1 set
structure · apex sleeve with laminating flange	·	1 each
structure · seal cap, gasket and six over-cent	·	1 set
structure · removable cover panel and captive	·	1 set
structure · grommets for every penetration of	·	8 each
water · PEX-A, 1/2 in	·	24 ft
water · crimp rings	·	16 each
water · four-port manifold with valves	·	1 each
water · full-port isolating valve	·	1 each
water · capped fixture tails	·	4 each
drain · 2 in ABS stack	·	11 ft
drain · 2 in P-trap	·	1 each
drain · floor-port boot and clamp	·	1 each
drain · solvent cement and primer	·	1 tin
power · 50 A four-wire feeder cord	·	12 ft
power · recessed inlet, 50 A	·	1 each
power · eight-space load centre with 50 A	·	1 each
power · branch breakers	·	4 each
power · 12/2 for the outlet ring	·	60 ft
power · 14/2 for the lighting drop	·	25 ft
power · receptacles, boxes and covers	·	6 each
power · cable clamps and strain reliefs	·	10 each
parts · $1,405 · labour · $201		

23 lines, and none of them exotic: PEX and crimp rings, ABS and cement, a load centre and four breakers, four sections of insulated chase, and a cap. Every one is something you can buy this afternoon from somewhere already near you.



Everything you buy for one.

One core, every dome

The core is not consumed by the dome it goes into. It unbolts from four joints and a lift, and it goes into the next one. When somebody outgrows a stem cell and builds the size up, the core they built on a bench goes with them.

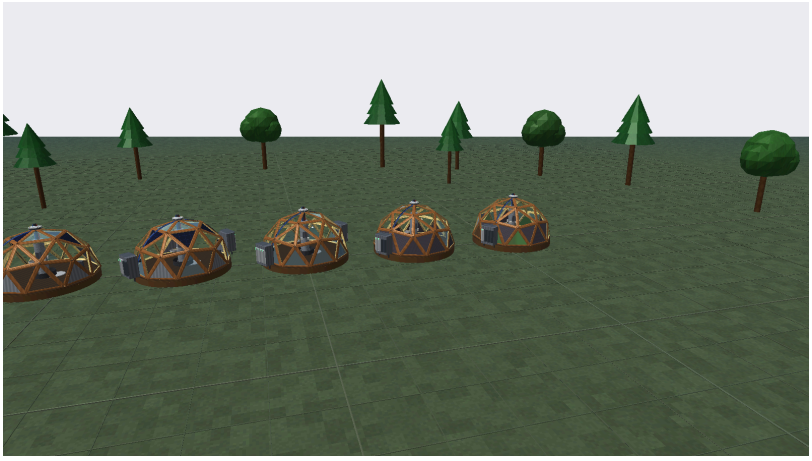
So the hours you just read are not hours per dome. They are hours once, for as long as you keep building.

CHAPTER 65

Fifteen Buildings From One Body

The catalogue a homestead wants second, priced from the same frame and the same core

The catalogue a homestead wants second, and third, and fifth.



The catalogue, priced.

The list

Every one of these is the same frame, the same shell and the same utility core. What differs is the fit-out — and the fit-out is the only line in the quote that moves.

Stem cell	$	12,036
Sauna seed	$	15,726
Jacuzzi seed	$	17,383
Treehouse seed	$	17,907
Bunker seed	$	18,009
Cold storage seed	$	18,068
Nursery seed	$	18,588
Gym seed	$	20,022
Studio seed	$	20,132
Guest house seed	$	21,914
Garage seed	$	22,621
Home seed	$	22,844
Advertiser seed	$	23,181
Workshop seed	$	24,279
Food seed	$	25,511

15 structures. The bare body is the Stem cell at $12,036; the most specialised is the Food seed at $25,511.

That spread — about two to one across the whole range — is the differentiation cost. Everything below it is the body, and the body is the same body.

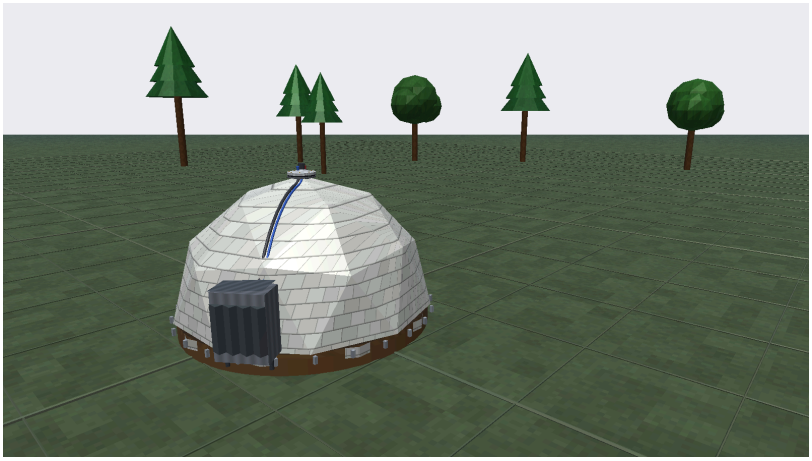
What the spread tells you

Water is what makes a building expensive. Sort that list and the ones with plumbing in them are at the top. A sauna, a jacuzzi and a food seed are all wet buildings, and the ones that are merely rooms are cheap.

Power is not. The core already carries a sub-panel and a spare breaker, so a building that wants more sockets wants almost nothing.

Structure is never the variable. No building on that list changes the frame. Chapter 63's argument, made visible: the expensive parts never learn what the building is for.

The ladder



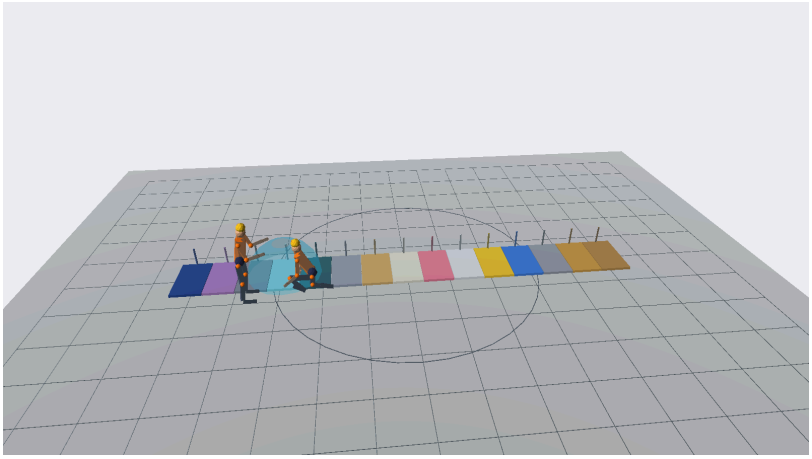
How far down the ladder goes.

Every one of these can be bought in stages, and the ladder matters more than the list does.

The rungs are, roughly: the bare body; the body with a floor; with a cap and one quilted layer; with the fit-out; with the polyps. You can stop at any rung and live there, and each one is a real building rather than a half-built one.

What makes that possible is Chapter 3 — the four features that keep the skin and the floor swappable. Without them the ladder is just a half-finished building with a tarpaulin on it.

One building, fifteen stations



One building, fifteen stations.

There is a manufacturing view of this, and it is worth a page because it is where the argument stops being about one person with a chainsaw.

The nine processes of Chapter 22 do not change between these 15 structures. Neither does the jig, the cut list, or the core. What changes is which station a particular dome stops at on its way through.

That is a **production line**, and the thing that makes it one is not machinery — it is that the parts list and the verb list are both short and neither grows with the catalogue.

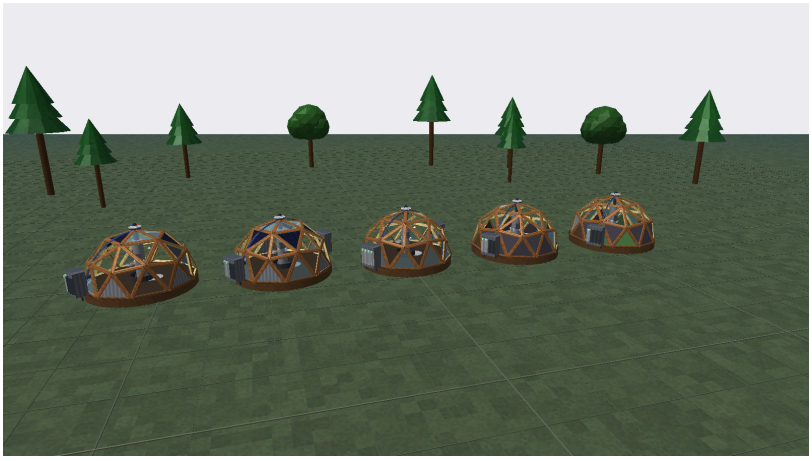
A workshop that builds one dome a month is building the same nine operations whether this month's is a sauna or a garage. That is the difference between a product line and a range of products.

What this catalogue is not

It is not a range anybody has built. The Stem cell is the one this project has solved end to end; the rest are configurations of the same model with their fit-outs costed, and they are **priced rather than proven**.

The difference matters. A food seed's price is real arithmetic over real line items. Whether a food seed is a good building — whether the ventilation is right, whether the floor drains where it should — is not something a cost model can tell you, and this book will not pretend otherwise.

Treat the list as what it is



The buildings a homestead wants second.

: evidence that the body generalises, and a set of starting points rather than a set of finished designs.

CHAPTER 66

Ten Reasons This Shape

Less skin for the same floor, nothing for the wind to push on, a parts list that does not grow with the house -- and the stem wall, brim and paint that fix what a bare dome gets wrong

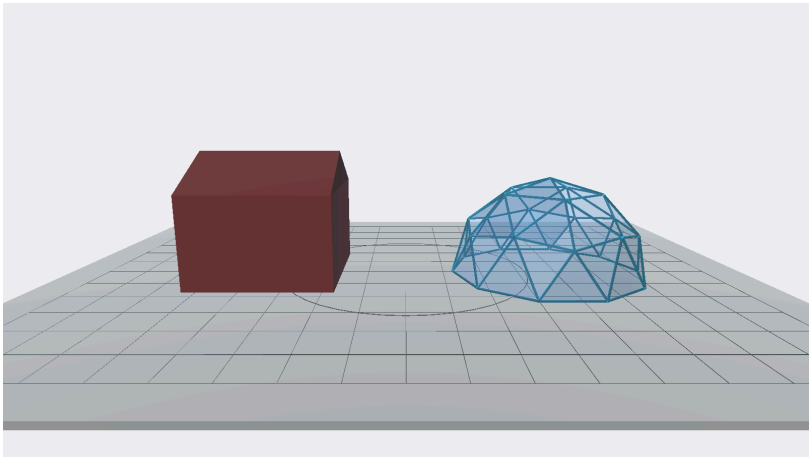
A house costs what somebody says it costs. This one costs what its parts cost, and you can count the parts: 120 sticks, 40 triangles, 9 operations, repeated. That is the whole building.

The campaign films that came before this book argued for the shape in ten points. Each one is a consequence of the geometry rather than a claim about it, and each is worked below on the same reference dome: 314 square feet of floor.

Same floor, less building

Start with the shape everybody already builds: four walls, a roof, corners. Give it 314 square feet of floor and you have to build, wrap, seal and heat **997 square feet** of outside. Every one of them costs money twice – once to put up, and again every winter for as long as anybody lives there.

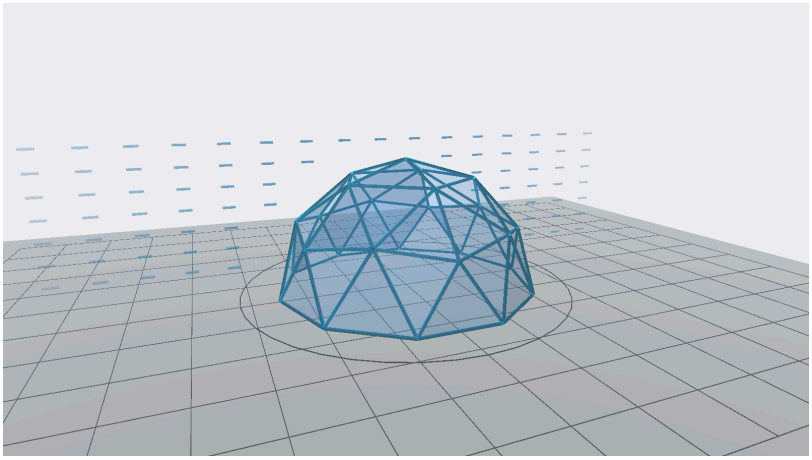
Put a dome beside it on exactly the same floor. It needs **583**. That is 42 percent less to build, to seal, to paint and to heat, forever, for the same room to stand in. Nobody invented that. It is what the shape does.



Same floor, less building.

And it stands up for a reason Chapter 6 spent a whole chapter on. Push on a square and it leans: the corners hinge, and every rectangular building on earth needs bracing to stop it. Push on a triangle and nothing happens, because changing its shape means changing the length of a side, which means breaking something. A dome is 40 triangles. There is nothing left to brace.

Wind is the same story. A flat wall catches everything thrown at it. A box has a drag coefficient of about 1.05; a dome, about 0.42 – roughly 60 percent less load out of the shape alone, before anything is bolted down. The wind arrives, finds nothing square to lean on, and goes over the top.



Nothing for the wind to push on.

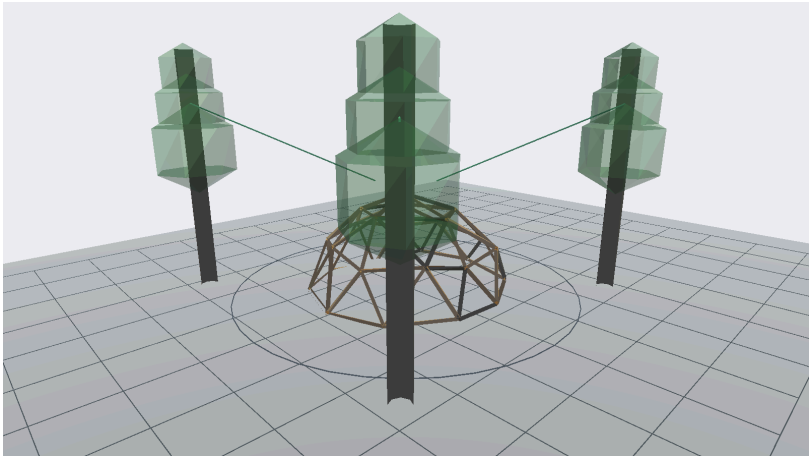
9 times the house, same parts list

This is the part that makes it a business instead of a hobby. A 10-foot dome has 79 square feet of floor. A 30-foot dome has 707. Both are 120 struts, 120 brackets, 960 screws and 9 operations. The sticks get longer and the parts list does not change at all. Learn to build one and you have learned to build every size of it.

The bracket is part of why. Every joint in the campaign's frame is a strip of flat steel with eight holes in it, folded once down the middle into a V: four holes a side, four screws into each strut. That is the entire connector, and it is why that frame needs no machine shop, no supplier and nobody's permission to exist.

And one already exists. It is not a rendering: it stands in a yard, framed from four trees cut and milled by hand, with a hundred and twenty brackets bent out of sheet metal on a bench because buying them was not an option, and it has been up through half a year of weather without moving.

> **THE NUMBER THAT DOESN'T HELP**
>
> **The first one took ten days, not 40 hours.** It was milled, not split, and it was the first – every mistake in it was made for the first time. The 40-hour clock in this book is a target worked from measured rates (Chapter 27), not a record anybody has set yet. The ten days are a record. Both numbers are in the book for that reason.

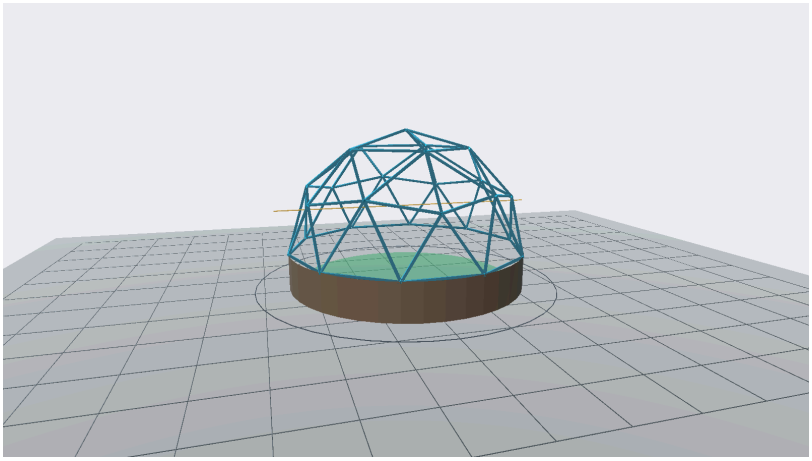


One already exists.

Fixing what a dome gets wrong

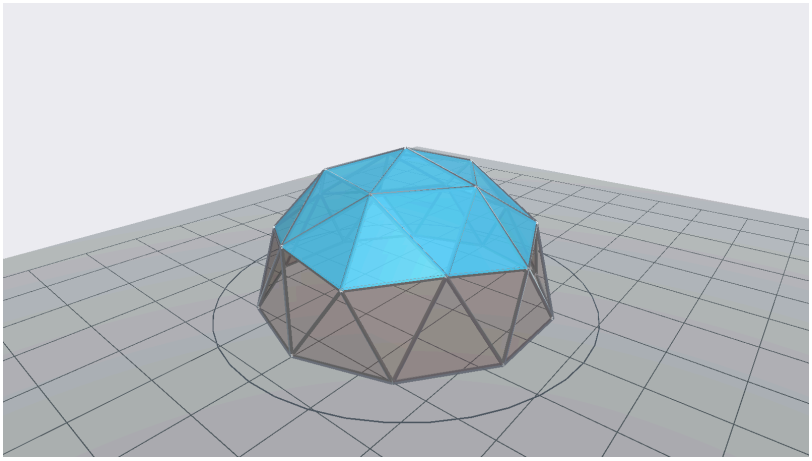
Now the honest complaint about domes, and the fix for it. A hemisphere has a beautiful volume and almost none of it is against the edge, where the ceiling comes down to meet the floor. Out of 314 square feet you can stand up in **174**. The rest is crawl space with a view.

So put the whole shell on a short wall. On a three-foot stem wall the ceiling profile lifts with it and usable floor goes to **272** square feet, for about $791 – near $8.1 for every square foot it buys back. It is the cheapest room in the building, and the first thing worth building into every one of them.



The pony wall.

Waterproofing is where the money goes, so here is the trick. Sort the 40 triangles by height and the bottom twenty come to exactly half the surface. Exactly – a property of the shape, not a rounding. So laminate the top twenty like a hat, 292 square feet of it, and leave the bottom ring to the siding: half the resin, and resin is the expensive part. Chapter 30 is the long version of this.

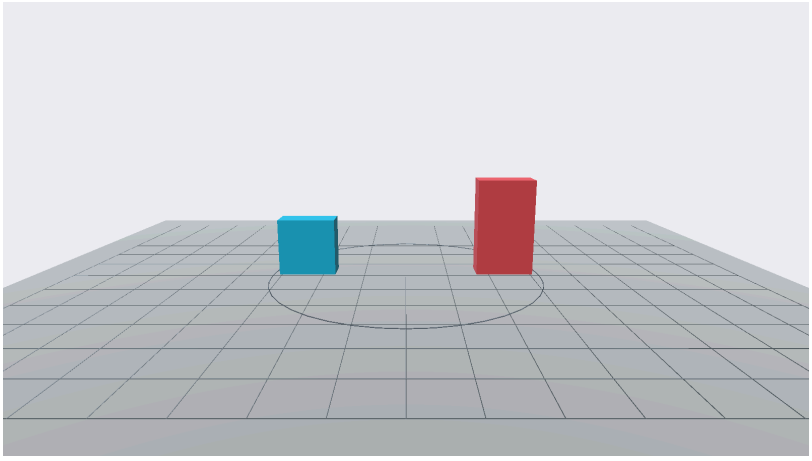


Glass the top, skip the bottom.

But a hat that stops flush with the wall is useless: the water runs down the face and into every joint below it. So the brim stands proud, ribbed to keep it stiff, and throws the rain clear of the bottom ring entirely. And once it overhangs, it is a gutter. A 18-inch brim gives 343 square feet of catchment, which on 40 inches of rain a year is **7,259 gallons** – about 19.9 a day, straight into a tank beside the door. One downpipe. No pump. The roof shape does the collecting.

What it costs to run

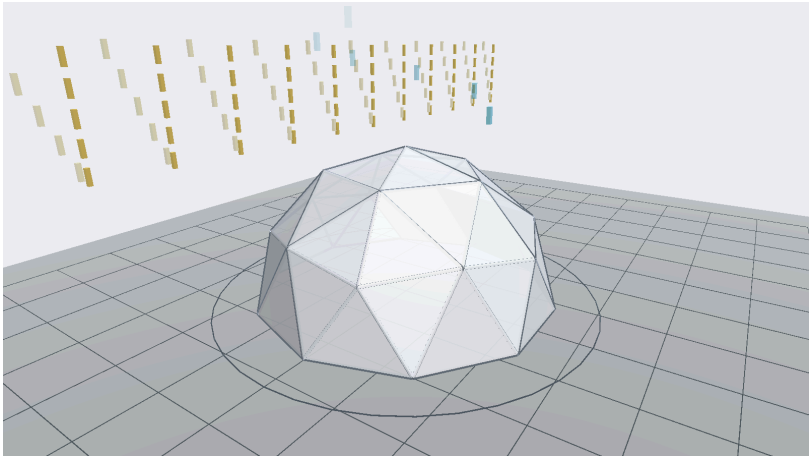
The number nobody puts on a listing. With the cavity insulated, this dome takes about **$129 a year** to heat and cool – not a month, a year. The same floor in a square house costs $220, because it has 414 more square feet of skin to lose heat through. That is $91 a year never spent, on a building that was already cheaper to put up.



What it costs to run.

And you can do better with paint. Radiative sky-cooling paint reflects nearly all of the sunlight that hits it and emits strongly in the band the atmosphere is transparent at, so the heat does not stop in the air – it goes to space. The shell runs roughly 48 degrees Fahrenheit cooler than a dark one, below air temperature in full sun. The roof stops being a heat source and becomes a heat sink.

> THE NUMBER THAT DOESN'T HELP
>
> **On a well-insulated shell, the paint is worth about $11 a year.** Very little of that heat was getting in anyway. On a thin wall it is more like $51. You do it for the comfort and for the air conditioner you do not have to buy, not for the bill.



A roof that refuses the sun.

One skeleton, many products

One skeleton is already four products: a home, a storage shed, a greenhouse, a storm shelter you could put in a back yard. Four price points, four markets, the same 40 triangles underneath every one. And the bones do not care what goes on them – a baseball, a basketball, a disco ball with a facet on every panel. 40 flat faces, all replaceable, all pointing somewhere: a house that can also be a sign, a venue or a greenhouse without changing a stick.

So the whole argument, in ten lines:

1. Less exterior for the same floor.
2. A parts list that does not grow with the house.
3. 9 operations, endlessly repeated.
4. Rigid because of its shape, not its bracing.
5. Almost nothing for the wind to push on.
6. It costs what the parts cost.
7. A stem wall buys back the wasted rim, cheaply.
8. Less skin means less to heat.
9. A painted roof that refuses the sun.
10. A brim that collects the rain it was already throwing clear.

Every one of them is a consequence of the shape.

CHAPTER 67

The Standard Article, and What the Money Is For

Twelve thousand dollars with the invoice on the page, the four ways to build the same dome, running it on nothing, and two campaigns' asks as lists

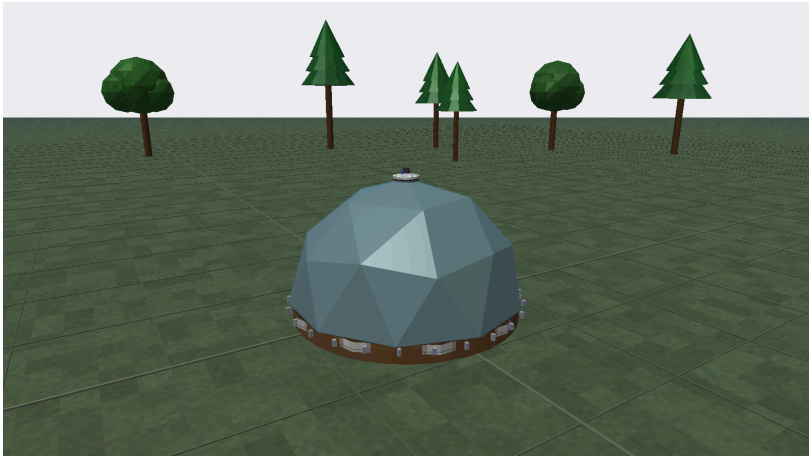
This is a house you can take apart. 19.42 feet across, 277 square feet of floor, and every piece of it fits in a pickup – a tiny house, with the part of a home that is hard to make done, and the part that is yours to decide left to you.

The frame is already standing on your land: two trees, split into 120 wedges, no hubs, no jig to buy, three saw settings for the whole building. Everything this book has said about the stick, the panel and the skin comes together here as one thing a person could order, and this chapter prices it – then prices the two campaigns that would make it more than once.

Twelve thousand, and here is the invoice

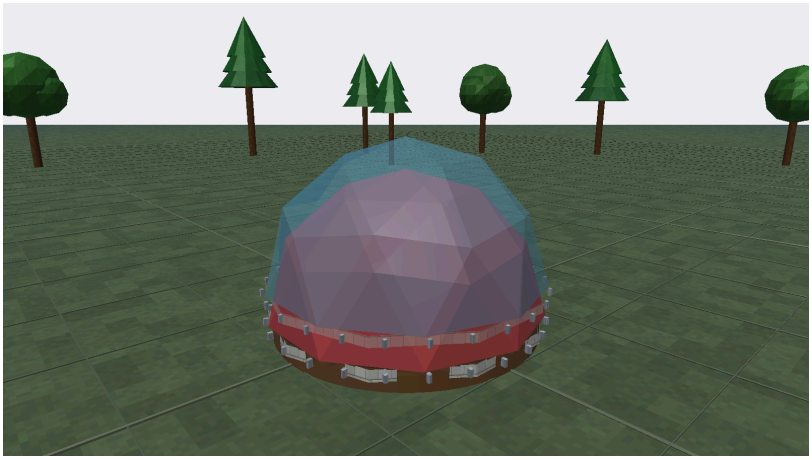
```
what it costs, all of it.

   Wedge-cut frame  ·  $692
   Shower-cap shell  ·  $3,265
   Utility column and seal cap  ·  $1,405
   Light, fan and cooling  ·  $485
   Utility panels  ·  $491
   Build labour  ·  $1,884
   ---------------------------------------------
   materials and labour  ·  $8,221
   shop overhead  ·  $1,480
   warranty reserve  ·  $329
   COST TO BUILD  ·  $10,030

   our profit, 20% marked up on cost  ·  $2,006
   WHAT YOU PAY  ·  $12,036

   per square foot of floor  ·  $43.45

and the ground is quoted beside it, not inside it:
   a platform to stand on  ·  $6,240
```

$12,036. $10,030 to build, and $2,006 is the maker's profit, marked up on cost and stated on the page rather than buried in it. The ground it stands on is quoted beside it, not inside it, and nobody marks it up.



Twelve thousand, and here is the invoice.

What makes the shell cheap is Chapter 30's idea: a dome is a head and it wears hats, and only the outermost one is waterproof. Under it go quilted layers of recycled clothing – about 132 t-shirts a layer, sewn at somebody's kitchen table and tagged with the name of whoever sewed it. Every layer added makes the next cap a size bigger, so there is no limit: a rigid shell fills up and stops, and this one does not.



The insulation is a thousand t-shirts.

And nothing here is finished, on purpose. A laminated hull instead of the cap ($21,513 for the dome with one), a steel mast through the column, a floor bought later, a rig that hangs the whole building between two trees ($4,011 for the three). Buy them when you want them.

Four ways to build the same dome

The first campaign priced the same reference dome four ways, every part bought new – nothing salvaged, nothing harvested, pressure-treated lumber off the rack, sheathing, foam, epoxy, cloth, screws, and a basic kitchen and bathroom:

build	at the till	per sq ft
Cheapest	$5,116	$16
Budget	$6,583	$21
Pristine, Hat	$6,943	$22
Pristine, Full	$9,496	$30

$6,943 for the middle of the range, about $22 a square foot finished. Bare frame and a hat on top, $5,116; pressure-treated and glassed all over, $9,496. The cheapest of them is under the price of a used car – and cut your own timber, and the largest line on the sheet goes to zero.

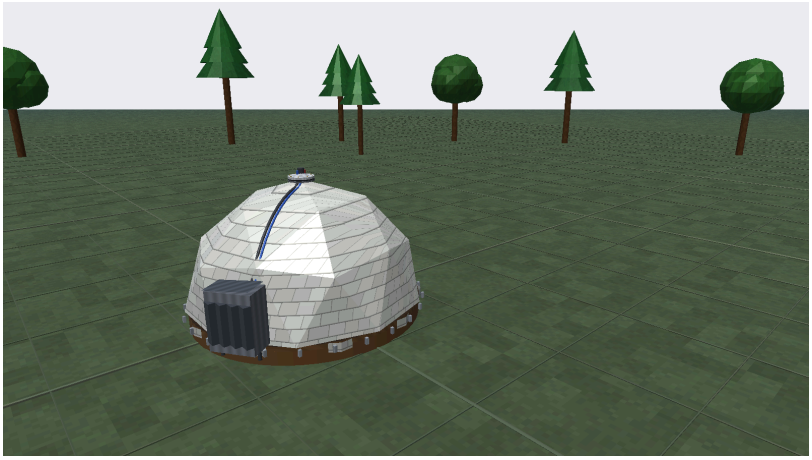
> THE NUMBER THAT DOESN'T HELP
>
> **These are not the same number as the invoice above, and both are right.** The campaign's ladder is materials at the till for a dome its buyer builds. The standard article is a dome somebody else builds, with labour, overhead, a warranty reserve and a margin on it. Quote the first as the price of a house and you have quietly left out everybody's wages.

Running it on nothing

If the envelope is that cheap to run, the off-grid arithmetic gets easy – and it is not sized against a guess. The draw is this dome's own heating and cooling year out of the performance model, plus the lights and the fan it ships with:

```
what it takes to run this thing on nothing.

the draw is not a guess. it is this dome's own
heating and cooling year, plus lights and a fan:
   demand  ·  4.59 kWh a day

   800 W of panel  ·  4.6 sun hours  ·  2.87 kWh a day
   covers  ·  63% of the draw
   10 kWh bank  ·  2.2 days with no sun
   refills from empty in  ·  3.5 days

   800 W of panel  ·  $760
   10 kWh of battery  ·  $4,100
   3.0 kW inverter  ·  $630
   the set  ·  $5,490

one honest note. the brief said three thousand for the
battery. at kilowatt hours that is $1,230,000 of cells
and two years of autonomy -- almost certainly watt hours,
or the inverter's rating, was meant.
```

Panels on the sunward bays or up a mast beside the dome cover most of a day, and the bank carries it through a couple of days with no sun. One more honest note: square cells waste the corners of a triangular bay. Triangular cells are the obvious fix, and they are not something anybody can order yet.



Running it on nothing.

What the money is for

Two campaigns, and both asks are lists rather than round numbers. The first was for the means to build domes faster than by hand:

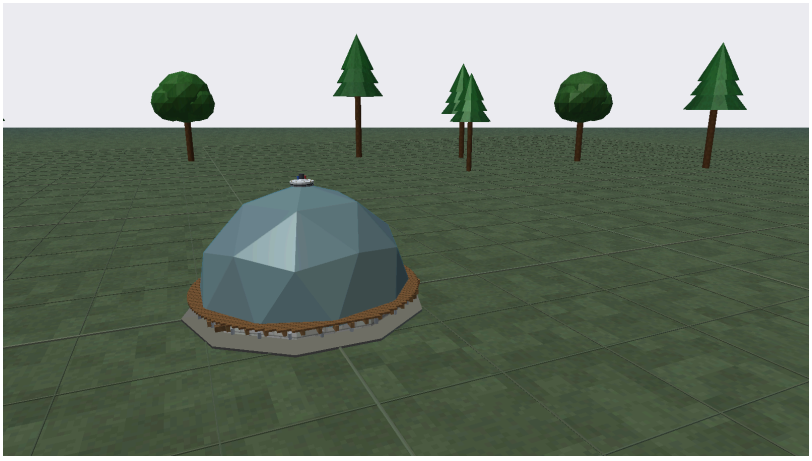
Flatbed truck, used	$	18,000
Trailer	$	6,000
Crane lift / telehandler, used	$	28,000
Sawmill, portable band	$	9,000
Table saw, mitre saw, sleds and jigs	$	3,500
Compressor, guns, hand tools	$	3,000
Laminating kit, extraction, PPE	$	4,500
Shop rent and power, six months	$	9,000
Insurance, licence, permits	$	5,000
equipment	$	86,000
material for the first domes	$	14,000

$100,000. $86,000 of it is equipment that still exists on day three hundred – a used flatbed, a trailer, a used telehandler because a dome goes up in panels and panels are heavy, a portable sawmill so the timber line really does go to zero – and the rest is material for the first units, so the thing can start paying for itself. Vague asks deserve to fail, so it was not vague.

The second was for the measurements and the tooling that turn a solved geometry into something makeable twice:

A full test platform and one standing dome	$ 4,200
Sensors in the wall, logging for a year	$ 1,850
Utility hub: three iterations to a design that ships	$ 9,500
CNC router, 4x8 bed	$ 14,500
CNC plasma table	$ 11,800
Laser cutter, for gaskets and templates	$ 8,900
Tooling to mould a strut with its hardware in it	$ 18,000
Composite triangle: steel core, moulded shell	$ 12,500
Automated processing for the repetitive work	$ 7,600
Seeding the quilt network	$ 5,000
An engineer, for the things that carry people	$ 6,800
Making and shipping what backers are owed	$ 9,350
the goal	$ 110,000

The top of that list is a test platform and one dome on it, left up through a full year with sensors in every layer of the wall. That turns the moisture question – the one this book's whole Part on the seam worries at – from an argument into a measurement, and it gets published whichever way it comes out. None of it is salary and none of it is marketing.



What the goal buys.

Three kinds of people

It only works if three different people turn up: somebody who sews, somebody with land who does not want to be a landlord, and somebody who wants a building. Chapter 50 is about why.

And every figure in this chapter came out of a model you can run yourself – including the ones that argue against it. Run them, and say which price is wrong.

PART 13

Variations

“The same frame can wear more than one roof, stand on more than one ground, and hang from nothing at all.”

The conventional hubbed dome for anybody who wants one, the author's first dome framed from whatever the chainsaw made, the joke and the serious argument behind it, and the floating dome -- a design possibility, priced and explicitly not rated.

CHAPTER 68

If You Build It on Hubs

The conventional two-frequency dome, for anybody with a hub system and no splitting maul: two lengths, one deduction, two bevels, and the four things that actually go wrong

This book splits logs and avoids the hub, and Chapter 58 explained why. But plenty of people will build a dome from bought stock on a bought connector, and the films that go with this book teach that build too. This chapter is that build, at the same two frequencies the rest of the book uses, so nobody has to go elsewhere for it.

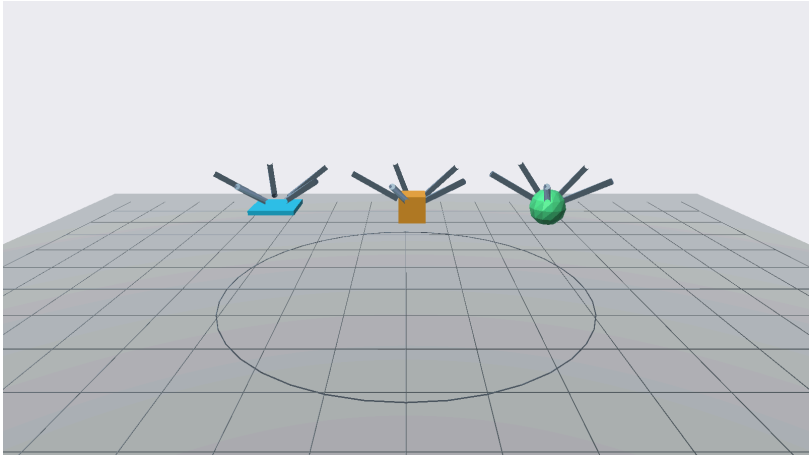
The five words

- A **hub** is a joint.
- A **strut** is a straight member between two hubs.
- A **panel** is the triangle they enclose.
- A **chord factor** is a strut's length divided by the radius – which is what lets one drawing serve every size of dome.
- **Frequency** is how many times each edge of the parent shape was divided: two, here, which is where "2V" comes from.

Two lengths, and the hub that decides every cut

Measure every edge of the finished sphere and they fall into two groups, and only two. The shorter runs from an original corner to a projected midpoint: **0.5465** of the radius. The longer runs between two projected midpoints: **0.6180** of the radius, which is exactly one over φ – the one place the golden ratio does show up in the answer. Their ratio is 1.1308, not φ. A hubbed hemisphere is 65 struts on 26 hubs.

Now the decision that comes before any cut: **choose the hub system.** A steel star plate, a bevelled timber joint and a drilled hub ball all build the same geometry, but they eat different amounts of strut. Choose one, build one joint for real, and measure how much shorter the members have to be. Everything downstream depends on that measurement, and no catalogue number is a substitute for it.



Choose the hub system before anything else.

Because every number in the geometry is **hub centre to hub centre** – that is what the sphere is made of – and the stick you cut has to be shorter by whatever the two connectors occupy:

$$\textit{cut length} = \textit{centre length} - \textit{deduction}$$

The deduction is the total across *both* ends, not per end. Halving it by mistake is one of the commonest ways a dome ends up refusing to close.

A strut is a chord, so it dips below the sphere between its ends, and it leaves the surface at **half the central angle** it spans: one line of arithmetic. That is 15.86 degrees for a short strut and 18.00 for a long one – the angle to tip a hub plate to, or to cut a timber end back to.

Bevels, hub types and the stock

Where two panels meet along a strut they fold, and if you are skinning with rigid sheets each edge wants planing to half the fold so the faces meet cleanly. There are exactly two fold angles in the whole dome, one per strut class, so two saw settings and no thinking at the bench: **9.01 degrees** on a long board and **11.23** on a short one.

The hubs are not all alike. Group them by how many struts arrive and at what angles – 6 five-way and 10 six-way in the hemisphere, plus the rim – and each type is its own part. Make a set of each, keep them in separate labelled boxes, and check one against a drawing before making the rest. Make the base hubs first, because they also take the anchor bolts.

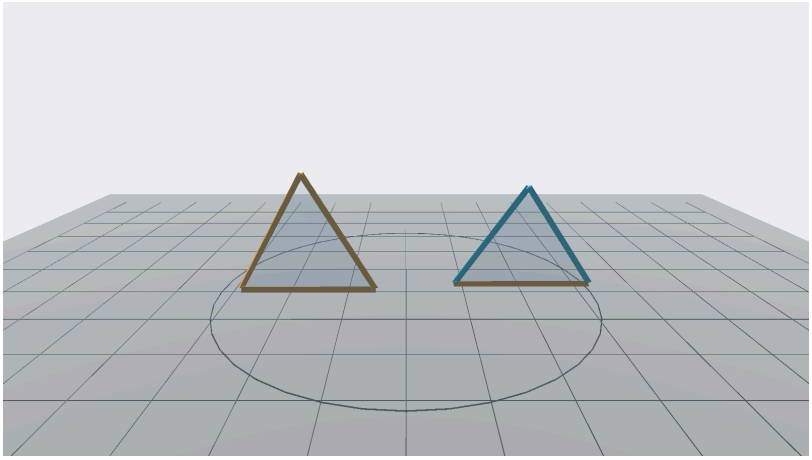
And buy the stock before you fix the radius. At some radii a standard length of stock yields exactly one strut and throws the rest away; a small change in radius can turn one piece per stick into two. Count kerf when you do the sum:

$$\textit{pieces per stick} = \textit{floor}((\textit{stock} + \textit{kerf}) / (\textit{cut} + \textit{kerf}))$$

Cut, jig, raise

1. **Cut one class at a time.** Set a stop block and cut every piece of that class without touching the setting. Keep the very first good piece as a master and compare the rest to it, not to a tape – a tape adds a new error every time. Label each piece as it comes off the saw. Then, and only then, move the stop.
2. **Build the triangles flat.** The hemisphere has two kinds of triangle: 30 with two short sides and one long, and 10 with three long sides. Lay one of each out on a flat floor, screw blocks down round it to make a jig, and assemble the rest inside the jig. Every panel comes out identical, and the raise becomes bolting known-good pieces together rather than fitting each one.
3. **Raise in rings from the base**, never up one side – Chapter 27.

PVC, timber and tube all share the same centrelines. They do not share end cuts, and they certainly do not share structural capacity.



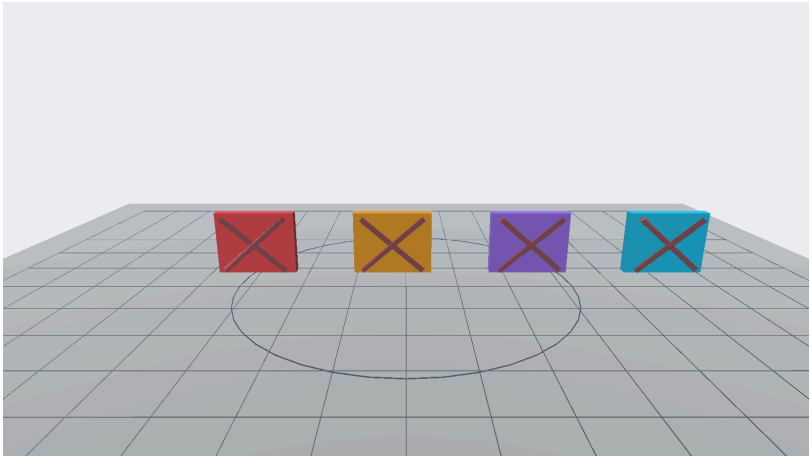
Build the triangles flat, first.

The four things that actually go wrong

Domes rarely fail because the arithmetic was wrong. They fail because:

1. the connector deduction was **guessed** instead of measured;
2. the base was **not level**;
3. somebody built **up one side** instead of in rings;
4. two similar-looking members **got mixed up**.

Each of those is a habit, not a calculation, and each is cheap to fix before you start and expensive after.



The four things that actually go wrong.

CHAPTER 69

The Frankendome

A hundred and twenty sticks of no particular description, brackets folded out of a washing machine, ten days, and half a year standing

Chapter 5 told you why I built the ugly one. This is how.

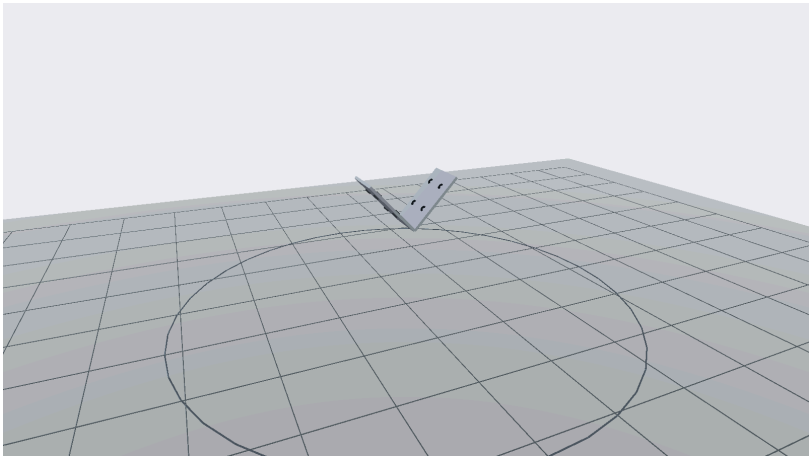
It was an experiment, not a recommendation, and the question was simple: how fast does a dome go up if you stop caring what the struts are? Not cheaper struts. Not rougher struts. Struts with no specification at all beyond being roughly the right length.

So the frame is whatever the chainsaw made. Round logs still in the bark. Quarter-sawn pieces off a milling jig. Wedges that were offcuts of something else. Square stock, rectangular stock, whatever came off the pile that day. They are not interchangeable in any engineering sense: every one of those sections has its centreline in a different place, so no two joints in the whole dome are the same joint.

A bracket out of a washing machine

The joints are the interesting part. Every one is a bracket I made, and every bracket started as a flat band sheared out of a scrapped washing machine. That casing is galvanised sheet in a gauge that is genuinely structural at this size, it is free, it is already flat, and there is a great deal of it in the world. Shear a band, drill the holes, and that is the whole component.

Then one fold, straight down the middle, and the band becomes a V. No welding, no castings, no bought connectors – and because you bend it yourself, the fold is simply whatever that particular corner turned out to be, which on a frame like this is never the angle on the drawing. **The bracket adapts to the joint instead of the joint having to match the bracket.** That is the trick, and it is why the method tolerates such rough stock.



One fold makes it a joint.

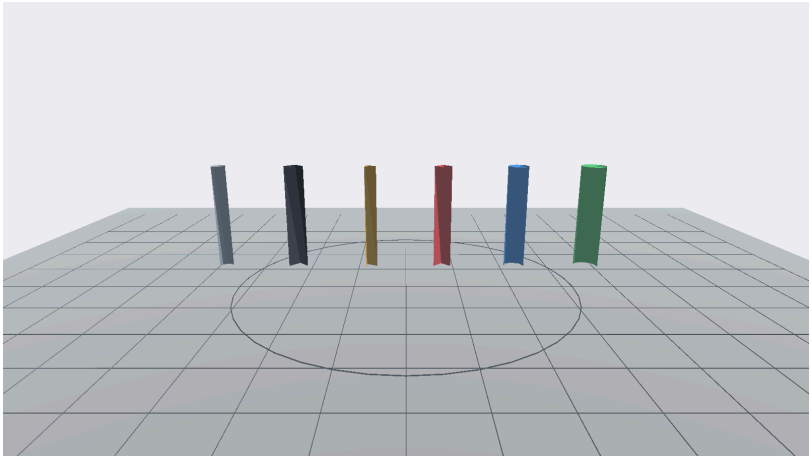
In place, the V straddles the corner, one leg down each strut, four screws through each leg: 8 screws a bracket. Four in shear along a leg is a genuinely stiff connection, and it fails gradually and visibly rather than suddenly, which for a structure you are experimenting with is exactly the behaviour you want.

The unit of work is one triangle: three sticks, three brackets, built flat on the ground. No ladder, nothing held overhead. 40 of those and the dome is a pile of finished triangles waiting to be stood up.

Hubless, so nothing has to be accurate

What makes it buildable by one person with no jig is that it is hubless. Every triangle carries its own three sticks, so the frame is 120 struts rather than the 65 a hubbed dome needs: 55 edges with two sticks lying against each other, and 10 rim edges with one. Open a joint and you can see it: two triangles meet along an edge and each arrives with its own complete stick, so the joint is a lap between two members rather than a point where several converge. **Nothing has to meet accurately at a point, because nothing meets at a point at all.**

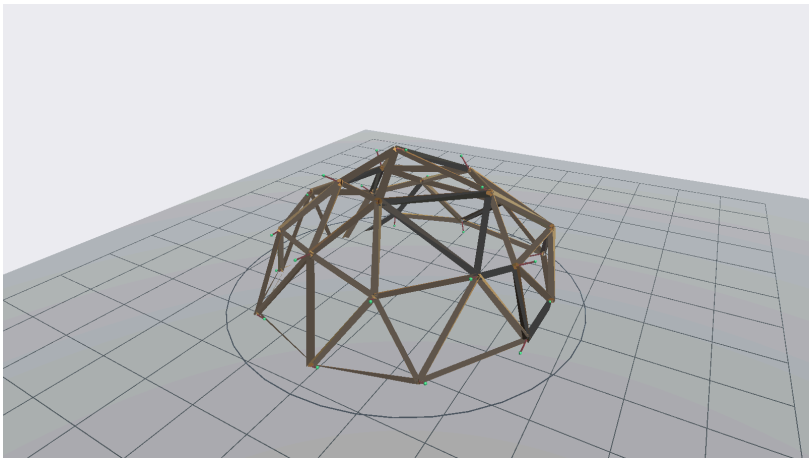
And because nothing has to be accurate, the sticks do not have to match. In the film every strut is coloured by the section it was cut to – full round, half, quarter, eighth, milled square, and salvaged steel only where wood would not serve – and the whole frame came out of the one woodpile.



Six sections, one woodpile.

Nothing lands where it should

Here is the honest picture. In the film, green points are where the geometry says every hub belongs and red lines are where the sticks actually put them. Not one joint lands on the sphere. Some are out by a quarter of an inch, some by more, and every error is a different error because every stick is a different stick.



Green is the geometry. Red is where the stick put it.

Then pull it together and stand it up, and the frame **settles**. Every triangle is individually rigid, so a joint that wants to sit proud is held by the two around it, and the error is shared out across the whole shell instead of piling up round a ring the way it does in a precise dome. That redundancy is not a nice property here. It is the only reason the thing stands at all – and it is the lumpy skull of Chapter 1, built.

What is left is a lumpy brain of a frame, near the sphere without being on it. The sheathing finishes the job: a skin bonded over the outside bridges every remaining gap and gives back the shell action the sloppiness cost. Once it is on, the structure behaves far more like the skin than like the frame – which is the whole reason the frame is allowed to be this bad.

The ledger

It went up between four trees, and that was not an accident. Cable overhead, tied back into the trunks, meant the frame never had to support itself while it was going together, and the trees kept taking a share afterwards. Covered in clear plastic, it stood through **6 months** of weather before I took it down on purpose – 18 times its own build time, which is the only durability number I have.

triangles	40
brackets, folded from scrap	120
screws	960
days	10
triangles a day	4
screws a day	96
months standing	6

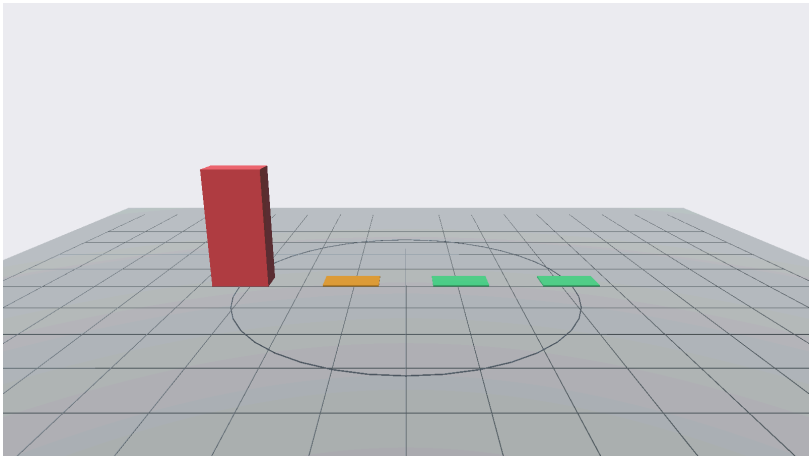
It stood on a round deck the same diameter as the dome, built and levelled first, on blocks so air moves underneath: 314 square feet, 14 joists and 231 feet of joist, on 37 blocks. Get the deck round and level and the dome has something honest to land on; get it wrong and every triangle above it inherits the error.

> **WE GOT THIS WRONG**
>
> **This dome was screwed together. It was not bolted.** If you build one to keep, put two bolts with nuts and washers through every edge as well – 130 bolts and 260 washers on top of the screws. It is the single best place to spend money on this structure, because bolts come out again and go into the next and better dome, so heavy hardware bought once is never obsolete. But it is not what held this one up, and saying otherwise would be inventing a structure I did not build.

So what does one really cost? Everything structural was free. The cost is the skin: 628 square feet of shell, 314 under the floor and 117 for the utility tower – about 1,059 square feet to laminate, which at two layers and some waste is 60.9 gallons of resin and 135 square yards each of cloth and mat.

timber, cut down myself	$0
brackets, from a washing machine	$0
screws	$48
fibreglass	$4,697
the structure	$4,745
per square foot of floor	$15.1

Almost all of it is the one material I could not make myself.



The bill.

Few processes, endlessly repeated

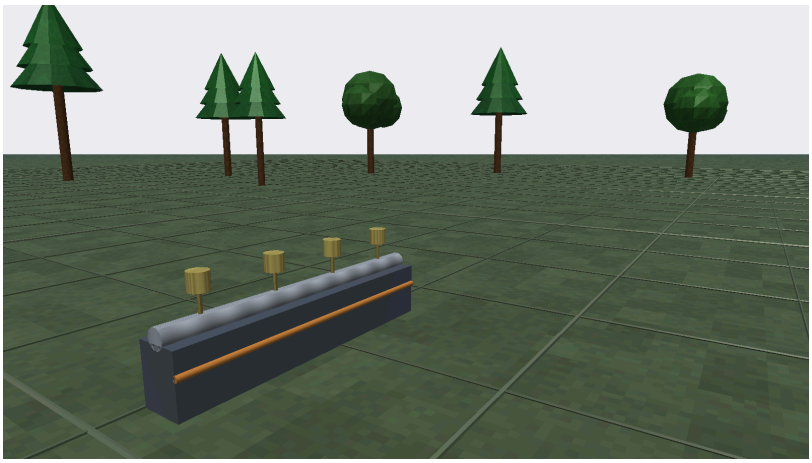
The argument that makes this a business rather than a hobby falls out of the ledger. A 10-foot dome and a 30-foot dome are the same 40 triangles, the same 120 struts, the same 120 brackets and the same 960 screws, through the same 9 operations. Only the sticks get longer, and they grow with the diameter while the floor grows with its square.

A conventional house is thousands of dissimilar operations, so improving any one barely moves the total. A dome is a few operations repeated over and over, so shaving seconds off one cut saves minutes on every dome you will ever build. That is what makes a factory worth building around this shape rather than around a rectangle.

What it trades, and what comes after it

You give up precision, and with it any claim to a calculated structure. You get back an enormous amount of speed, a material cost near zero, and a method that tolerates stock nobody else would use. For a storage building, a workshop or a shelter, where the covering matters more than the tolerance, I think that is a completely valid way to build. For anything inspected, lived in full time, or carrying snow you have not calculated for, it is not. Both things are true at once.

And the member after the split log is not another split log. It is a member that arrives with its screw holes, its threaded inserts and its gasket ridge already moulded in, a steel core where the strength has to be – so the hardware that joins two of them is a standard, reusable part that comes off with a driver, and its load rating comes off a test rig rather than an argument. It does not exist yet. Chapter 67's goal is mostly for that, and what breaks will be published.



The member we want to make instead.

CHAPTER 70

The Evil Dome Empire

A house as a platform, a business plan built on ink cartridges, the one catastrophic weakness in it -- and then the serious part

What if a house was not really a finished product? What if it was a platform?

Build the geodesic skeleton once. Keep the bones. Then replace, repair, upgrade, mutate, Frankenstein, and occasionally bolt questionable things onto it for the next fifty years.

That is what fascinates me about geodesic construction. The dome is not just a weird round house for hippies, Bond villains, mathematicians and people who own suspicious amounts of polycarbonate. It could be an entirely different building paradigm. Think of building a PC: keep the chassis, upgrade the components. And apparently I have decided that this is now my problem to investigate.

The world's poorest supervillain

Programmer, plus electronics technician, plus fabricator, plus engineering student, plus scavenger, plus Alabama hobo-redneck, plus aspiring science madman. A perfectly stable combination.

I like programming things. I like wiring things. I like woodworking and metalworking. I like taking things apart. I like finding machinery somebody threw away and immediately thinking: excellent, the organs are still good. Give me discarded motors, bearings, sheet metal, wiring harnesses, switches, sensors, lumber, gears and random military surplus, and I become the world's poorest supervillain.

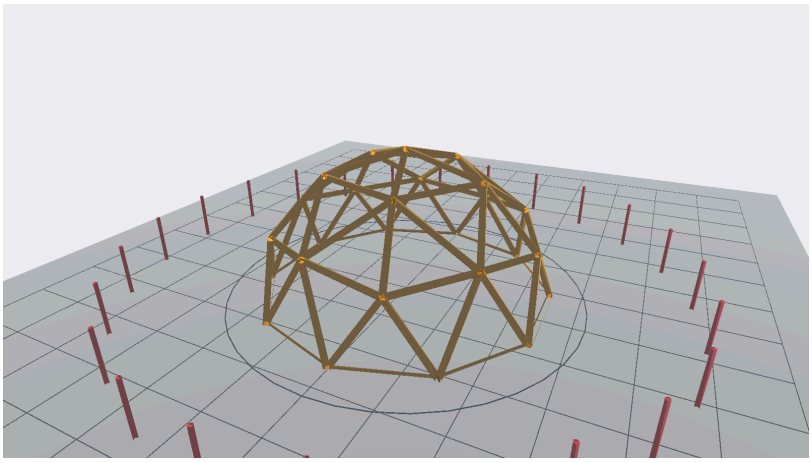
That mentality is most of the Frankendome of the last chapter. When the material is cheap, salvaged or already considered garbage, **failure becomes affordable.** Drill the wrong hole? I do it often. Cut something too short? Again? Excellent. Wire it wrong? Educational. Discover your revolutionary invention is actually a terrible idea? Congratulations: you have conducted science, and nobody has to hold a funeral for the four-hundred-dollar precision component you just murdered.

Eventually you stop seeing junk. You see components temporarily trapped inside the wrong machine. As a kid I loved Operation Junkyard. In hindsight I was clearly receiving vocational training.

The business plan

Which brings us to my completely reasonable long-term business plan: **the Evil Dome Empire.** Chapter 50 gave its three phases. Here is the revenue model.

Need the Imperial Arctic Insulation Package? That will cost you. Fancy greenhouse panels? Money, please. Solar-integrated exterior shell? Straight into the treasury. Luxury window module? Your tribute has been accepted. It is printers and ink cartridges, except the printer is your house. And eventually I shall sit inside an enormous spherical fortress overlooking thousands of domes, and rule benevolently, but firmly, over the Hoboclass.

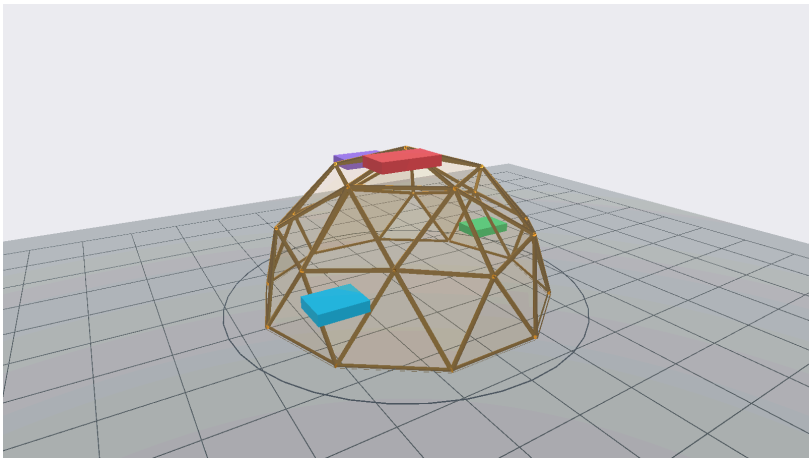


The Evil Dome Empire.

Unfortunately, my flawless strategy has one catastrophic weakness. **The Hoboclass owns power tools.**

I will unveil a magnificent four-thousand-dollar Imperial Replacement Wall Module, and somebody will stare at it for twelve seconds and say: brother, that is plywood, insulation and sheet metal. Three hours later their dome has a salvaged RV window, half a solar panel, two cheap brackets, a door off a chicken coop, and something they found beside the highway bolted to it.

My monopoly: destroyed. My empire: humiliated. My customer: now completely self-sufficient. And standing proudly in front of me, the natural predator of proprietary modular architecture: **Frankendome.**



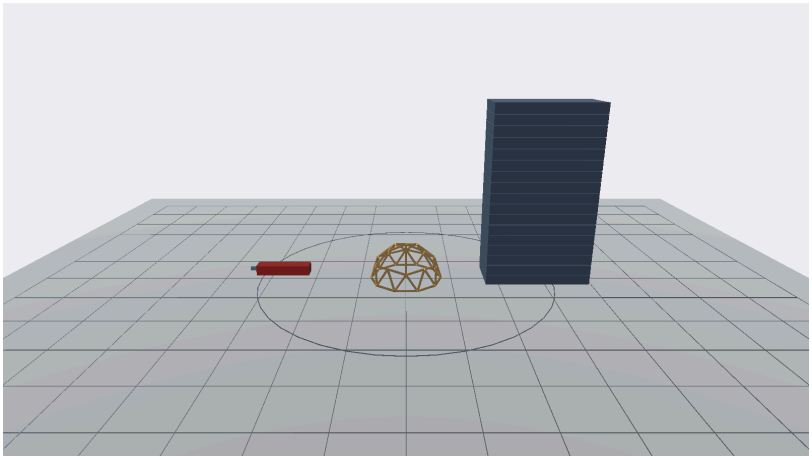
Three hours later.

(One more revenue stream, while the Empire still exists. Look at the dome like a landlord: 40 panels is 40 flat, replaceable surfaces, each one facing somewhere. That is not a roof. That is inventory. Live on a road and the wall is already pointed at the traffic: span one picture across a dozen triangles and the house earns while you sleep in it. Try that with a rectangle.)

Now the serious part

Everything up to here was a bit. This part is not, so I am going to drop the voice and say it.

Institutional capital is buying single-family housing at a scale that is genuinely difficult to picture. They have trillions of dollars and entire departments. I have the leftover remains of one prototype and the cheapest chainsaw in the store. I am not pretending that is a fair fight. I am telling you it is the actual balance of forces, and I am doing it anyway.



The actual balance of forces.

And here is the thing I want you to sit with. You know somebody who is never going to afford a house. Not because they are lazy and not because they did anything wrong. Because the arithmetic stopped working. You know exactly who they are. You thought of them just now.

So the ask is small: put this in front of one person who can carry it further than I can. Homesteaders, builders, developers, makers, preppers, people who crowdfund, creators of every description – reach is reach, and I am not in a position to be picky. One share from somebody with an audience does more than a month of me in a field with a chainsaw.

If it reaches somebody who wants to put weight behind it, it turns into tools: a CNC router, a laser cutter (said in the correct voice), cameras so the documentation stops being the bottleneck, and the power and space to run all of it. None of that is a salary. Every bit of it turns straight back into parts, and into footage of parts. I am not going to beg for it and I am not going to dance for it. I am going to keep building either way. It would simply happen faster with tooling.

> **THE NUMBER THAT DOESN'T HELP**
>
> **Three films, three numbers.** The Frankendome film asked for $50,000. The pad network asked for $100,000. The house you can take apart asked for $110,000. They are not one figure rounded three ways. They are three different plans made at three different times, and each one is costed in its own chapter of this book – Chapter 50 and Chapter 67 – so nobody has to guess which is which.

Who it is for

And honestly, that is the point. I do not just want to build domes. I want to document the geometry, the fabrication, the structure, the energy systems, the modular panels, the failures, the experiments, the software, the electronics and the weird prototypes well enough that somebody else can take the idea and run somewhere I would never have thought to go.

A dome could become a house. A workshop. A greenhouse. A cabin. A disaster shelter. An off-grid power station. A modular research lab. A timber-frame monster made from trees you milled yourself, because apparently buying lumber normally was not entertaining enough.

And you should not need a hundred-thousand-dollar shop to experiment. Maybe you have a circular saw, a drill, some reclaimed lumber, YouTube, and questionable confidence. Plus the most important engineering instrument ever created: *I wonder if this will work.*

That is who this is for. Normal people. DIY builders, homesteaders, programmers, fabricators, tinkerers, people learning as they go. People whose workshop looks like a tornado hit a hardware store. People who look at a pile of scrap and accidentally start designing infrastructure.

I want to find out how cheap, modular, repairable, scalable and adaptable geodesic construction can actually become. If conventional construction has no interest in that, it is fine: we will build weird triangular houses in the woods until somebody notices.

The mission remains simple. Spread geodesic domes through civilisation like a highly efficient modular plague. Establish the Evil Dome Empire. Rule the Hoboclass. Accidentally teach the Hoboclass to manufacture all my products themselves. Destroy my own recurring revenue. Continue anyway.

Because in the end: Frankendome always wins. Follow the experiments – every one of them, including the ones that fail, is published as it is made.

CHAPTER 71

So a Mother Could Raise Her Child

A single mother, a repossessed trailer, the school lunches I would not take -- and the sentence of Buckminster Fuller's that made this a mission

The later cuts of the Frankendome film kept going back to the list of likes, so here it is properly.

I like programming things. I like wiring things. (No, I don't. I lied. And yet there I am, doing it – which is why somebody should forward this to whoever makes the humanoid robots, because I need one, for humanity.) I like woodworking and metalworking. What do I *really* like? Cash money: resources, power and influence. How will I get it? Forcibly ruling hobos, by tricking them into having a good time. The darkness here, you cannot fathom it. I promise, don't try.

And the real one, the one that is true without a joke attached: I like taking things apart.

Why women specifically

In Chapter 50 I said that if there are people with the means to fund something like this, women especially, they are who I would most like to reach. I do not care which women: saints, sinners, anybody. (For what it is worth, this structure would make a very cheap livestreaming studio.)

Why women specifically? Because my mother is an angel and a saint, and growing up with a single parent was hard. We had a trailer that had to be repossessed. She had to file for bankruptcy.

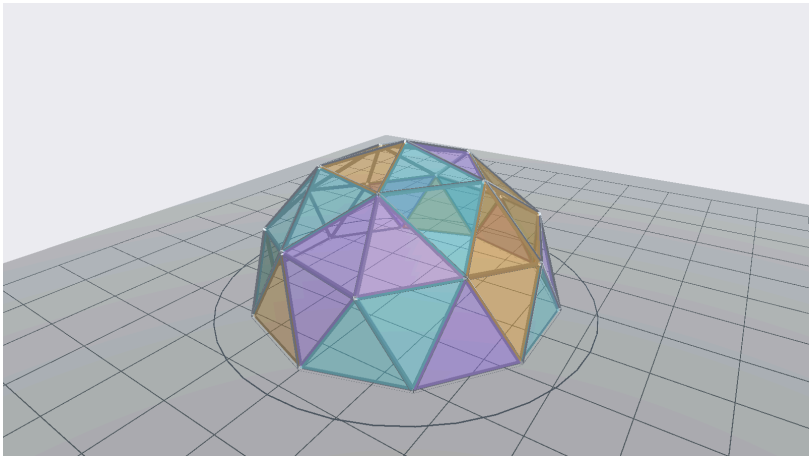
How did I handle it? As a teenager I avoided talking to girls, because I knew I could not in good conscience afford to take anybody out.

And the part nobody says out loud: we were so poor that I did not take the reduced school lunches, because of how absolutely death-blowingly embarrassing that would have been. Getting it is not embarrassing. Getting it and *needing* it is. And a lot of people need it.

And then I heard Fuller

Buckminster Fuller, who gave us the geodesic dome and a great deal else, once described his goal as housing affordable enough that a mother was not so daily and endlessly burdened by it that she could not attend to the thing that actually matters, which is raising her child. That is not a direct quotation. It is what I took from him.

When I heard it, I knew immediately that this was the thing I had been preparing to make my life's work. Hooked, determined, sold, without needing any further proof. That is what I thought then and it is what I think now. It is enormously larger than me, for all the right reasons, and it makes profound sense on the mathematics – which is the other half of this book.



And then I heard Fuller.

The proof already exists

And if you want proof that the shape works, it already shipped. The Inuit built compression domes out of the only material they had and lived in them through some of the harshest conditions on the planet. What further proof do you need? Ask me, and I will try to give it to you – with a model you can run, which is what the rest of this book is.

> **WHAT THE CLOCK IS REALLY FOR**
>
> The 40-hour clock in this book is not a speed record. It is the answer to a single question: how much of a working week does a roof cost, if the roof is built like this? For a single parent, the difference between a week and a mortgage is the difference Fuller was talking about.

CHAPTER 72

Hang It Between Two Trees

The most speculative corner of the method: a mast, three cables, and no ground at all -- priced, and explicitly not rated

This is the most speculative corner of the method, and it exists because the floor of Chapter 38 clamps to a mast rather than being framed into the base ring.

Once the floor hangs off something in the middle, the ground stops being structural. And once the ground stops being structural, you can take it away.

The same mast and floor, hung on three cables to saddles on two trees.

What changes

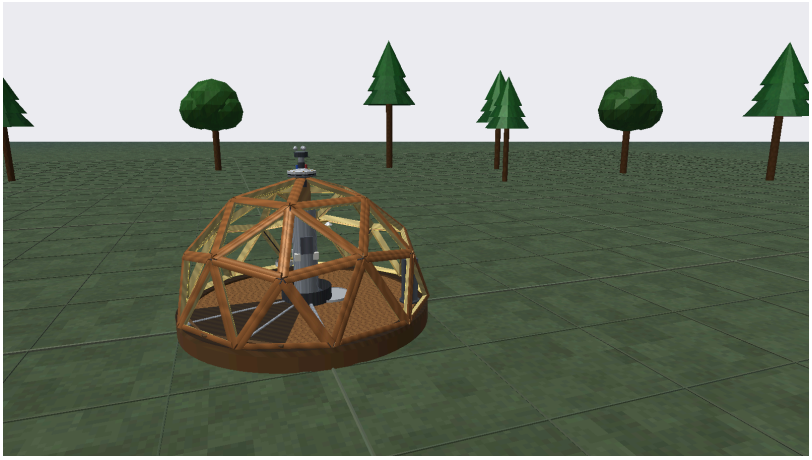
Everything the earlier chapters assumed about the ground quietly reverses.

The base ring stops being what holds the building up; the mast is. The floor stops being something the dome stands on and becomes something the dome **carries** — clamped to the mast, hanging under the shell, level because the mast is level.

The utility column keeps its job. Services come up a flexible drop instead of a rigid rise, which is a fitting rather than a redesign.

The parts

A steel core through the utility column, timber-clad, floor to apex, with a lifting ring under the seal cap. Three galvanised cables from that ring to three saddles on two trees. A brake winch and a block.



The mast through the column, and the floor that clamps to it.

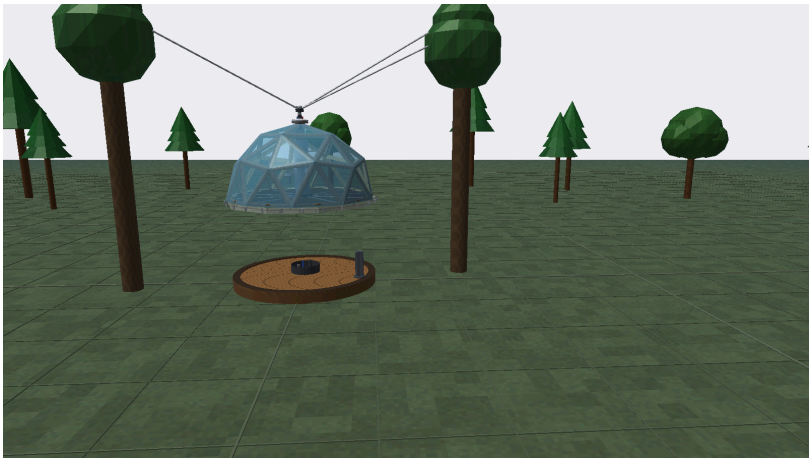
The rig is $1,490. With the mast and the dome's own floor the whole upgrade is $4,011, against a frame weighing 2,478 pounds in green pine.

Saddles, not holes. Nothing is drilled into the trees. A saddle spreads the load around the trunk and comes off without leaving a wound, which matters because a tree with a bolt in it is a tree with a problem in ten years.

Why you would

Flood, mostly. Getting a floor above water is the whole reason in some places and no amount of foundation solves it.

After that: slope without earthworks, airflow underneath that keeps timber dry, and the space below, which becomes storage or shade.



Hang it between two trees.

The sentence that has to go with it

That is a design possibility, not an engineered structure.

The loads on the trees, the cables, the saddles and the mast need an engineer before anybody stands under it. The model prices the parts; it does not rate them, and the difference between those two things is the whole of somebody's safety.

Trees are also not static. They move in wind, they grow, they shed limbs, and two of them move independently of each other. A rig that is correct on a still day in June is a rig that has not yet been tested.

Nothing in this chapter has been built. It is here because the floor design makes it possible and because a book that only prints the parts of a method that are finished is not documenting a method, it is advertising one.

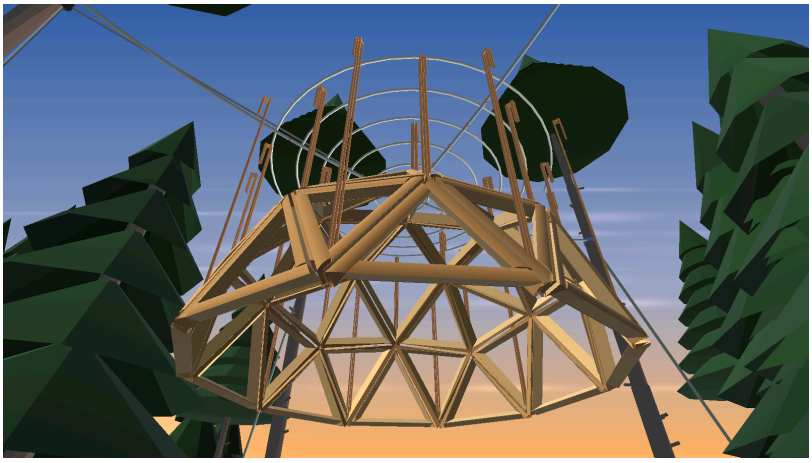
CHAPTER 73

Build It in the Air

An X of wire rope between four living trees, a 12 V winch, and forty triangles hung on three cords each, then gathered to one point, skinned, and left up -- with the owner's weighed struts, every load worked, and the numbers that argue back

Most domes are built from the ground up: a ring, then a band, then the next band, with a scaffold inside and somebody on a ladder holding a triangle against two others while somebody else drives the fastener. This chapter builds it the other way. A web of cable goes up between four living trees, and every triangle of the dome is hauled up to it by a battery winch, swung out to its own place, and left hanging there on three cords, touching nothing. Forty of them. Then they are joined where they hang, their cords are gathered to one point at the top, a skin goes over them, and the dome may simply stay up, off the ground, held by the trees.

Chapter 72 hung a finished dome from a mast between two trees and called it priced, not rated. This is a different idea and a more practical one. The trees are not a novelty here; they are the structure. They are the posts, the anchors, the shade and the drainage, and they were standing there before anybody arrived with a saw.



Forty triangles, each on its own three cords.

Every number in this chapter is worked out by `two_v_demo.sky_web`. It rests on two tables. The first is what the owner weighed or owns, and it is measured: struts of 15 to 22 kg, and a 1,500 lb, 12 V winch. The second is everything a person assumed or estimated, 53 figures, each with a reason. Every load below uses the heaviest strut.

```
WEIGHED OR OWNED BY THE OWNER
strut_kg_lo        15 kg     weighed    the owner's lightest strut
strut_kg_hi        22 kg     weighed    the owner's heaviest strut; every load here uses it
winch_lb         1500 lb     owned      the owner's 12 V battery winch, as rated
upgrade_lo          4 x      planned    the smaller planned winch, times the present one
upgrade_hi          8 x      planned    the larger planned winch, times the present one

DECLARED
tree_square_m           9 m      assumption the four trees stand at the corners of a square round the pad
anchor1_m               8 m      assumption where the X is tied on each tree, the first web
anchor2_m              12 m      assumption where an experimental second web would be tied
sag_ratio             0.1 -      assumption each rope of the X sags 10% of its span: deep enough not to pull the trees over
sag_ratio_taut       0.05 -      assumption the flatter alternative: 5% of the span
rope_3_8_kn          67.2 kN     nominal    3/8 in 6x19 IWRC EIPS wire rope, catalogue breaking strength
rope_1_2_kn         118.3 kN     nominal    1/2 in, same construction
rope_5_8_kn         183.3 kN     nominal    5/8 in, same construction
rope_3_4_kn         261.5 kN     nominal    3/4 in, same construction
rope_7_8_kn         354.1 kN     nominal    7/8 in, same construction
rope_1_kn           459.9 kN     nominal    1 in, same construction
design_factor           5 -      standard   wire rope is worked at a fifth of its breaking strength
spiral_pitch_m        0.5 m      assumption the spiral's turns half a metre apart, so a thrown pigtail always finds a strand
web_margin_m          0.8 m      assumption room between the web's lowest point and the top panel
pigtail_tie_m         0.5 m      allowance  cord for the hitch at each end of a pigtail
bridle_h_m              1 m      assumption the harness eye, this far above a triangle's balance point
tree_dbh_m           0.45 m      assumption an anchor tree's trunk at the base
green_mor_mpa          35 MPa    estimate   bending strength of green pine wood, low end
green_e_gpa             7 GPa    estimate   stiffness of green pine wood, low end
green_crush_mpa        15 MPa    estimate   crushing strength of green pine along the grain
post_in               5.5 in     nominal    a milled 6x6 post, as sold
post_e_gpa              9 GPa    estimate   stiffness of a dry milled pine post
tree_factor             3 -      assumption the margin a living tree should keep: they vary, and rot hides
next_tree_m             8 m      assumption the next tree out from each anchor tree, where its back-guy ties at the base
hook_drop_m           0.6 m      assumption block, hook and harness eye below the X's centre
winch_m_min             4 m/min  estimate   a small 12 V winch's line speed at a light load
winch_eff            0.35 -      estimate   battery to load: motor, gears and drum together
battery_wh           1200 Wh     assumption one 12 V, 100 Ah deep-cycle battery
battery_use           0.5 -      standard   a lead-acid battery is drawn to half, no further
float_gap_m          0.12 m      assumption how far apart the hung triangles float before they are joined
raise_step_m          0.5 m      assumption how far one corner of the dome is winched up at a time
rope_ea_mn              5 MN     estimate   1/2 in wire rope's stretch stiffness: metal area about 0.4 of the circle, E about 100 GPa
pigtail_kn              9 kN     nominal    1/4 in double-braid polyester cord, catalogue breaking strength, approx
master_drop_m         0.5 m      assumption the free ring where the cords meet, this far under the apex
cord_min_deg           15 deg    assumption a cord flatter than this squeezes more than it lifts; that joint hangs on its struts instead
lash_per_seam           3 -      assumption lashings along each seam, if it is lashed rather than screwed
lash_wraps              8 -      assumption turns of cord in one lashing
cotton_gsm            340 g/m2   nominal    10 oz cotton duck
silicone_gsm          150 g/m2   estimate   silicone left in the cloth after one soak, dry
skin_layers             2 -      assumption layers of the silicone skin
skin_overlap          0.1 -      allowance  laps and edges, on top of the dome's area
mix_parts               4 -      estimate   naphtha to silicone, by volume, to make a soak thin enough
silicone_kg_l        1.04 kg/L   estimate   density of 100% silicone sealant
tube_ml               300 mL     nominal    one 10.1 oz cartridge
tube_usd                8 $      estimate   one cartridge of 100% silicone
naphtha_usd_l           4 $/L    estimate   VM&P naphtha by the gallon
cotton_usd_m2           7 $/m2   estimate   cotton duck by the roll
sailcloth_usd_m2       25 $/m2   estimate   woven polyester sailcloth, mid weight
air_kg_m3             1.2 kg/m3  standard   air at sea level
gust_m_s               25 m/s    assumption a strong gale's gust, 90 km/h
gust_hi_m_s            40 m/s    assumption a storm's gust, 144 km/h
dome_cd               0.5 -      estimate   drag of a skinned dome, side on
transpire_l_day       150 L/day  estimate   water a mature pine draws from the ground on a summer day
```

Change any line, and every number after this one changes with it. Your trees will not stand where these do.

> ***Two corrections.*** *The first version of this chapter, and of the film it comes from, got two things wrong. It drew a hand winch; the winch is a 12 V battery winch rated 1,500 lb, 6.7 kN. And it weighed the frame with the solver's estimate, 7.2 kg a strut and 869 kg in all. The owner's struts, weighed, are 15 to 22 kg: 2.1 to 3.0 times as heavy. A triangle is three struts, so 45 to 66 kg, and the frame 1,800 to 2,640 kg, about 5,820 lb at the heavy end. Every load in the old version was too small by that much, and several of its conclusions moved.*

Four trees and an X

Four trees round the pad, near enough a square, 9 m a side. Two wire ropes run corner to corner, so each spans the diagonal: 12.7 m. They are tied 8 m up, on tree-saver straps, never wrapped bare round the bark, because a rope that saws through the cambium kills the tree that is holding your house up.

A rope with weight on it pulls sideways on its ends, and the flatter it hangs, the harder it pulls. With real struts, that settles the design. Sagging 10 percent of its span, 1.3 m, with each triangle's cords on the rope nearest them, the worse rope pulls 27 kN: **5/8 in** wire rope, worked at a fifth of its breaking strength. Pull it flatter, to 5 percent, and the pull doubles to 51 kN and wants 3/4 in rope. A deep sag is not a choice. It is the design.

Then the trees, which take that pull 8 m up, as a lever. A 45 cm pine, green, breaks in bending at about 35 MPa. Unguyed, the design web works its trunk to a margin of 1.5, where a living tree, which can hide rot, should keep 3. The 5 percent web works it to 0.77: past breaking. So every anchor tree is guyed.

The guy goes where the owner put it: back and down to **the base of the next tree out**, 8 m behind. Nothing to buy, nothing to dig, no concrete. The guy pulls 36 kN, 5/8 in rope, and it pulls on that next tree at its root flare, where a tree is strongest and has no lever at all. The trees round a clearing become a network, each one holding the next.

Guyed to the base of the next tree out.

A tree is a better post

Guyed at the tie, an anchor tree stops being a lever and becomes a post: squeezed from above by 33 kN, held at the top by its guy, and fixed at the bottom by its own roots. A post fails by buckling, and a post's resistance to buckling grows with the fourth power of its thickness.

Cut that tree, mill it, haul it to a yard, a store and back, and you have a 6x6 post. Stood 8 m tall it buckles at about 90 kN. The living trunk it came from buckles at about 4.4 MN: 49 times more, and it carries this web with 134 times to spare. The tree was already the better post. It also comes with roots, which no lumberyard sells.

A milled 6x6 beside the trunk it would be sawn from.

One triangle, start to finish

The spiral goes on before any of this. It is woven out from the crossing and seized to the X wherever it crosses, 24 seizings, and it covers the dome's footprint, a circle 5.9 m across, but above it. Its turns are 50 cm apart: 5.9 turns, 55 m of line. The X carries the weight. The spiral gives a thrown cord somewhere to land.

1. **Rig the winch.** A 12 V winch on one tree, run off a battery. Its line runs up the trunk to a block at the tie, along under the rope to a pulley at the crossing, and down to a hook. The heaviest triangle pulls 0.65 kN: the winch is 10 times what it needs. A lift costs about 3 Wh, so all forty take 126 Wh, about 21 percent of what one 100 Ah battery should give.

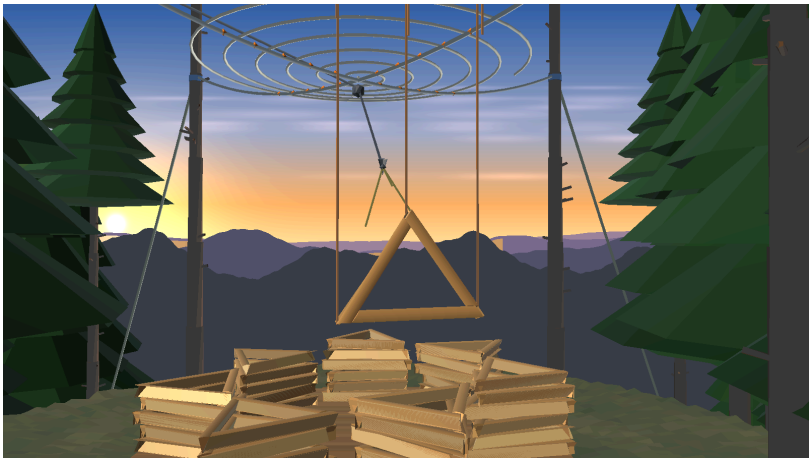
2. **Harness it.** Three legs, one to each corner, meeting in an eye 1 m above the point the triangle balances on. Hooked by that eye, it rises flat. The dome has two shapes of triangle, so make two harnesses: legs of 1.35 / 1.42 / 1.42 m for the one with two short sides, and 1.45 / 1.45 / 1.45 m for the equal one. Label them.

3. **Hoist it flat.** Up to the crossing. The hook tops out at 6.1 m, and a triangle takes about 1.5 minutes. Flat matters: a triangle that tips on the way up catches the line, the X, and anyone underneath.

4. **Swing it out.** A tag line from the ground swings it from under the crossing to its own place, turning it as it goes from flat to the angle it will sit at in the finished dome. The hook stays on.

5. **Throw the pigtails.** Every triangle carries a cord tied at each corner. Thrown up over the spiral and tied back, the three of them hold it where it is. Each holds about 216 N, on a cord 42 times stronger. The pigtails run from 1.3 to 4.5 m: 120 of them, 363 m of cord.

6. **Slip the harness.** Only now. The hook goes back to the centre for the next one.



Three pigtails over the spiral, then it slips the harness.

Rim first, crown last, then join

The hook comes down at the centre, so the middle has to stay clear until the end. Hang the rim first, then the band above it, and the crown last. The crown is also where the hook runs out: with a metre of harness above them, 5 crown triangles would need the hook higher than it can climb. They are hung a little low, their pigtails go on, the harness comes off, and they are hauled the last 44 cm on their own cords.

When the last one is up, the rim hangs 3.0 m above the grass and the crown at 5.9 m, and nothing is holding anything else up. Joining is easy from here: each triangle swings a little on its cords, so its neighbour can be drawn up to it and the seam closed, a pair at a time, with nothing to lift.

The seams need not be screwed. A dome works mostly in compression: its triangles lean on one another, and a joint mainly has to stop them sliding apart. Cord can do that. Lash each of the 55 seams in 3 places, 8 turns round both struts, and it takes 165 lashings and about 1,101 m of cord. Cord stretches and creeps, so lashings want re-tightening, and they suit best a dome whose shell – fibreglass, say – takes the pulling. Under a hard shell, the lashings only hold the shape.

Every cord to the apex

Once the dome is joined, the pigtails come off the web and are gathered to the top, so the whole dome hangs from one point, and the shell can go over the cords and leave them behind, with one line out through the cap. This is the owner's idea, and it is a good one, with one thing to get right: **how the cords run.**

Laid along the struts under the shell, a cord does not do what it seems to. A cord pressed on a curved surface lifts nothing where it lies: whatever it pulls up at its corner, it presses back down along its length. The corners are squeezed, not lifted, and the whole weight arrives at the cap.

Run straight through the air *inside* the dome, from each joint to one free ring just under the cap, the cords really do share the load. Each joint is lifted by its own cord. The dome has 26 joints: 20 cords and the apex itself, which hangs straight from the line. The 5 joints of the crown ring sit almost level with the ring, where a cord would squeeze far more than it lifts, so they hang from the apex on their own struts; the apex line carries 7.7 kN. The rim cords lean in at 40 degrees and squeeze the rim ring with 1.3 kN, which a dome takes happily: being squeezed is what it is for.

Cords from every joint to one free ring under the cap.

The price is in the room. Through the air means through the living space: a rim cord passes, halfway to the wall, only 1.2 m above a floor laid at the rim. That is fine while it is a frame in the trees. In a home, those cords are lowered away once the dome is where it will stay, or it is hung by its rim instead.

With every cord at one point, one winch can lift the whole dome evenly, up and down. But that winch now carries everything: frame and skin, 2,695 kg, 26.4 kN. Today's winch lifts 25 percent of it. The planned winch 4 times bigger lifts it with almost nothing to spare (1.01 times); 8 times bigger lifts it 2.0 times over, and that is the one to buy. Hang the winch itself at the crossing: pulled from a tree, its line turns at the pulley and loads the X with 37 kN instead of the dome's weight. With every kilogram at the crossing the ropes pull 34 kN, still 5/8 in, with the guys at 47 kN.

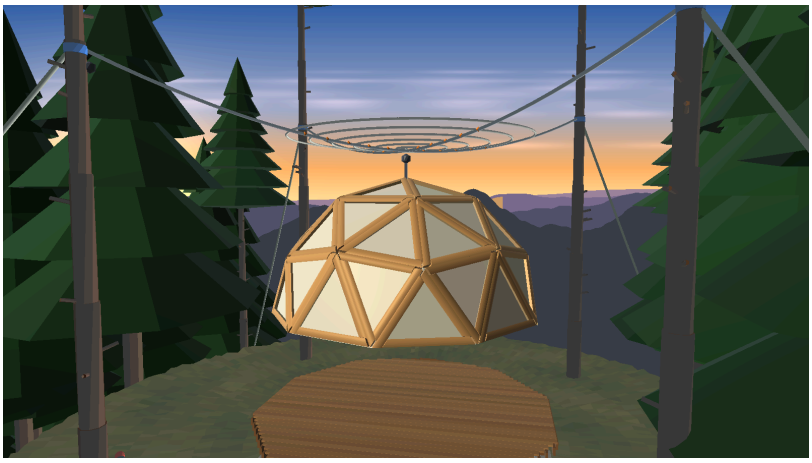
A dome can do this, and a house cannot, because its weight sits on its own axis: hung from one point, it hangs level by itself. But only if its parts are even. Placed at random, struts of 15 to 22 kg tip it about 1.0 degree, and one time in twenty 2.2 degrees, 22 cm across the rim. Every heavy strut on one side: 14 degrees. **Weigh the struts and spread the heavy ones round the dome.** Sorted, it hangs dead level.

A silicone skin

Sailcloth would cover it, at about $25 a square metre. The owner's covering is cheaper and simple. Dissolve 100% silicone sealant in naphtha, soak a cotton cloth in it, and let the naphtha evaporate. The silicone stays behind in the fibres: a waterproof, chemically quiet skin.

Two layers over the dome's 51 m², with laps, take about 55 cartridges of silicone, 65 litres of naphtha (at 4 parts to one, an estimate) and 112 m² of cotton: about $1,486, against $2,809 in sailcloth, and 55 kg on the frame.

Silicone sticks to silicone, so layers stack into one monolithic skin: the one watertight layer of Chapter 28, made thicker. Almost nothing *else* sticks to cured silicone, so decide what goes under it first. Cotton wicks wherever the coat is broken, at cut edges and scuffs, so seal edges and inspect for wear. And naphtha is very flammable, and its vapour is heavier than air: soak outdoors, in the open, far from any flame or spark.



The silicone-cotton skin, laid from the crown down.

Building with the trees

Step back and look at what the trees are doing. They are the posts, stronger than anything sawn from them. Their crowns shade the dome in summer. Their roots draw water out of the ground round it: a mature pine perhaps 150 litres on a summer day, 600 for the four, an estimate. Nothing touches the soil, so there is no foundation to pour, no damp rising, no frost heave, and air moves under and round the building. The guys tie to the next trees out, so even the anchors are trees.

A skinned dome catches wind, and a hanging one swings. Side on it shows 13.8 m². A 90 km/h gust pushes it with about 2.6 kN; it weighs 26.4, so it swings 6 degrees. A 144 km/h storm, 6.6 kN, 14 degrees. Its weight keeps it still and its round shape sheds wind the same from every side, but tie it down to the trees on three sides before a storm all the same.

A web also needs room to give, because trees move apart in the wind. The design web goes taut once its trees lean apart 34 cm; a 5 percent web after only 8.5. After that the rope stretches, adding about 3.9 kN for every further centimetre. So put a heavy spring or tensioner at every end of the X.

The trees ask some things back:

- **Lightning.** They are the tallest things on the hill, and now they carry steel rope to the dome. Bond the ropes together and ground them.
- **Growth.** Trunks thicken. Loosen tree-saver straps every year; never let one girdle the bark.
- **Deadwood.** Clean the crowns before anything hangs beneath them.
- **Rot.** It hides. Have an arborist look at all eight trees every season, and after every big storm.

Higher, if the trees allow

A second web, higher up the same trees, is experimental. It depends on the trees, and it has to be planned before they are chosen; the owner's are not set up for it. Strung at 12 m, the same pull is half again the lever: unguyed, the trunks would be worked to a margin of 1.0, at the edge of breaking, and the guys pull 46 kN.

If the trees allow it, it goes like this. The first web is in the way: hung from the second, the dome would have to pass 3.2 m up through it. So the dome is handed over first, on long cords from the second web, and the first web comes out. Then it rises a corner at a time, 50 cm a pull, 32 pulls. The heaviest corner pulls 7.1 kN, more than today's winch and well within the bigger one.

The joined dome, raised a corner at a time.

What it will and will not carry

> ***The shell, yes. A floor full of life, on bigger rope.*** *The shell, frame and skin, hangs from this X on 5/8 in rope with the trees guyed. Add a floor and what a household puts on it, 3.3 tonnes by the cell wall's own figure, and it is 6.0 tonnes pulling 75 kN: 1 in rope, and every guy and tree checked again. It can be done. It is a structure an engineer and an arborist should sign, and it is priced here, not rated.*
>
> ***Never under the load.*** *Dropped from the top of the hoist, 6.1 m, a 66 kg triangle lands at 39 km/h with 4.0 kJ. Mark a ring under the hook and keep it empty. Wear a helmet. Climb on a proper climbing system, tied in twice, or hire an arborist for the ties. The web is only as safe as the worst knot in it.*

For the record: two ropes for the X, 55 m of spiral, 45 m of guys to four more trees, 363 m of pigtails, one battery winch and, one day, a bigger one at the top. No scaffold, no crane, no foundation. The ground can wait. Maybe for good.

PART 14

Style It

“The frame is forty hours. Everything you will look at for the next forty years goes on after it.”

The cabin's frame dressed every way the Dome Creator can dress it -- twelve styles rendered, every menu swept, every skin on every frame, and every item on the price list -- each one a priced rebuild of the same dome.

CHAPTER 74

The Cabin in the Creator

The forty-hour cabin's dome rebuilt in the Dome Creator at its own size -- the same struts and panels -- priced line by line, with every place the Creator's figure and the wedge method's own differ, said out loud

Everything up to here has been about the frame: the tree, the split, the member, the pinwheel, the forty hours. The frame is the part that has to be right. It is not the part you look at. Over the next five chapters the frame stays exactly as it is, and everything that goes on it – the skin, the cladding, what it stands on, what goes on its floor – changes, every way the software in this book can change it, with a price on every version.

The software is the Dome Creator, which chapter 59 introduced: a customizer whose menus multiply to 32,105,299,968 different shells. Those chapters toured its own twelve designs, which are bigger buildings than this one. Here it builds this one. Given the cabin's frequency and the cabin's size, it rebuilds the same dome: 19.42 feet across, a sphere of 2.96 metres, 120 struts and 40 panels, the count chapter after chapter of this book has been cutting.

The cabin, bare, in the Creator.

The same dome, in another tool

The Creator prices a building the only honest way software can: it measures quantities off the model it just built – the length and volume of every strut, the area of every panel, the footprint of the foundation – and multiplies each by a unit price somebody typed into its catalogue. Chapter 78 prints that whole catalogue. Nothing it says is a quote. A price is a declared input; a quantity is a measurement; the total is the one times the other, and every row in these chapters is a fresh rebuild of the cabin, not an adjustment to a total.

So the starting point is the cabin, bare: a frame of Timber (SPF) in Quarter Wedges, Plywood panels, Cedar Shakes over them and a Gravel Pad under it, the finish of the Creator's own split-log design, with nothing on the floor.

The bare cabin, line by line

The Creator prices that cabin at **$3,947**:

- the frame, $1,037, weighing 795 pounds;

- the panels, $919;
- the cladding, $1,634;
- the foundation, $357.

The whole shell weighs 2,577 pounds, and the Creator's construction record puts it at 50 hours of work. Hold onto those numbers. Every other cabin in this Part is this one with something changed.

Where the two tools disagree

They are the same dome, and they do not report the same figures, and it is worth saying exactly where and why before trusting either.

The floor. The Creator measures 296 square feet: the full circle of the dome's radius. The wedge method measures 277: the floor inside the base ring, where the members' own thickness has already taken its share. The panels agree – 550 square feet of shell in the Creator, 550 in the wedge method – because a panel is a panel in both.

The frame. The Creator's frame line is $1,037. The wedge method's own frame, from chapter 82, costs $692. The Creator frames in Quarter Wedges at a fixed width and prices timber by the kilogram; the wedge method splits eighth-log wedges out of trees it counts, prices those trees, and does not charge for timber twice. The wedge method's figure is the one this book stands on. The Creator's is printed beside it so the two tools can be compared honestly, and so a reader who changes the frame material in the next chapters is comparing like with like.

The hours. The Creator's 50 hours cover the whole shell: foundation, frame, panels, cladding. The wedge method's 37.5 hours are the frame alone, with an allowance for mistakes. Neither is wrong. They are answers to different questions, and the next chapters use the Creator's, because the next chapters are about everything but the frame.

From here on, the frame stays, and the rest moves.

CHAPTER 75

Twelve Ways to Dress One Frame

The cabin's frame in the skin, colours and foundation of each of the Creator's twelve designs: rendered, priced, weighed and timed

The Dome Creator ships twelve designs, and chapter 60 toured them at their own sizes – a workshop, a glass loft, a trunk lodge, a hangar. Each of those designs is really two things: a frame, and a way of dressing it. This chapter keeps the cabin's frame and takes only the dressing: each design's panels, its colours, its cladding and the foundation it stands on. The frame stays the cabin's – two-frequency, 120 struts, Timber (SPF) – so every difference below is the dressing alone.

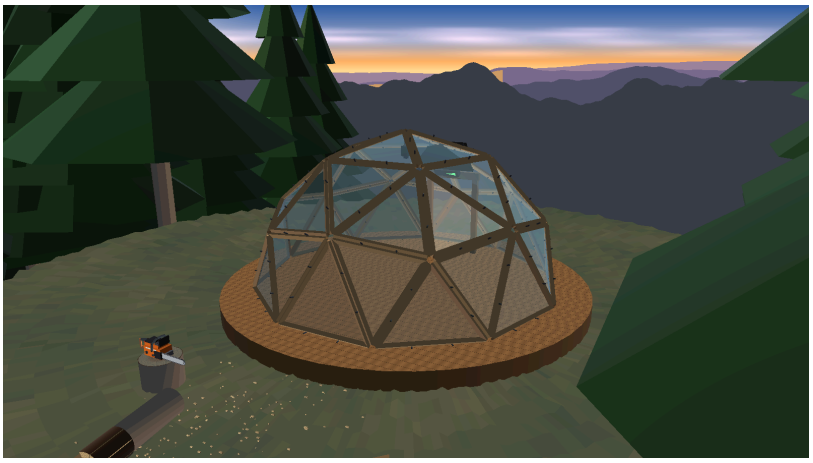
A style is a skin and a footing

Four things carry over from a design: the panel that fills the triangles, the colours of panels and frame, up to three layers of cladding over the panels, and the foundation. Nothing else does. Every row is a fresh build of the cabin, priced, weighed and timed by the Creator's own model; every picture is the Creator's own mesh in its own shader, standing where the deck stands in the rest of this book.

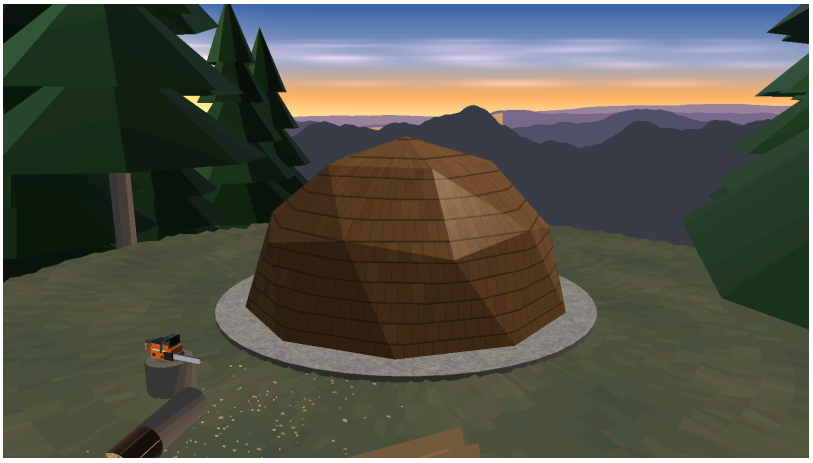
Timber and glass

The cabin as the Timber Workshop.

The Timber Workshop's dressing is plywood on a concrete slab: a working shell, plain inside and out, and the slab is most of its price. The Glass Studio Loft fills every triangle with glass and stands on a wood deck, and the glass is the most expensive common panel in the catalogue. The Split-Log Homestead is the cabin bare – plywood under cedar shakes on a gravel pad – the starting point of the whole Part.



The cabin as the Glass Studio Loft.



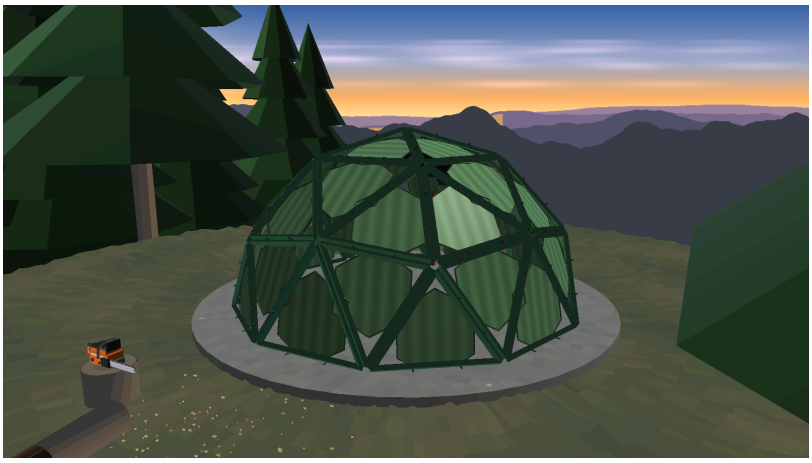
The cabin as the Split-Log Homestead.

Trunks, twinwall and hexes

The cabin as the Whole Trunk Lodge - 20 ft.

The Whole Trunk Lodge's dressing is canvas on gravel, which makes it one of the cheapest cabins here, and one of the lightest: canvas weighs almost nothing. The Grow Dome is twinwall polycarbonate on a grass pad – a greenhouse, and priced like one. The Hex Cell Pavilion fills the triangles with hexagonal composite on a slab.

The cabin as the Grow Dome.

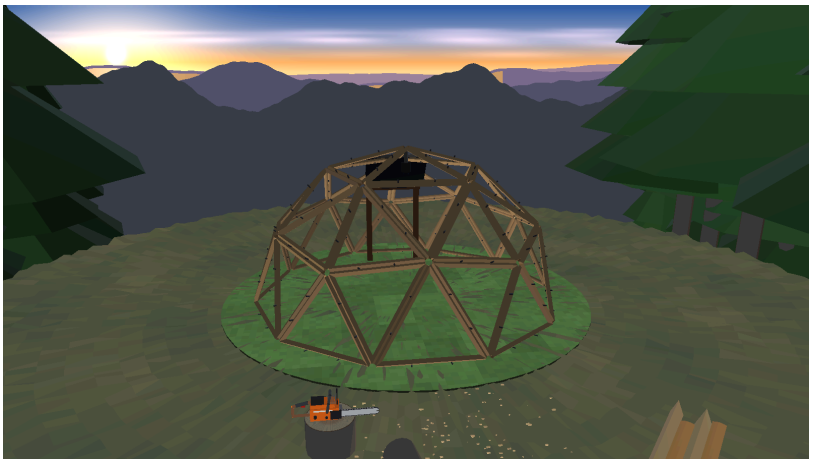


The cabin as the Hex Cell Pavilion.

Open frames and concrete

The cabin as the Continuous Steel Arc Hangar.

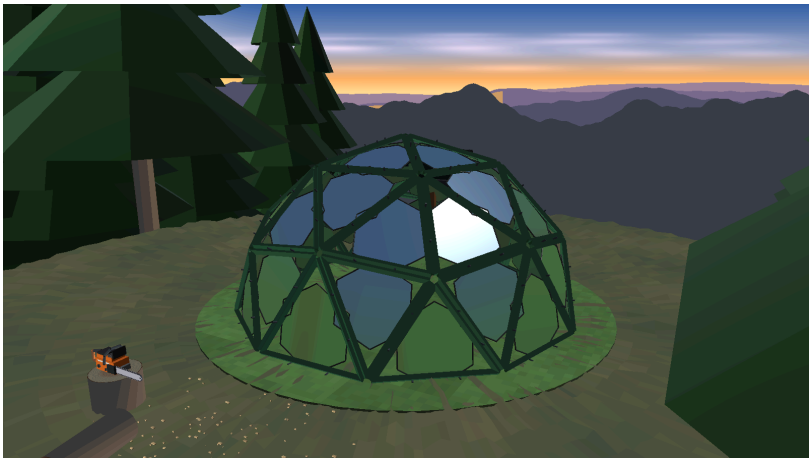
Two of the Creator’s designs leave the triangles open: the steel arc hangar and the rebar garden. On the cabin’s frame that means a frame with nothing in it, which is a pergola rather than a room. On a grass pad that is the cheapest cabin in this chapter; on the hangar’s concrete slab, the slab is most of the price. The Concrete Monocoque Form is the other extreme: form panels under a poured concrete shell on a slab, the heaviest cabin in the book by a long way.



The cabin as the Rebar Garden Dome.

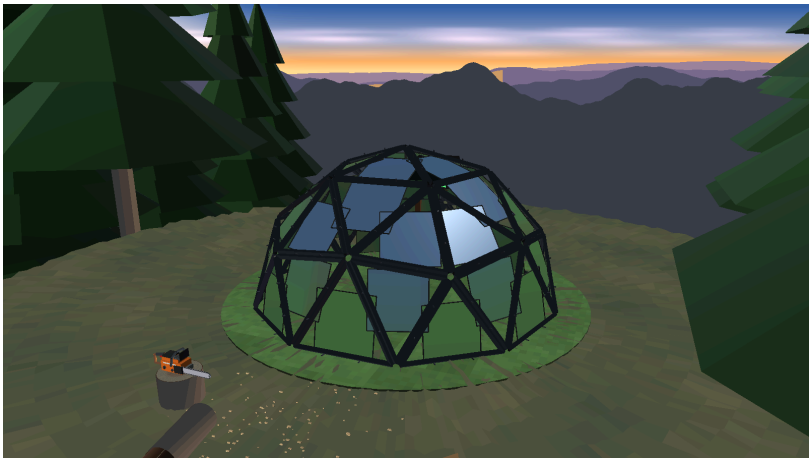
The cabin as the Concrete Monocoque Form.

Mirrors and a treehouse



The cabin as the Woodland Hex Mirror.

The two woodland designs fill the triangles with mirror – hexagonal tiles in one, square in the other – so that the dome all but disappears into the trees around it. Mirror is the dearest panel the Creator stocks, and it shows. The Treehouse Canopy Dome puts the cabin up a tree: hex composite panels on a platform, and the platform costs more than everything above it put together, $7,526 on its own.



The cabin as the Woodland Square Mirror.

The cabin as the Treehouse Canopy Dome.

All twelve, side by side

The same frame, dressed twelve ways, cheapest first:

Style	Panels	Cladding	Foundation	Total	Weight	Hours
Rebar Garden Dome	Open	-	Grass Pad	$1,116	820 lb	31
Whole Trunk Lodge - 20 ft	Canvas	-	Gravel Pad	$1,802	877 lb	47
Grow Dome	Polycarb Twinwall	-	Grass Pad	$2,546	1,012 lb	47
Split-Log Homestead	Plywood	Cedar Shakes	Gravel Pad	$3,947	2,577 lb	50
Continuous Steel Arc Hangar	Open	-	Concrete Slab	$4,404	820 lb	31
Timber Workshop	Plywood	-	Concrete Slab	$5,323	1,564 lb	47
Hex Cell Pavilion	Hex Composite	-	Concrete Slab	$7,060	1,777 lb	47
Glass Studio Loft	Glass Window	-	Wood Deck	$8,266	2,509 lb	59
Woodland Square Mirror	Square Mirror	-	Grass Pad	$11,076	2,284 lb	47
Treehouse Canopy Dome	Hex Composite	-	Treehouse Platform	$11,219	1,777 lb	47
Concrete Monocoque Form	Concrete Form Panel	Poured Concrete Shell	Concrete Slab	$11,606	16,359 lb	50
Woodland Hex Mirror	Hex Mirror	-	Grass Pad	$11,842	2,284 lb	47

The cheapest dressing, the Rebar Garden Dome's, comes to $1,116; the dearest, the Woodland Hex Mirror's, to $11,842 – 10.6 times as much, on the same frame. And line by line, so you can see which part of each style is doing the spending:

Style	Frame	Panels	Cladding	Foundation	Total
Rebar Garden Dome	$1,037	$0	$0	$79	$1,116
Whole Trunk Lodge - 20 ft	$1,037	$409	$0	$357	$1,802
Grow Dome	$1,037	$1,430	$0	$79	$2,546
Split-Log Homestead	$1,037	$919	$1,634	$357	$3,947
Continuous Steel Arc Hangar	$1,037	$0	$0	$3,367	$4,404
Timber Workshop	$1,037	$919	$0	$3,367	$5,323
Hex Cell Pavilion	$1,037	$2,656	$0	$3,367	$7,060
Glass Studio Loft	$1,037	$4,852	$0	$2,377	$8,266
Woodland Square Mirror	$1,037	$9,960	$0	$79	$11,076
Treehouse Canopy Dome	$1,037	$2,656	$0	$7,526	$11,219
Concrete Monocoque Form	$1,037	$2,349	$4,852	$3,367	$11,606
Woodland Hex Mirror	$1,037	$10,726	$0	$79	$11,842

Two things stand out. The frame line is the same in every row, because the frame never changed – that is the point of the chapter. And the expensive line moves from style to style: the panels are the biggest line in 4 of the twelve, the foundation in 4, the cladding in 2 and the bare frame in 2 – the ones so light that the frame is most of what there is. The next chapter takes the menus one at a time to show exactly how much each one can move.

CHAPTER 76

One Menu at a Time

Every frame material, every panel, every cladding and every foundation on the bare cabin in turn -- which menu moves the money, and by how much

A style changes four things at once. To see what each one is worth, change them one at a time. Every table in this chapter starts from the bare cabin of chapter 74 – $3,947 – and changes exactly one menu, rebuilding the cabin for every row.

Change one thing

Each table shows the line that menu controls, the weight of the cabin it makes, and the cabin's new total. The rows run cheapest first.

The frame

A frame material is not only a price per kilogram; it is a shape and a thickness, because nobody builds a steel dome out of solid steel wedges. The Creator weighs a frame as solid struts of the shape and width it is given, so each material here is made the way one of the Creator's own designs makes it – and where no design uses a material, it is given a tube, and the table says so.

Frame material	Made as	Width	Taken from	Frame line	Frame weight	Cabin total
Bamboo	Round Tube	5 cm	assumed: no design uses it	$394	223 lb	$3,305
PVC Pipe	Round Tube	4 cm	assumed: no design uses it	$705	280 lb	$3,615
Timber (SPF)	Quarter Wedge	10 cm	the cabin's own	$1,037	795 lb	$3,947
Rebar Steel	Rebar Rod	2.4 cm	the Rebar Garden Dome's	$1,783	1,590 lb	$4,694
Aluminum	Round Tube	4 cm	the Grow Dome's	$2,321	540 lb	$5,231
Galvanized Steel	Round Tube	4 cm	assumed: no design uses it	$2,501	1,570 lb	$5,411
Whole Tree Trunk	Full Tree Trunk	20.21 cm	the Whole Trunk Lodge's	$3,991	8,059 lb	$6,902
Structural Steel	Square Tube	5 cm	the Woodland Square Mirror's	$5,381	3,099 lb	$8,292

At one end, Bamboo brings the cabin to $3,305; at the other, Structural Steel takes it to $8,292 – a swing of $4,987. Remember the cabin's actual frame is split eighth-log wedges from trees the wedge method counts and prices itself, at $692; this table is what the same dome costs framed some other way, by the Creator's arithmetic.

The panels

Fifteen ways to fill forty triangles:

Panel	Its line	Weight	Cabin total
Plastic Sheeting	$77	1,851 lb	$3,104
Canvas	$409	1,890 lb	$3,436
Plywood	$919	2,577 lb	$3,947
Polycarb Twinwall	$1,430	2,025 lb	$4,458
Metal Panel	$1,532	2,397 lb	$4,560
Acrylic Window	$3,320	1,496 lb	$4,713
Shingle Panel	$2,145	3,635 lb	$5,173
Concrete Form Panel	$2,349	3,861 lb	$5,377
Hex Composite	$2,656	2,791 lb	$5,684
Insulated SIP	$2,809	3,185 lb	$5,837
Glass Window	$4,852	2,509 lb	$6,246
Precast Concrete	$6,384	12,193 lb	$9,412
Solar Panel	$9,194	3,185 lb	$12,221
Square Mirror	$9,960	3,298 lb	$12,987
Hex Mirror	$10,726	3,298 lb	$13,754

Plastic sheeting is the floor of the table and mirror is the ceiling: Plastic Sheeting at $3,104, Hex Mirror at $13,754. Solar panels sit near the top, and are the only row here that pays something back; precast concrete is the heaviest by far, which matters to a frame designed around wood.

Cladding and foundation

Over the panels, one layer of cladding:

Cladding	Its line	Weight	Cabin total
None	$0	1,564 lb	$2,313
Plastic Film	$61	1,580 lb	$2,374
House Wrap	$92	1,586 lb	$2,405
EPDM Membrane	$562	1,766 lb	$2,875
Foam Insulation	$613	1,699 lb	$2,926
Asphalt Shingles	$766	2,915 lb	$3,079
Cedar Shakes	$1,634	2,577 lb	$3,947
Green Roof	$2,452	7,757 lb	$4,764
Poured Concrete Shell	$4,852	15,076 lb	$7,165

And under the floor, the foundation:

Foundation	Its line	Weight	Cabin total
Bare Ground	$0	2,577 lb	$3,591
Grass Pad	$79	2,577 lb	$3,670
Gravel Pad	$357	2,577 lb	$3,947
Stone Pavers	$2,179	2,577 lb	$5,769
Wood Deck	$2,377	2,577 lb	$5,967
Concrete Slab	$3,367	2,577 lb	$6,958
Treehouse Platform	$7,526	2,577 lb	$11,117

The foundation menu swings $7,526 on its own. The pad the wedge method builds in chapter 38, and the pentagon floor of chapter 83, are both cheaper than the dearest rows here, because they are built rather than bought.

Where to spend

Ranked by how much each menu can move the cabin's price on its own: the panels ($10,649), the foundation ($7,526), the frame material ($4,987) and the cladding ($4,852).

Colour does not appear on that list because it costs nothing at all: every frame colour and every panel colour prices the same. The cheapest decision in the whole building is also the one people agonise over most.

So, for a cabin built the wedge way: the frame is already settled and cheap, and the panels are the menu that moves the money most. Spend it on the panels people will look at and touch – and think hard before buying an expensive foundation for a building this light, which is the argument chapter 38 makes from the other direction.

CHAPTER 77

Every Skin on Every Frame

The permutations in full: every panel on every frame material, every cladding on every foundation, each a priced rebuild of the cabin -- and how many buildings the menus multiply to

The last chapter changed one menu at a time. This one changes two at once, in full: every panel on every frame material, and every cladding on every foundation. Each cell is the whole cabin rebuilt and priced – not a sum of the one-at-a-time rows, though for menus that do not touch each other it comes out the same.

Every combination, priced

Read across for the frame material, down for the panel; the number is the cabin's total, in dollars. Each frame material is made in its own section, as in chapter 76.

Panel on frame

Panel \ frame	Galv	Steel	Rebar	Alum	Timber	Trunk	PVC	Bamboo
Plywood	5,411	8,292	4,694	5,231	3,947	6,902	3,615	3,305
Glass Window	7,710	10,590	6,992	7,529	6,246	9,200	5,913	5,603
Acrylic Window	6,178	9,058	5,460	5,997	4,713	7,668	4,381	4,071
Polycarb Twinwall	5,922	8,802	5,204	5,742	4,458	7,412	4,126	3,815
Plastic Sheeting	4,569	7,449	3,851	4,388	3,104	6,059	2,772	2,462
Insulated SIP	7,301	10,181	6,583	7,121	5,837	8,791	5,505	5,194
Shingle Panel	6,637	9,517	5,919	6,457	5,173	8,127	4,841	4,530
Metal Panel	6,024	8,905	5,307	5,844	4,560	7,514	4,228	3,917
Solar Panel	13,686	16,566	12,968	13,505	12,221	15,176	11,889	11,579
Canvas	4,901	7,781	4,183	4,720	3,436	6,391	3,104	2,794
Hex Composite	7,148	10,028	6,430	6,967	5,684	8,638	5,351	5,041
Hex Mirror	15,218	18,098	14,500	15,037	13,754	16,708	13,421	13,111
Square Mirror	14,452	17,332	13,734	14,271	12,987	15,942	12,655	12,345
Concrete Form Panel	6,842	9,722	6,124	6,661	5,377	8,332	5,045	4,735
Precast Concrete	10,876	13,757	10,159	10,696	9,412	12,367	9,080	8,770

That is 120 cabins. The cheapest is Plastic Sheeting with Bamboo, at $2,462; the dearest, Hex Mirror with Structural Steel, at $18,098. Read down any one column and the frame stays put while the panel moves the price; read along any one row and the panel stays put while the frame does. The frame material sets the floor of the price; the panel decides how far above it you go.

Cladding on foundation

Read across for the foundation, down for the cladding layer over the cabin's plywood panels:

Cladding \ foundation	Bare Ground	Grass Pad	Gravel Pad	Concrete Slab	Wood Deck	Stone Pavers	Treehouse Platform
None	1,956	2,035	2,313	5,323	4,333	4,135	9,483
Plastic Film	2,018	2,097	2,374	5,385	4,394	4,196	9,544
House Wrap	2,048	2,127	2,405	5,415	4,425	4,227	9,575
Foam Insulation	2,569	2,648	2,926	5,936	4,946	4,748	10,096
Asphalt Shingles	2,722	2,802	3,079	6,089	5,099	4,901	10,249
Cedar Shakes	3,591	3,670	3,947	6,958	5,967	5,769	11,117
EPDM Membrane	2,518	2,597	2,875	5,885	4,895	4,697	10,044
Green Roof	4,408	4,487	4,764	7,775	6,785	6,587	11,934
Poured Concrete Shell	6,808	6,888	7,165	10,175	9,185	8,987	14,335

The bare cabin’s own combination is one cell of this table. The treehouse platform column sits above every other, whatever is on the roof; the green roof and the poured concrete shell rows sit above every other, whatever the dome stands on.

How many domes that is

These two tables are a few hundred buildings. The Creator’s menus – four frequencies, six frame styles, eight strut shapes, eight frame materials, sixteen panels, three stacked cladding layers, seven foundations and the rest – multiply to 32,105,299,968 shells. Some settings do nothing in some combinations, a hub style on a frame with no hubs, for one, and taking those out leaves 16,554,295,296 genuinely different shells. Furnish the floor and the room menus multiply to 4,398,046,511,104 fit-outs on top, and the four dials – radius, strut width, recess and foundation size – add 446,823 settings each.

Nobody will price them all, and nobody needs to: the tables in this Part are built so you can find your own combination between two you can see. And the Creator itself, in the launcher, prices any one of them the moment you pick it.

CHAPTER 78

The Price List, Item by Item

Every unit price the Creator uses, as declared -- frame, panels, cladding, foundations, fixings, wire and pipe -- and every piece of equipment it can put on the floor, with each design's fit-out priced as a package

Every dollar in the last four chapters is one of the numbers on this page multiplied by a quantity the Dome Creator measured. Here are the numbers.

Declared, not measured

None of these prices was measured. They are what the Creator's catalogue says, typed in by a person, each a reasonable figure for its item at the time it was written. Your supplier's price is the right one. Change a price in the catalogue and every table in this Part changes with it the next time the book is built – that is the whole reason they are printed: so that you can see which numbers you would change.

The shell

Frame materials, priced by the kilogram, with what a hub costs where the frame has hubs:

Frame material	Density	Price	A hub
Galvanized Steel	7,850 kg/m3	$2.80/kg	$14.00
Structural Steel	7,850 kg/m3	$3.20/kg	$24.00
Rebar Steel	7,850 kg/m3	$1.90/kg	$5.00
Aluminum	2,700 kg/m3	$7.50/kg	$24.00
Timber (SPF)	480 kg/m3	$2.00/kg	$7.00
Whole Tree Trunk	650 kg/m3	$0.90/kg	$9.00
PVC Pipe	1,400 kg/m3	$3.40/kg	$5.00
Bamboo	720 kg/m3	$1.60/kg	$4.00

Panels, by the square metre (and the square foot):

Panel	Price	Weight	Note
Plywood	$18.00/m2 ($1.67/sq ft)	6.6 kg/m2	
Glass Window	$95.00/m2 ($8.83/sq ft)	15 kg/m2	window
Acrylic Window	$65.00/m2 ($6.04/sq ft)	6 kg/m2	window
Polycarb Twinwall	$28.00/m2 ($2.60/sq ft)	1.7 kg/m2	
Plastic Sheeting	$1.50/m2 ($0.14/sq ft)	0.15 kg/m2	
Insulated SIP	$55.00/m2 ($5.11/sq ft)	12 kg/m2	
Shingle Panel	$42.00/m2 ($3.90/sq ft)	16 kg/m2	
Metal Panel	$30.00/m2 ($2.79/sq ft)	5 kg/m2	
Solar Panel	$180.00/m2 ($16.72/sq ft)	12 kg/m2	190 W/m2
Canvas	$8.00/m2 ($0.74/sq ft)	0.5 kg/m2	
Hex Composite	$52.00/m2 ($4.83/sq ft)	8.5 kg/m2	
Hex Mirror	$210.00/m2 ($19.51/sq ft)	13 kg/m2	
Square Mirror	$195.00/m2 ($18.12/sq ft)	13 kg/m2	
Concrete Form Panel	$46.00/m2 ($4.27/sq ft)	18 kg/m2	
Precast Concrete	$125.00/m2 ($11.61/sq ft)	92 kg/m2	

Cladding layers, laid over the panels:

Cladding	Price	Thickness
Plastic Film	$1.20/m2 ($0.11/sq ft)	10 mm
House Wrap	$1.80/m2 ($0.17/sq ft)	10 mm
Foam Insulation	$12.00/m2 ($1.11/sq ft)	50 mm
Asphalt Shingles	$15.00/m2 ($1.39/sq ft)	30 mm
Cedar Shakes	$32.00/m2 ($2.97/sq ft)	40 mm
EPDM Membrane	$11.00/m2 ($1.02/sq ft)	15 mm
Green Roof	$48.00/m2 ($4.46/sq ft)	90 mm
Poured Concrete Shell	$95.00/m2 ($8.83/sq ft)	80 mm

Foundations, by the square metre of footprint:

Foundation	Price	Weight
Bare Ground	$0.00/m2 ($0.00/sq ft)	0 kg/m2
Grass Pad	$2.00/m2 ($0.19/sq ft)	4 kg/m2
Gravel Pad	$9.00/m2 ($0.84/sq ft)	160 kg/m2
Concrete Slab	$85.00/m2 ($7.90/sq ft)	300 kg/m2
Wood Deck	$60.00/m2 ($5.57/sq ft)	28 kg/m2
Stone Pavers	$55.00/m2 ($5.11/sq ft)	130 kg/m2
Treehouse Platform	$190.00/m2 ($17.65/sq ft)	70 kg/m2

And the small parts that hold a panel in its frame, each with the minutes the Creator allows to fit one:

Part	Each	Minutes to fit
V-Bracket	$4.50	6
L-Bracket	$2.80	4
Corner Gusset	$3.20	5
Foam Seal (m)	$1.10	1.5
Silicone Bead (m)	$0.80	2
Hinge	$6.50	8
Latch	$4.00	4
LED Strip (m)	$9.00	5

A screw is counted too, at $0.06 apiece.

Wire, pipe and walls

The fit-out runs on four rates: wire at $1.40 a metre, run in conduit; PEX at $2.10 a metre for each of hot and cold; drain at $5.00 a metre; and $40 to rough in each fixture. A floor section can be assigned to any of 15 uses – office, bathroom, kitchen, lounge, wood shop, metal shop, electronics, 3d printing, assembly, paint booth, storage, grow room, studio, fitness, classroom – and partitioned with markings, low walls or full walls.

Everything on the floor

The Creator stocks 39 pieces of equipment in 12 categories, each with a price, a weight and the power it draws:

Category	Item	Price	Weight	Power
Access / Handling	Extension Ladder	$260	40 lb	-
Access / Handling	Mirror Panel Rack	$540	181 lb	-
Access / Handling	Rolling Scaffold	$880	265 lb	-
Concrete / Rebar	Cement Mixer	$920	203 lb	750 W
Concrete / Rebar	Rebar Bender	$1,850	298 lb	1,100 W
Concrete / Rebar	Shoring Jack	$145	53 lb	-
Electrical	Battery Bank	$3,800	397 lb	-
Electrical	Charge Controller	$850	31 lb	-
Electrical	Power Column	$260	55 lb	-
Electrical	Power Meter LCD	$160	11 lb	-
Electrical	Wall Outlet	$18	2 lb	-
Fabrication	Air Compressor	$780	150 lb	1,500 W
Fabrication	Metal Cutoff Saw	$640	163 lb	1,800 W
Fabrication	Welding Cart	$2,100	254 lb	3,200 W
Kitchen / Bath	Bathroom Sink	$180	55 lb	-
Kitchen / Bath	Kitchenette	$600	176 lb	800 W
Kitchen / Bath	Shower Stall	$420	132 lb	-
Kitchen / Bath	Toilet	$240	99 lb	-
Lighting	Shop Light	$110	26 lb	80 W
Lighting	Tripod Light	$70	18 lb	50 W
Office	Office Chair	$120	26 lb	-
Office	Office Desk	$220	77 lb	150 W
Office	Sofa	$450	99 lb	-
Office	Whiteboard	$90	33 lb	-
Safety	Fire Safety Station	$220	31 lb	-
Safety	First Aid Station	$180	20 lb	-
Site Power	Portable Generator	$1,250	209 lb	-
Storage	Crate Stack	$60	55 lb	-
Storage	Filing Cabinet	$150	66 lb	-
Storage	Shelving Unit	$140	88 lb	-
Storage	Trash Bin	$25	11 lb	-
Water / Climate	Dehumidifier	$520	68 lb	620 W
Water / Climate	Utility Sink	$390	93 lb	-
Water / Climate	Ventilation Fan	$460	62 lb	380 W
Water / Climate	Water Storage Tank	$740	165 lb	-
Workshop	Machine Station	$2,400	485 lb	1,100 W
Workshop	Pegboard Bench	$260	132 lb	-

Workshop	Tool Chest	$320	121 lb	-
Workshop	Worktable	$180	99 lb	-

Each of the Creator’s designs comes furnished, and each furnishing is a package you could put in the cabin as it stands – priced here from the list above, item by item:

Fit-out of	Items	Price	Weight	Power
Woodland Square Mirror	4	$1,360	362 lb	380 W
Hex Cell Pavilion	5	$1,480	443 lb	0 W
Grow Dome	11	$1,675	756 lb	980 W
Woodland Hex Mirror	4	$1,730	425 lb	770 W
Treehouse Canopy Dome	5	$1,750	423 lb	950 W
Glass Studio Loft	11	$1,860	677 lb	280 W
Whole Trunk Lodge - 20 ft	9	$1,910	657 lb	1,050 W
Split-Log Homestead	12	$2,595	855 lb	1,050 W
Rebar Garden Dome	5	$3,620	717 lb	1,480 W
Timber Workshop	12	$3,995	1,173 lb	1,380 W
Concrete Monocoque Form	6	$4,160	902 lb	1,850 W
Continuous Steel Arc Hangar	6	$5,870	1,071 lb	6,500 W

The fit-out is where a small building’s money quietly goes. A cabin whose whole shell cost a few thousand dollars can carry a fit-out that costs as much again, and draw more power than a small solar array makes. Chapter 82 prices the utility column, the power bench and the water bench the wedge method actually builds; this list is everything else you might put on the floor.

PART 15

The Tools

“Every number in this book came out of a program about a second before it reached the page. Here are the programs.”

What ships with this book: the solver, the worlds, the cost model and the calculators, what each is for and how to drive it -- and the arithmetic that turns one number into every picture in these pages.

CHAPTER 79

The Software That Wrote This Book

Eight three-dimensional worlds, a cost model and a solver -- what each one is for, what it can do, and how to open it

Not one figure in this book was typed into a sentence.

Every dimension, count, angle, percentage and price is a token in the manuscript that gets filled in, at the moment the book is made, by the same software that draws the three-dimensional models. When this book says the dome is 19.42 feet across, that number came out of a solver about a second before it reached the page.

That software comes with the book. 11 programs — 5 worlds you can fly around in, 4 that are arithmetic with no picture, and the desks where the book and the films are made.

Why a builder would open any of it

The short answer: **because your tree is not my tree.**

Every number in these pages is for one specific dome, cut from one specific size of log, at one split count, with one orientation. Change any of those and every downstream figure changes — the chords, the cuts, the seam key, the timber, the price. The tables at the back are a snapshot of one solve.

The tools are how you take your own.

Three kinds of thing

Worlds are three-dimensional environments with a window, a camera and keys. This is where you go to *look* at something — to turn one decision and watch what it does to the joint.

Models are arithmetic with no picture. They answer a question and print a table: what does this cost, how long is that stick, what does the platform take off to.

Desks are where you make something: this book, the films, the calculators.

What each one is for

```
WORLDS -- things you fly around in

  The raw-wedge solver
      You want to see the dome this book is about, and turn
      one decision at a time to watch what it does to the
      joint.
      start it:
        py -3.12 geodesic_raw_wedge_dome_dihedral.py
      - solves the whole 2V frame from one number: the long
        member
      - four wedge orientations, and the seam changes under
        each
      - two seam keys, three things to put in the channel
      - explodes the panels apart without moving the geometry,
        so you can see into a seam
      - switches off the wireframe, the node markers, the
        debug lines and the sacrificial head stock -- which is
        what turns a screenshot into an illustration
      - exports a full-size panel drawing as SVG, to print and
        lay a stick on
      - carries the cost console, so the price is over the
        geometry rather than in another program
      keys: W A S D walk   mouse look   E explode the panels
            apart   I the mathematical wireframe on and off
            H the head-end stock on and off   J the
            fabrication jig   C the cost console

  The fabrication jig
      You are about to build the fixture and you want to see
      it assembled, one step at a time, before you cut the
      board.
      start it:
        py -3.12 geodesic_raw_wedge_dome_dihedral.py   (then J)
      - twelve steps from a bare bench to a finished panel
      - each step says what it is enforcing, not just what it
        is
      - the red and green fences that make a rotated stick not
        fit
      - the two cut planes: the flush-cut fence and the
        receiving face
      - the sacrificial tail drawn separately from the
        finished member
      - camera stations at each corner, because the corner is
        where the method either makes sense or does not
      keys: J show the jig   [ ] step through the twelve   V
            walk the camera round the corners   P which panel
            the jig is set for

  The Dome Creator
      You want a dome that is not this one: a different
      frequency, a different material, glass instead of ply, a
      different foundation -- and the price for each.
      start it:
        py -3.12 dome_creator.py
      - frequency 1V to 4V, and the truncation
      - eight frame materials, from bamboo to structural steel
      - eight strut cross sections including the quarter wedge
      - sixteen panel types: ply, glass, acrylic, twinwall,
        SIP, shingle, metal, solar, canvas, mirror, precast
      - nine cladding layers over the panels
      - seven foundations
      - a bill of materials for whatever you just built
      keys: F frequency   M frame material   P cycle the panel
            under the cursor   L cladding layers   G
            foundation   B print the bill of materials

  The seed world
      You want to see the product version: the utility column,
      the shower cap, the quilted layers and the mast, on the
      dome this book prices.
      start it:
```

```
        py -3.12 -m two_v_demo.app --lesson seed_pitch
      - the utility column and the gasketed cap at the apex
      - the shower cap, and the hats that stack under it
      - the mast through the column and the floor clamped to
        it
      - the floating rig: three cables to saddles on two trees
      - the composite member, drawn at an exaggerated section
        and labelled as exaggerated
      keys: 1-9 the chapters of the campaign film   mouse
            orbit   wheel zoom

  The pad and the park
      You are thinking about the ground: what a pad is, what
      it costs, and what a network of them looks like.
      start it:
        py -3.12 -m two_v_demo.app --lesson dome_park
      - a pad built one course at a time
      - amber for what the host owns, cyan for what the tenant
        owns
      - several domes on several pads, and the shared services
        between them
      - the rotating pad, which is a design possibility and is
        marked as one
      keys: mouse orbit   1-9 the chapters of the park film

MODELS -- arithmetic, no picture

  The geometry
      You want the numbers: chords, cuts, angles, areas,
      counts -- for your dome, not for this one.
      start it:
        py -3.12 -m wedge_book.numbers
      - two chords and two cut lengths at any diameter
      - the bite the pinwheel takes, which does not scale
      - face areas, seam runs, member volumes, timber totals
      - the whole builder's reference, regenerated

  The cost model
      You want to know what one costs, line by line, with
      every assumption named and changeable.
      start it:
        py -3.12 -m seed_model
      - every line carries its own source constant
      - materials, labour, overhead and warranty kept apart
      - the maker's markup stated as markup on cost, not
        margin on price, because those are different numbers
      - the pad priced separately, because it is the host's
        bill
      - fit-outs, shells and resins swapped without editing
        anything

  The platform take-off
      You are pricing the ground and you want boards and piers
      counted rather than a rate per square foot.
      start it:
        py -3.12 -m pad_deck
      - five constructions: gravel, slab, ring, deck, deck
        with ply
      - counted line by line, not rated
      - sized to the dome plus a walking margin

  The shell model
      You are deciding what to put over the frame, and you
      want the R-value and the price of each layer.
      start it:
        py -3.12 -m soft_shell
      - one watertight cap, and the quilted layers under it
      - each hat a size up from the one beneath
      - against a laminated hull, priced four ways
      - the concerns it will not hide, printed with the answer

DESKS -- where you make something

  The book studio
```

```
    You want to change this book: its outline, its prose,
    its figures, or the dome it is written about.
    start it:
      py -3.12 -m wedge_book.build
    - the outline is code, so reordering is an edit and not
      a rewrite
    - every number is a token resolved at build time
    - an audit that fails the build if a figure in the prose
      is one the code cannot produce
    - the pictures render from the solver and from the films

The film exporter
    Thirty-seven narrated films are built from the same
    geometry, and you can re-render any of them with your
    own numbers in it.
    start it:
      py -3.12 -m two_v_demo.app --lesson wedge
    - one lesson is one Python module: chapters, narration,
      scenes
    - landscape and phone cuts from one render
    - captions, chapter thumbnails and description copy as a
      set
    - a still from any second of any film
```

One codebase, several faces

Most of those programs are faces of one codebase, and they all read the same geometry module, so a strut length in one cannot disagree with the same strut in another. Four of them do most of the work behind Parts 9 to 12:

- **The Dome Creator** – the walkable site with a live bill of materials, where every design in Chapter 60 comes from.
- **The Dome Forge** – one dome taken apart like the layers of a paint program: frame, panels, drains, veins, ring, pipe, tank, rain, each a layer you can hide, fade, reorder or tune. Its default stack answers the oldest complaint there is, that domes leak: the panels dish inward and drain to one point each, a vein runs along the inside of every seam standing off the skin, and anything that gets past a joint lands in a channel instead of on the floor, runs downhill to a collector ring, and goes down a pipe into a cistern. The leak becomes the water supply.
- **The Assembly Line** – the factory model that prices every screw, every wage and every overhead, and turns a bill of materials into a planning price (Chapter 61).
- **The film engine** – which is what drew every plate in this book. Every frame it makes is a pure function of time, every scene is drawn by code, and every figure on screen is computed by a module that proves itself before a frame renders. Generative video cannot be asked to be correct. This can.

So the point of a film made this way is not that you have to trust it. It is that you do not have to: open the tools, change a preset, run it again, and every dome, dollar and labour hour has a path back to code.

The one that matters most

If you open exactly one thing, open the solver, and press the key that explodes the panels apart.

Everything difficult about this method is in the gap between two panels — the key, the angle, why the point faces inward, where the error goes. It is invisible in an assembled dome and obvious the moment you can see into a seam, and no drawing in this book, including the section in Chapter 20, teaches it as quickly as turning it in your hands for a minute.

How the numbers get from there to here

Worth stating once, because it is the unusual part.

The manuscript is Markdown with holes in it. A sentence that needs a figure names the number it wants – `dome.diameter_ft` in double braces – rather than printing one. At build time a resolver fills every one of them from the solver, the geometry and the cost model, and an **unknown token is a hard error** — the book refuses to build rather than printing a blank.

The reason is not showing off. A number typed into a paragraph is correct on the day it is typed and silently wrong from the first time anybody changes anything. Nothing fails. The book just quietly becomes untrue in places nobody remembers.

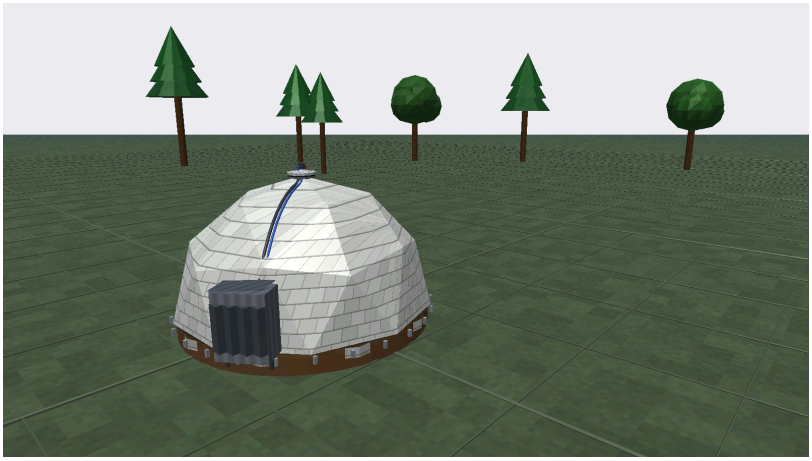
So the arithmetic in this book is a test suite. It runs, it checks itself against the geometry, and it fails loudly.

It caught four things

This is not theoretical. An audit reads the finished prose and checks every figure in it — every number carrying three or more decimals, and every dollar figure — against what the code can actually produce. It found:

- a set of reference designs that **scaled the pinwheel's bite with the dome**, which is the exact error Chapter 16 spends a page warning about. Every stick of the smallest dome was an inch too long;
- a triangle altitude printed as 52.520 where the solve says 52.516 — in the sentence that tells a reader whether plywood fits;
- four prices from before the shell lost a waterproof layer, still being quoted as current;
- a chapter quoting the cost of a whole upgrade as the cost of two of its three parts, double-counting the third.

Every one of those was written by somebody being careful. That is the point: care is not a method for keeping numbers true across time



What one costs, line by line.

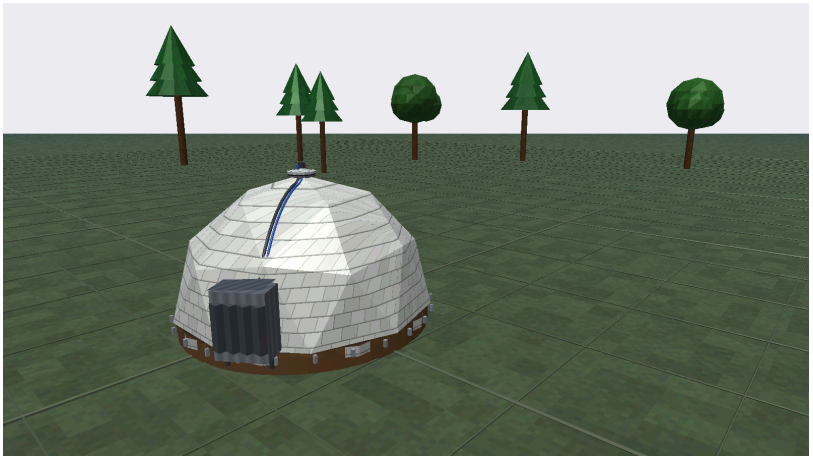
, and a program that re-checks them on every build is.

Change the tree and rebuild

The whole of it — geometry, prices, figures, this book — regenerates from one set of inputs.

Change the trunk diameter to what you actually felled. Change the split count to what your saw and your log will give you. Rebuild. Every chapter that mentions a figure now says your figure, without anybody going looking, and the pictures are redrawn from your solve.

That is what "the software comes with the book" means here. It is not a companion download. It is the thing the book is a printout of.



Three things this says against itself.

CHAPTER 80

From One Number to a Lit Pixel

How every picture in this book was drawn: a strut given a body, a camera placed with two angles, three matrices, one divide, one comparison and three dot products -- thirty times a second

Every picture in this book started as arithmetic. There is no model file behind any of them, no artist and no drawing: one number, a handful of formulas, and a list of instructions for turning the answers into coloured pixels.

Part 2 did the first half – the shape. One irrational number places twelve points (Chapter 7); a division puts them on a sphere; halving every edge makes more points in slightly the wrong place, and one push outward puts them right, which is what makes the shape geodesic (Chapter 8). This chapter does the other half – the picture – because a builder who knows how the drawing is made knows exactly how far to trust it. Seven stations, end to end:

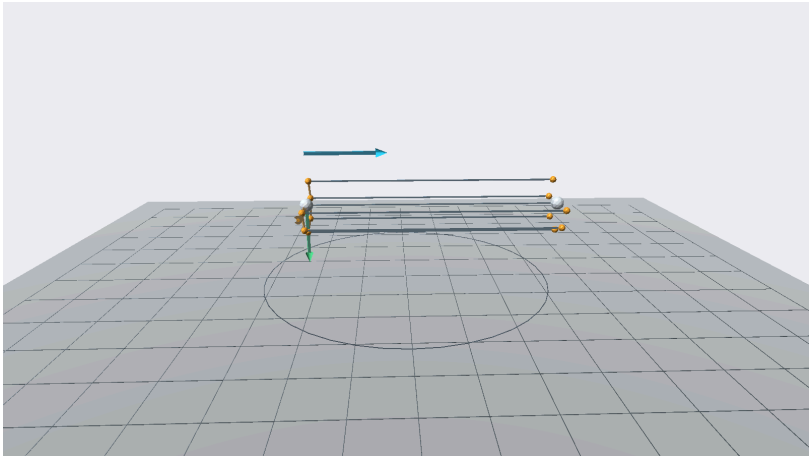
1. one number places twelve points;
2. a division puts them on a sphere of radius one;
3. every edge is halved;
4. the new points are pushed out to the sphere;
5. the shape becomes a mesh, because a graphics card draws only triangles;
6. a camera is placed, and the whole world is moved in front of it;
7. a divide, a stretch and a lighting sum turn all of it into a picture.

Every worksheet below was computed by the same code that drew the film it comes from, while the frame was being drawn.

A line with no thickness

The first surprise is that the shape cannot simply be drawn. A graphics card draws triangles, and that is very nearly its whole vocabulary – not even lines. A strut in the model is two points with no thickness, so each one has to be given a body: a tube of triangles wrapped round the line where the strut goes. To build a ring round a line you need two directions square to it, and the cross product hands you them.

```
a strut in the model is two points. A graphics card cannot
draw a line with thickness -- it draws triangles and nothing
else. So the line has to be given a body:
  d = (b - a) / |b - a|     the direction of the strut
  pick any t that is not parallel to d, then
  s = d x t   and   u = d x s      x is the cross product
  the cross product of two directions returns a third at
  right angles to both -- exactly what a ring needs
walk a ring of 8 points around the axis:
  ring(i) = a + r ( s cos(2 pi i / n) + u sin(2 pi i / n) )
  r is the strut radius, n the number of sides
do the same at b, then stitch the two rings together:
  8 side quads = 16 triangles
  2 end caps   = 12 triangles
  one strut    = 28 triangles
  x 65 struts  = 1,820 triangles of frame
```



Building one strut.

Every triangle also needs a front and a back. Without one it cannot be lit, because lighting depends on the angle between the surface and the light, and it cannot be hidden, because nothing could tell the inside of the dome from the outside. The direction a face points – its **normal** – comes from the cross product of two of its edges, and cross products care about order: list the corners one way round and the normal points out, the other way and it points in. One cross product answers three questions at once:

```
a triangle has to know which way it faces, or it cannot be
lit and cannot be hidden. Take its three corners a, b, c:
  n = (b - a) x (c - a)     cross product of two of its edges
  n = (-5.2153, +4.0021, +2.3011)
  n stands at right angles to the surface: the NORMAL
  divide by its length to make it exactly 1 long, which is
  what the lighting maths assumes:
  n_hat = (-0.748783, +0.574604, +0.330384)
the same cross product hands you the area for free:
  area = |n| / 2 = 3.482504 square world units
corner ORDER decides the sign. Listed counter-clockwise as
  seen from outside, n points outward:
  n_hat . (direction away from centre) = +0.999468
  list them the other way round and the dome turns inside
  out: lit from within, and thrown away as back-facing
```

That check is worth automating. A dome with one triangle wound backwards has a hole in it that only appears from one angle.

Then the elegant structure goes away. Corners, edges, faces, the classes of strut – none of it survives the trip to the card. What the card gets is one flat list of numbers, ten per vertex: three for where it is, three for which way it faces, four for its colour, the fourth being opacity. Three vertices in a row make a triangle, and that is the whole format.

```
by now the model is not points and edges any more. It is one
flat list of numbers. Ten of them per vertex:
  position x y z    where this corner is
  normal   x y z    which way its surface faces
  colour   r g b a  what shade it is; a is opacity
  10 floats x 4 bytes = 40 bytes per vertex
count the dome standing to the left of this panel:
  floats    = 117,000
  vertices  = 117,000 / 10 = 11,700
  triangles = 11,700 / 3  = 3,900
  bytes     = 468,000 = 457.0 KB
no corner is shared between triangles here: three fresh
  vertices per face. That costs memory and buys the freedom
  to give every face its own flat normal and its own colour
and that buffer is rebuilt and re-uploaded 30 times a second
```

That is smaller than a single photograph, and it is rebuilt and sent across 30 times a second.

Placing the camera

Before anything can be photographed, everything has to agree where it is. That agreement is **world space**: one chosen origin and three directions at right angles. Here, X and Y run along the ground and Z points up, the convention surveyors and architects use; the units are whatever we decide, because the geometry only cares about ratios.

A camera is placed the way a person thinks about one: how far round it is (yaw), how high up (pitch), and how far back. Turning those into a position is straight trigonometry:

```
the camera orbits the model. Three numbers place it:
  yaw      = 34.0 deg    how far around
  pitch    = 24.0 deg    how high up
  distance = 15.0      how far back
  target   = (0.00, 0.00, 2.25)   what it looks at
turn two angles and a distance into a position:
  eye_x = target_x + distance cos(pitch) cos(yaw)
  eye_y = target_y + distance cos(pitch) sin(yaw)
  eye_z = target_z + distance sin(pitch)
  cos(24 deg) = 0.913545   sin(24 deg) = 0.406737
  eye = (11.3605, 7.6627, 8.3510)
  check: |eye - target| = 15.0000 = the distance we asked for
```

Now the idea that surprises people most: **a graphics card has no camera.** It draws whatever sits in front of a fixed eye at the origin, looking down one axis, always. So instead of moving the camera to a good spot, we move the entire world so that the good spot lands on the origin. That is the **view matrix**, and it is three directions at right angles and one offset:

```
a graphics card cannot move a camera. It draws whatever sits
in front of a fixed eye at the origin, looking down one axis.
So we do not move the camera to the world. We move the world
to the camera. Build three directions from the eye:
  f = (target - eye) / |target - eye|      forward
  r = f x up_hint, normalized              right
  u = r x f                                true up
  f = (-0.757364, -0.510848, -0.406737)
  r = (-0.559193, +0.829038, +0.000000)
  u = (-0.337200, -0.227444, +0.913545)
  all square to each other: r . f = +5.6e-17
stack them as rows, eye offset in the last column:
  [ -0.5592,  0.8290,  0.0000,  0.0000 ]
  [ -0.3372,  -0.2274,  0.9135,  -2.0555 ]
  [ 0.7574,  0.5108,  0.4067,  -15.9152 ]
  [ 0.0000,  0.0000,  0.0000,  1.0000 ]
  that last column is -(row . eye): turn first, then slide,
  both in one multiplication instead of two
```

The number with “e-17” in it is not a mistake. It says the two directions are perpendicular to within the accuracy a computer can represent, which is as perpendicular as anything gets.

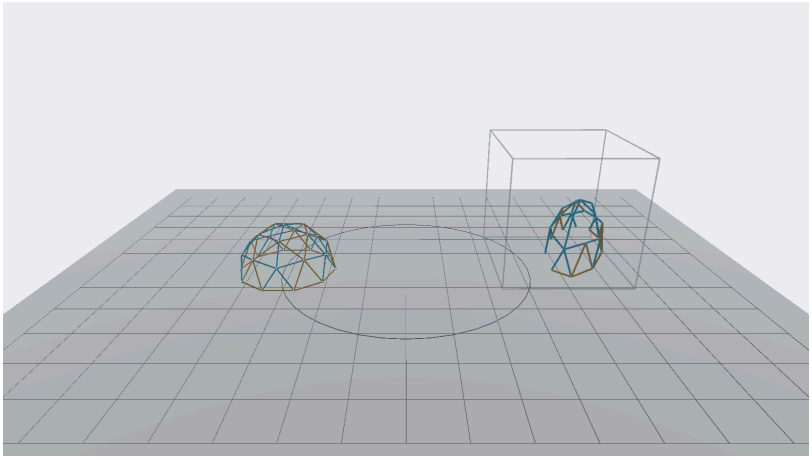
The camera cannot see everything. It sees a pyramid with its tip cut off – a **frustum** – inside a field of view of 48 degrees vertically, nearer than a far limit of 120 and further than a near one of 0.08. Anything outside is thrown away before it costs anything to draw. The projection matrix sets that up, and it is the matrix people find hardest, because it does almost nothing:

```
perspective is one idea: things look smaller the further off
they are, so divide by distance. The matrix sets up that
divide -- it does not perform it.
  field of view = 48.0 deg  (how wide the lens is)
  aspect = 1920 / 1080 = 1.777778
  near = 0.08, far = 120.0  (outside this range, nothing is drawn)
  f = 1 / tan(fov / 2) = 2.246037
  [ 1.2634,  0.0000,  0.0000,  0.0000 ]  <- x, scaled by f / aspect
  [ 0.0000,  2.2460,  0.0000,  0.0000 ]  <- y, scaled by f
  [ 0.0000,  0.0000,  -1.0013,  -0.1601 ]  <- z, remapped into near..far
  [ 0.0000,  0.0000,  -1.0000,  0.0000 ]  <- the whole trick
  the last row copies -z into w, the fourth slot. That is
  the only reason anything ever looks smaller with distance
  row 2's -1.001334 and -0.160107 are (far+near)/(near-far)
  and 2 far near/(near-far): they squeeze depth into 0..1
```

The projection matrix shrinks nothing. It loads each point's depth into a fourth slot, *w*, so that the next step can do the shrinking.

The divide that makes distance work

Now the divide. Every point's x, y and z are divided by *w*, its distance from the camera, so things twice as far away come out half as big. **That is perspective.** Not a special effect: a division that had to happen anyway to get the numbers into range. Everything the camera can see lands inside a cube from minus one to plus one, and one stretch turns the cube into the frame – flipped once, because screens count their rows from the top.

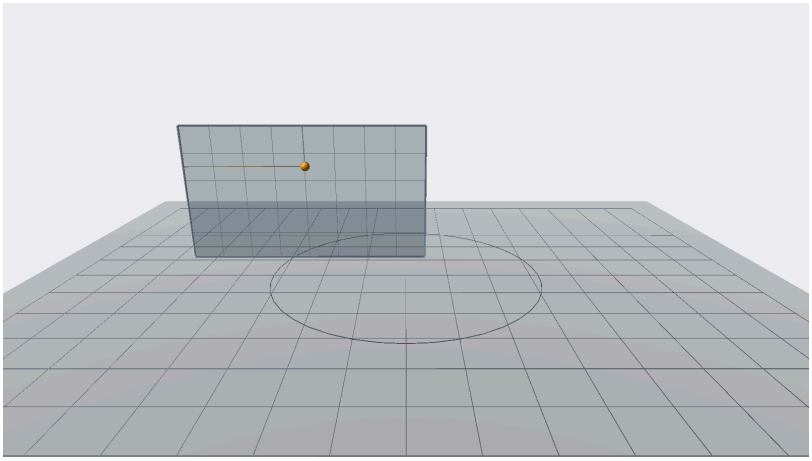


The divide that makes distance work.

Follow one point the whole way – the very top of the dome:

```
follow one point -- the very top of the dome -- all the way:
  world = (0.0000, 0.0000, 5.0000, 1)
  the 1 on the end marks it a position, not a direction
multiply by the view matrix (world -> the camera's frame):
  view  = (+0.0000, +2.5122, -13.8815)
  its z of -13.8815 is depth in front of the lens; compare
  the straight-line distance |eye - point| = 14.1070
multiply by the projection matrix (camera -> clip space):
  clip  = (+0.0000, +5.6426, +13.7399, w=+13.8815)
  and there is the depth again, sitting in w = 13.8815
now the divide. This is the moment perspective happens:
  ndc = clip / w = (+0.000000, +0.406485, +0.989800)
  everything still visible now lies inside a -1..+1 cube
stretch that cube across the frame:
  px = (ndc_x * 0.5 + 0.5) x 1920 = 960.0
  py = (1 - (ndc_y * 0.5 + 0.5)) x 1080 = 320.5
  y is flipped because screens count their rows downward
```

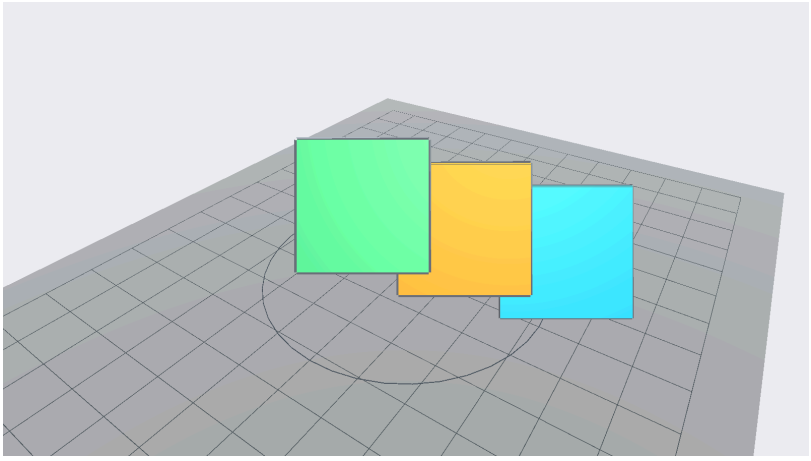
The apex lands exactly halfway across the frame, which is the check: the camera is aimed at the dome's centre, so its top has to sit on the vertical centre line. Every label you have seen floating over a strut in this book's plates was placed by exactly this calculation, run on the middle of that strut.



One vertex, all the way.

What hides what, and how bright

Two things want the same pixel. Which one wins? Real graphics cards do not sort – sorting thousands of triangles every frame is expensive, and two triangles can overlap in ways that have no correct order at all. Instead every pixel keeps a second number beside its colour: the depth of whatever is drawn there. A new fragment arrives, one comparison decides it, and it works whatever order the triangles come in.



What hides what.

There is a catch, and it explains a bug every 3D artist has seen:

```
the far side of the dome must not paint over the near side.
Nothing is sorted. Every pixel simply remembers its depth:
  if the arriving fragment is nearer, keep it; else drop it
  depth is stored 0 to 1, from near=0.08 to far=120.0
the remap is not even, because it comes out of that same
  divide by w:
    0.08 units away  ->  depth 0.000000
    0.16 units away  ->  depth 0.500334
    1.00 units away  ->  depth 0.920614
    5.00 units away  ->  depth 0.984656
   15.00 units away  ->  depth 0.995330
  120.00 units away  ->  depth 1.000000
  the first 0.08 to 0.16 units alone spend 50.0% of the range
  so precision is lavish up close and thin far away
  two far-off surfaces whose depths round to the same number
  flicker against each other: that is z-fighting, and pushing
  'near' further out is the usual cure
```

WE GOT THIS WRONG

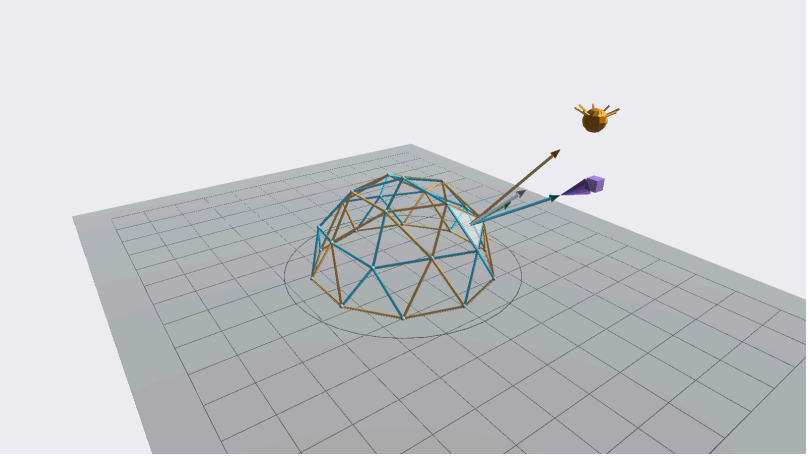
This project got that catch wrong for years without noticing. The film renderer never asked for a depth precision, so it took the driver's default: 16 bits. With these near and far limits that cannot tell apart surfaces about a metre apart at sixty metres out. Domes drawn on a dark plate never showed it. The first film to put decks a few centimetres above their ground came out in horizontal stripes across the whole site – z-fighting, at true scale, with nothing wrong in the geometry. The renderer now asks for 24 bits. Pushing the near limit out is the cure the table recommends; asking for more bits is the other, and it is the one taken, because every published film depends on the lens it already has.

And a free saving: on any closed shape, about half the surface faces away from you at any moment, hidden behind the near half. So before doing any work on a triangle the card checks its winding *on screen*, and one that comes out clockwise is facing away and is dropped – no depth test, no lighting, no pixels. That is why the corner order mattered four sections ago.

Then brightness. All the shading in this book's plates comes from three directions and the angles between them: where the surface faces, where the light is, and where the eye is. The dot product of two directions of length one is the cosine of the angle between them – one facing, zero side-on, negative behind – so a single dot product answers the question a painter answers by eye: is this surface turned toward the light?

```
shading asks one question per pixel: how much of this light
does this surface throw toward this eye? Three directions:
  n = the surface normal   (+0.6284, +0.4089, +0.6617)
  l = toward the light     (+0.4448, +0.5437, +0.7117)
  v = toward the eye       (+0.7254, +0.5070, +0.4656)
the dot product of two unit directions is the cosine of the
  angle between them: 1 facing, 0 side-on, negative behind
  diffuse = max(n . l, 0) = 0.972818
  that one number is Lambert's law -- square to the light is
  brightest, edge-on is dark, and nothing lights from behind
highlights use the direction halfway between light and eye:
  h = (l + v) / |l + v|
  specular = max(n . h, 0) ^ 42 = 0.644452
  the exponent 42 is gloss: higher means a tighter, harder highlight
  rim = (1 - max(n . v, 0)) ^ 2.5 = 0.000140
  rim brightens the silhouette, where the surface curves away
add them in the weights this film's shader really uses:
  lit = colour x (0.3 + 0.74 x diffuse)
        + cool tint x rim x 0.18
        + warm tint x specular x 0.22
  a face of base brightness 0.15 comes out at 0.2948
```

Everything you read as shape in those pictures is three dot products and a power. No shadows, no bounced light, no ray tracing.



Three dot products and a power.

Glass takes one thing back. The depth test let drawing order stop mattering; transparency makes it matter again, because a see-through surface has to be mixed with whatever is behind it, so whatever is behind has to be drawn first. The renderer works in two passes – everything solid, then everything transparent with depth-writing switched off – and the mixing is one line: the new colour times its opacity, plus the old colour times one minus the opacity. Opacity is that fourth number in every vertex, carried since the upload and used only now.

And then it does it again

Everything above – the tubes, the upload, the camera, three matrix multiplications and a divide per corner, a depth comparison and a lighting sum per pixel – happens once for one picture. Then it all happens again for the next, 30 times a second:

```
and then it all happens again for the next picture:
  30 frames a second = 33.33 ms per frame
  clear the screen, rebuild the batch, upload it, draw it,
  draw the text panel over the top, swap the buffers
  3,900 triangles inside that budget = 8.547 microseconds each
the vertex shader runs once per vertex, thousands at a time:
  11,700 vertex runs per frame
the fragment shader runs once per pixel covered:
  a full frame is 1920 x 1080 = 2,073,600 pixels
  = up to 62.2 million shader runs a second
  which is exactly why the lighting had to be dot products
and because every scene is a pure function of (chapter,
  progress), the same second of film renders identically on
  any machine, every time
```

That budget, 33.33 milliseconds, is the reason for nearly every decision in this chapter: why lighting is dot products rather than ray tracing, why half the triangles are thrown away before shading, why depth is one comparison instead of a sort. And because every scene in the films is a pure function of which chapter it is and how far through, nothing accumulates – the same second of film renders identically on any machine, every time, which is why the plates in this book can be checked by anybody who runs them.

That is the entire chain, from nothing: φ placed twelve points, two divisions made them a geodesic sphere with two strut lengths, half of it counted into a cut list; each strut became a tube, each triangle got a normal, all of it flattened into a list of numbers; two angles placed a camera, three matrices moved the world in front of it, one divide made distance work, one stretch found the pixel, one comparison decided what hides what, and three dot products decided how bright it was. No step in that chain is beyond somebody with a calculator. The cleverness is not in any one formula. It is that they compose, and that every single number can be checked.

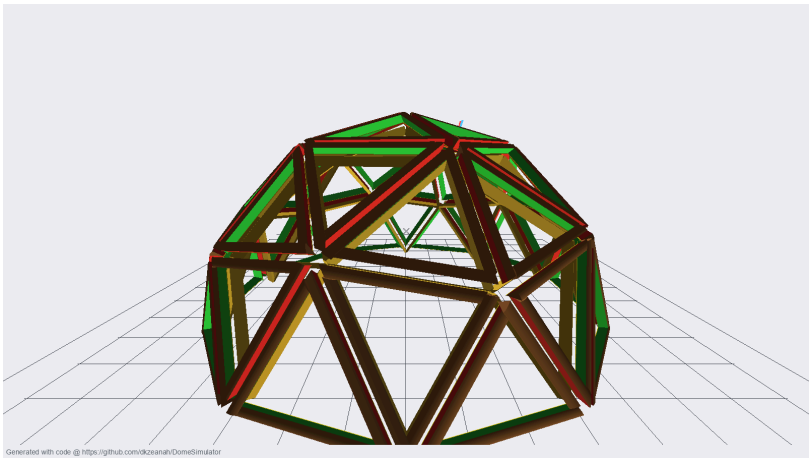
CHAPTER 81

What Each World Shows You

One picture from every three-dimensional environment in the project, with what it is for and what to do in it

One picture from every three-dimensional environment that ships with this book, what it is for, and what to do once you are in it.

The raw-wedge solver



The reference build, opened eight inches.

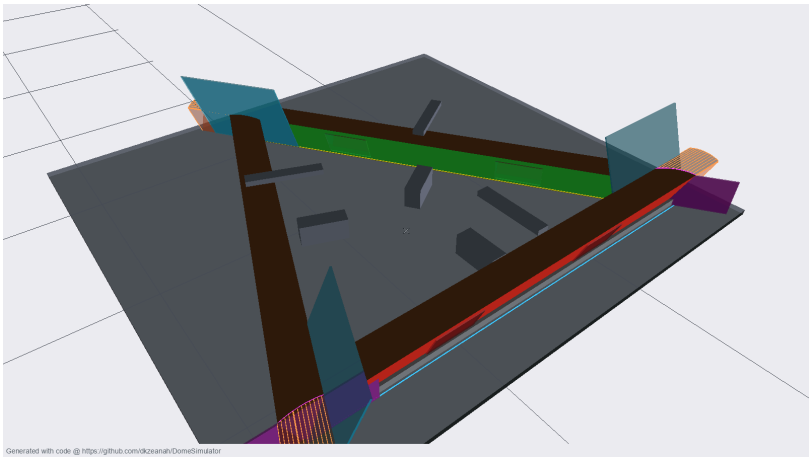
What it is. The dome this whole book is about, solved from one number: the long member. 120 members, 40 panels, 55 seams, every angle derived.

What to do in it. Press the explode key. The panels slide apart along their own face normals — nothing in the geometry moves, it is the same solve drawn with a gap — and for the first time you can see into a seam and watch the key sitting in it.

Then turn one decision at a time. Change the wedge orientation and watch the key's size change. Change the split count from 8 to twelve and watch the seam channel shrink to something that will not take a hose.

What it is not. It has no doorway, no foundation and no utility column, and it builds 2V hemispheres only. Those live in the other worlds.

The fabrication jig



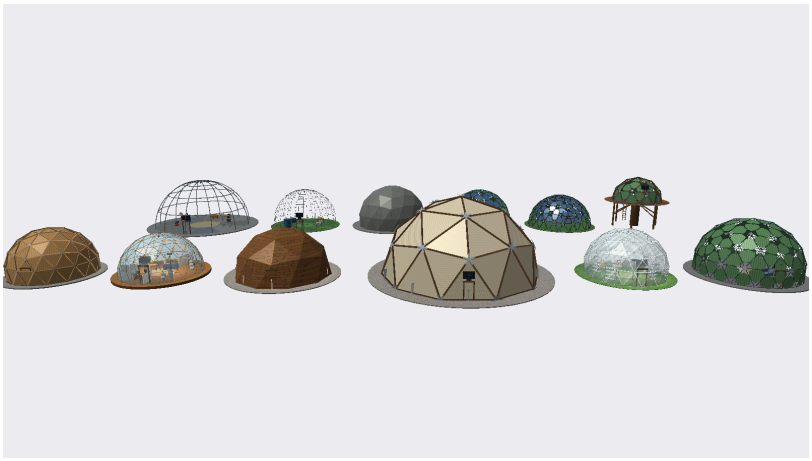
A panel closed on the jig.

What it is. The fixture of Chapter 24, in twelve steps, each one saying what it *enforces* rather than just what it is.

What to do in it. Step through all twelve before you cut a board. The corner stations are the ones worth flying to — the corner is where this method either makes sense or does not, and seeing the butt land on a radial face rather than on an end is the moment it clicks.

Then set it for the other panel family and step through again. The fixture does not change. That is the claim, and you can check it.

The Dome Creator

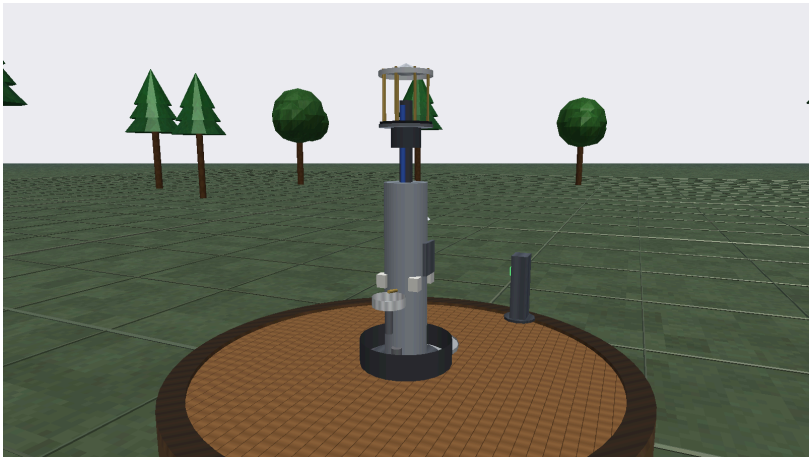


Twelve designs, drawn by the tool that builds them.

What it is. Every dome this project can describe that is *not* this one. Frequencies 1V to 4V, eight frame materials, eight strut sections, sixteen panel types, nine claddings, seven foundations — and a bill of materials for whatever you just assembled.

What to do in it. Build the dome you were going to build before you read this book, price it, then build this one and compare. That comparison is more persuasive than any argument in these pages, and it is yours rather than mine.

The seed world



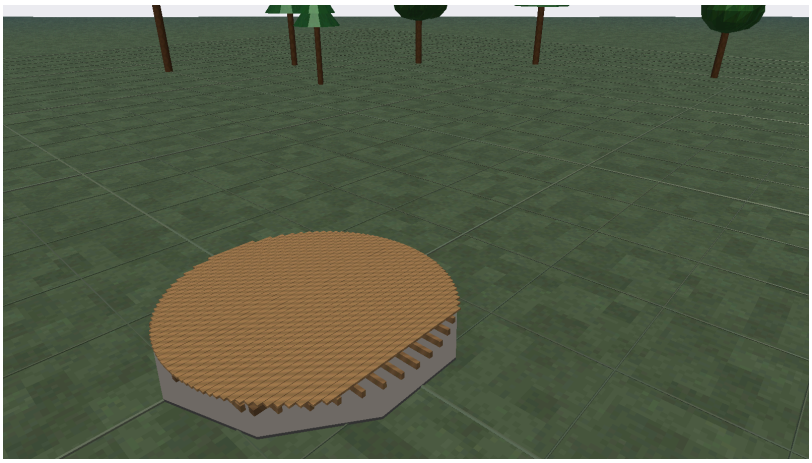
Power and water up the middle and out through the apex.

What it is. The product version: the utility column, the gasketed cap at the apex, the shower cap, the quilted layers, the mast and the floor.

What to do in it. Look at the column. It is the one object that touches the ground, the wall and the sky, and almost every decision in Part 5 and Part 7 of this book is downstream of it existing.

Then add layers to the shell and watch the cap grow. Chapter 30 is arithmetic on a page; here it is a thing changing size.

The pad and the park

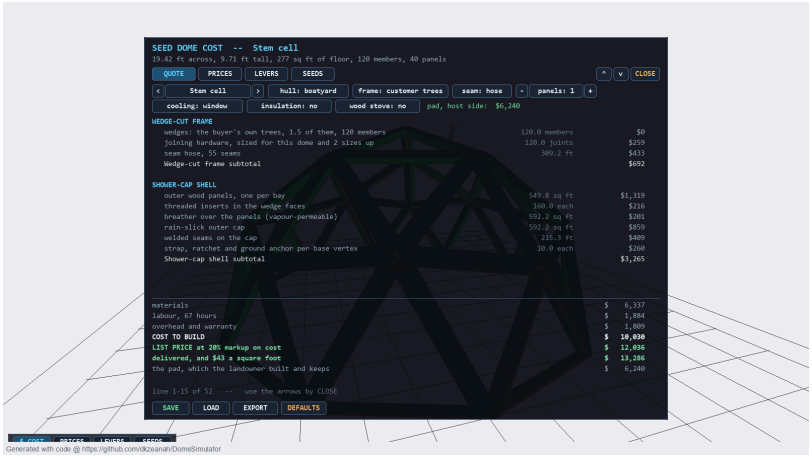


The platform, built one course at a time.

What it is. The ground of Chapter 39: a pad built course by course, and a network of them with the shared services between.

What to do in it. Watch the colours. Amber is what the host owns, cyan is what the tenant owns, and the clean split of Chapter 45 is something you can see rather than something you have to be told.

The cost console



The geometry and the price in one program.

What it is. The cost model, open over the solved dome. Every line carries its own source constant, and changing an assumption moves every figure downstream of it immediately.

What to do in it. Find a price you disagree with — there will be one — and change it. The whole quote re-derives. That is the right way to argue with this book: not by doubting the total but by editing the input you think is wrong and seeing what it was worth.

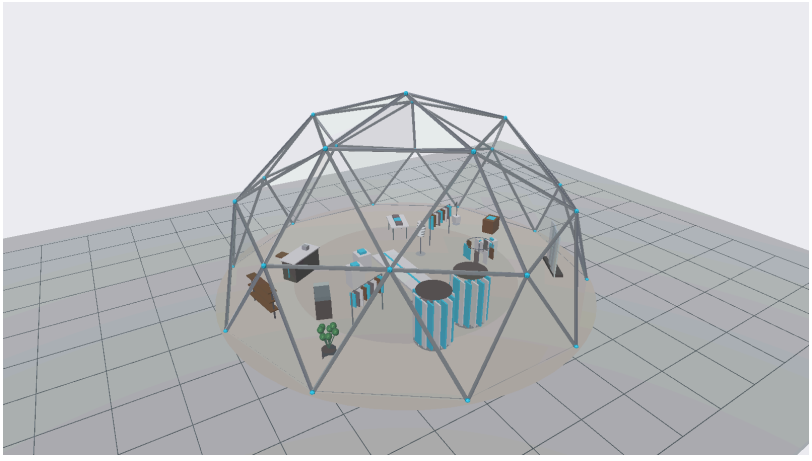
The number to look at first. $12,036 for the standard article, built to $10,030, at $43.45 a square foot of floor. The maker's markup is stated as a markup on cost rather than a margin on price, because those are different numbers and the one a reader can check with a calculator is the first.

Where to start

If you have an afternoon: the solver, exploded, then the jig's twelve steps.

If you have a tree: the cost console, with your own trunk diameter typed into it.

If you have ground: the park.



The same shell, furnished.

CHAPTER 82

Every Part, Priced

The column, the pad, the hardware, the power bench and the water bench -- the whole bill, with a source on every line

Six bills. Every line carries the constant it came from, so any of them can be argued with by editing one number rather than by doubting the total.

What is separate from what, and why

The dome is $12,036, and it is priced in Chapter 63 and the chapters around it. Nothing on this page changes it.

The pad is the host's bill, not the buyer's — Chapter 39. It is here because somebody has to build it and because a book that prices only half of an arrangement is not pricing the arrangement.

Everything else on this page is an upgrade. A dome works without the seam module, without the solar bench and without the water bench; it sits on a pad's pedestal and drinks from the pad's stub. These are what you fit when you want it to work with nothing plugged into it.

Added up, the systems on this page come to $20,525.

The utility column, complete

Including the steel, which is the part most bills leave out.

column housing, insulated chase	12.0 ft	$6.00	$72
steel core, service port to apex	12.0 ft	$18.00	$215
base flange onto the service port	1.0 each	$95.00	$95
apex sleeve and gasketed seal cap	1.0 each	$240.00	$240
apex lifting ring	1.0 each	$260.00	$260
floor hub that clamps the mast	1.0 each	$310.00	$310
radial steel floor spokes	10.0 each	$34.00	$340
sub-panel, breakers and outlets	1.0 each	$310.00	$310
water manifold and rise	1.0 each	$240.00	$240
drain stack and floor-port tie	1.0 each	$165.00	$165
Utility column, complete			**$2,247**

Two lines there are doing more than they look.

The steel core runs the full height, floor port to apex ring, and it is what the floor clamps to and what the lifting rig hangs from — Chapter 72. It is a pipe. It is also the single component that makes a floating dome possible, and it costs $622 in a $12,036 building.

The radial floor spokes are the 10 steel arms from the hub out to the base ring. They are what make the floor a component rather than framing, which is Chapter 3's third feature.

The power bench

solar array	800.0 W	$0.95	$760
MPPT charge controller	1.0 each	$195.00	$195
battery bank, LiFePO4	10.0 kWh	$410.00	$4,100
DC energy meter and shunt	1.0 each	$62.00	$62
Power bench			**$5,117**

A 8-way split log frame does not need electricity, but the condensing plates and the barrier fan of Chapter 29 do, and so does everybody who lives anywhere.

The array is sized at 800 W and the bank at 10 kWh, which is the owner's stated arrangement rather than a calculation from a load list. The metering is in the bill deliberately: a system you cannot read is a system you cannot size next time.

The water bench

under-floor tank	120.0 gal	$1.35	$162
first-flush diverter	1.0 each	$68.00	$68
three-stage filter	1.0 each	$210.00	$210
inline UV lamp	1.0 each	$165.00	$165
on-demand pump and accumulator	1.0 each	$145.00	$145
pulse-output water meter	1.0 each	$78.00	$78
Water bench			**$828**

This is what turns 309 feet of seam gutter into something you can drink.

The first-flush diverter is the cheapest item and the one that matters most. It throws away the first gallons off a dry roof — the ones carrying whatever has landed on the cap since the last rain — and it is the difference between roof water and drinking water.

The UV lamp is the other one. A filter makes water clear; a lamp makes it safe. Neither does the other's job.

Joining hardware, stainless

A2 stainless bolt, nut, two washers	440.0 sets	$0.92	$405
threaded insert in a sawn face	880.0 each	$0.74	$651
flat V bracket, four holes a flap	120.0 each	$5.80	$696
Joining hardware, stainless			**$1,752**

Edge to edge, into threaded inserts rather than into end grain — Chapter 3.

Stainless throughout is not a luxury here. Chapter 38 notes that wet timber and plain steel together are a battery; on a frame whose whole argument is that it lasts decades and gets taken apart, a corroded fastener is the one failure that turns a demountable building into a permanent one.

The V bracket

The flat V bracket is the connector this method settled on: two flaps meeting at the panel's angle, **4 holes in each flap**, folded from stainless on a bench brake.

The flaps run out to as much as **half the member's length** — 35.1 inches on a long member — which puts about 8.8 inches between holes. That spacing is the point of the thing. Screws clustered at the end of a stick are all working the same short grain; spread along it, each one is into fresh wood, and the joint stops depending on any single fixing.

At $5.80 each, 120 of them is not the expensive part of anything.

The pad’s own materials

compacted gravel base over fabric	369.2 sq ft	$1.20	$443
deck boards, 2x6 laid flat	902.2 ln ft of 2x6	$1.17	$1,053
joists at 16 in centres	310.1 ln ft of 2x6	$1.17	$362
beams, doubled 2x6	179.3 ln ft of 2x6	$1.17	$209
precast piers	20.0 each	$12.00	$240
hangers, structural and deck screws	369.2 sq ft	$0.55	$203
two coats of penetrating sealer	369.2 sq ft	$1.10	$406
Pad materials (framed deck on piers, sealed)			**$2,916**

The host’s bill, counted board by board rather than rated per square foot — which is why it can be checked against a merchant’s quote.

What this page does not include

Labour, which is Chapter 63’s territory and is 67 hours on the standard article.

Tools. A chainsaw, a circular saw, a drill and a bench brake, none of which are consumed by one dome.

And freight, permits, and the ground itself, all of which are local and none of which this book can know.

CHAPTER 83

Every Piece, Numbered

The calculator that fills a triangle and frames the floor: every piece numbered, every cut set up on the saw before the blade turns, ten ways to fill one panel, seven ways to frame the platform under the dome from five 20 ft half trunks -- every span and pier checked, and the patterns that fail shown failing

Two things in this building are made of a great many small pieces of wood, and both are where a builder loses days. One is the inside of a triangle, when it is filled with wood instead of covered with a sheet. The other is the floor the whole dome stands on. Neither is hard, piece by piece. What makes them slow is that no two pieces are alike: each one has its own length along each edge, its own two end cuts, its own place, and a builder who works them out at the saw works them out forty times, and gets some of them wrong.

This chapter is about the calculator that works them out first. It is in the launcher, called **Triangle Infill**, and it opens in a window of its own: a triangle, or the floor, in three dimensions, every piece in it numbered. Click a piece and it gives you that piece: which stick it comes out of, every length to mark, and every cut, set up on the saw the way you will make it – which face goes down on the table, what goes against the fence, the mitre and which way it swings, the bevel and which way it tilts – and then it plays the cut, the blade coming down and the offcut falling away, so you can see what is about to happen before it happens to a real stick.

Seventeen wedges in one triangle, every one numbered.

Everything in it is solved from two numbers you already have, the dome's longest stick and the log it is split from, except what cannot be solved. That is written down, every line with its reason, in two tables: 34 figures for the triangles – saw kerfs, the sizes sheets and boards are sold in, densities, prices – and 33 for the floor.

Figure	Value	Kind	Why
live_psf	40 lb/ft2	standard	floor live load for a dwelling (IRC table R301.5)
dead_psf	10 lb/ft2	estimate	decking, framing and fittings spread over the floor
fb_psi	875 psi	assumption	allowable bending stress: No. 2 green softwood order of magnitude. Your logs are ungraded, so this is a guess, not a rating
e_psi	1.2e+06 psi	estimate	stiffness (modulus of elasticity) of green softwood
deflect_ratio	360 L/x	standard	a floor may sag no more than span/360 under live load
soil_psf	1500 lb/ft2	standard	presumptive bearing of ordinary soil (IRC table R401.4.1); each pier's gravel pad is sized from its load with it

pad_min_in	16 in	assumption	smallest gravel pad under a stump pier, square
pier_dia_in	12 in	assumption	stump piers cut from a 12 in log
clearance_in	6 in	assumption	air under the lowest member when the floor is on piers
dome_overhang_in	6 in	assumption	how far the dome's frame and skin stand past its sphere radius at the base (the simulator's 8 in-trunk dome shows 3.8 in)
node_tol_in	0.5 in	assumption	member ends closer than this are one joint
min_piece_in	12 in	assumption	joists shorter than this are left out; block the gap instead
lumber_max_in	192 in	nominal	the longest common lumber: 16 ft
lumber_kerf_in	0.125 in	nominal	a mitre-saw blade
saw_height_in	6 in	nominal	tallest piece a 12 in sliding mitre saw cuts standing on edge
saw_depth_in	12 in	nominal	widest crosscut of a 12 in sliding mitre saw
trim_in	1 in	assumption	the first cut on a log takes this much past its raw end
deck_2x6_span_in	24 in	standard	2x6 deck boards laid across their supports
deck_54_span_in	16 in	standard	5/4 x 6 deck boards laid across their supports
deck_ply_span_in	24 in	standard	23/32 in tongue-and-groove plywood or OSB subfloor
deck_puncheon_span_in	48 in	assumption	half logs flat side up, 10 in or more across
deck_waste	0.1	assumption	decking cut-off and waste
usd_4x4_ft	1.6 $/ft	estimate	ground-contact treated 4x4
usd_6x6_ft	4.5 $/ft	estimate	ground-contact treated 6x6
usd_2x6_ft	1.25 $/ft	estimate	ground-contact treated 2x6
usd_2x8_ft	1.7 $/ft	estimate	ground-contact treated 2x8
usd_2x10_ft	2.3 $/ft	estimate	ground-contact treated 2x10
usd_2x12_ft	3.1 $/ft	estimate	ground-contact treated 2x12
usd_deck_ft	1.1 $/ft	estimate	2x6 or 5/4 x 6 deck board
usd_log	0 $	assumption	a 20 ft log from your own trees; set a price if you buy them
usd_hanger	2.5 $	estimate	one galvanised joist hanger with its nails
usd_pier_pad	6 $	estimate	gravel for one pier pad
usd_gravel_yd3	40 $/yd3	estimate	a cubic yard of crushed gravel, delivered

The floor's table is the one that matters most. It holds the load a floor must carry, 40 pounds a square foot for living on and 10 for the floor itself, which are standards; and it holds how strong the wood is taken to be, 875 pounds per square inch in bending, which is a guess. Change any line, and every number after it in this chapter, and in the calculator, changes with it.

A triangle is a tray

A dome whose longest stick is 72 inches is 22.6 feet across, and its forty triangles come in two shapes: 30 with a long base of 83.69 inches and two short sides of 74.01, and 10 with three long sides. Fill one of each well and you have filled the dome.

The opening is not a flat hole. Each frame member is a wedge, an eighth of a 12 inch log, lying on one sawn face with its point toward the middle, the way chapter 20 builds it. Its other sawn face rises at forty-five degrees and leans back over it. So the opening is a tray: smallest at the flat side, where the members' points are, widening up the slope to the open side, 4.24 inches above. Every piece that fills it has to rest on that slope, which is why no piece in it has a square end.

And the triangles next door fold away from this one, at 18.03 and 22.46 degrees. A filling stops at the frame. A covering does not: it runs over the frame and meets the next panel's covering at the seam, so it has to lean to meet it.

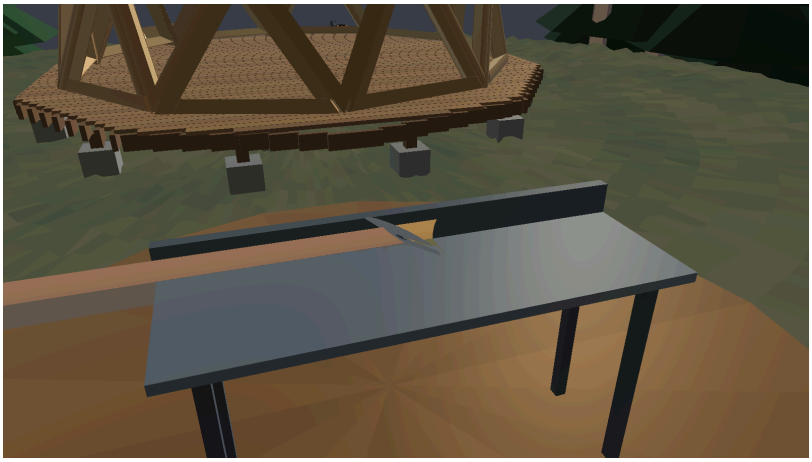
One wedge, start to finish

Take the obvious way to fill a triangle with wood from the same log as its frame. Split it into wedges, lay them in stripes parallel to the long base, and flip-flop them: one point up, the next point down, so each pair nests into a strip with two flat faces. A 12 inch log fills the long-based triangle with 17 of them, from 75.6 inches long down to 5.9. Each one stops where it meets the forty-five degree wall at both ends, so every end is a compound cut, and no two are the same length.

The calculator follows each one from the stick to the panel. Here is piece 5, start to finish.

Its lengths. A wedge has three long corners – the point, and the two bark corners – and because its ends lean, each is a different length: 58.38 inches along the point, 50.56 along one bark corner, 43.08 along the other. Those three numbers are what you mark. A fourth number, the length to take from the pile, is the longest of them plus a little to trim.

Cut one. Lay the stick on its sawn face on the mitre saw, the bark corner against the fence, the keeper to the left of the blade. Swing the mitre to 34.4 degrees and tilt the blade 45, and take just enough off the raw end to clear it everywhere. What falls away is a tapered sliver, 8.15 inches at its longest and 0.50 at its shortest.



Cut one: sawn face down, bark corner to the fence.

That setting is not a rule of thumb. The calculator found it by trying every face of the stick on the table, with each end toward the fence, and working out for each what the saw would have to do: how far to swing, how far to tilt, and which way. A single-bevel mitre saw tilts one way only, and swings about fifty-five degrees either side. Of every way that works on that saw, it keeps the one that lies flat and steady and puts the point where you can see it.

Cut two. Turn the stick end for end, its point against the fence, and measure from the first cut: 58.38 inches along the point, and the other two corners to their own lengths. Join the marks across the top face, and the blade follows the line, at the same 34.4 and 45. This offcut is 20.77 inches long. That is a piece in its own right, and it goes back on the pile for a shorter stripe.

Cut two, and an offcut long enough to keep.

Check it, and put it in. Measure the three lengths against the list, chalk the number on, and it goes into the panel in its stripe, its point down or up as the list says.

Sixteen more like it, each with its own numbers, and the panel is full.

When the saw cannot reach

Not every cut fits a mitre saw. Cover the panel instead of filling it, with a sheet of oriented strand board, and the calculator cuts it in 2 halves to fit a sheet, and gives each edge a bevel so it meets the next panel's covering: half the fold at that seam, which is 9.0, 11.2 and 11.2 degrees.

Those cuts run the length of a sheet, and no mitre saw takes a sheet. So the calculator says so, and gives the cut another way: the corners measured in from the sheet's own corner, a chalk line snapped between each pair, and a circular saw run along the line with its blade tilted to the bevel. The same happens to any cut steeper than the saw can swing or tilt: it is marked on the stick and cut by hand, and the calculator says where each mark goes.

Ten ways to fill one triangle

Wedges in stripes are one way. The calculator does many, and lays out each with every piece numbered and cut. Here are 10 of them in the long-based triangle, and how many pieces each takes.

Material	Pattern	Lay	Fill or cover	Pieces
Wedges (1/8 log)	Stripes parallel to one edge	Flip-flop	fill	17
Wedges (1/8 log)	Chevron to the centre line	Flip-flop	fill	18
1/16 slices	Stripes parallel to one edge	Point-down / point-up	fill	60
Wedges (1/8 log)	Concentric pinwheel rings	Flip-flop	fill	15
Milled 1x6 boards	Stripes parallel to one edge	Laid flat, gapped	fill	9
Fence pickets	Louvers (vent slats)	-	fill	13
Log rounds (whole-trunk slices)	Honeycomb	Layers aligned	fill	139
OSB sheet (7/16)	One piece	-	cover	2
Corrugated sheet metal	Lapped strips	-	cover	3
Greenhouse film (6 mil)	One piece	-	cover	1

One of the ten: the same triangle, another fill.

Log rounds are the odd one out, and the most work. They are slices across a whole trunk, 1.5 inches thick, laid in a honeycomb 3 layers deep. 18 go in whole and 38 are cut where they meet the wall. That leaves gaps no round fits, and the calculator fills them the way a carpenter would: lay another round over the gap, trace the gap onto it from underneath, and cut to the line. There are 83 of those, so the

panel takes 139 pieces in all.

Every pattern can be turned over, too. The face the room will see is the flat side, where the frame's points are, and in a flip-flopped fill every other wedge shows its point there. It is worth looking at before you cut anything, because it is the face you will live with.

From underneath, the face the room will see.

Five trunks, one pentagon

The floor is the same kind of work on a bigger scale, and it starts with wood already on the ground: five half trunks, twenty feet long. Stood on edge, round back out and flat face in, and joined at the corners, they make a pentagon whose backs measure the full 20 feet.

How big a floor is that? The largest circle that fits inside a pentagon touches the middle of each side, at the apothem: the side over twice the tangent of thirty-six degrees, 13.76 feet here. Measured to the flat faces, inside the half trunks, the circle is 13.26 feet in radius. The dome needs 11.78 feet: its sphere, plus 6 inches for its frame standing out at the foot. It fits, with 1.48 feet to spare at the narrowest, and the five corners of the pentagon are left over outside it, which is room for steps, a woodpile, or a bench in the sun.

Five half trunks, round backs out, round the dome's centre.

Every corner is the same cut. A pentagon turns seventy-two degrees at each corner, and each trunk takes half of that: its end is cut 36 degrees off square. Along the round back a trunk keeps its full 240.00 inches. Along its top, where the flat face starts, each cut takes 4.36 inches, so the top is 231.28 inches long. No mitre saw takes a log that size. Mark the line across the top, plumb it down both sides with a level, and cut to the lines with the chainsaw.

Thirty-six degrees off square, marked and plumbed for the chainsaw.

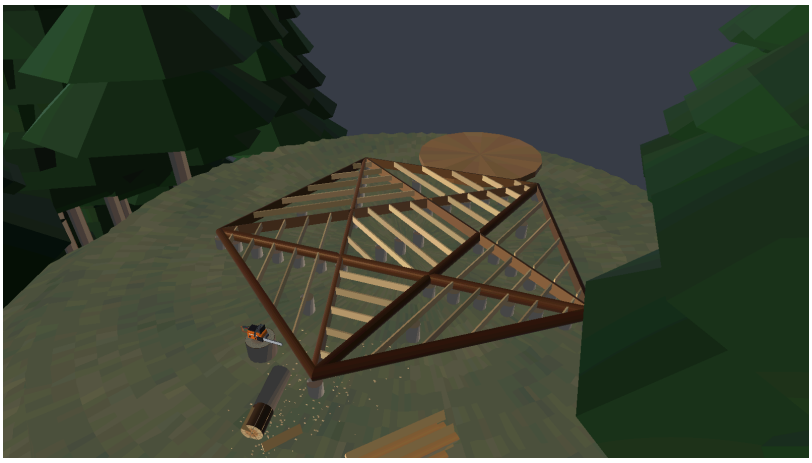
A star, and why it needs joists

What goes inside? The first idea is a five-point star, corner to corner, because a star braces a pentagon across every corner and is made of only five lines. Each line crosses two others, so each is cut in three, and in half trunks the star is 15 pieces: the outer ones 148.3 inches, the inner ones 91.7, mitred where they meet.

But the star alone fails, and it fails at the deck, not the frame. It leaves the floor in large triangles, and something has to bridge them. The widest gap a deck board would have to cross under the star alone is 119 inches. A two by six deck board may cross 24.

The star alone: the gaps a deck board cannot cross.

So every space between the points is filled with joists: 2x8 on edge, 24 inches apart, each one running across its space the short way, cut to meet the star or the frame where it lands. There are 55 of them, and now the widest gap is 22.5 inches, inside the 24 a deck board may cross.



Joists between the points.

A joist shows the other thing the saw has to be told. It meets a star line at an angle, so its end is cut off square, and a two by eight standing on edge is 7.25 inches tall: more than the 6 inches a twelve-inch sliding saw cuts standing up. So it lies flat on the table, and the angle moves from the mitre to the bevel: mitre zero, bevel 18 degrees. Its other end, 54.0° off square, is past the 46 degrees the blade tilts, and is marked and cut by hand.

Where the weight goes

A floor has to carry its load somewhere, and the calculator follows every pound of it. 40 pounds a square foot for living and 10 for the floor, over the whole pentagon, is 34,417 pounds.

Each patch of floor goes to the member nearest it, because that is what the deck spans to. The joists hand their load to whatever they hang on, half at each end. The star and the frame hand theirs to the piers under their joints. Add up what reaches the piers and it is 34,417 pounds: every pound in is a pound out, which is the first thing to check about any load path, and the calculator checks it.

> **THE SPAN CHECK**
>
> A member between two supports bends under its load, and the wood takes it as long as two numbers stay small enough. The bending moment, the load per inch times the span squared over eight, divided by the section's strength in bending, has to stay under 875 pounds per square inch. And the sag, five times the load times the span to the fourth, over 384 times the wood's stiffness and the section's stiffness, has to stay under the span over 360. Both are worked out for every span of every member, for whatever section it is: a half trunk on edge, a quarter, a whole log or a board.

Where a span fails, the calculator puts a pier under it, as many as it takes. Under the star and joists it took 8, and now the hardest-working span in the floor works at 94 percent of what it can carry.

That makes 43 stump piers, rounds of log standing on tamped gravel, and each pad is sized from its own load, at 1,500 pounds a square foot of ordinary soil. Most need only the smallest pad, 16 inches square. The heaviest carries 2,902 pounds, and needs 18. The 60 joist ends that hang on a member rather than a pier get a joist hanger where they meet sawn lumber, or a notch and a pair of spikes where they meet a log.

Seven ways to frame the floor

The star with joists is one way. The calculator builds seven, from the same trunks and the same joists, checks each the same way, and adds piers where each needs them. On piers, with a two by six deck:

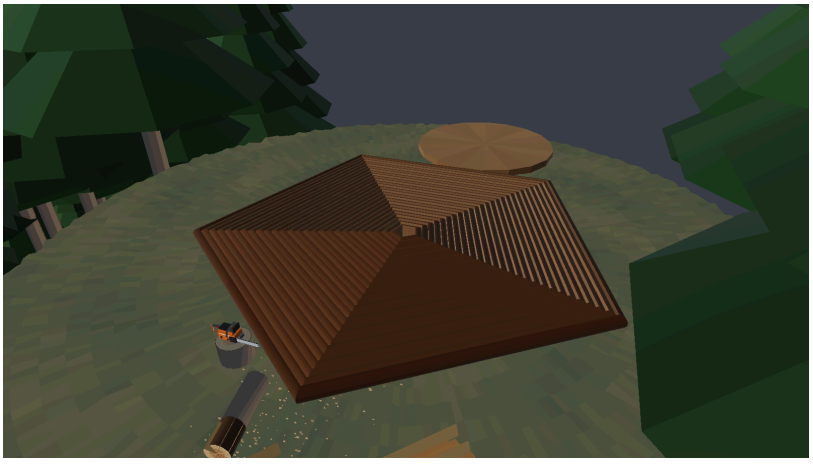
Pattern	Pieces	Piers	Hardest span	Deck gap	Logs	Bought
Star + joists	75	43	94%	22.5 in ok	7	$2,598
Star alone	20	10	85%	118.5 in FAILS	6	$1,798
Wheel: 5 spokes + rings	45	26	96%	22.5 in ok	5	$2,571
Sunburst: 10 spokes + rings	75	46	92%	22.5 in ok	7	$2,640
Fan: 5 spokes + joists square to the walls	56	26	97%	22.5 in ok	5	$2,672
Beams + joists	74	42	68%	22.2 in ok	6	$2,704
Nested rings	130	130	59%	16.8 in ok	38	$2,518

They all carry the floor; they do not all carry it the same way. Beams with joists between, the way a deck is usually framed, works its hardest span at only 68 percent. Only the nested rings have more to spare, 59 percent, and they take 130 piers to get it. The fan sits at the other end: five spokes meet at one centre pier, and that pier carries 7,551 pounds, on a pad 28 inches square. The wheel's centre carries 5,090. Bring the spokes to one point and you bring the load there too.

On a levelled gravel bed nothing spans at all. Every member bears along its length, the piers go away, and the gaps can be packed with gravel, earth or cob instead of decked.

Pattern	Pieces	Logs	Bought
Star + joists	75	5	$1,563
Star alone	20	5	$1,166
Wheel: 5 spokes + rings	45	3	$1,694
Sunburst: 10 spokes + rings	75	5	$1,592
Fan: 5 spokes + joists square to the walls	56	3	$1,716
Beams + joists	74	4	$1,680
Nested rings	130	34	$342

That is where the nested rings belong, the pattern this whole floor began with: 25 rings of half trunks inside the frame, each ring's back pressed against the flat face of the ring outside it, every corner the same 36 degree cut. There is a measuring rule that makes it easy: the length of the next ring's piece, along its round back, is the length of the flat face of the ring you just set. It costs about 342 dollars of gravel and nothing else bought. But it eats logs, 34 of them, against 5 for the star with joists.



Nested rings on a gravel bed, the gaps packed.

And the materials can change, member by member. Keep the star and joists, change what the star is made of, and the floor changes round it:

The star in	Piers	Added mid-span	Logs	Bought
Half trunk on edge (round back out)	43	8	7	$2,598
Raw trunk (whole log)	43	8	12	$2,564
Quarter trunk (sawn faces up and in)	48	13	6	$2,628
6x6	48	13	3	$3,352
4x4	58	23	4	$2,968

Whole logs carry what half trunks carry but take 12 logs. Six by sixes are the strongest bought section and the dearest, about 3,352 dollars. Four by fours need 58 piers, 23 of them propping spans a four by four cannot carry.

The dome on its floor.

What these numbers are not

The loads are a dwelling's, and they are a standard. The wood's strength is not. These are your logs, not graded lumber, and 875 pounds per square inch is a careful guess: no bonus for size, none for joists sharing a load, stricter than a lumber span table. It is why some of these floors carry more piers than a builder would expect. The prices are estimates, and the logs are counted at nothing because they are your own trees.

So this is a floor that is checked and priced, not rated. Before people live on it, an engineer signs it, with the wood graded or tested. What the calculator gives the engineer, and you, is every piece numbered, every cut set up, every span and every pier worked, and the patterns that fail shown failing, so the conversation starts from the right floor.

Glossary

2V
Two-frequency. Each face of an icosahedron is divided in two along every edge, giving the dome in this book: two strut lengths and two triangle shapes.

3V
Three-frequency. Each face of an icosahedron is divided into nine triangles, three to an edge, before the points are pushed out to the sphere. It has three strut lengths.

apex
The top point of the dome, where the crown star's spokes meet. The only opening in the weather surface is here, closed by the seal cap.

base ring
The ten-sided ring of members the dome stands on, resting on the rim of the pad.

base section
One of the boards a panel dome's bottom panels sit on. On a five-eighths dome some must be tapered.

breather
A vapour-permeable sheet over the panels. It sheds water that gets past the cap and lets moisture out, which a plastic sheet would not.

chord
The straight line between two points on a curve. A wedge's flat face is a chord sawn across the bark side of its slice of log.

desiccant
A material that pulls water vapour out of air and holds it, until heat drives it back out.

dew point
The temperature at which the water in a body of air starts coming out as liquid. The lower it is, the drier the air, whatever its temperature or relative humidity.

dihedral angle
The angle between two neighbouring flat faces where they meet along a seam. It is shallow on a dome, which is why the seam leaves a channel.

equilateral
A triangle with three equal sides. Ten of the forty faces are equilateral; the other thirty are the star triangles.

face
One flat triangle of the dome's surface, bounded by its own three wedges and closed by one panel. The stem cell has forty.

five-eighths cut
A three-frequency sphere sawn off at the ring of points nearest five-eighths of the way down. Its base zig-zags between two heights.

framed panel
A triangle of the dome built as its own small frame of sawn sticks on a jig, then screwed to its neighbours.

galvanic corrosion
What happens when two different metals touch in water: they make a battery, and the less noble one dissolves.

geodesic dome
A dome made of flat triangles arranged on a sphere, so that every piece of the frame is short, straight and carries its load along its length.

hemisphere
Exactly half a sphere. The stem cell is a 2V hemisphere, twice as wide as it is tall.

hub
A metal connector where struts meet in most dome kits. This method has none: the wedges meet each other directly.

icosahedron
The twenty-sided solid a geodesic dome is built from. Its twelve corners become the only places where five members meet.

junction
A point in the frame where members meet: five-way at a star's centre, six-way elsewhere, four-way on the base ring.

linseed oil
Oil pressed from flax seed that sets by taking up oxygen rather than by drying. Rags soaked in it can catch fire on their own.

member
One straight structural piece of the frame. Here every member is a split wedge.

moisture content
The weight of water in wood as a share of the weight of the dry wood. Below nineteen per cent, wood counts as dry and rot cannot get going.

molecular sieve
A zeolite desiccant whose crystal lattice has holes the size of a water molecule; the 13X grade holds about a fifth of its weight in water and gives it back at around two hundred degrees.

pad
The serviced platform a dome stands on, built once by the host with one service port at its centre. It stays when a dome leaves.

panel
The flat triangle that closes a face, fixed to the outside face of its three wedges. Swapping panels changes what the building is.

Peltier plate
A thermoelectric heat pump: current through it moves heat from one face to the other, and reversing the current reverses the direction.

pentagon
A five-sided figure. Around each five-way junction the star's outer edges form one.

pinwheel
The way a face's three wedges meet at its corners: each end butts against the side of the next, turning one way round the point, so nothing is cut to a point.

radial deck
A floor framed like a wheel: a joist from the centre to every corner of the dome's base, with a rim along its edges.

radial split
A split along a radius of the log, from the heart of the log to the bark, like cutting a pie. Eight of them make eight wedges.

rip cut
A chainsaw cut along the length of the log. Each log section becomes eight wedges by rip cuts through its heart, along the radius.

seal cap
The gasketed lid over the apex opening, held by over-centre catches. Services leave the building under it.

seam
The line where two neighbouring faces meet. Their two wedges lie back to back along it.

seam channel
The V-shaped groove two back-to-back wedges leave along a seam. It runs to every junction and carries cables, air and water.

service port
The single opening in the pad's deck through which power, water and drain come up to the utility column.

shell
The weather surface over the frame and panels. In this book the outermost layer, the cap, is the only watertight one.

stem cell
The standard dome in this book: bare frame, panels, shell, utility column and seal cap, before anything decides what it will be.

trammel
A stick with a point at each end, swung like a compass to draw an arc of a set radius.

utility column
The upright chase at the centre of the dome that carries power, water and drain from the pad's port up to the apex.

utility panel
A small cabinet outside the dome's footprint for noisy, hot or weather-facing equipment, fed from under the seal cap.

wedge
A member split from a round log like a slice of pie: triangular in section, its round bark face outward and its point -- the heart of the log -- toward the centre of the dome.

wedge block
A strip ripped to a bevel on both faces that fills the V between two square-edged panels.

yakisugi
Charring the surface of wood with a flame, then brushing and oiling it. The char leaves nothing for fungus or insects to eat.

zome
A dome-like building swept from a spiral rather than subdivided from a sphere.

About the Author

Donovan Zeanah

I joined the U.S. Navy as an undesignated sailor and went on to serve six years as a Mass Communication Specialist, three of them living in Japan. Along the way I learned to program computers and was certified as an avionics bench technician, troubleshooting and repairing electronics. I am now in school on the G.I. Bill, working toward my A&P certificate — and, beyond it, a future as an aerospace engineer.

To me, a dome is the physical embodiment of the spirit of coding a home. It is modular and reusable. Its interfaces are decoupled, so one part can change without disturbing the rest. It is flexible, hyper-personalized, and built as a framework meant to be extended rather than a finished product. DomeSim, the open-source software this book was computed with, is that idea written as code; this book is the same idea in timber.

This is my passion, and my aim is to change the housing market for the better.

The whole project — the solver, the films and the source of this book — is open at github.com/dkzeanah/DomeSimulator, and every number in these pages can be traced back to the line of code that produced it.

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